Engineering Chemistry · Unit_03 · Energy storage devices, hydrogen energy, and sensors

DRAFT — UE26CY151A · Unit_03 — Energy storage devices, hydrogen energy, and sensors · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

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· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_03 · Energy storage devices, hydrogen energy, and sensors

1 UE26CY151A · Unit_03: Energy storage devices, hydrogen energy, and sensors

This unit master organizes source material for Engineering Chemistry, Unit_03, officially titled Energy storage devices, hydrogen energy, and sensors. It includes 10 video transcripts and 12 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26CY151A, Unit_03 (Energy storage devices, hydrogen energy, and sensors). Bucket folder present: yes. Videos: 10; slide decks: 12.

EV0000001

2 Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I)

This lesson is indexed from the source video titled Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000002 · EV0000003

Graphs and diagrams

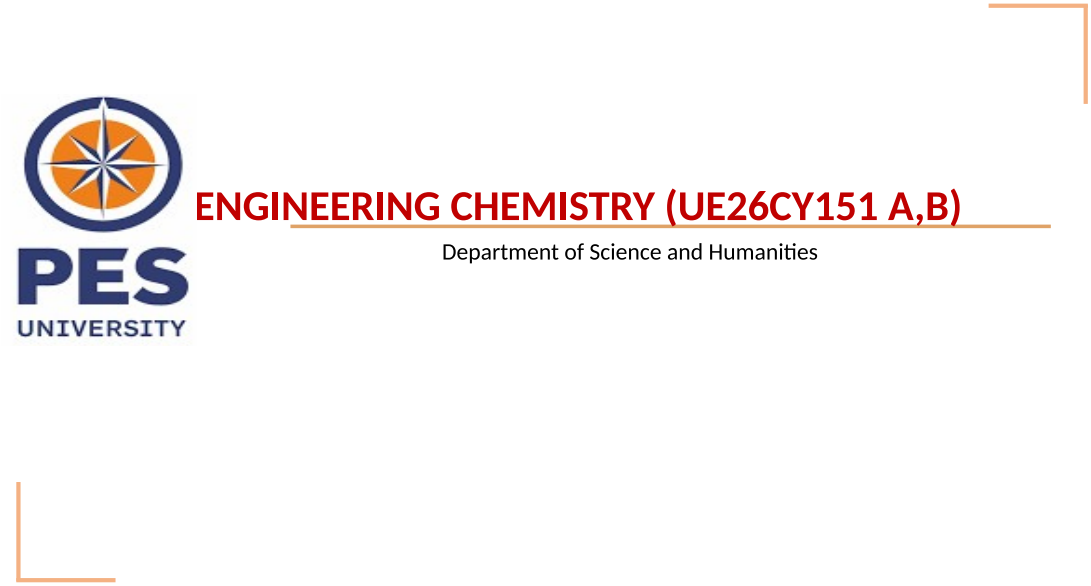

Figure G000001. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 1.

EV0000187

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000002. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 2.

EV0000190

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

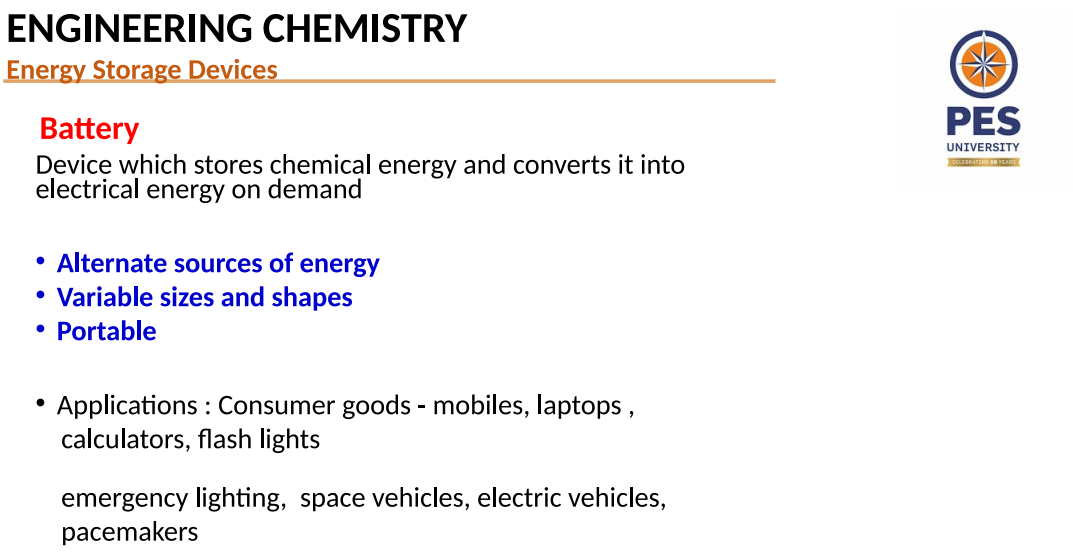

Figure G000003. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 3.

EV0000193

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

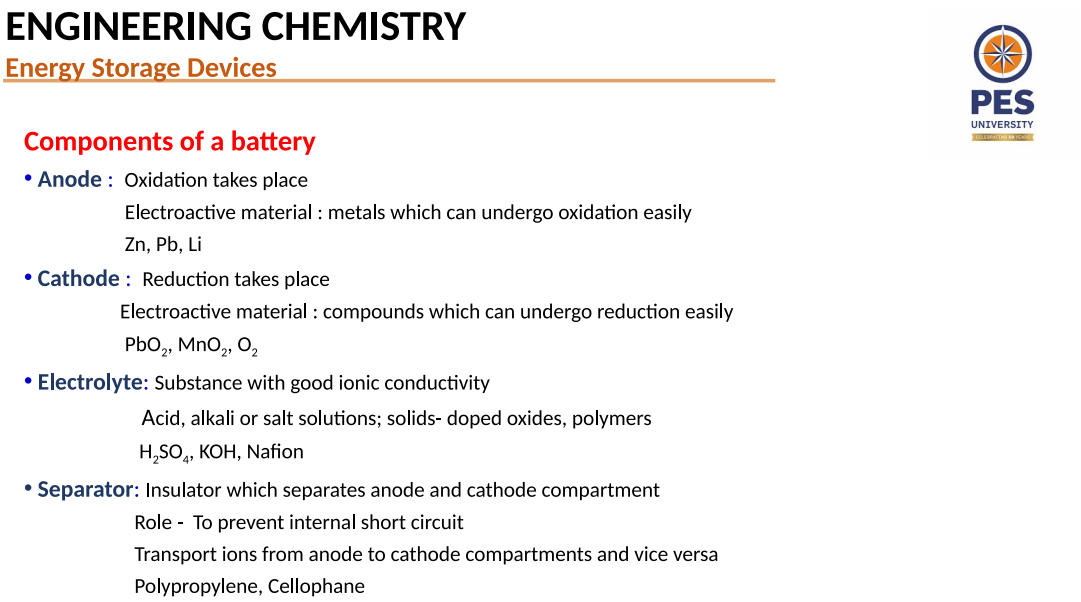

Figure G000004. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 4.

EV0000196

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000005. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 5.

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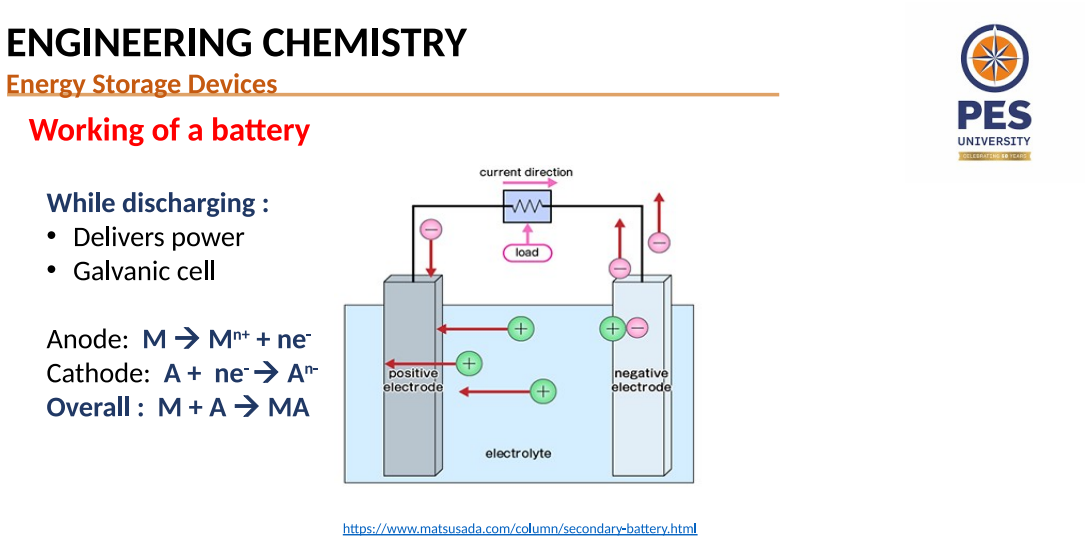

Figure G000006. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 6.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

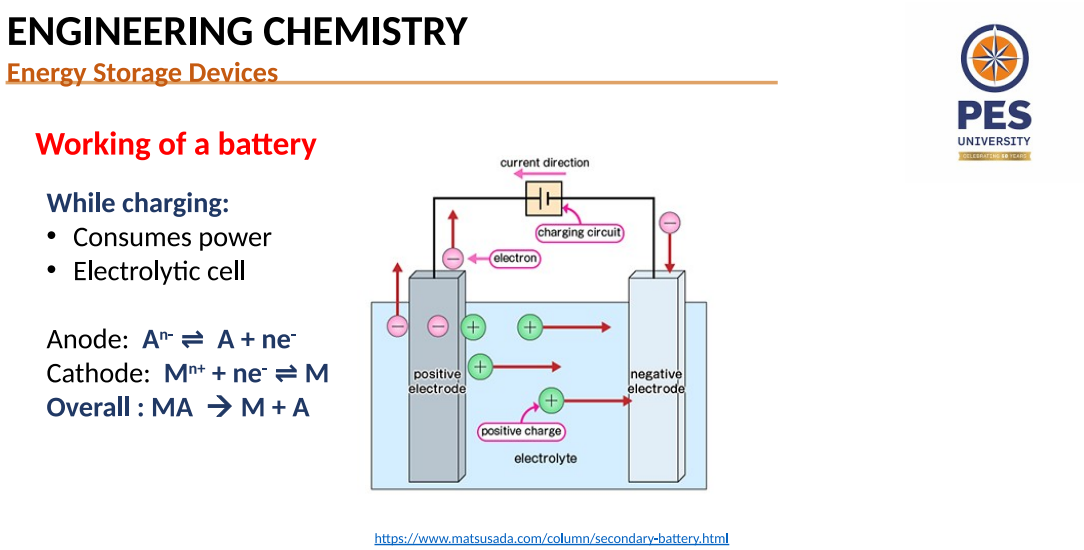

Figure G000007. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 7.

EV0000205

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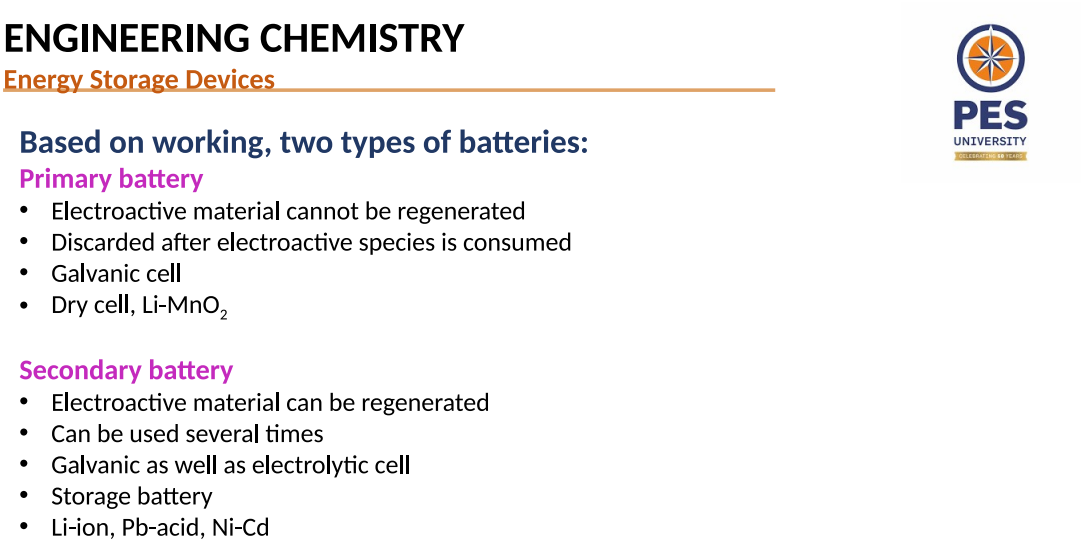

Figure G000008. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 8.

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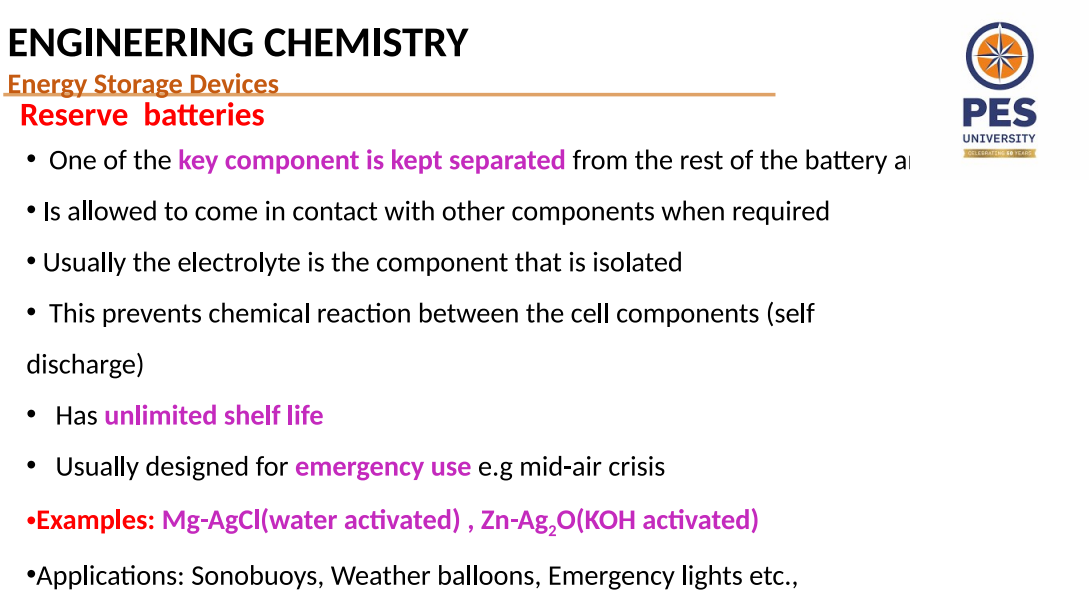

Figure G000009. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 9.

EV0000211

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000010. Batteries Components and working of a battery classification of batteries primary secondary and reserve batter · source slide page 10.

EV0000214

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000186 · EV0000187 · EV0000188
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Class 27 content: Batteries: Components, working & classification EV0000189 · EV0000190 · EV0000191
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Battery Device which stores chemical energy and converts it into electrical energy on demand Alternate sources of energy Variable sizes and shapes Portable Applications : Consumer goods - mobiles, laptops , calculators, flash lights emergency lighting, space vehicles, electric vehicles, pacemakers EV0000192 · EV0000193 · EV0000194
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Components of a battery Anode : Oxidation takes place Electroactive material : metals which can undergo oxidation easily Zn, Pb, Li Cathode : Reduction takes place Electroactive material : compounds which can undergo reduction easily PbO2, MnO2, O2 Electrolyte: Substance with good ionic conductivity Acid, alkali or salt solutions; solids- doped oxides, polymers H2SO4, KOH, Nafion Separator: Insulator which separates anode and cathode compartment Role - To prevent internal short circuit Transport ions from anode to cathode compartments and vice versa Polypropylene, Cellophane EV0000195 · EV0000196 · EV0000197
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Schematic diagram of a battery EV0000198 · EV0000199 · EV0000200

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working of a battery [Image text, OCR] current direction load positive negative electrode electrode electrolyte (b)in case of discharge While discharging : Delivers power Galvanic cell Anode: M ☒ Mn+ + ne- Cathode: A + ne- ☒ An- Overall : M + A ☒ MA https://www.matsusada.com/column/secondary-battery.html EV0000201 · EV0000202 · EV0000203
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working of a battery [Image text, OCR] current direction charging circuit electron positive negative electrode electrode positivecharge electrolyte (a)in case of charge While charging: Consumes power Electrolytic cell Anode: An- ⇌ A + ne- Cathode: Mn+ + ne- ⇌ M Overall : MA ☒ M + A https://www.matsusada.com/column/secondary-battery.html EV0000204 · EV0000205 · EV0000206
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Based on working, two types of batteries: Primary battery Electroactive material cannot be regenerated Discarded after electroactive species is consumed Galvanic cell Dry cell, Li-MnO2 Secondary battery Electroactive material can be regenerated Can be used several times Galvanic as well as electrolytic cell Storage battery Li-ion, Pb-acid, Ni-Cd EV0000207 · EV0000208 · EV0000209
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Reserve batteries One of the key component is kept separated from the rest of the battery and Is allowed to come in contact with other components when required Usually the electrolyte is the component that is isolated This prevents chemical reaction between the cell components (self discharge) Has unlimited shelf life Usually designed for emergency use e.g mid-air crisis Examples: Mg-AgCl(water activated) , Zn-Ag2O(KOH activated) Applications: Sonobuoys, Weather balloons, Emergency lights etc., EV0000210 · EV0000211 · EV0000212
10. Source slide page 10 (extracted text): THANK YOU Department of Science and Humanities EV0000213 · EV0000214 · EV0000215

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.388] Hello everyone, welcome back to this course on engineering chemistry. Today we'll start with unit 3, so that is module 5, and we are going to learn about energy storage devices. [00:00:14.308] So in this module we will talk about batteries, components of working of a battery and battery characteristics. Then we learn about modern batteries like zinc air battery, lithium batteries, and specifically lithium ion battery. Then we study about reserve batteries and then fuel cells. [00:00:34.180] principle working and efficiency of fuel cells. We also look at three different types of fuel cells. Hydrogen oxygen alkaline fuel cells carbon monoxide oxygen solid oxide fuel cell and methanol oxygen polynomial membrane fuel cell. Then we talk about oxygen sensors [00:00:53.348] And then supercapacitors and finally Ragoni plot. So today's class we are going to study about batteries, components of a battery, working of a battery, and then we will look at some battery characteristics. So let us define battery. [00:01:12.516] Battery is a device which stores chemical energy and converts it into electrical energy on demand. So basically we can say that the battery behaves like a galvanic cell. So if you remember when we talked about galvanic cells. [00:01:32.708] We said that they are devices which convert chemical energy stored in the reaction into electrical energy. So we can say that batteries are basically sources of energy and they generate power. [00:01:51.972] Now with the energy demand going up every day in the world and the non-renewable sources depleting fast, we have to

look for alternate sources of energy and batteries happen to be very good alternate sources of energy. [00:02:09.732] Another property of battery is that they come in variables, shapes, and sizes. You can have batteries which are very tiny like the ones which are used in hearing aids, or you can have batteries which are very huge like [00:02:24.132] They could weigh up to 200 tons and they are used in submarines. And also the shapes of the batteries can be varied depending on the application. [00:02:34.980] And the third most important point about batteries is that they are portable. You can carry them into space, you can use them on the land, and you can also take them underwater. So portability wherever we go, we can carry these batteries with us. So these are [00:02:51.536] Three important properties of batteries which make them very very useful. [00:02:58.628] We all know that there are a lot of applications of for batteries like in consumer goods. We have mobiles, laptops, calculators, flashlights on one hand. And on the other hand, we have very specific applications like in emergency lighting. [00:03:15.088] space vehicles, electric vehicles, and pacemakers. Let us look at the various components of a battery. [00:03:27.588] We know that a battery is a galvanic cell, therefore we will require the two electrodes, anode and cathode. So at the anode, now we know oxidation takes place. [00:03:43.620] What material would I choose for the anode? It has to be something which undergoes oxidation very easily. Now we find that metals are such [00:03:56.836] materials which can undergo oxidation easily. Therefore, we find most of the batteries they use metals as the electroactive material at anode. And most commonly used materials for batteries are zinc, lead and lithium. [00:04:16.964] Now metals we see are important materials which are used as anodic materials and all of you are familiar that lithium is a very very popular anodic material. Most of the batteries now are being made with lithium. [00:04:34.948] Okay. And we also need a cathode where reduction takes place. We know that oxidation and reduction should happen simultaneously. So at the cathode we have to use material which can undergo reduction easily. [00:04:50.692] And please remember this that a metal can never undergo reduction. It can undergo oxidation.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.264] But it can never undergo reduction for the simple reason that it's already in the lowest reduced form. So metal cannot be used as the cathode. So what is the alternative? Metal in the form of iron. That means in the form of compounds. So [00:05:18.144] Metal itself is used as the anode, but at the cathode we use compounds of metals which can undergo reduction easily. [00:05:28.676] And most of the times we find lead oxide is a good example, like in your lead acid battery. Then MnO2 is also an oxide which is used. Not only oxides, we can use chlorides, nitrates, any of the compounds, but most commonly used are oxides. [00:05:45.936] And we can even use oxygen as the cathode active material because we know that oxy can undergo reduction easily. [00:05:56.612] Now coming to the third component that is electrolyte. What would serve as an electrolyte? Substances which have very good ionic conductivity. Ionic conductivity means we need ions which will move very easily from one electrode to the other. [00:06:15.076] So what is usually used as electrolyte? We can have acids, we can have alkali, we can have salt solutions, we can also have molten salts. [00:06:25.988] And also because molten salt can produce ions which are free to move. And we can also have solids which are doped oxides. [00:06:37.412] And then we can have polymers also which can be used as electrolytes. So for example, we can have H2SO4, we can use H2SO4 or KOH or Nafion. NAFion is a polymer membrane which is used in some batteries. [00:06:56.356] Then we come to the last component that is a separator. What is a separator? It's usually an insulator which is going to separate the anode

and the cathode compartment. [00:07:07.876] So what is the role of a separator? It is to prevent the anode and cathode coming in contact with each other because if that happens, there's going to be internal short circuit. So this insulator is present. [00:07:22.448] To separate the anode and the cathode. But at the same time, it should be able to transport the ions because apart from the oxidation and reduction which is happening at the anode and cathode, respectively, this transport of ions in the [00:07:38.720] uh through the electrolyte is very very important for the battery to function. Therefore the electro the separator [00:07:46.180] should be able to transport the ions from anode to cathode and cathode to anode. So what is used usually is polypropylene or that could be cellophin. There are many types of separators which are employed based on the design of the battery. [00:08:02.852] Now, let us draw a schematic diagram of a batter using all these components. Now we said that at the anode say we have the [00:08:15.268] anode active material. But this anode active material may be in the form of a powder or a paste and therefore we need what we call as a current collector. [00:08:30.404] Always there will be a current collector. The current collector itself is not involved in the reaction. It is the it is the electroactive species which is involved in the reaction. And then on the cathode health also we have a similar [00:08:50.180] Electroactive species, but this is the cathode electroactive species, and this will be the current collector. So the current collectors are the one which are providing the electrical contact, but what is undergoing reaction is the cathode material. [00:09:07.332] active material and here we have the anode active material. Now this whole anode and cathode they are placed inside a housing [00:09:23.012] So we have the anode and cathode inside a housing. [00:09:28.484] Then what do we need next? We use the anode material, we use the cathode material, and then we need the electrolyte. So the electrolyte is going to be present inside here. So this is the electrolyte which is present there. [00:09:44.164] So we have used the electrolyte also and now we need the separator. So what is the separator doing? It is present between the anode and cathode and it is going to make sure the anode and cathode do not come in contact. So we can say this is the separator.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.132] Okay and then we also have terminals here, which you are aware because we all use a lot of batteries. So these are the terminals. So this is how a battery looks like all the components which we studied just now. [00:10:21.216] we have placed inside the battery. So overall the schematic diagram depending on the design of course we have uh different positions of this anode, cathode, electrolyte separator, etc, but overall this is the scheme. So we can say that this is the terminal. [00:10:38.404] So inner battery, therefore, so both are terminals here. So we have this electroactive species at the anode which is undergoing oxidation, and here we have the electroactive species of the cathode which will undergo reduction. And this whole [00:10:58.276] thing is now kept inside a housing it is called it's a container into which this anode and cathode reactive materials along with the current collectors are placed and this is filled with electrolyte and there's a separator separating the anode and the cathode. [00:11:14.656] So this is what your battery looks like. Now let us look at working of the battery. [00:11:23.044] Now the battery can discharge because we know it is going it is a source of energy, it is source of power. So the battery can undergo discharge. What do you mean by discharge? It is delivering power. [00:11:37.248] So we are getting power from the battery. When the battery discharges, we it is delivering power. So therefore it acts like a galvanic cell. Again, galvanic cell uses the chemical energy and converts it into electrical energy. Therefore, when the battery is discharging and giving us power, the battery [00:11:54.992] Bacteria is behaving like a

galvanic cell. Now what happens at the anode and cathode? Let us see. Now you can see the representation here exactly what we do just now. So we have at the anode we have the metal which is going to undergo oxidation. [00:12:11.872] So at the anode we have elect the metal undergoing oxidation giving out mn plus plus n electrons. So we can see that this is the anode here. So negative electrode, we have the metal which is undergoing oxidation and n electrons are being produced here. [00:12:31.712] They move through the external wire and at the cathode we have some species say A, which is going to take up these n electrons, and we get A N minus. So overall reaction what happens is M plus A is giving you [00:12:48.928] MA. So basically what's happening here? We have the elect the electrons being generated at the anode. So this becomes the negative electrode and then [00:12:59.536] They move through the external wire thereby generating power. The power there's a flow of current and whenever electrons flow, then we say it is it is going to it is giving you current. So that is how we have a load here, then that load uh [00:13:16.656] The power will be provided to that load. And then here the at the cathode the electrons are being taken in on absorbed and that gives rise to an minus. Now these ions Mn plus and An minus which are generated at the anode and the cathode. [00:13:33.792] They are going to move in the opposite direction. So the electrolyte is here, and these are there are current collectors. So the electrons are going to move through the current collectors. Again, you have to remember electrons never [00:13:48.816] come through the solution. They go through the current collectors, pick up these electrons and through the external wire we see that the electron movement will happen like this. So this is what happens when the battery is discharging. Now let us look at [00:14:05.828] a battery which can charge also. Now this is not possible for all types of batteries. Only certain batteries we can charge them uh back to uh the original uh condition. So here [00:14:23.264] While charging the battery is consuming power. The battery consumes power and now it is going to behave like an electrolytic cell. So what is an electrolytic cell? When we provide power and the [00:14:38.672] Reaction takes place, a non-spontaneous reaction takes place. So this is going to act as electrolytic cell because we are providing power. You can see there's a battery which is connected here, and now what happens at the anode. [00:14:52.880] oxidation has to take place. Now what happened is your MA because of your you're giving this power over here the MA now undergoes electrolysis and it breaks into Mn plus and An minus.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.424] Now we know that anions will move towards the anode and cations will move towards the cathode. So let us see what is happening here. The anion AN minus is moving [00:15:21.808] towards the anode and at the anode it forms a and n electrons are given out. So you can see at the anode this is what is happening. And at the cathode what happens? The cations move towards the cathode and here [00:15:38.720] you have they take up the electrons. The cations will take up the electrons. [00:15:44.080] And why is this negatively charged? Because you have connected it to the negative terminal of the battery, and this is positive because you connected it to the positive terminal of the battery. So cations will move towards the negative electrode that is the uh cathode here. It is going to take up electrons. [00:16:01.024] And then it is going to form M. So what we see here is during the charging process A and M have been regenerated. So both the electroactive species now we got back. [00:16:18.080] Because of the charging process. So what is the overall process down here? MA is again. [00:16:25.828] become M plus A. So that is what is happening. How did what happened what happened to this MA actually, how did we get this MA? Because MA underwent electrolysis

produced [00:16:39.136] A n minus and m n plus which move towards the respective electrodes, and that is how we got back M and A. And therefore, M A is giving you M plus A is the overall reaction. So what is important here is that. [00:16:55.300] The battery now has because of the charging, we have been able to regenerate the electroactive species, so it is ready for the next cycle where it can discharge and deliver power. So when it behaves as a [00:17:13.764] When it is discharging, it is behaving like a galvanic cell. When it is charging, it behaves like a electrolytic cell. [00:17:23.396] Now, based on the working which we learned just now, there can be two types of batteries. One is this primary battery. [00:17:32.080] So primary battery here the electroactive material cannot be regenerated. So it's a one-time use battery and it will work only as long as the electroactive species are present. Once the electroactive species are consumed, the battery has to be discarded. [00:17:49.808] It. So here the battery can act only as a galvanic cell, that means we cannot charge the battery to get back the electroactive species. So it can act only as the galvanic cell. And examples for primary cell would be dry cell and lithium MNO2. [00:18:07.908] Now we have other types of batteries which can be recharged and they are called secondary batteries. So in this the electroactive material can be regenerated. So by the charging process, we can regenerate the electroactive species and they can be used several times. You can [00:18:26.160] Discharge the battery, use the power, then again charge it back so that you get the electroactive species. So this cycle can be repeated several times. So therefore we can say it behaves as a galvanic cell as well as electrolytic cell. When it is discharging, it is a galvanic cell, when it is charging, it is an electro. [00:18:43.824] Electrolytic cell, and they are also called as storage batteries. Secondary batteries are also called storage batteries because we are regenerating the electroactive species and storing them inside, and then whenever [00:18:57.648] There is demand the battery is discharging and giving us power. Example would be the most famous lithium ion battery, then we have the lead acid battery, and then we have those nickel cadmium batteries which were called as the nicad batteries. [00:19:15.428] Now let us come to something very very important that is about battery characteristics. Now we know that there are a lot of applications for the battery, and each application has a specific demand. [00:19:30.980] So we need to specify what are the battery characteristics which are required for a particular application to give a specific performance. So there are certain characteristics which we look at. [00:19:47.440] Suppose we want to use a battery for a particular application, there are certain characteristics we will look for and that we are going to study under the heading of battery characteristics. Now the first and foremost let us study about the voltage of a

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:03.088] battery. Which is very very important. When we want to take a battery or select a battery for a particular application, we have to look for what is the voltage we [00:20:13.216] require for that particular application. So what is voltage now? We know that voltage is basically the EMF generated from the cell. Now EMF of the cell is going to depend on free energy. Free energy of the [00:20:28.260] Free energy. So free energy is what we know delta minus delta G is equal to NFE. So the voltage is going to depend on the free energy, and the free energy itself depends on the reaction which is used. For each reaction, there's a different free energy delta G. So delta G is going to depend on the reaction. [00:20:46.032] Now what reaction is going to happen inside the battery obviously is going to depend on what are the electrodes we have chosen to make that particular battery. [00:20:57.316] Now for voltage of a battery or EMF of a cell, we have a very simple equation which we studied,

and that is the Nernst equation. Now Nernst equation we know that E cell is equal to E naught cell minus 2.303 R T over NF log of Q. [00:21:16.292] So, if we look at this Nernst equation, we can understand what are the factors on which the voltage of a battery depends. The first and foremost we see is E cell, so E naught cell. [00:21:30.244] Now, E na cell depends on what? The difference between the cathode electrode potential and the anode electrode potential. Remember we are always talking of standard electrode reduction potentials. So standard reduction potential of the cathode and the [00:21:48.640] And what the difference between the two will decide what is E0 cell of the battery. And higher the difference, more is going to be E0 cell, and therefore in return E cell will be a very high value. [00:22:05.252] So if we have a battery, if we want to design a battery which should a very good voltage, we have to make sure E naught cell is very high. How will we make a make sure E naught cell is high? By choosing the right kind of electrodes. [00:22:22.372] We can have a high E naught cell value. Now coming to the next that is temperature. Now temperature T with increase in temperature. [00:22:36.176] You can see from here that the voltage of the cell is going to decrease. So temperature is another factor, and the third factor is the reaction quotient Q. [00:22:46.784] Now log q you can see here now as the value of q. What is q basically reaction quotient means concentration of the products divided by the concentration of the reaction. Remember there's a reaction which is happening inside the battery, so the concentration concentration of the products are continuously. [00:23:04.496] Increasing and that of the reactant is decreasing when the battery when the forward reaction is happening and the battery is generating power. So this value Q keeps changing continuously, but the voltage [00:23:19.344] will not change to that extent because then it is in a log term. So because log q is involved in this term. So the second term does not affect the voltage of the battery to a great extent. It's mostly decided the voltage of the battery is mostly decided [00:23:37.008] By E0 cell. Nevertheless, as the battery reaction continues, we will find that there is a change in the E cell value, but it's only marginal. That is about voltage. Now let us come to the second. [00:23:54.468] Battery characteristic again very important and that is current. So what is current? It's a measure of the rate at which the battery is discharging. Now here when we say rate, remember we are talking of the [00:24:11.684] flow of something or the rate at which something is happening. And what is that something here? The rate at which the battery is discharging means what? The rate at which the current is being generated. [00:24:24.548] The rate at which the current is flowing. So here we are talking of how fast the electrons are moving from one electrode to the other. [00:24:37.380] So, current will is a measure of the rate at which the battery is discharging. So, basically the rate at which the electrons are being generated or flowing through the external wire to the load. Now the unit is [00:24:55.268] usually ampere or depending on the battery or the size of the battery or the current which is generated it can be in milliampere as well. Now what does this current depend on? Now we know that V is equal to IR

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:13.572] So if I want the voltage of a battery is a fixed quantity. So [00:25:19.136] If I want to get a good current from that battery, we cannot change the voltage because voltage depends on the choice of electrodes. So we have already chosen the electrodes, the voltage is fixed value. Now I want to get the maximum current from this battery. So what should I do? [00:25:36.848] I should keep the resistance to minimum. Now we know that resistance can be kept minimum by having an electrolyte which has very good conductivity, so that the ions will move very [00:25:51.440] Fast and another way of keeping the

resistance low is keeping the inter electrode distance less, meaning that the anode and cathode have to be placed. [00:26:01.696] close to each other so that the inter electrode distance is less and also the electrolyte should have good conductivity so that the ions will move easily from one electrode to the other. Apart from that [00:26:15.456] One more important thing which should happen is that so all this we are doing basically for a rapid electron transfer reaction because if the electrons are generated fast, then the current will be more. If [00:26:30.336] you there's a very slow generation of electrons then obviously the current flow is will not be that high. So and the electrons remember are moving through the external wire ions have to move through the electrolyte and overall that will decide the [00:26:45.536] current which is available from the battery. So one more important factor which plays an important is a role it plays a role in this getting a good current or getting a rapid electron transfer is by having a large amount of electroactive species. [00:27:03.364] You know from kinetic study that more is the concentration of the reactant, more is going to be higher is going to be the rate of the reaction. The same thing applies here. If we have a large quantity of electroactive species, the reactants are more in the [00:27:22.308] quantity and therefore the electron uh it can undergo oxidation or reduction uh faster the rate. [00:27:30.496] will be fast here and therefore the electrons will be generated fast. So we say rapid electron transfer is going to happen and therefore we can get a good flow of electrons. Therefore the current which is the rate at which the electrons are flowing is [00:27:47.968] Going to be very high. So these are the factors on which current generated from the battery depends on. [00:27:56.772] So, these are two of the characteristics which we discussed today. In the next class, we are going to look at many more interesting battery characteristics. Thank you.

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Full transcript: 4017 normalized words, 112 timestamped segments, 1691.050667 seconds duration. Low-confidence segments flagged: 0.

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3 Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II))

This lesson is indexed from the source video titled Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000026 · EV0000027

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.292] Hello, everyone. Welcome back to this course on engineering chemistry. We continue with module five on energy storage devices. We are talking about batteries. In the previous class, [00:00:14.244] We discussed two

battery characteristics that is voltage and current. So in today's class we are going to continue learning about battery characteristics. [00:00:26.244] And some of them which we are going to study today are capacity, energy storage density, cycle life, energy efficiency, shelf life, tolerance to service conditions, power density and energy density. [00:00:41.700] Let us first learn about capacity of a battery. What is the definition? The charge or amount of electricity that can be obtained from the battery is known as capacity of the battery, and the unit in which it is expressed is ampere hours. [00:01:01.732] So, the amount of charge that is contained inside the battery is called as the capacity of the battery. [00:01:10.660] Now, one way of determining the capacity of the battery is given by the Faraday relation, which is C is equal to Wnf. [00:01:22.596] Now, C is the capacity in this equation. W is the mass of the electroactive material. M is molar mass of the electroactive material. [00:01:35.748] And n is the number of electrons being exchanged in the reaction. So suppose at the anode we have lead. So lead is becoming PV. [00:01:47.616] 2 plus oxidation should happen at dianode. So lead is becoming PB2 plus plus 2 electron. So what would be n in this case? N is going to be 2, and F is the Faraday that is 96500 coulombs per mole. So this expression. [00:02:07.588] We can use to determine the capacity of a battery if we know the amount of the electroactive material present at the electrode. So we can see very clearly from this expression that capacity depends on size of the battery. [00:02:26.820] How can we say that? Because W is figuring in this expression. And what is W? Just now I told you W is the amount of the electroactive material. So more the amount of the electroactive material. [00:02:43.044] Bigger is going to be the battery. Therefore, we can say that the size of the battery determines the capacity of the battery. Now, another [00:02:57.156] condition on which the capacity of the battery depends is the discharge of the battery. What does it mean? How much current is being drawn from the battery. So there's another expression which gives us the capacity [00:03:15.556] which shows very clearly that capacity also depends on the discharge conditions. The capacity is also given by a current into time. It is not very different from what we have studied before. [00:03:29.636] Because capacity is a charge contained in the battery. And you know current multiplied by time is also charge. So this expression is justified that capacity is also given by current into time. [00:03:45.764] So, if we are drawing a certain amount of current, then the capacity will determine how long the battery is going to last. So, T is the time period for which the battery is going to last when we [00:04:02.916] did use a certain current I from the battery. So C is equal to I into T. So now you have to remember there are two expressions for capacity. [00:04:13.600] Wnf by M. So if you know what is the amount of the electroactive material being used, then we can use this expression. If we know how long the battery lasts when we are drawing a certain amount of current, then we can use this expression to determine capacity of the battery. [00:04:32.964] Now, if you plot voltage versus time for a battery, suppose I'm drawing a current I from the battery, then what happens to the voltage as we as time [00:04:49.124] passes. So we find that you get a graph like this. So voltage versus time. So what are we measuring here? [00:04:59.040] We are seeing how the voltage varies with time for a particular battery. So if I want to determine the capacity of this battery, initially there's a small drop in the voltage, and then we see that there is a

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.296] almost a flat portion and after that it comes to a pos uh part the voltage is now too small for the battery to [00:05:24.944] Perform the required function. So we say that say at this time this is the E minimum which is

required for a particular uh instrument to operate or a device to operate, we say that the battery was dead after this time. So [00:05:41.264] we can say that this is the E minimum which is required for the battery and beyond this the battery is dead. So this is how all batteries will behave. [00:05:53.008] Initially, there will be a drop in the voltage, and then we see that it gives a steady voltage for a long time until it reaches a point. [00:06:01.376] After which we declare that battery as dead, if it's a rechargeable battery, we need to recharge it. So this is the cutoff voltage because beyond this, every device requires a certain voltage. So beyond this, the [00:06:14.880] Device is not going to function using this battery, so either it needs to be replaced or it needs to be recharged. So we can say that the capacity of the battery, because what we are looking at when we are drawing a certain amount of current, how long the battery will last. So we can say that this flag. [00:06:32.528] portion of this curve can give us an indication as to what is the capacity of the battery. So if this [00:06:42.724] flat portion is very long, we say that the battery has a good capacity because at this particular current it is giving you uh it it lasts for a longer time. If this portion is the flat portion is less then we say that the capacity is not so good. [00:07:03.812] Now, let us come to the next battery characteristic that is energy storage density. Energy storage density is defined as the amount of charge or electricity per unit weight which the battery can hold. It is telling the amount of charge. [00:07:23.140] Which the battery can hold, but it is now it is become density. We are calling it density, so that is per unit weight which the battery can hold. Now when we say weight, this weight includes weight of all the components of the battery, like [00:07:38.992] the electrodes, the electrolyte, the case which is used, the current collectors, the terminals, etc. So all the components of the battery have weight of those have to be taken into account if we are trying to find out what is the electricity storage density of a battery. [00:07:57.636] So that means if we want to get a good electricity storage density, we have to use the right kind of material for all the components. So if we get a very high electricity storage density, [00:08:13.920] We say that the designing of the battery is very good. Now what we have to understand when we say capacity, it is the weight of W there stands for weight of the [00:08:26.832] Electroactive material that is taking part in the reaction at the cathode or the anode. But here when we say weight, we refer to weight of the battery on the whole, meaning all the components of the battery have to be taken into account. So now we can see very clearly. [00:08:44.676] Why lithium is such an important anodic material? Why is it such a popular anodic material? Because we know it's a lightweight metal. Just seven grams of lithium can give you one faraday of charge. [00:09:00.592] Whereas if we use lead, we know that lead is also a very popular anonic material. [00:09:06.704] uh like uh lead acid batteries etc which are which are used quite uh a lot so but here we see that we require 104 grams of lead to produce the same amount of charge that is one farity so therefore lithium has become [00:09:23.364] a very popular aerodic material. [00:09:29.092] Now coming to the next battery characteristic, cycle life. Now some batteries we know can be charged again and again. [00:09:40.260] Such batteries are referred to as secondary batteries. These secondary batteries can be again and again charged, but they also have a lifetime. Now if you look at the working of a secondary battery, we know that [00:09:56.464] The electroactive when it is when when the battery is discharging, the electroactive material is consumed. But when the electroactive material is completely consumed, we can again

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.264] recharge the battery and when we recharge the battery this electroactive material is getting regenerated. That means the

cycle can go on and on because you can charge and discharge, charge and discharge as many times as we want, but we know that this is not true. [00:10:24.304] Every secondary battery also has a life. So that life is referred to as a cycle life. How will we define it? The number of charge-dischar cycles that are possible before the battery fails. [00:10:41.328] How many times can we charge and discharge the uh battery? So we know it's about if the cycle life is 500, what does it mean? That 500 times the battery can be charged and discharged. So it is the number of cycles which the battery. [00:10:58.788] The battery can go through before it fails. So we know very clear since primary batteries are only one-time use batteries, it is this particular characteristic is not applicable to primary batteries, it's only applicable to secondary batteries. Now what is the reason for this failure? [00:11:17.860] Why do secondary batteries have a limited cycle life? Theoretically, you can charge and discharge, charge and discharge, but this does not go on forever. We know that there is a limited life for a particular battery, even though it is a secondary battery. One reason. [00:11:37.988] For this is corrosion of the contact points. We know that we are using as electrolytes in the form of acids, alkali, salt solution, all those are very corrosive materials, and therefore the contains with time could get corroded. So this is [00:11:57.296] Reason why we see that the batteries have a limited life. Now, another important reason which we should know is that shedding of active materials from the plates. Now what does this mean? [00:12:12.960] Let us draw a battery here and see what exactly happens inside the battery. Now we know that there is a current collector here, and what we expect in the battery during the [00:12:28.576] reaction. Suppose the battery is discharging, all this is the electroactive species present around it, and this is the current collector. So this electroactive species is present around the current collector and as time and when the battery is discharging, these I this is [00:12:46.256] Going to come out and electrons will be generated and the electroactive species is getting consumed. Now, this is when the battery is discharging. Now, suppose the battery is charging, you are doing the reverse now, all this material which now got discharged. [00:13:05.408] has to come back and form a nice layer around the around the current collector. [00:13:13.168] Now that is not going to be very easy. Initially, yes, there will be nice layers of this electroactive material, and you will get something like in this form. But as time progresses, you go on using the battery for a long time. We find that this material instead of being [00:13:30.800] on the current collector it will start to shut down. So now what happens? This electroactive material during the [00:13:40.868] charging process has not gone back to the current collector and therefore what will happen the this material which has shut down is not available for the next discharging cycle. So as a result you will find that the battery becomes weaker and [00:13:58.480] weaker and at one point of time it cannot function anymore. So this is one reason why the battery fails shedding of active material from the plates. The the electroactive material is [00:14:13.776] Able to undergo reaction as long as it is around the current collector. It is available for the discharging. [00:14:21.968] But once it is not on the current collector and a shutdown like this, the electroactive material slowly goes on decreasing, and we find that the battery will become weaker and weaker. So this is the second point which we discussed now. So first we talked about corrosion, next [00:14:39.632] Now let us look at the third point, shorting between electrodes due to irregular growth and change crystal growth and changes in morphology. What do we mean by this? [00:14:51.216] Now the same thing let us look here. Now we have this current collector like this, and we have the uh electroactive material around the current collector. So same thing happened that the

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:03.408] When the battery is discharging, we find this electroactive material gets away from the current collector. When you are charging, again as I said in the previous point, this material has to come back to the current collector. Now, many times what happens is instead of [00:15:21.072] Forming nice layers around the current connector, we find that with time these dendrite like growths start. [00:15:31.940] the battery shows tree like growth or they are called dendrite like growth. Now when this happens, now this is happening say at the [00:15:41.796] Anode, the same thing can happen at the cathode also. Now if it happens at the cathode and the anode, these tree like groups, the dendrite formation, there will be morphological changes in the battery, and you know that if the the the anode and cathode come in contact, they will [00:15:59.568] Even there is a separator in between, but they may pierce through the separator and what we find is what will happen when cathode and another come in contact with each other, there is going to be short circuiting and the battery is going to fail. [00:16:14.160] So this is another reason for failure of the battery. The electroactive material did not get back onto the current collector. Instead, we find dendrite like growth. So these are called dendrites. [00:16:31.204] Or you can simply say tree like structures. So when these grow then they grow on the anode as well as cathode, we find that the bat there could be short circuiting, and remember to get good current from the battery, we [00:16:47.440] Have to maintain low inter electrode distance. So the distance is very low. So when these dendrites come in contact with each other, then this is going to result in short circuiting. So these are the three reasons why the bat every secondary battery has a cycle life. [00:17:08.580] Coming to the next battery characteristic, energy efficiency of a battery. Now this is given in the form of percentage. Percentage of energy efficiency would be energy released on discharge by the energy required for charging multiplied by 100. So since we are talking of discharge as well as [00:17:28.528] Charging this is again applicable to secondary batteries. So since we are talking, we are talking again of discharge as well as charging. So this battery characteristic. [00:17:43.040] Or this definition of energy efficiency is applicable to secondary battery. So what is the amount of energy that you're getting on discharging the battery and what is the energy it would require to charge multiplied by 100. [00:17:56.912] So, ideally, this should be 100%, but it is not 100%. It is never 100%. And this efficiency depends on the efficiency of the electrode reactions. If there is any hampering in the electrode reactions, or they are going slow, or there is some kind [00:18:14.608] Problem the anode efficiency, the cathode efficiency is not good, then we find that the energy efficiency also falls down. Coming to the next characteristic that is shelf life. [00:18:28.004] Many times we keep batteries at home, and especially the dry cell, we find that when we want to use the battery, it is not working anymore. So, in that case, [00:18:40.996] That what is it, what is responsible for the battery being dead when it is not at all being when it has not even been used. That characteristic is called shelf life. Shelf life, how will we define shelf life? [00:18:57.920] The maximum time, the maximum time for which you can store the battery, and yet it will give you the specific performance. [00:19:08.836] How long can you keep the battery without using it? And yet it gives you the specific performance which is required. But many times we find that batteries are dead by the time we want to use, and of course, this is we are talking of [00:19:26.336] a primary battery like dry cell, we uh find that [00:19:30.468] They are not useful anymore and they cannot be used anymore and have to be discarded. What could be the reason for this? The reason is there is something called as self-discharge, which happens inside the battery. The moment the components are assembled. [00:19:47.268] the reaction in the battery starts. So therefore, or there could be other secondary reactions also happening which are going to

[00:19:56.576] uh consume the electroactive material and therefore when we leave it for a long time without using it then the battery all the material could have been used up or there could also be corrosion and leaking of the battery. So all these reasons are uh possible for

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:14.592] the shelf for a battery having shelf life. Of course it is very important for a primary battery to have a good shelf life. Why because so that even if it is stored for some time we can still use it as primary batteries are one time use batteries. [00:20:32.336] For secondary batteries also have a shelf life, but the good thing about secondary batteries is they can be recharged. So even if they have a shelf life, you can still recharge and use the batteries, of course. [00:20:46.224] Problems like corrosion and all will of destroy the battery. So even primary or secondary batteries have shelf life, but the characteristic becomes very important for a primary battery more than for a secondary battery. Now coming to the next point, tolerance to service conditions. [00:21:07.172] The battery has to be tolerant to different service conditions. For example, if there is a variation in temperature. Now, cars, sometimes with a lead acid battery, which has to be used in the car has to [00:21:23.524] Be uh available, I mean it should operate at very low temperatures or at the high temperature at which the [00:21:31.824] The motor gets heated up, I mean the motor is going to heat up the machine, and therefore uh it has to tolerate that high temperature. The temperature range has to be very large for, for example, the car batteries, and apart from that, they should also be tolerant to say vibration. [00:21:50.640] or shock because when the when you're moving the car when the car is moving on the road or in any terrain there will be lot of vibration and shocks which the battery will also have to face. So therefore the battery should be designed such a way that it should be tolerant [00:22:08.336] To whatever service conditions you are going to use it for, whatever application it is going to be used for, the service conditions, the battery should be able to tolerate those service conditions. [00:22:23.396] Now coming to again a very important battery characteristic and that is energy density. Now energy density is defined as the amount of energy which is available from the battery divided by mass of the battery, or you can say weight of the battery. Again, when we [00:22:41.168] We say density, we should remember the weight includes weight of all the components of the battery. So what is the energy available to us by [00:22:51.556] Divided by weight of the battery. So energy density is given by current into voltage multiplied by time divided by weight of the battery. [00:23:04.176] So mass of the battery here is W, that is mass of the battery or weight of the battery, where all the components taken together. And the what is the current the battery is giving us multiplied by the uh voltage at which it is operating, multiplied by the time for which the battery will. [00:23:21.860] work. So energy density is I into E cell into the time divided by W. So if from the expression you can make out that the energy density unit is going to be what hours per kg [00:23:40.196] Watt hours per kg, or we can say kilowatt hours per kg. Depending on what is the application, the unit can be varied, but nevertheless, it is power into time divided by weight of the [00:23:57.632] battery. So that is energy density. Coming to the last battery characteristic that is power density. Pow density is defined as the power available from the battery divided by mass of the battery or weight of the battery. Again since we are talking of density [00:24:15.472] It should be the weight of all the components of the battery or mass of all the components of the battery. So what would be the uh formula now? Power is always current into voltage. So what is the current being given now? Uh being you have used there multiplied by voltage of [00:24:33.136] The battery divided by weight of the battery, mass of the battery. So I

into E cell divided by W. That would be power density of the battery and the unit will be watts per kg. So what per kg is going to be the unit. [00:24:50.704] I said for energy density, it can also be kilowatt per kg, watts per gram, depending on the application and depending on the size of the battery, and depending on what power it is generating. We can choose the correct unit, but the unit is going to be that of power divided by

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:08.272] weight of the battery. So watts per kg. Now only thing you have to remember what is the difference between power density and energy density. Both look very similar. [00:25:21.764] But power density, as we know, energy is the work done. Energy is always work done, and power is at what rate the energy is being produced. So therefore. [00:25:36.528] When would you require a high power density when you when you uh say for example when you're when the car is going uphill. [00:25:45.296] When the car is going uphill, you need a high power density. And when would you require a good energy density? How would you know whether the energy density is good? Well it lasts for a long time before it requires charging. The battery. [00:26:02.800] it lasts for a long time before it requires charging. So that will give the energy density. Power density will be uh when it the when the say it is in a car and how long so energy density is how long the car would go before it requires charging and power density [00:26:20.432] How uh steep a hill can it climb? That would be determined by its power density. So the both are different characteristics. Nevertheless, we have to know. [00:26:32.464] See what is the application. Some applications could require a very high power density, uh but some applications could require a low uh power density. And [00:26:44.516] Also, energy density or power density, the requirement could be that we need a certain amount of energy density for a long period of time or [00:26:55.216] Uh a high energy density for a very short period of time. For example, in the flash of a camera, we need very high energy density, but for a very short period of time. Whereas if you're running the car, if you're running an electric vehicle, then you would require a high energy. [00:27:12.944] Density, a constant energy density for a long time. So that's why the requirement for energy density as well as power density could be for a short time very high value or a certain value for a long time. So that is the difference between energy density, power density. [00:27:30.512] density and all these battery characteristics are very important and it's not necessary that all these characteristics uh should be at the uh high value uh for every for an application. Sometimes we require [00:27:46.992] Um low power density is fine. For example, in a pacemaker, we require we don't require very high power, we require just few micro watts. But then there are other requirements like it should be able to tolerate the body temperature, then it should have it should be very uh accurate voltage. [00:28:06.560] And it should be very, very reliable. It should last for a long time. So every application has a certain uh requirement and we choose the battery characteristics which are required for that particular application. So application. So in the next class. [00:28:24.304] We are going to look at some modern batteries and then we will see how all these battery characteristics have been used for these. [00:28:37.088] or what are these battery characteristics or that particular uh battery. So in the next class we will talk about zinc air battery. Thank you

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Full transcript: 4171 normalized words, 115 timestamped segments, 1729.642667 seconds duration. Low-confidence segments flagged: 0.

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4 Numericals_Energy storage devices

This lesson is indexed from the source video titled Numericals_Energy storage devices. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000050 · EV0000051

Graphs and diagrams

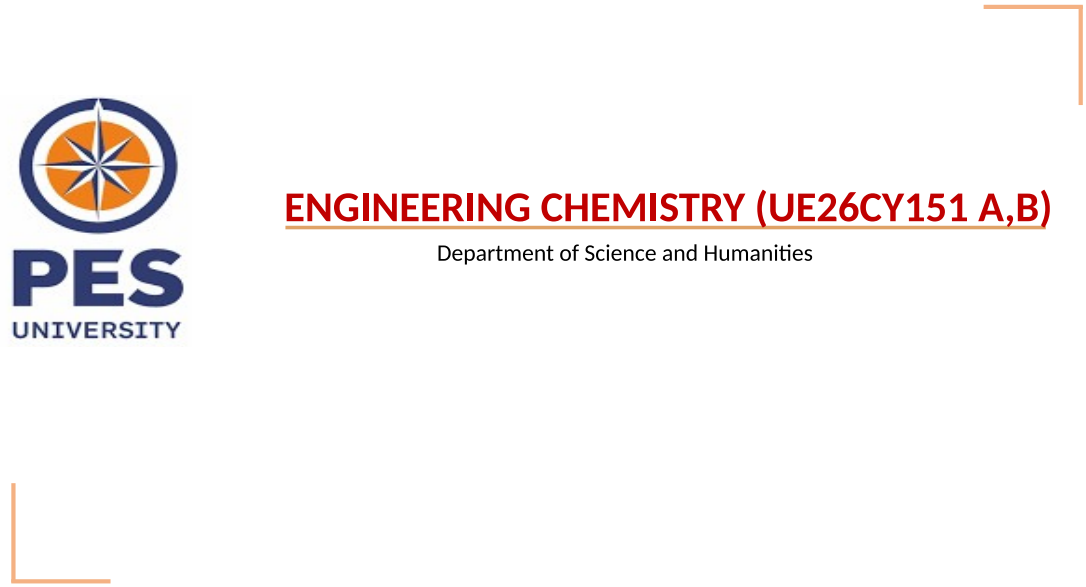

Figure G000011. Battery characteristics Numericals based on battery characteristics · source slide page 1.

EV0000217

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

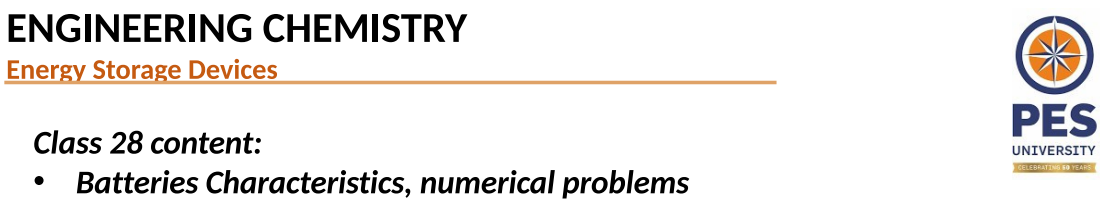

Figure G000012. Battery characteristics Numericals based on battery characteristics · source slide page 2.

EV0000220

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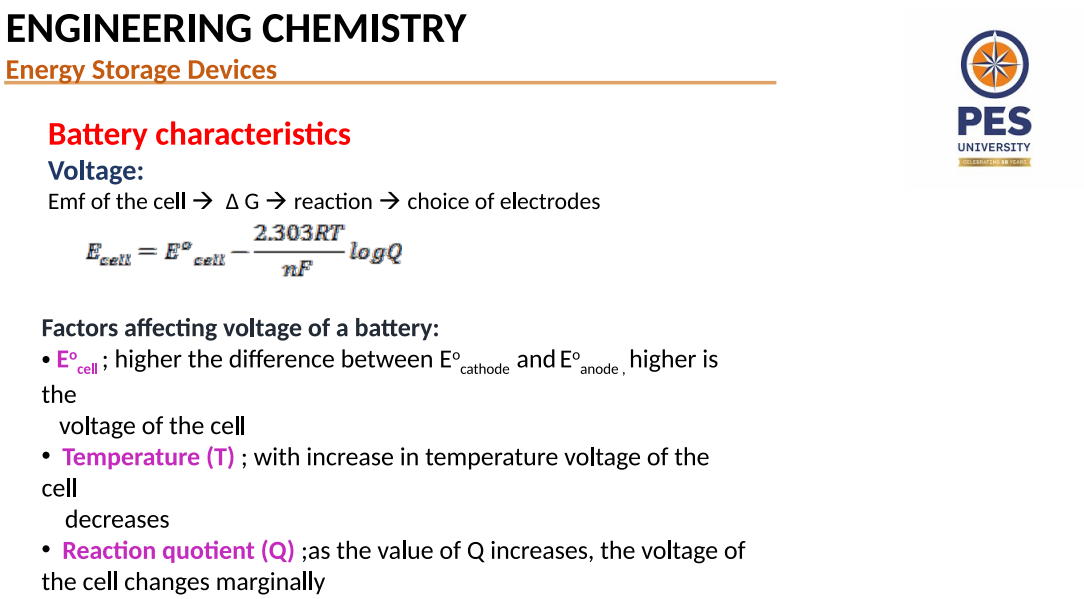

Figure G000013. Battery characteristics Numericals based on battery characteristics · source slide page 3.

EV0000223

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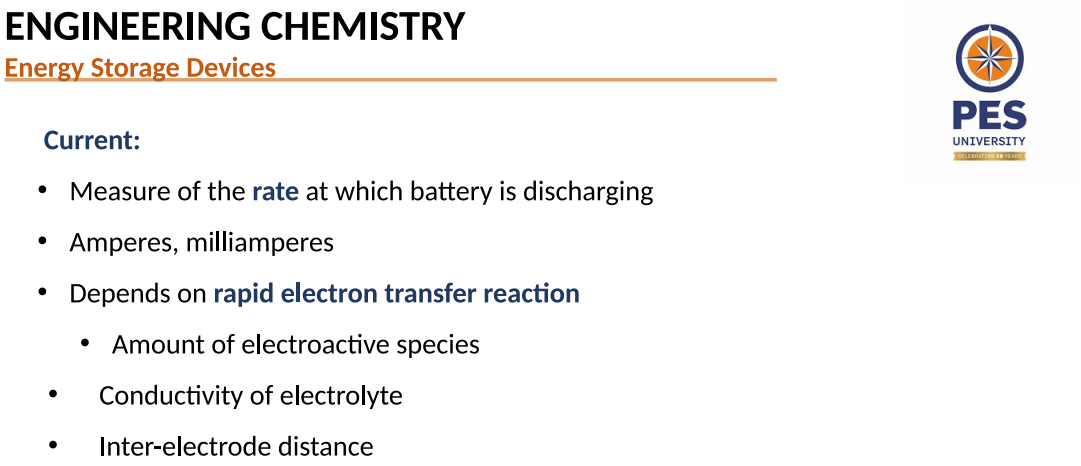

Figure G000014. Battery characteristics Numericals based on battery characteristics · source slide page 4.

EV0000226

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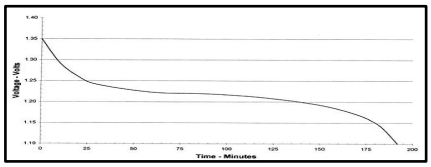
ENGINEERING CHEMISTRY

Energy Storage Devices

Capacity:

- Charge or amount of electricity that may be obtained from the battery and is given in **Ah**
- Faraday's relation $C = \frac{wnF}{M}$ C is the capacity , w is the mass and M is the molar mass of active material
- Capacity depends on : **Size of the battery**

- **C = I t**
- The time period, t, for which the battery will last when a constant current, I, is drawn depends on a) Discharge conditions of the battery b) Capacity of the battery
- **Longer the flat portion** of the curve better is the capacity



http://www.hangtimes.com/robert_hoff_on_esv_s.html

Figure G000015. Battery characteristics Numericals based on battery characteristics · source slide page 5.

EV0000229

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

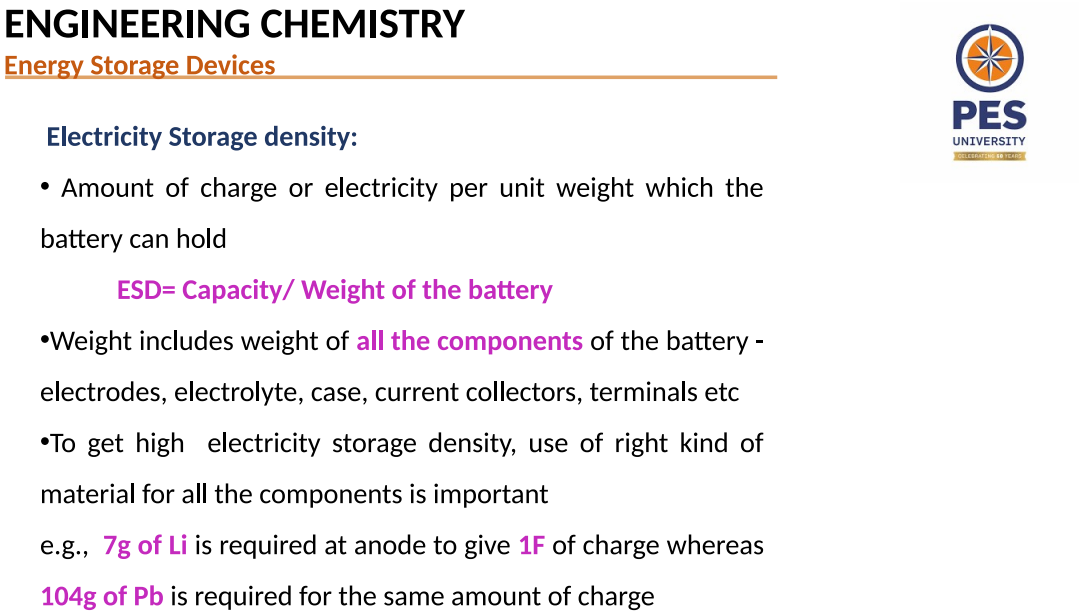


Figure G000016. Battery characteristics Numericals based on battery characteristics · source slide page 6.

EV0000232

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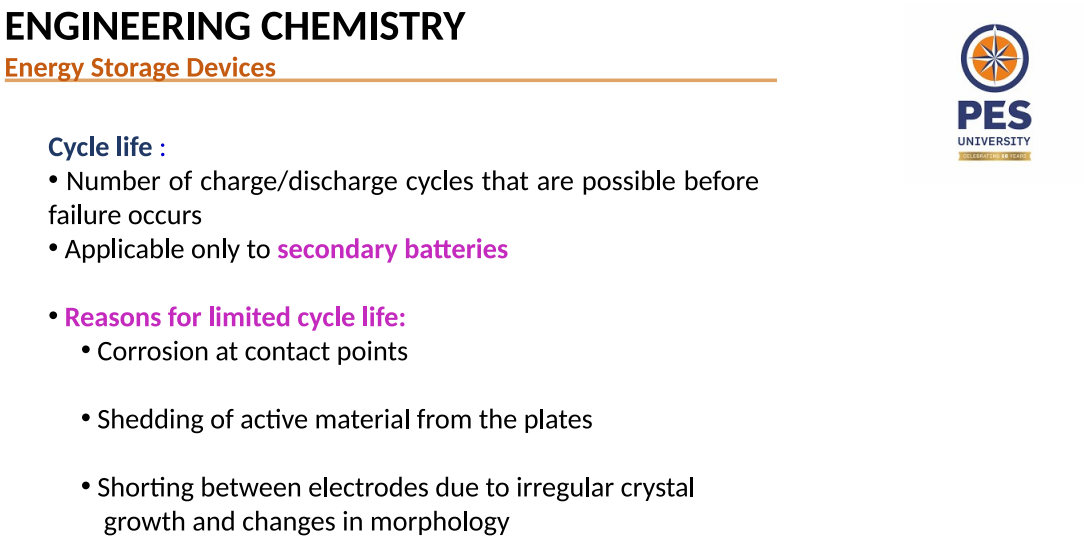

Figure G000017. Battery characteristics Numericals based on battery characteristics · source slide page 7.

EV0000235

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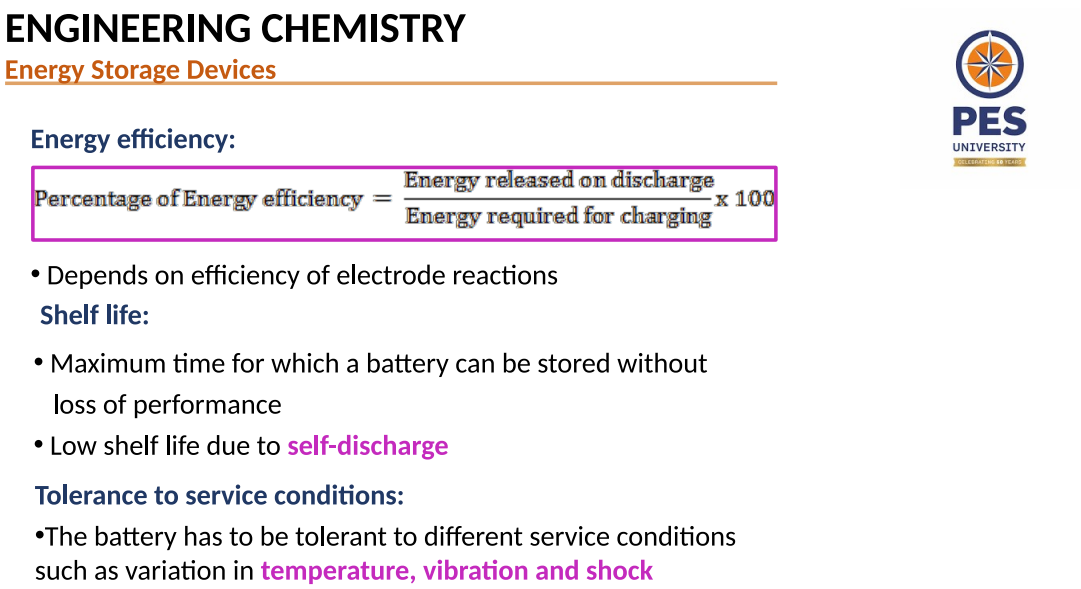

Figure G000018. Battery characteristics Numericals based on battery characteristics · source slide page 8.

EV0000238

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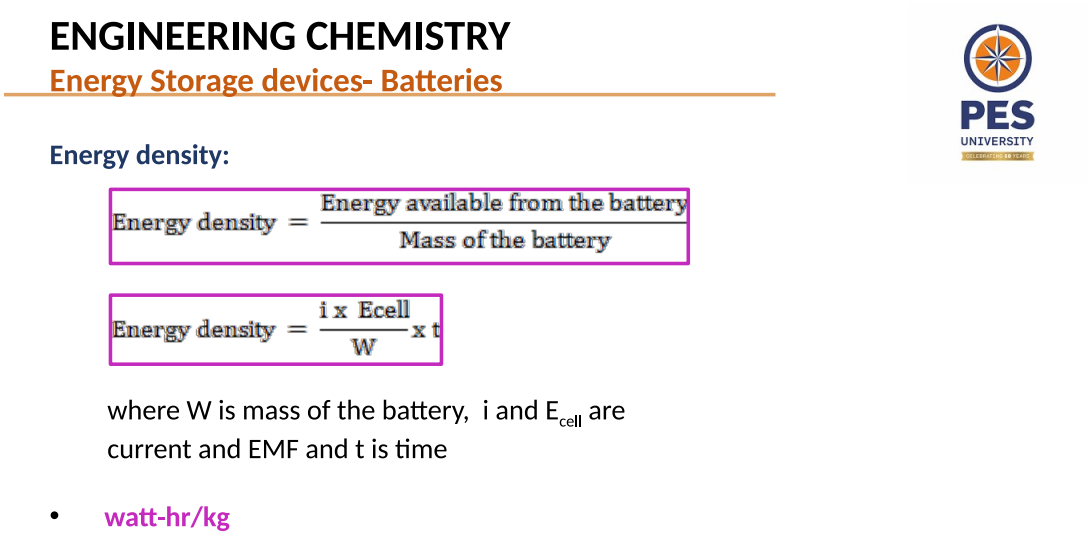

Figure G000019. Battery characteristics Numericals based on battery characteristics · source slide page 9.

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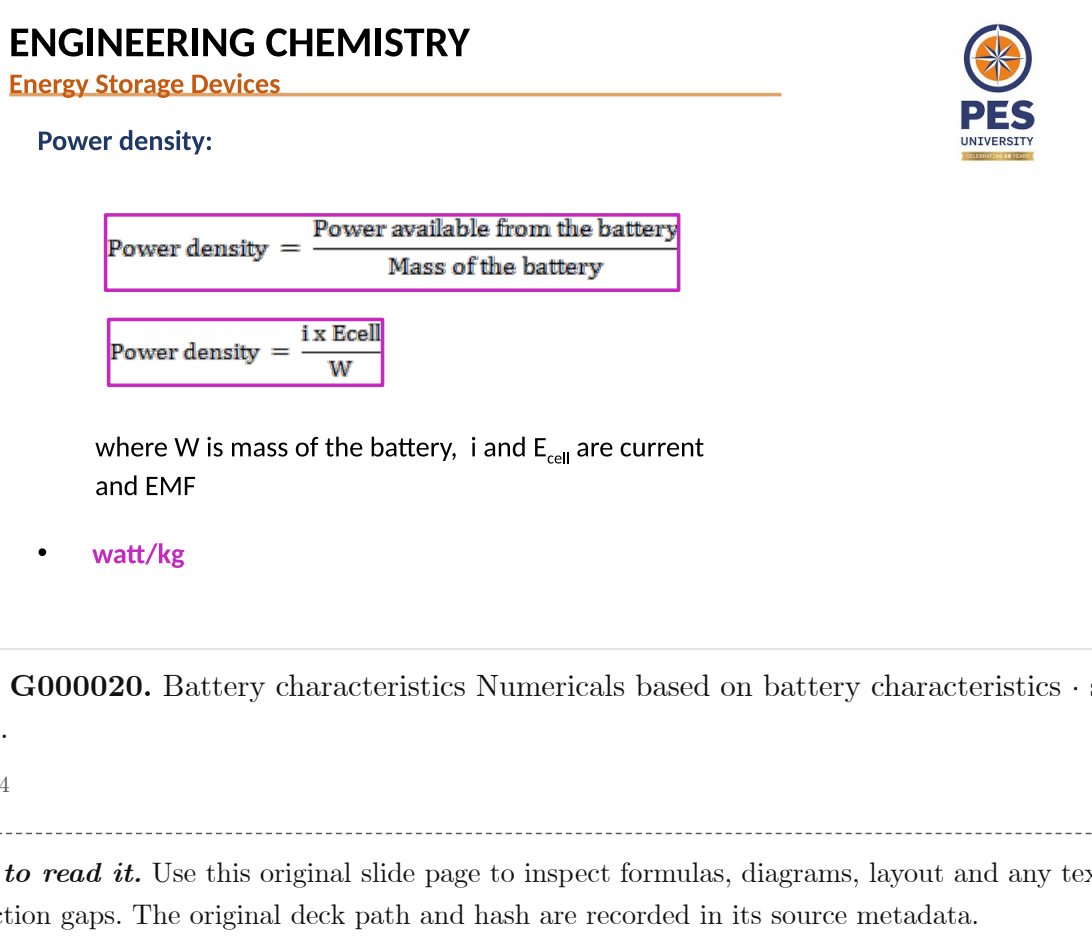

Figure G000020. Battery characteristics Numericals based on battery characteristics · source slide page 10.

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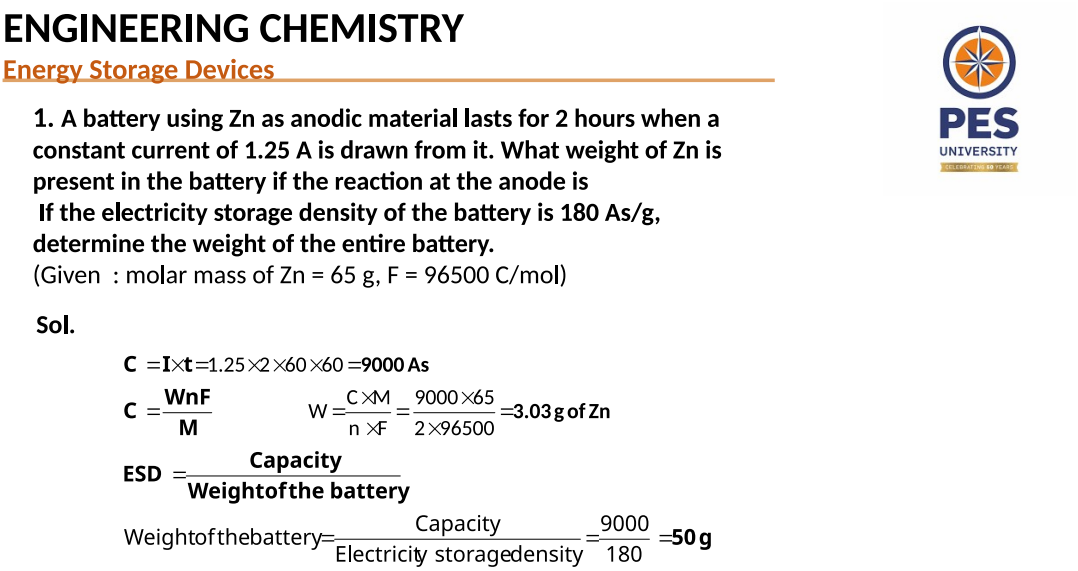

Figure G000021. Battery characteristics Numericals based on battery characteristics · source slide page 11.

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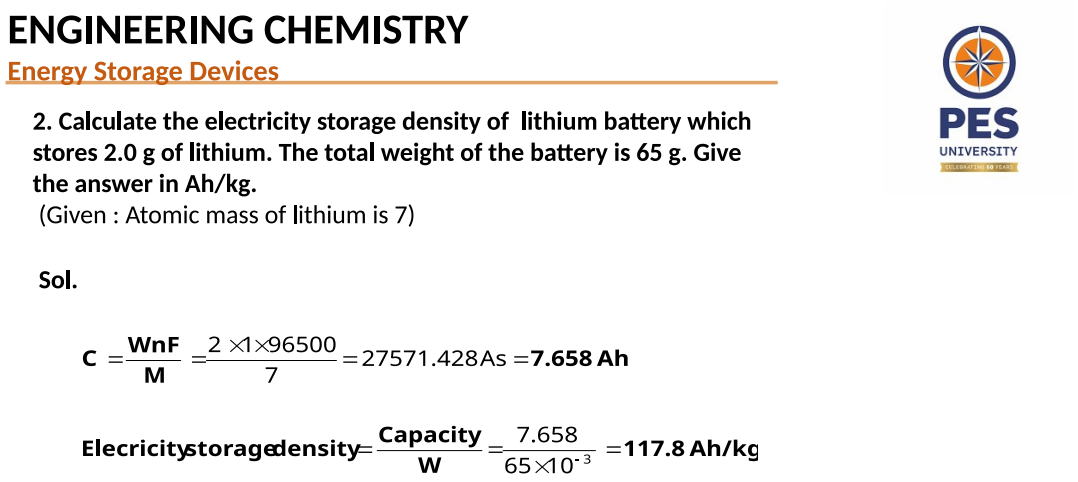

Figure G000022. Battery characteristics Numericals based on battery characteristics · source slide page 12.

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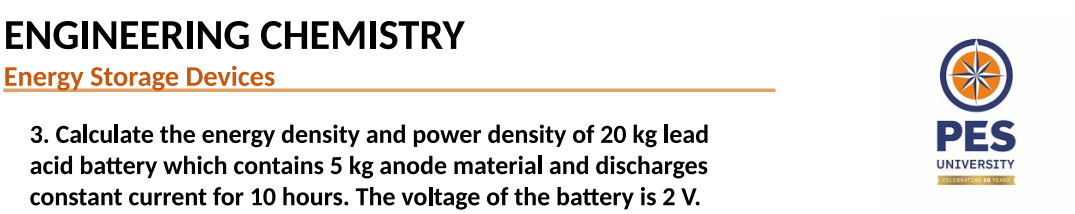

Figure G000023. Battery characteristics Numericals based on battery characteristics · source slide page 13.

EV0000253

> ***How to read it.*** Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000024. Battery characteristics Numericals based on battery characteristics · source slide page 14.

EV0000256

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000216 · EV0000217 · EV0000218
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Class 28 content: Batteries Characteristics, numerical problems EV0000219 · EV0000220 · EV0000221
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Battery characteristics Voltage: Emf of the cell ☒ Δ G ☒ reaction ☒ choice of electrodes Factors affecting voltage of a battery: Eocell ; higher the difference between Eocathode and Eoanode , higher is the voltage of the cell Temperature (T) ; with increase in temperature voltage of the cell decreases Reaction quotient (Q) ;as the value of Q increases, the voltage of the cell changes marginally EV0000222 · EV0000223 · EV0000224
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Current: Measure of the rate at which battery is discharging Amperes, milliamperes Depends on rapid electron transfer reaction Amount of electroactive species Conductivity of electrolyte Inter-electrode distance EV0000225 · EV0000226 · EV0000227

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Capacity: Charge or amount of electricity that may be obtained from the battery and is given in Ah C is the capacity , w is the mass and M is the molar mass of active material Faraday's relation Capacity depends on : Size of the battery C = I t The time period, t, for which the battery will last when a constant current, I, is drawn depends on a) Discharge conditions of the battery b) Capacity of the battery Longer the flat portion of the curve better is the capacity [Image text, OCR] Figure1-DISCHARGECURVEFORONE600mAhCELL 1.40 1.35 1.30 1.25 1.20 1.15 1.10 0 25 50 75 100 125 150 175 200 Time-Minutes http://www.hangtimes.com/robert_hoff_on_esv_s.html EV0000228 · EV0000229 · EV0000230
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Electricity Storage density: Amount of charge or electricity per unit weight which the battery can hold ESD= Capacity/ Weight of the battery Weight includes weight of all the components of the battery - electrodes, electrolyte, case, current collectors, terminals etc To get high electricity storage density, use of right kind of material for all the components is important e.g., 7g of Li is required at anode to give 1F of charge whereas 104g of Pb is required for the same amount of charge EV0000231 · EV0000232 · EV0000233
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Cycle life : Number of charge/discharge cycles that are possible before failure occurs Applicable only to secondary batteries Reasons for limited cycle life: Corrosion at contact points Shedding of active material from the plates Shorting between electrodes due to irregular crystal growth and changes in morphology EV0000234 · EV0000235 · EV0000236
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Energy efficiency: Depends on efficiency of electrode reactions Shelf life: Maximum time for which a battery can be stored without loss of performance Low shelf life due to self-discharge Tolerance to service conditions: The battery has to be tolerant to different service conditions such as variation in temperature, vibration and shock EV0000237 · EV0000238 · EV0000239
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage devices-Batteries Energy density: where W is mass of the battery, i and Ecell are current and EMF and t is time watt-hr/kg EV0000240 · EV0000241 · EV0000242
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Power density: where W is mass of the battery, i and Ecell are current and EMF watt/kg EV0000243 · EV0000244 · EV0000245
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices 1. A battery using Zn as anodic material lasts for 2 hours when a constant current of 1.25 A is drawn from it. What weight of Zn is present in the battery if the reaction at the anode is If the electricity storage density of the battery is 180 As/g, determine the weight of the entire battery. (Given : molar mass of Zn = 65 g, F = 96500 C/mol) Sol. EV0000246 · EV0000247 · EV0000248
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices 2. Calculate the electricity storage density of lithium battery which stores 2.0 g of lithium. The total weight of the battery is 65 g. Give the answer in Ah/kg. (Given : Atomic mass of lithium is 7) Sol. EV0000249 · EV0000250 · EV0000251

13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices 3. Calculate the energy density and power density of 20 kg lead acid battery which contains 5 kg anode material and discharges constant current for 10 hours. The voltage of the battery is 2 V. (Given : Atomic mass of lead is 207.2, number of electrons transferred in the redox reaction is 2, F= 96500 C/mol) Sol. EV0000252 · EV0000253 · EV0000254
14. Source slide page 14 (extracted text): THANK YOU Department of Science and Humanities EV0000255 · EV0000256 · EV0000257

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.612] Hello everyone. Welcome back to this course on engineering chemistry. [00:00:07.268] We will continue with module five on energy storage devices and we'll try to solve some numericals today on this topic. So basically we'll look at some numericals on battery characteristics and some on the efficiency of fuel cells. [00:00:26.788] Let us take up the first numerical on battery characteristics. So the question says a battery using zinc as the anodic material lasts for two hours when a constant current of one point two five amperes is drawn from it. [00:00:43.140] So what weight of zinc is present in the battery if the reaction at anode is zinc, giving you zin 2 plus plus two electrons. [00:00:52.740] After that, you also need to find out what is the weight of the entire battery. So here first we have to find out the weight of zinc [00:01:04.580] And then we have to find out the weight of the entire battery. And for this electricity storage density is given to us and molar mass of zinc is known, 65 grams, and F, that is Faraday's constant, is to be taken as 96500 Coulombs per mole. Let us see the solution. [00:01:25.764] Now capacity of a battery is equal to [00:01:29.988] I into T, that is current into time. So what is the current given here? 1.25. So that's what we have taken here. And two hours is the time. It is working for it is lasting for two hours. So the capacity would be now [00:01:46.496] 2 multiplied by 60 into 60. That is we are converting it into seconds. So the capacity of the battery would be current into time, so that comes out to be 9000 ampere second. We need to be very very careful with the units. Okay now another [00:02:06.148] formula for finding out capacity is also Wnf by M, where W is the weight of the electroactive material. In this case it is zinc [00:02:18.784] That is the anodic material. N is the number of electrons getting exchanged. So what would what would it be? We know from the reaction it is two electrons. F is given to us, and molar mass of zinc is also given to us. So therefore C is equal to Wnf by M. From here we can find out the weight of the electroarms. [00:02:36.528] material by rearranging this equation. So it would be W becomes equal to C into M divided by N into F. So all we have to do is substitute capacity which we already know from here. That is the reason why we found out the capacity. [00:02:54.624] So, capacity, there are two formulae, and depending on what the data is given to you, you have to decide whether you can use this formula or this formula. In this case, we could determine capacity from here, and this formula is used to determine weight of the electroactive material. So, therefore, capacity is known. [00:03:12.272] into 652 multiplied by divided by two multiplied by nine six five zero zero and we have found out the weight of zinc it is three point zero three grams of zinc [00:03:24.420] Now coming to the next part where we want to find out the weight of the entire battery. And what is given to us electricity, storage, density. Now electricity, storage, density, how did we define the [00:03:36.400] Amount of charge that can be stored in the battery per unit weight of the battery. And then there we said weight of the battery, when we say per unit weight of the battery, that is weight of the entire battery, including all the components. So ESD [00:03:53.616] That is electricity

storage density because we said the amount of charge that can be stored in the battery. And what is the amount of charge that can be stored in the battery? That is equal to the capacity of the battery. So capacity divided by weight of the battery. So now [00:04:07.716] If we want to find out the weight of the battery using the same formula, we rearrange and we know the capacity, we have calculated capacity, we also know electricity storage density, which is given to us. Only thing you need to be very, very careful with the units. [00:04:22.928] So here it is 9000 ampere second and here it is 180 ampere second per gram. So your answer is going to be 50 grams. The unit is where you have to be very careful. So this is how we are going we solve this problem. [00:04:40.932] Let's go to the next problem where you've been asked to calculate the electricity storage density of a lithium battery, which has two grams of lithium as the anodic material, and the total weight of the battery is sixty-five grams. [00:04:56.976] You have to give your answer in ampere hour per kg. This is the requirement of this numerical atomic mass of lithium is seven. So what let us look at this. Now when we say electricity storage density, we know that electricity storage density is capacity per unit weight.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:17.380] Now, capacity. Capacity is either current into time or it can be determined by WNF by M. So because no current or time is mentioned here, we have to go with the other formula that is C is equal to Wnf by [00:05:34.928] the weight of the battery. So then we will get the electricity storage and see the only thing we need to be careful in this particular question is the final answer should be in ampere hour per kg. Let us look at the solution. [00:05:52.100] So we have to use this formula for capacity, WNF by M, because the weight is of lithium, which is used as the anodic material, is given to us. It is 2 grams. [00:06:04.752] So it's a lithium battery, so it is using two grams of lithium as the anodic material. Lithium is always we know lithium is [00:06:14.852] monovalent so plus one electron therefore here we have one electron that is being exchanged in this and 9650 divided by seven that is the atomic mass of lithium. So what we get is this value 257 [00:06:32.032] 27571.428. But this is in Coulombs. Remember capacity is always what when we calculate it will be in Coulombs and Coulombs is always ampere second. [00:06:44.336] So therefore if you want to convert it into ampere hours what do you need to do? Divide it by three six zero zero and that would give you an answer seven point six five eight ampere hours is the capacity. Now once we know the capacity, electricity storage density can be found out easily capacity by uh [00:07:01.968] Weight of the battery. Now, what is the weight of the battery? So you should not be confused with these two weights which are given here. [00:07:09.360] Please read carefully. This is the weight of the electroactive material which is used to calculate capacity of the battery, and this weight is weight of the entire battery. So that we are going to use here when we calculate electricity storage density, and that would be we calculated capacity here, use that, and this. [00:07:29.668] 65 grams. So since your answer should be in kg, ampere hours anyway we have here we have converted the capacity in in uh into ampere hours, ampere seconds into ampere hours. So we have this value. Now this needs to be in kg, so 65 multiplied by [00:07:47.440] 10 raised to the power of minus 3. So when you do the calculation, your answer is 117.8 ampere hour per kg. [00:07:57.732] Let us see another type of question where we need to find out the energy density and the power density. And what is given here it is a lead acid battery which weighs twenty kg. So when you say twenty kg lead acid battery [00:08:13.252] It is the total weight of the battery and it contains 5 kg of lead as the anode material. So here 5 kg is weight of the anodic material, the electroactive material which is being used and it discharges

constant current. [00:08:29.824] for 10 hours. Understand the current value is not given here. The time is known to us. So and also the voltage of the battery is also known to us. [00:08:40.448] So atomic mass of lead is given to us as 2.207.2. Number of electron transferred in the redox reaction is 2. Even if it is not given, we should understand lead is P B2 plus plus 2 electrons. These are simple uh [00:09:00.868] Uh elements for which may you should know what is the uh simple uh reaction when it undergoes oxidation. Since it's the unknown material, we write the reaction this way. So it has now two electrons and f uh is to be taken as 96500 coulombs per mole. So what do we need? We need the energy. [00:09:18.640] Energy density and we need the power density. What do we have? The data given to us 5 kg lead. So that means I can straight away determine the capacity here using C is equal to Wnf by M. Let us look at the [00:09:36.640] solution. Yes, so we will use C is equal to Wnf by M that is 5 into 10 raised to the power of 3. Why did we do this? Because we are now expressing in grams. 2 is the number of electrons that is getting exchanged over here and 9650 is [00:09:54.384] zero zero is known to us. Two zero seven point two is given to us. So now we can determine [00:09:59.904] What is the value in ampere second? This value which you got from here is ampere seconds. Now ampere seconds when you divide by 3600, then you get the answer in ampere hours. So it's better to write here itself.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:17.028] 60 into 60. So that would give us 1293.7 ampere hours. So why did I write it here? Because straight away you can't write this and divide by 3600. It doesn't make sense. [00:10:31.328] Uh if you write this as ampere seconds and then you show it as ampere hours that would make more sense. Doesn't matter, we can always write it like this. So here what we are doing is trying to get the answer in ampere hours straight away. So we have to divide by sixteen to sixty, that's exactly what is done here and we can [00:10:49.040] at 1 to 93.7 ampere hours. Now knowing the capacity, can we determine energy density? Now energy density the formula is I into T into V, that is current into time into voltage. [00:11:07.184] divided by weight of the battery. Now this is the total weight of the battery which we are going to use here that is 20 kgs is given to us. [00:11:16.768] Now we can straight away get the answer in kg. If it was asked in a grams, then you to need to convert this into grams. Now here, since there is no uh unit restriction as such, so we could continue with twenty here as kg. So we have I into t into volts. [00:11:34.448] But we neither we don't we know the time but we don't know the current. But is there any other way I can use this formula? Yes, we know that I into t is the capacity of the battery. [00:11:48.784] And that is exactly what we calculated here. So we will use this 1293.7 that is capacity into voltage. Volta is given to us as two volts here. So [00:12:01.040] That is employed, and then we divide it by 20 weight of the battery, and we get 129.4 watt hours per kg is the energy density. So it just says constant current. Understand, we don't know the value of current, you need not know also, because once you know the capacity. [00:12:18.672] You can straight away in this formula use capacity. So, uh, can we say another formula for energy density would be capacity multiplied by voltage divided by weight of the battery. So this is how you get the answer. Now coming to power density, that is. [00:12:35.396] will be energy density divided by time because it is current into voltage divided by [00:12:50.980] Current into voltage divided by [00:12:53.312] The weight of the battery. That is the formula for power density. Current into voltage divided by weight of the battery. Now, if you look at energy density, what is it? Current into time into voltage by weight of the battery. So all other things are there except the time. So when you want to find out [00:13:10.992] Power density, all you have to do is if you know the

energy density divide it by time and you get your formula I into v by w that's exactly what we have here. So energy density by time. The time is known to us as 10 hours. So since [00:13:25.840] This answer is also in hours. So that is why we don't convert it into seconds. Instead we wanted it in hours because the time is in hours, they get the units get uh hour gets cancelled. [00:13:37.936] here and therefore we'll end up with 129.4 divided by 10 and the answer would be this is also hours this is also what hour per kg so therefore you end up with watts per kg 12.9 watts per kg is your answer [00:13:57.008] uh for power density. So this is how you can determine energy density and power density. So it's always recommended that all the formula be written in one place and then see the data and use the formula accordingly. Now coming to [00:14:16.324] The efficiency of a fuel cell. Now efficiency of a fuel cell, simple problems, calculate the efficiency of a hydrogen oxygen alkaline fuel cell. [00:14:26.768] And what is given to you E naught cell is given to you and delta H formation of water is equal to minus two eighty five point eight kilojoules per mole. So per mole of water formed you have minus two eighty five point eight kilojoules of energy. [00:14:44.400] Energy being given out. I mean 285.8 kilojoules of energy being given out because it's an exothermic reaction, we are using the minus sign. So what would be the reaction here for formation of water? We know that in an alkaline fuel cell.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:01.072] with hydrogen and oxygen. So you have H two plus half O two is giving you [00:15:07.888] H2O since we need only one mole, so this is how we write, and we know if you we know that if we write it as anode and cathode reaction, the number of electrons is going to be two electrons which will be exchanged in this reaction. Let us look at the solution. [00:15:28.356] Now efficiency percentage is given by we should write it as percentage. [00:15:34.688] Efficiency percentage is equal to delta G by delta H multiplied by 100. Now since here E naught cell is given to you and not delta G, we need to first calculate delta G. So delta G would be minus N F E. So we will substitute for [00:15:52.432] n that is the number of electrons since it is talking of one mole of water getting produced so number of moles of electrons being exchanged is 2 so 2 into 96500 that is f multiplied by 1.2 that is e naught cell is given to you and we also know now delta [00:16:10.000] delta H. So this would give us the value of delta G and delta H is given to us minus 285.8 and because it's in kilojoules we will convert it into joules. So when you substitute here and because we have to find out the percentage multiplied by 100 [00:16:27.568] So when you do that, you uh substitute the values, get the answer, you will get it as 81%. So for this particular hydrogen alkaline hydrogen oxygen alkaline fuel cell, the efficiency turns out to be 81%. [00:16:43.908] So it is just substituting into the formula. If delta G was directly given to you, all you have to do is just substitute the value of delta G, delta H and multiplied by 100 would give you the [00:16:55.952] uh percentage of efficiency but if delta g is not given instant e naught cell is given to you then you need to sub find out delta g by using delta g is equal to minus nfe. So we have delta g is equal to [00:17:13.216] Minus N F E. So we substitute that and this is how we uh solve the problem. Now let's come to the last problem again on efficiency. Now here E naught [00:17:25.476] cell for H2O2 alkaline fuel cell is 1.23 volts and the efficiency is 83%. We have to calculate the heat evolved during the reaction. So basically we want to find out what is delta H. [00:17:44.356] Let us look at the solution. Now we are using the same formula, it's not so difficult. Efficiency is again in the form of percentage. Efficiency percentage is delta G by delta H multiplied by 100. Now efficiency percentage is given to you as 83%. So 83 is equal to again delta. [00:18:04.336] The G is not given to you, but we know the E naught cell that

is 1.23 volt. So we will substitute 1.23 here. N again we know is is going to be [00:18:15.440] Two for this particular uh reaction, what is the reaction happening here? Alkaline fuel cell H two plus half O2 is giving you. [00:18:24.720] H2O. So for this reaction, N is equal to 2. So when we substitute 2 into 96500 into 1.23 divided by delta H value. The delta H value is to be found out multiplied by 100. Now 83% is known to us. [00:18:42.448] So when you rearrange this, then we find that delta H will be equal to minus 286. 000. That's you will get it in joules, but then you want to determine in kilojoules. So minus 286 kilojoules per mole. [00:19:02.340] of water. Now one thing which you need to understand here the delta H value [00:19:11.108] could be given in the form of delta H formation of one water molecule that is water, one mole of water, or it could also be given as delta H combustion of hydrogen. Both the reactions [00:19:28.608] When you say delta H combustion of hydrogen, when you say Delta H combustion [00:19:35.296] Of hydrogen, it means in enough oxygen when one mole of hydrogen burns, what is the delta H or the heat which is given out? Or you can say delta H formation of one mole of water. In that case also the reaction remains the same, therefore the delta H will be same. So whether you [00:19:52.944] say delta H formation of water or you say delta H combustion of hydrogen the value will remain the same and both of them here we see it is one mole therefore delta H is equal to minus 286 kilojoules per mole.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:10.096] So, this is how we are going to determine the heat evolved during the reaction. You need to know the efficiency and you also need to know what is the E naught cell, or directly if you know the delta G also, you will be able to use this formula to calculate [00:20:27.760] Delta H. So in this case it is minus two eighty six kilojoules per mole. [00:20:36.772] Thank you. So in the next class, we are going to start with the next unit that is about corrosion chemistry. Thank you.

EV0000066 · EV0000067 · EV0000068 · EV0000069

Full transcript: 3266 normalized words, 84 timestamped segments, 1249.216 seconds duration. Low-confidence segments flagged: 0.

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5 Zinc -air battery

This lesson is indexed from the source video titled Zinc -air battery. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000070 · EV0000071

Graphs and diagrams

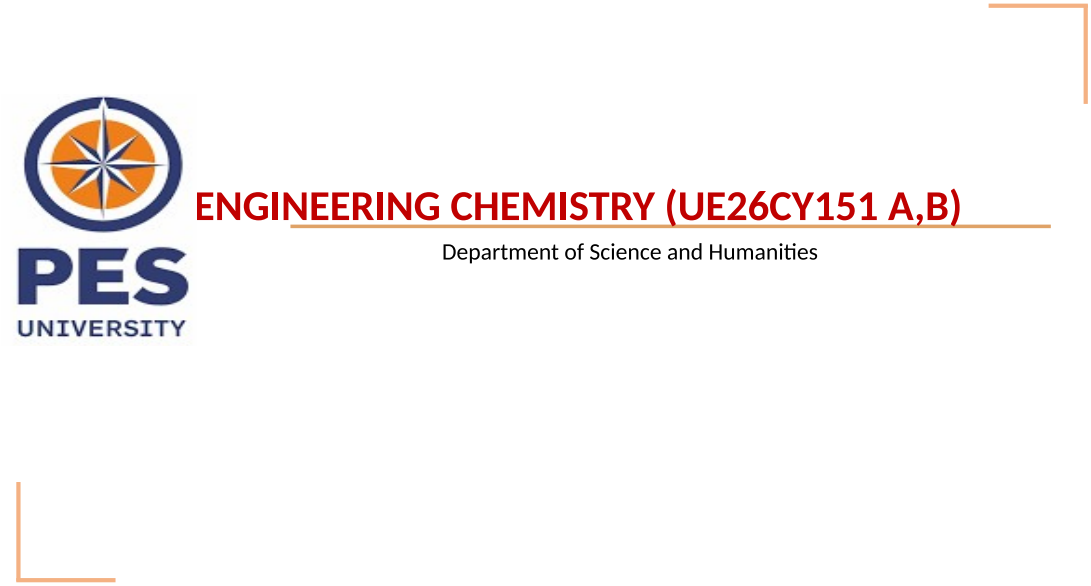

Figure G000025. Modern batteries Zn air battery Li metal batteries · source slide page 1.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000026. Modern batteries Zn air battery Li metal batteries · source slide page 2.

EV0000262

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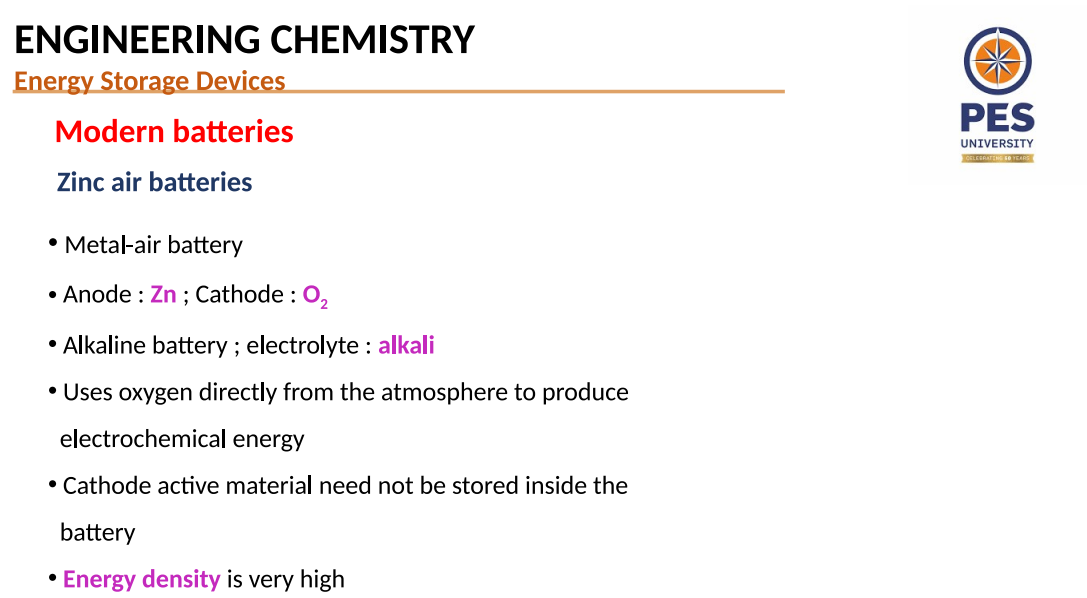

Figure G000027. Modern batteries Zn air battery Li metal batteries · source slide page 3.

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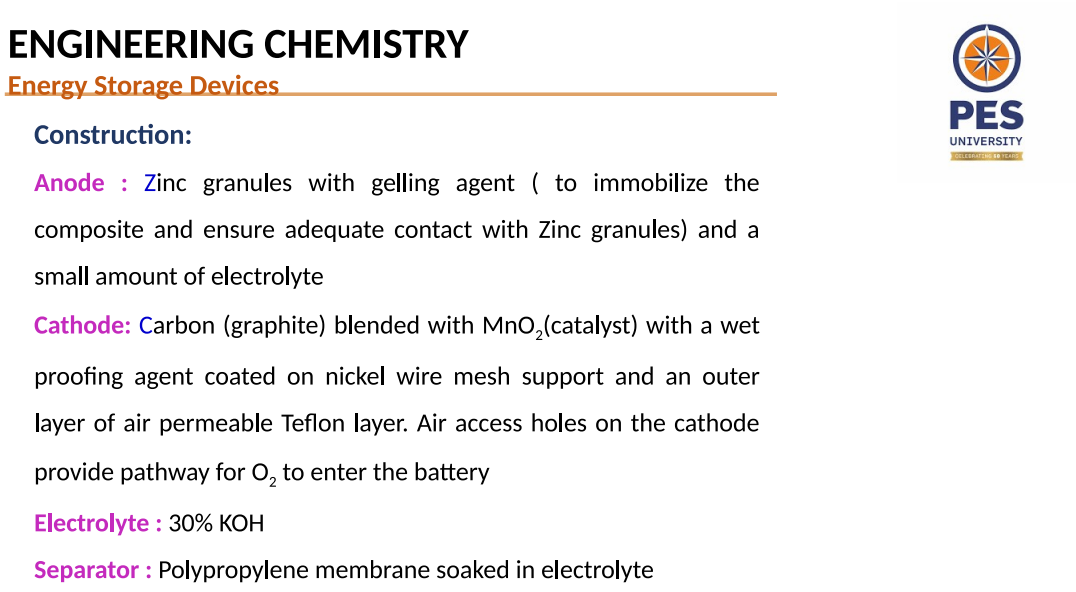

Figure G000028. Modern batteries Zn air battery Li metal batteries · source slide page 4.

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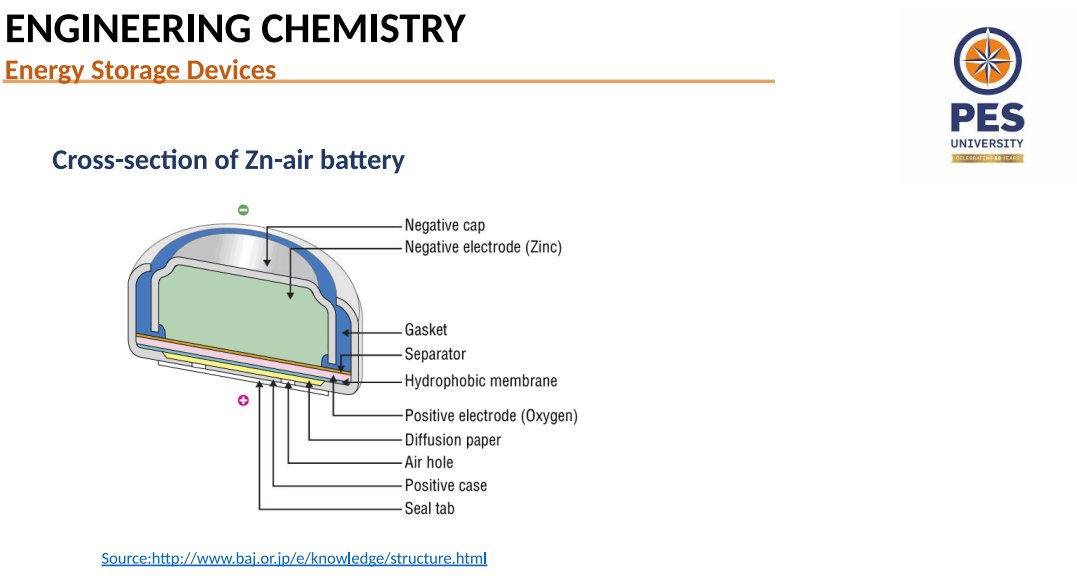

Figure G000029. Modern batteries Zn air battery Li metal batteries · source slide page 5.

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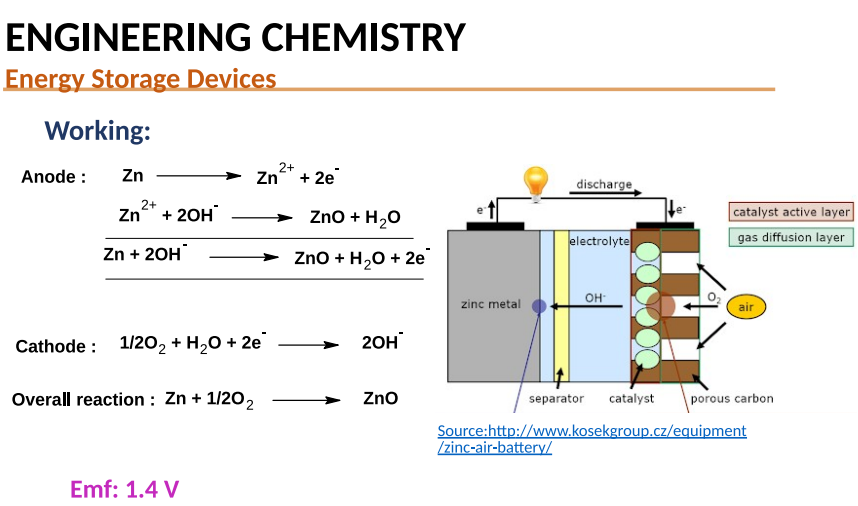

Figure G000030. Modern batteries Zn air battery Li metal batteries · source slide page 6.

EV0000274

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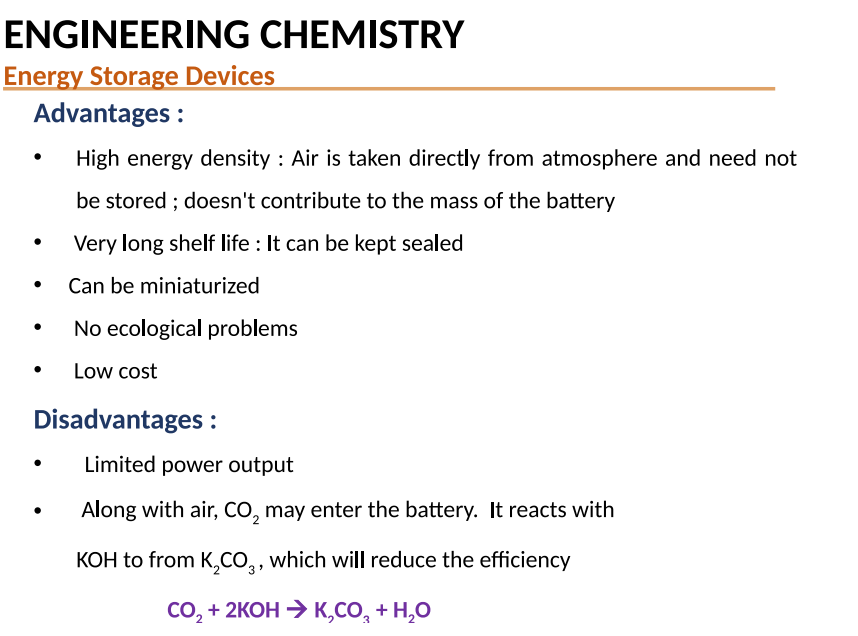

Figure G000031. Modern batteries Zn air battery Li metal batteries · source slide page 7.

EV0000277

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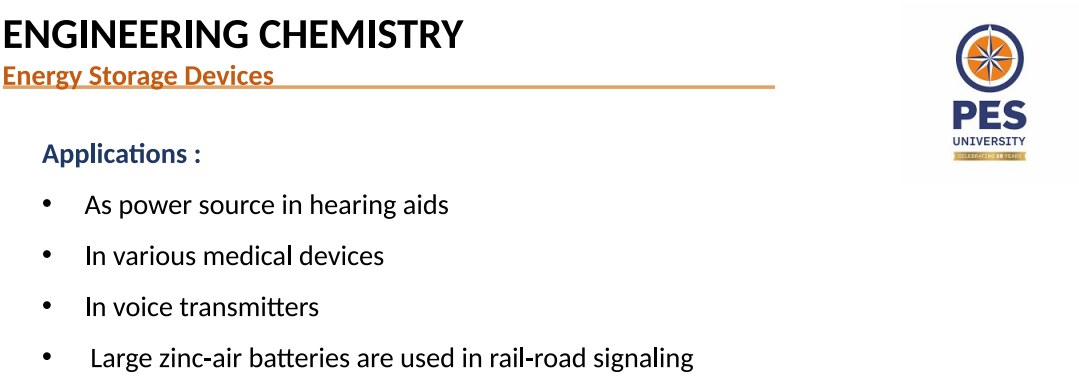

Figure G000032. Modern batteries Zn air battery Li metal batteries · source slide page 8.

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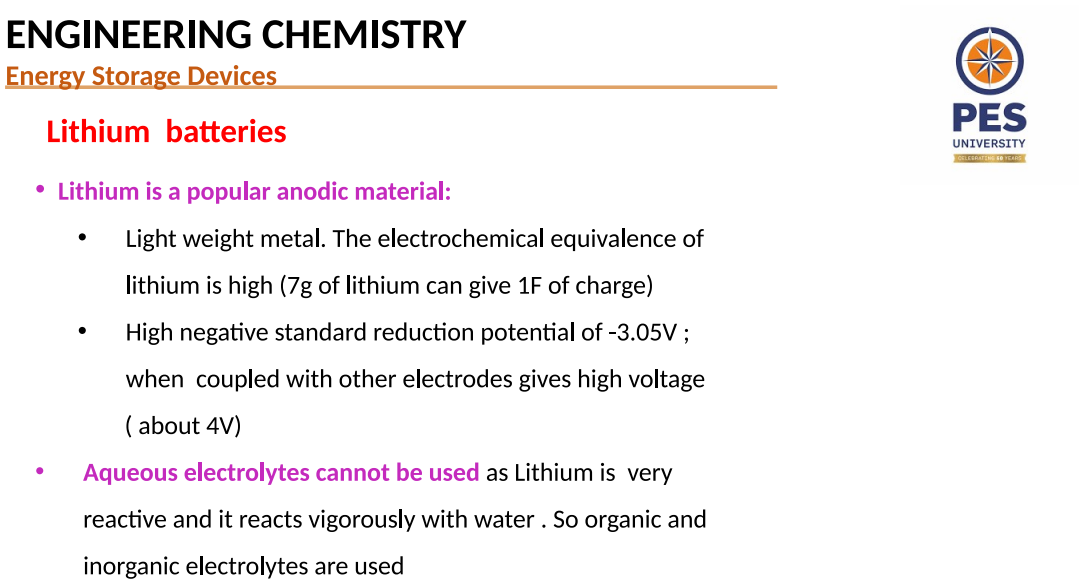

Figure G000033. Modern batteries Zn air battery Li metal batteries · source slide page 9.

EV0000283

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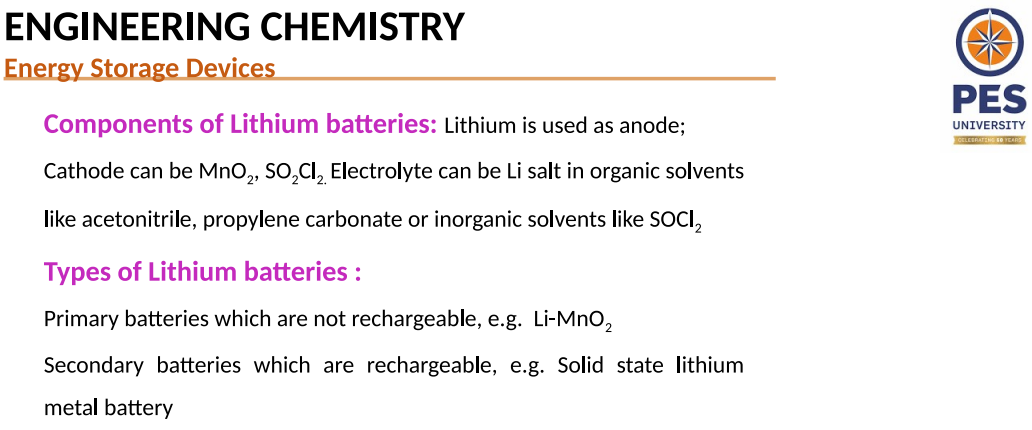

Figure G000034. Modern batteries Zn air battery Li metal batteries · source slide page 10.

EV0000286

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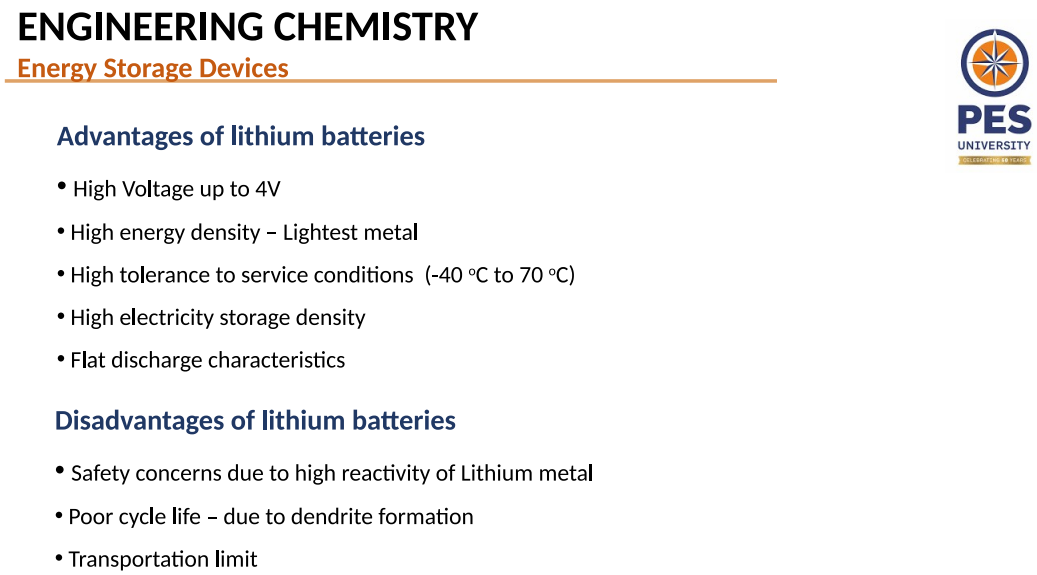

Figure G000035. Modern batteries Zn air battery Li metal batteries · source slide page 11.

EV0000289

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000036. Modern batteries Zn air battery Li metal batteries · source slide page 12.

EV0000292

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000258 · EV0000259 · EV0000260
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy storage devices - Batteries Class 29 content: Modern Batteries- Zn –air battery, Li metal batteries EV0000261 · EV0000262 · EV0000263
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Modern batteries Zinc air batteries Metal-air battery Anode : Zn ; Cathode : O2 Alkaline battery ; electrolyte : alkali Uses oxygen directly from the atmosphere to produce electrochemical energy Cathode active material need not be stored inside the battery Energy density is very high EV0000264 · EV0000265 · EV0000266
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Construction: Anode : Zinc granules with gelling agent (to immobilize the composite and ensure adequate contact with Zinc granules) and a small amount of electrolyte Cathode: Carbon (graphite) blended with MnO2(catalyst) with a wet proofing agent coated on nickel wire mesh support and an outer layer of air permeable Teflon layer. Air access holes on the cathode provide pathway for O2 to enter the battery Electrolyte : 30% KOH Separator : Polypropylene membrane soaked in electrolyte EV0000267 · EV0000268 · EV0000269

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Cross-section of Zn-air battery [Image text, OCR] Negativecap Negative electrode(Zinc) Gasket Separator Hydrophobic membrane + Positiveelectrode(Oxygen) Diffusionpaper Airhole Positivecase Sealtab Source:http://www.baj.or.jp/e/knowledge/structure.html EV0000270 · EV0000271 · EV0000272
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working: [Image text, OCR] discharge catalyst active layer electrolyte gas diffusion layer zinc metal OH- air separator catalyst porous carbon Zn→Zn2++2e O2+2H2O+4e-→4OH- Zn2++4OH-→[Zn（OH）4]2- or [Zn（OH）4]2-→ZnO+H2O+2OH O2+H2O+2e-→HO2-+OH Source:http://www.kosekgroup.cz/equipment/zinc-air-battery/ Emf: 1.4 V EV0000273 · EV0000274 · EV0000275
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages : High energy density : Air is taken directly from atmosphere and need not be stored ; doesn't contribute to the mass of the battery Very long shelf life : It can be kept sealed Can be miniaturized No ecological problems Low cost Disadvantages : Limited power output Along with air, CO2 may enter the battery. It reacts with KOH to from K2CO3 , which will reduce the efficiency CO2 + 2KOH ☒ K2CO3 + H2O EV0000276 · EV0000277 · EV0000278
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Applications : As power source in hearing aids In various medical devices In voice transmitters Large zinc-air batteries are used in rail-road signaling EV0000279 · EV0000280 · EV0000281
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Lithium batteries Lithium is a popular anodic material: Light weight metal. The electrochemical equivalence of lithium is high (7g of lithium can give 1F of charge) High negative standard reduction potential of -3.05V ; when coupled with other electrodes gives high voltage (about 4V) Aqueous electrolytes cannot be used as Lithium is very reactive and it reacts vigorously with water . So organic and inorganic electrolytes are used EV0000282 · EV0000283 · EV0000284
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Components of Lithium batteries: Lithium is used as anode; Cathode can be MnO2, SO2Cl2. Electrolyte can be Li salt in organic solvents like acetonitrile, propylene carbonate or inorganic solvents like SOCl2 Types of Lithium batteries : Primary batteries which are not rechargeable, e.g. Li-MnO2 Secondary batteries which are rechargeable, e.g. Solid state lithium metal battery EV0000285 · EV0000286 · EV0000287
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages of lithium batteries High Voltage up to 4V High energy density – Lightest metal High tolerance to service conditions (-40 oC to 70 oC) High electricity storage density Flat discharge characteristics Disadvantages of lithium batteries Safety concerns due to high reactivity of Lithium metal Poor cycle life – due to dendrite formation Transportation limit EV0000288 · EV0000289 · EV0000290
12. Source slide page 12 (extracted text): THANK YOU Department of Science and Humanities EV0000291 · EV0000292 · EV0000293

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.804] Hello everyone, welcome back to this course on engineering chemistry. We continue with module 5 on energy storage devices.

That is, we are talking about batteries. So today we are going to look at modern batteries and specifically the zinc air battery. [00:00:21.444] For the Z care battery we look at the construction, working, advantages, disadvantages, and applications. [00:00:31.204] So zinc air battery can be classified as a metal air battery because as you can see in the name the zinc and the air, so it is metal air batteries. [00:00:42.884] So it's very obvious that at the anode we are going to use zinc as the electroactive material, and at the cathode, it is going to be oxygen from the air. So zinc air batteries will use zinc as the anode and oxygen as the cathode. [00:01:03.236] Another important fact about this battery is that it's an alkaline battery. So the electrolyte is going to be an alkali. So we know what is the anode active material, what is the cathode active material, and we know that the electrolyte is an alkali. [00:01:20.164] The most important thing about this battery is that it is using oxygen directly from the atmosphere to produce electrochemical energy. So, what does it mean that [00:01:37.808] the act uh the electroactive material of the cathode need not be stored inside the battery. So that reduces the weight of the battery considerably and hence one important [00:01:52.324] Property of this battery is that the energy density is going to be very very high. The energy density for such batteries is very high. In the previous class, we talked about energy density, that is energy available per unit weight. Now, since the weight of the battery [00:02:10.096] Battery has been considerably reduced. The reason being that one of the electroactive material that is the cathode electroactive material which is oxygen need not be stored inside the battery and can be taken directly from air. So one important [00:02:27.344] Salient feature in fact of this battery is that it has high energy density. Let us look at the construction of such a battery. [00:02:40.356] For in a battery, in the first class when we talked about batteries, I told you there are four components, basic components of a battery, the anode, [00:02:49.264] The cathode, the electrolyte, and the separator. So in this zinc air air battery, let us see what is there at the anode, cathode, electrolyte, and separator. We have already studied just now that zinc granule zinc is the anode active material. In what form is it present? [00:03:06.896] Zinc is taken in the form of granules, and why would that be to increase the surface area so that the reaction can happen at a faster rate. So zinc granules, they are [00:03:19.920] uh taken along with the gelling agent. The role of this gelling agent is to hold the granules together. So zinc granules with gelling agent and a small amount of electrolyte is also used. So there are three parts here one or three components here one is the [00:03:36.592] uh zinc granules and the other is the gelling agent and then we have a small amount of electrolyte which is added. So that is the composition of the anode. Now coming to the cathode. Now we know that at the cathode [00:03:51.136] It is activity material is oxygen, but then we know that there needs to be a current collector. [00:03:59.040] Just having the electroactive material is not sufficient. Yes, oxygen will be taken from the air, but we need to have electroactive material. Uh electroactive material is taken from the air, but we need to have a current collector. That is uh [00:04:15.536] Responsible for picking up the electrons or providing the electrons for reduction in this case because it is the cathode. So what do we have at the cathode? We have carbon. [00:04:27.472] That is in the form of crepite. So the moment you see carbon, it is understood it is for the conductivity. So we have carbon which is blended with MnO2, which is a catalyst. Now, why is this catalyst required for [00:04:44.772] uh this particular uh battery. That is because [00:04:48.944] At the cathode, oxygen has to undergo reduction, which is a slow process. In order to expedite the process, in order to make the reaction happen faster, we need to use a catalyst. So carbon is blended with MNO2, and now

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.608] Now a wet proofing agent is also added. Why would that be required? We are taking oxygen from the atmosphere. So along with oxygen, some moisture may enter the battery. So therefore, in order to uh [00:05:23.136] Remove the moisture, not allow the moisture into the battery, we need to add a wetproofing agent, and all this is now coated on a nickel wire mesh support which is acting as the current collector. And apart from this, we also have an outer. [00:05:40.816] layer of air permeable teflon layer. Now what would this be uh what would be the role of this teflon layer it's very obvious that teflon layer will [00:05:54.384] pr prevent the moisture from entering into the battery, but at the same time it has to be air permeable also because it has to allow the oxygen to enter the battery. [00:06:06.304] battery. So this is what we are going to have at the cathode. Though there is no cathode active material still there has to be a [00:06:16.932] cathode inside the battery. Of course this cathode is going to form a very thin layer in the a thin small part of the battery and the remaining part will be the uh anode. So there are [00:06:30.320] Uh through this air permeable Teflon layer, so we have air access holes on the cathode side, which are going to provide the pathway for oxygen to enter the battery. [00:06:42.528] Now coming to the electrolyte, as we said, it is an alkaline battery, so 30% KOH is used as the uh electrolyte, and the separator here it can be uh polypropylene, which is soaked in electrolyte. So we need to see [00:07:01.444] Why it has to be like this. The separator is soaked in electrolyte. We'll learn about it in a few minutes now. [00:07:10.020] That is the construction. Now let us see how it is placed inside the battery. So what where exactly do we have all these components which we saw just now in the zinc air battery? So let us start with the [00:07:24.932] Positive that is the cathode. So here we have the this side is the cathode, and you can see some holes over here. So through these holes, oxygen, air hole, you can see. So oxygen is going to enter. [00:07:40.608] Through these holes. So you have oxygen entering from this side. So this has to be cathode because oxygen uh undergoes reduction. So that is going to happen at the cathode. So you have a thin cathode over here where oxygen is going to enter, and at the same time, we know that there has to be [00:07:59.696] Uh uh uh the uh hydrophobic membrane that is your teflon layer, which we uh mentioned just now. So there is a teflon layer, there is a [00:08:08.896] Carbon and catalyst also to in order to for one carbon is for conductivity and catalyst is to speed up the reaction. So that is your cathode, and then we have the separator which is [00:08:23.204] Soaked in electrolyte. Separator soaked in electrolyte. Now [00:08:31.972] Okay, so this is where you have the uh separator. So you can see the separator here, and the rest of the battery is the negative electro uh negative uh [00:08:44.512] active material that is anode active material and that is zinc. So this whole area is zinc. So we can see that we because oxygen is being taken from the [00:08:57.508] Atmosphere directly, so it need not be stored inside the battery. Therefore, the entire battery space is actually filled with the uh anode active material and that is zinc. And then we have a [00:09:12.512] cap here the metal cover which is the act acting as the negative terminal and we have a cup over here which is going to act as the positive terminal. Now since these two are made of metal there has to be a gasket or something which is going to prevent [00:09:30.192] The gasket is going to [00:09:32.560] Prevent the anode and cathode coming in contact with each other. At the same time, it is going to seal the two metal plates together, the metal cup and the metal cover together. This plastic gasket is placed in between, and therefore this. [00:09:50.320] This is the all the components which we saw just now. [00:09:54.480] They are all placed like this inside the battery. So we have cathode here, then the separator which is soaked

in electrolyte. So and then we have this anode active material that is zinc. Then we have the anode terminal and the cathode terminal, and the in between there is a

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:12.112] gasket which is for preventing the anode and cathode from coming in direct contact with each other. Let us look at the working of this battery. Now since we have zinc as the anode material there can be only one reaction which can happen. [00:10:29.088] So that is zinc is giving you Zn2 plus plus 2 electron, but since we have added some uh there is an electrolyte over there, the reaction does stop here, does not stop here, and we have zinc plus 2 OH minus is giving you ZnO plus H2. So overall reaction which happens at the anode is zinc. [00:10:48.784] Plus 2OH minus is giving you ZnO plus H2O plus 2 electrons. We should put the two electrons here because we are talking of the anodic reaction. Now coming to the cathodic reaction, what is happening in the cathode? Oxygen has to undergo reduction. [00:11:06.448] The reaction is half O2 plus H2O plus 2 electron is giving you 2OH minus. So when we add this anode reaction and the cathode reaction, we get the overall reaction and it comes out to be a very simple reaction. [00:11:22.720] Plus half O2 is giving you ZnO. So the overall reaction for a zinc air battery is simple. Zinc is reacting with oxygen to give you Zn oxide. So what we saw the picture of the zinc air battery, what is actually [00:11:40.368] looks like. But if you look at the schematic diagram which uh when I talked about battery we uh I showed you the diagram where we talked about working of the battery we have the anode we have the cathode and we have the electrolyte and the separator. So here we'll see you can see here zinc [00:11:58.116] Is the anode material, and at the cathode we have this uh catalyst which is placed here, and we have this. [00:12:05.504] Holes through which oxygen can enter. So how what is if you are going to be looking at working of this battery, and here there is a separator, of course, uh the electrolyte. This is just a schematic representation and not the actual representation because the electrolyte is not. [00:12:23.184] Present in bulk like this, but it is present in the separator only. The separator soaked in electrolyte. So what is the working if you see from this oxygen is entering here and reaction happens, OH minus. [00:12:38.416] is produced here and this move towards the moves towards the zinc electrode and then we have this reaction happen at the animal. So electrons are being produced here, they are getting absorbed here and therefore we get power from this battery. So this is a uh [00:12:56.048] Course batteries all batteries are galvanic cells, so this is how this uh the battery works. This is about the working of the battery, and we have [00:13:04.484] EMF is 1.4 volts. Now one important fact which we should know about zinc care battery that why is it that we don't have so much of electrolyte and we need to use electrolyte in a very uh small amount that is [00:13:20.912] The separator soaked in electrolyte. The reason for that, if you look at the final reaction, zinc plus half O2 is giving you Z and O. OH minus is not figuring in the final reaction. That means the amount of OH minus used. [00:13:37.968] is not going to uh affect the [00:13:42.384] working of the battery. So of course OH- is required here at the anode and also at the uh cathode we see it is getting produced there is an exchange of OH- but the amount of OH minus is not important. Whatever is produced at the cathode is being used at the anode. So [00:14:00.928] Just a small amount of QH just to facilitate the movement of the ions is what is required. Hence, we need not use a large amount of uh electrolyte and the separator soaked in electrolyte is sufficient, a small amount of electrolyte. [00:14:18.576] Which is soaked inside the separator is enough for the battery to function and therefore [00:14:24.548] One important characteristic of this battery is that these batteries can be miniaturized. So what do we mean by

the batteries can be miniaturized is that they can be made in very small sizes. [00:14:40.740] Because the amount of electrolyte is not important. If you look at lead acid battery, we know that in the final equation we have H2SO4. So PB plus P BO2 plus H2SO4 is giving you P V SO4 plus H2O. So that amount. [00:14:58.512] Of H2SO4 matters, and therefore, lead acid batteries always come in large sizes. So, therefore.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.340] You have to make them large, it cannot be made small. Whereas here OH minus the electrolyte is not there in the final equation. So that means final reaction. So amount of electrolyte can be very small, and therefore these batteries can be made into very small sizes. [00:15:25.732] The word uh the term we use to describe that is that these batteries can be miniaturized. Now coming to advantages, they have very high density. [00:15:36.112] high energy density as I explained uh right in the beginning and that is because air which is the cathode active material is not getting stored inside the battery it is taken directly from the atmosphere therefore the weight of the battery is sufficiently reduced and therefore this result [00:15:53.840] in a very high energy density of the zinc air battery. Second, it has a very long shelf life. [00:16:04.452] We cannot say unlimited shelf life because there could be some other reasons like corrosion, etc., which can happen when you store the battery for a long time. But nevertheless, the shelf life is very long because as long as the battery is kept sealed. [00:16:23.172] The air cannot enter. So that means oxygen cannot enter the holes which are there on the battery. If they are sealed, oxygen cannot enter the battery and therefore the reaction cannot happen. Therefore, you can store these batteries for a very very long time and therefore they are. [00:16:41.060] They have very long shelf life. They also do not pose any uh ecological problems. There is a zinc and oxygen which are being used in the battery, and it is a low cost battery because zinc. [00:16:56.560] uh oxygen is anyway taken from directly from the atmosphere and zinc when used in manufacture in mass mass scale the cost becomes very uh low so it's a low cost battery. Now there are certain disadvantages the [00:17:13.188] output the power output is very limited. [00:17:16.912] And the second uh important uh point which we need to know is that along with air, when we open the battery, then there is carbon dioxide also in the atmosphere. So along with oxygen, there the carbon dioxide can also enter the uh battery. [00:17:34.576] And since we have used alkali here as the electrolyte, what the electrolyte is KOH, therefore, there could be a reaction between this carbon dioxide and KOH, which is going to result in formation of K2CO3. [00:17:52.832] And that is going to block the pathways and therefore reduce the efficiency of this battery. So we can say that this battery depends on environmental conditions. Now coming to applications of this battery. [00:18:11.792] You can see the first application is in hearing aids. Why? Because these batteries, as I told you just now, can be miniaturized, they can be made into very small sizes. So the batteries which are used in hearing aids are zinc air batteries, they also used in [00:18:29.488] in various medical devices or in voice transmitters and large zinc care batteries are used in railroad signaling. [00:18:40.868] So this is about the Zincare battery, and in the next class we are going to look at the batteries which are used profusely in the present day, and that is lithium batteries. Thank you.

EV0000082 · EV0000083 · EV0000084 · EV0000085

Full transcript: 2789 normalized words, 79 timestamped segments, 1140.181333 seconds duration. Low-confidence segments flagged: 0.

EV0000070 · EV0000071 · EV0000072 · EV0000073

6 Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1)

This lesson is indexed from the source video titled Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000086 · EV0000087

Graphs and diagrams

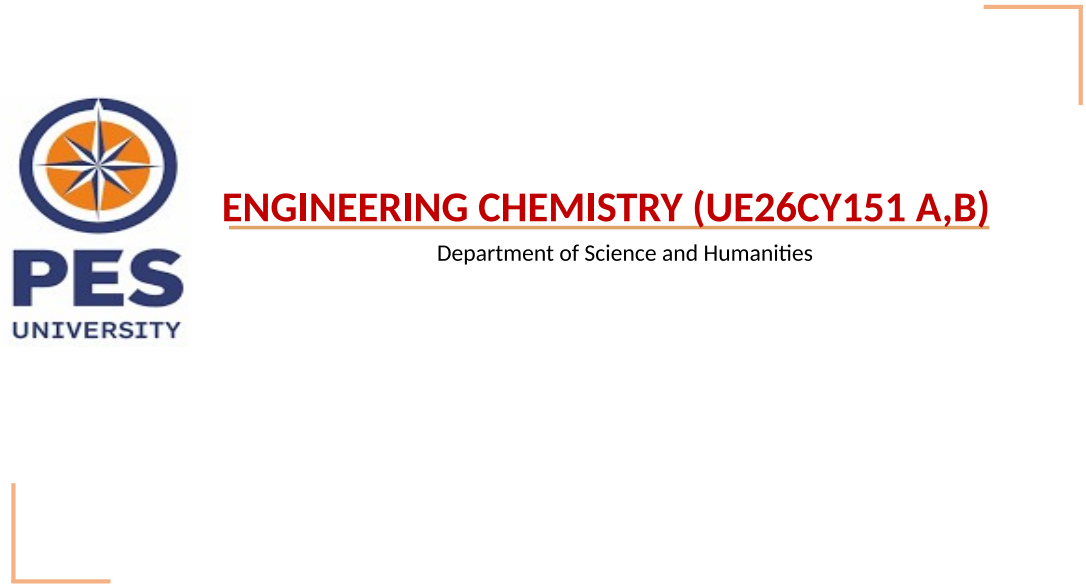

Figure G000047. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 1. EV0000325

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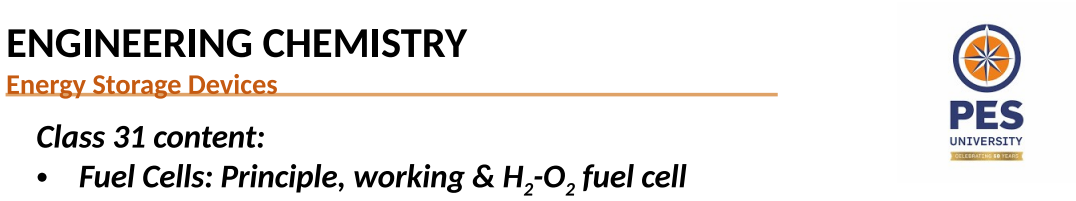

Figure G000048. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 2.

EV0000328

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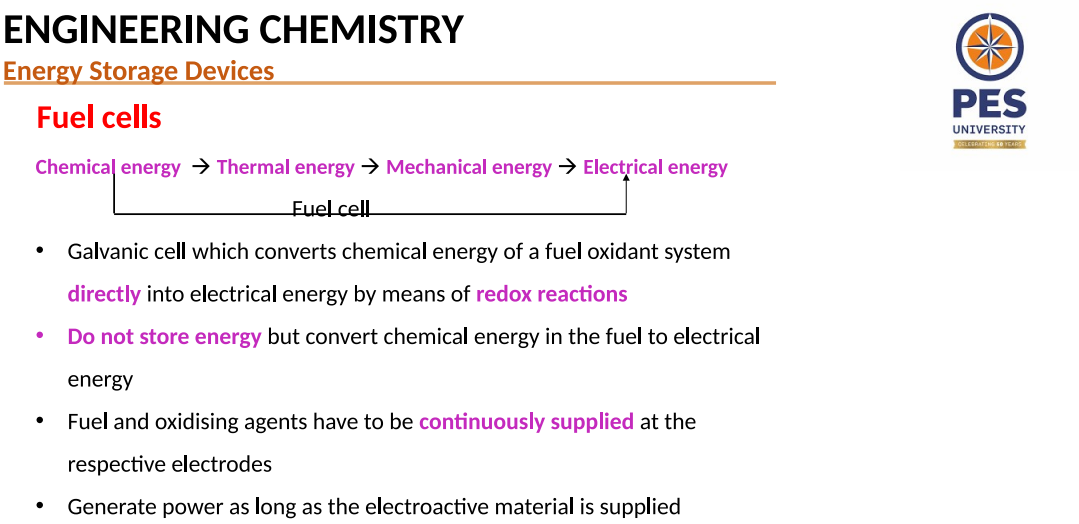

Figure G000049. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 3.

EV0000331

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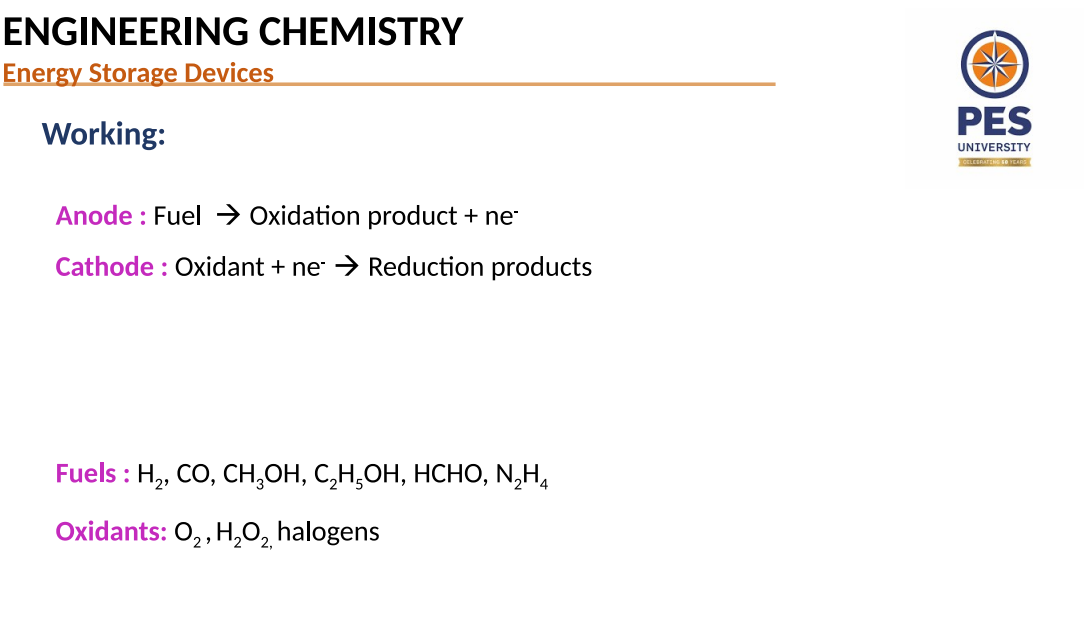

Figure G000050. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 4.

EV0000334

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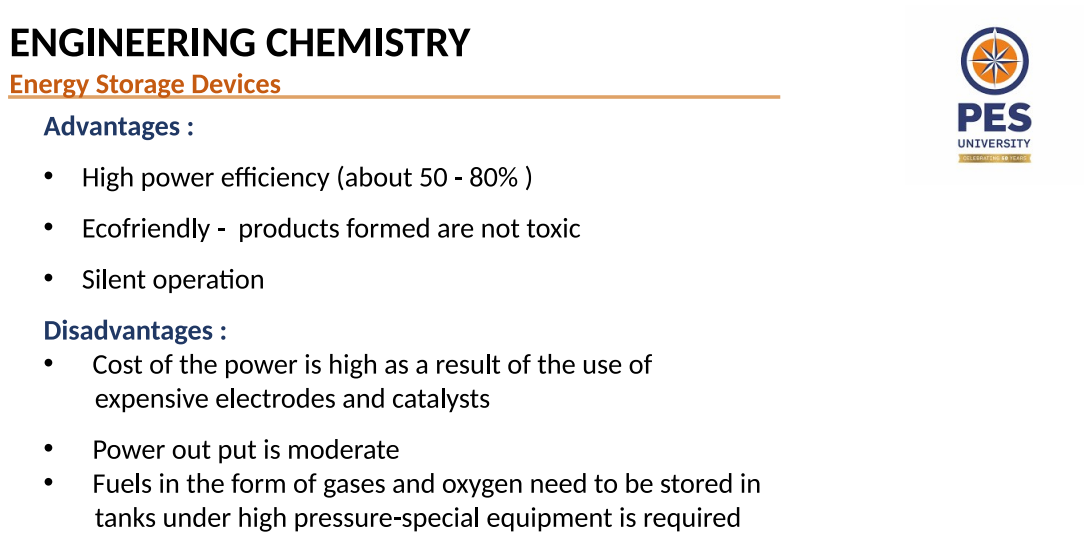

Figure G000051. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 5.

EV0000337

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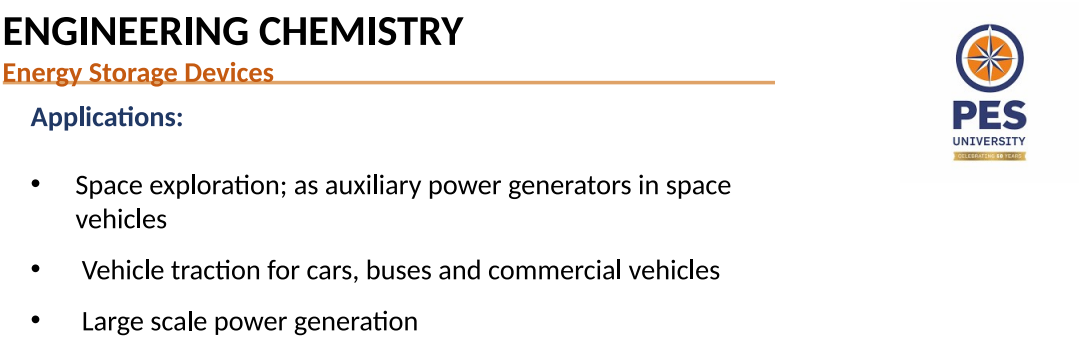

Figure G000052. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 6.

EV0000340

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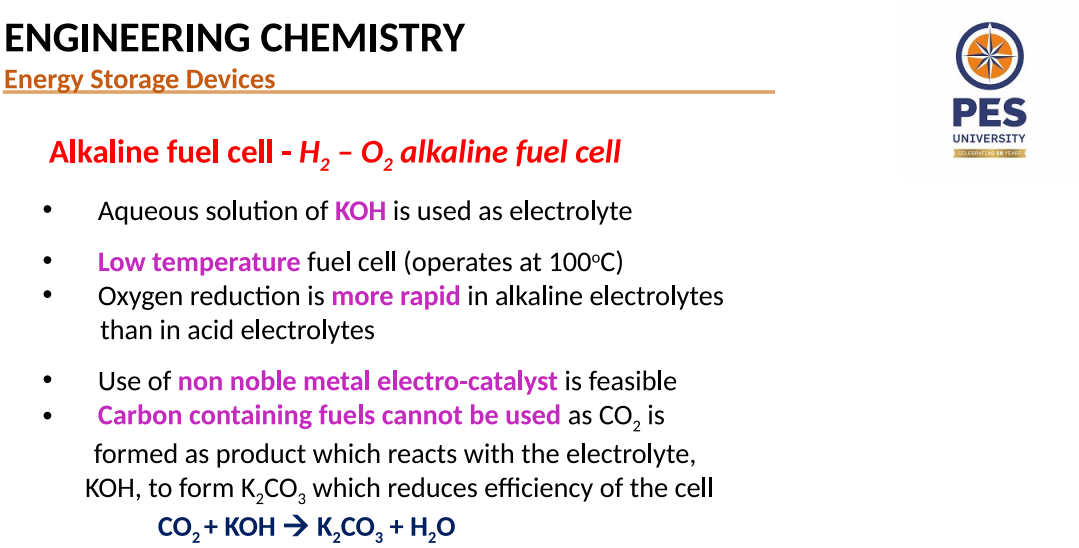

Figure G000053. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 7.

EV0000343

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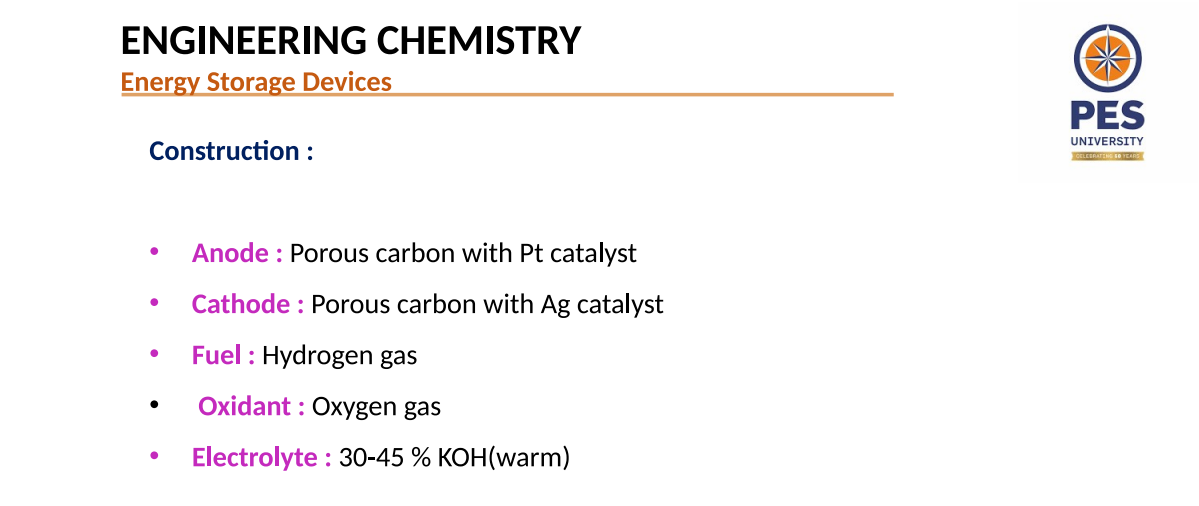

Figure G000054. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 8.

EV0000346

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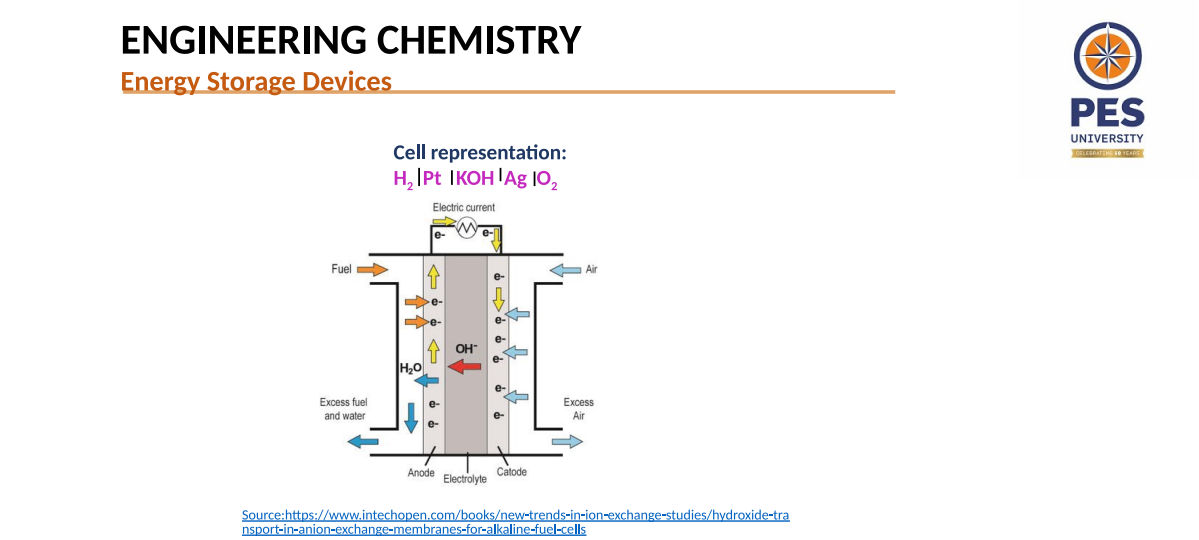

Figure G000055. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 9.

EV0000349

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ENGINEERING CHEMISTRY

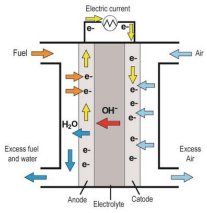
Energy Storage Devices

Working:

At Anode : $H_2 + 2OH^- \longrightarrow 2H_2O + 2e^-$

At Cathode : $1/2O_2 + H_2O + 2e^- \longrightarrow 2OH^-$

Over all reaction : $H_2 + 1/2O_2 \longrightarrow H_2O$



- H_2 gas diffuses through anode ,gets adsorbed on the electrode surface ,reacts with OH- to form water
- At cathode O_2 diffuses through electrode, is adsorbed and reduced to OH-
- Product is water which dilutes the KOH
- Cell operates at 100°C, so that water from KOH escapes as steam
- The water was used by astronauts for drinking on Apollo spacecraft

Emf = 1.23 V

Figure G000056. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 10.

EV0000352

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ENGINEERING CHEMISTRY

Energy Storage Devices

Advantages:

- Operates at low temperature
- Alkali is used as electrolyte hence non noble metal catalyst can be used so less expensive

Disadvantages:

- Reactants must be free from C, because on oxidation CO_2 is formed .The alkali reacts with CO_2 to form carbonates which reduce efficiency of the cells; pure fuel and oxidant which are free of carbon compounds must be used
- Liquid electrolytes pose handling problems

Figure G000057. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 11.

EV0000355

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000058. Fuel cells Principle and working H2 O2 alkaline fuel cell · source slide page 12.

EV0000358

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000324 · EV0000325 · EV0000326
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Class 31 content: Fuel Cells: Principle, working & H2-O2 fuel cell EV0000327 · EV0000328 · EV0000329
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Fuel cells Chemical energy ☒ Thermal energy ☒ Mechanical energy ☒ Electrical energy Fuel cell Galvanic cell which converts chemical energy of a fuel oxidant system directly into electrical energy by means of redox reactions Do not store energy but convert chemical energy in the fuel to electrical energy Fuel and oxidising agents have to be continuously supplied at the respective electrodes Generate power as long as the electroactive material is supplied EV0000330 · EV0000331 · EV0000332
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working: Anode : Fuel ☒ Oxidation product + ne- Cathode : Oxidant + ne- ☒ Reduction products Fuels : H2, CO, CH3OH, C2H5OH, HCHO, N2H4 Oxidants: O2 , H2O2, halogens EV0000333 · EV0000334 · EV0000335
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages : High power efficiency (about 50 - 80%) Ecofriendly - products formed are not toxic Silent operation Disadvantages : Cost of the power is high as a result of the use of expensive electrodes and catalysts Power out put is moderate Fuels in the form of gases and oxygen need to be stored in tanks under high pressure-special equipment is required EV0000336 · EV0000337 · EV0000338

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Applications: Space exploration; as auxiliary power generators in space vehicles Vehicle traction for cars, buses and commercial vehicles Large scale power generation EV0000339 · EV0000340 · EV0000341
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Alkaline fuel cell - H2 – O2 alkaline fuel cell Aqueous solution of KOH is used as electrolyte Low temperature fuel cell (operates at 100oC) Oxygen reduction is more rapid in alkaline electrolytes than in acid electrolytes Use of non noble metal electro-catalyst is feasible Carbon containing fuels cannot be used as CO2 is formed as product which reacts with the electrolyte, KOH, to form K2CO3 which reduces efficiency of the cell CO2 + KOH ☒ K2CO3 + H2O EV0000342 · EV0000343 · EV0000344
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Construction : Anode : Porous carbon with Pt catalyst Cathode : Porous carbon with Ag catalyst Fuel : Hydrogen gas Oxidant : Oxygen gas Electrolyte : 30-45 % KOH(warm) EV0000345 · EV0000346 · EV0000347
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Cell representation: H2 Pt KOH Ag O2 [Image text, OCR] Electriccurrent e Fuel Air e e- OH- H20 e- Excessfuel e Excess andwater Air Anode Electrolyte Catode Source:https://www.intechopen.com/books/new-trends-in-ion-exchange-studies/hydroxide-transport-in-anion-exchange-membranes-for-alkaline-fuel-cells EV0000348 · EV0000349 · EV0000350
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working: H2 gas diffuses through anode ,gets adsorbed on the electrode surface ,reacts with OH- to form water At cathode O2 diffuses through electrode, is adsorbed and reduced to OH- Product is water which dilutes the KOH Cell operates at 100oC, so that water from KOH escapes as steam The water was used by astronauts for drinking on Apollo spacecraft Emf = 1.23 V EV0000351 · EV0000352 · EV0000353
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages: Operates at low temperature Alkali is used as electrolyte hence non noble metal catalyst can be used so less expensive Disadvantages: Reactants must be free from C, because on oxidation CO2 is formed .The alkali reacts with CO2 to form carbonates which reduce efficiency of the cells; pure fuel and oxidant which are free of carbon compounds must be used Liquid electrolytes pose handling problems EV0000354 · EV0000355 · EV0000356
12. Source slide page 12 (extracted text): THANK YOU Department of Science and Humanities EV0000357 · EV0000358 · EV0000359

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module 5 on energy storage devices on fuel cells. So today we are going to learn about fuel cells principle and working advantages, disadvantages and applications. [00:00:19.908] So, what are fuel cells? In a thermal power plant, we know that combus of fuel is done and power is generated from there. But this happens in several steps. So when combustion of fuel takes place, the chemical energy is stored inside the fuel. [00:00:39.152] When the fuel burns, there is heat generated. So we converted chemical energy into thermal energy. This thermal energy is used to run turbines. So we have

converted now again thermal energy to mechanical energy. These turbines now are used to generate power. So mechanical energy got converted. [00:00:58.352] To electrical energy, but at each step there is a loss of efficiency. So finally, the efficiency is just 35 to 40 percent. [00:01:09.892] The idea in a fuel cell is can we convert the chemical energy directly into the electrical energy? If that is done. [00:01:20.208] We will definitely have a better efficiency, and that is exactly what is done in a fuel cell. The chemical energy is directly converted into electrical energy, and we find that the efficiency is about 75 to 80 percent. So how is this done? [00:01:38.948] The combustion of fuel, that is, say for example, hydrogen and oxygen. Hydrogen is the fuel and oxygen is the oxidant is done in the form of galvanic cell. [00:01:52.592] Now we know that when we want to make a galvanic cell, then we need a redox reaction. So what is the redox reaction? Let us check whether the combustion of fuel is a redox reaction. So let's say we have hydrogen plus oxygen is giving you water. [00:02:12.132] So we know that when hydrogen undergoes combustion in oxygen, we get water. So is this a redox reaction? [00:02:20.528] We can see very clear hydrogen is getting oxidized to water and oxygen is getting reduced to water. So therefore, we know that combustion of a fuel it is a redox reaction. And we know in a galvanic cell we can use this redox reaction. [00:02:39.168] How we can make the reaction happen in two different compartments, meaning that we will not allow the hydrogen and oxygen to uh come in contact, but we will allow the hydrogen to undergo oxidation at the anode and oxygen to undergo reduction at the cathode. Now [00:02:56.816] Since hydrogen is undergoing oxidation [00:03:00.048] At the anode electrons will be generated and these electrons are going to move through the external wire towards the cathode where there is oxygen and that is going to absorb the electrons and undergo reduction. So basically we have allowed [00:03:16.912] the combustion reaction to happen in two separate compartments at anode and cathode and thereby we are generating electrical power. So the chemical energy in the reaction has been converted directly into the redox reaction. So this is the concept of a fuel cell. [00:03:36.388] So whatever is the chemical energy of the fuel oxygen system is getting directly converted into electrical energy by means of a redox reaction. So we can understand that here, what is the difference between a regular battery and a fuel cell? [00:03:53.200] Now in a regular battery we know that the reactants, the electroactive material are stored inside the battery. [00:04:01.376] And they are they undergo reaction, and that is how we get power from the battery. But a fuel cell, though it is also a galvanic cell, is different from a battery. Why? Because the fuel and the oxidant, that is the electroactive material, have to be supplied. [00:04:19.024] Continuously from outside as at the respective electrodes. So meaning that we do not store energy inside the battery, but the chemical energy of the fuel is converted into electrical energy. [00:04:36.720] So we can say that though batteries are energy storage devices, whereas fuel cells are energy conversion devices. So in a [00:04:50.084] Fuel cell, the fuel and the oxidizing agents have to be continuously supplied. They will be able to generate power only as long as the electroactive material is supplied at the anode and the cathode.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.900] Let us look at the construction. What would be what would the fuel cell look like? We understood the concept. Let us see what do we need to make a fuel cell. [00:05:20.000] Definitely, because we are talking of an cathode, we need two electrodes. So the two electrodes have to be chosen very carefully based on the reaction which is taking place. But apart from the electrodes, one very important uh [00:05:35.840] part component of the battery is going to be the catalyst. [00:05:41.252] Because the

reactions which take place during the fuel oxidation or the oxygen getting reduced or the oxygen getting reduced, they are very slow reactions. So we need to [00:05:58.564] make the reaction faster and for that catalyst has to be present. So this the two electrodes which you use the anode and the cathode they will be impregnated with the catalyst. So this whole system is very very important for the [00:06:16.800] uh fuel cell. So the catalyst also is very important for the fuel cell because the reactions are very slow. And apart from that of course we need the electrolyte like any galvanic cell. So you have electrodes you have the uh electrolyte which is uh present uh [00:06:35.840] for the ionic conductivity. So the electrolyte is uh just like any other galvanic cell, we have electrolyte which is used for ionic conductivity and the electroactive material they are supplied at the electrode. They are not being stored inside the [00:06:51.920] Uh cell in a fuel cell, but they have to be supplied externally. So we have electrodes, catalyst. [00:07:00.976] The electrolyte and the electroactive material which are being supplied from outside. So let us see how do we place all these components inside the battery. So the electrodes which we have they are the anode and the [00:07:20.740] Catholic. Now these electrodes are [00:07:25.188] Separating the fuel cell into three compartments. They make it into three compartments. The anode compartment, the cathode compartment, and the electrolyte. So we have these electrodes here. These are the [00:07:41.872] Two electrodes which are present, of course, and we know that the catalyst has to be present on the electrode. So it is the electrode is impregnated with the catalyst. Now on this side, we are going to allow the [00:07:58.848] fuel to get in. So here we have fuel coming inside and excess fuel is let out from here. Similarly on this side we have the inlet and the outlet for the oxidant. So in this side [00:08:17.984] We have on this side we have the oxidant coming inside, and here we have the excess oxidant coming out of the cell, and in between we have the electrolyte. So you can see very clearly there are three compartments. Now, since the fuel is being supplied from this electrode, I would say that this. [00:08:35.664] is the anode because the fuel undergoes oxidation and since the oxidant is coming in from here, this I would call it as the cathode. [00:08:46.404] and in between we have the electrolyte. So anode there is oxidation, electrons are being generated and we have power [00:08:58.020] Which we are getting from the battery. So electrons move through the external wire, being absorbed at the cathode. So this would be a schematic representation of the fuel cell. Now let us write the cell representation. So what do we see? We always start with the anode, like in any galvanic cell. [00:09:15.792] So we have the fuel which is in contact with the electrode, which is again in contact with the electrolyte. Then we have the electrode at the cathode, and then we have the oxidant. So this is how we write the cell representation for a fuel cell. [00:09:34.276] Now, let us find out the working of a fuel cell. We know that at the anode, the fuel is undergoing oxidation, so electrons are being generated, and when the fuel undergoes oxidation, obviously the product is going to be the oxidation product of the fuel. Similarly, [00:09:54.192] At the cathode we have the oxidant which is taking up the electrons and it will give rise to the reduction product whatever is the

EV0000090 · EV0000091 · EV0000092 · EV0000093

Transcript block 3 (source interval 000600-000899 seconds): [00:10:03.584] uh oxygen corresponding reduction product is going to be formed at the cathode. So this is how general working of any fuel cell happens. Of course we can have a variety of fuels which we use. [00:10:17.088] And also a variety of oxidants that can be used. So what are the fuels that we generally use? Hydrogen is the most popular. [00:10:26.240] fuel which is used. And then we also have carbon monoxide, which is also good fuel, which will get oxidized to carbon dioxide. And then in we have methanol, which is

catching up these days, and methanol oxygen fuel cell is also becoming very popular. And in place of methanol, we can have ethical [00:10:43.888] Then there is formaldehyde and hydrazine. So all these can be used as fuel. So basically, they are all. [00:10:52.420] They can undergo oxidation very easily. So and at the oxidant, for the oxidants we have air or oxygen, which is most commonly used, and then we have hydrogen peroxide, which is also a [00:11:07.024] uh oxidizing agents so it can be used and then halogens we know are good oxidizing agents. So these are the fuels and the oxidants that can be used in a fuel cell. [00:11:20.036] Now, what are the advantages of the fuel cell? Of course, we understand the efficiency is very high, much higher than what we see in the uh thermal power station. [00:11:33.828] The efficiency is about 50 to 80 percent in the case of a fuel cell, and most importantly, they are eco-friendly because the products which are formed are not toxic. So in the case, suppose we have hydrogen and oxygen, we get [00:11:51.072] uh water in the if we are using any carbon monoxide we get carbon dioxide. If we are using methanol we are going to get carbon dioxide and water. So they are all uh eco-friendly products which are getting formed unlike when uh this uh [00:12:06.532] They undergo combustion, the which are used in uh vehicles, we know that there's a lot of pollution happening due to the [00:12:14.980] Products being given out into the uh atmosphere. And then another thing, another important fact is that the advantage is that it they have a very silent operation. Unlike in a thermal power plant, there is there is a there is no noise when such a [00:12:32.144] uh power generation is happening in the fuel cell. But there are certain disadvantages also. The cost of power is very high. Why? Because we have to go for expensive [00:12:44.144] uh electrodes and also catalysts. More uh important are the catalysts because they have to be present for the reaction to take place. And these catalysts, electrocatalysts especially, are expensive. [00:12:57.760] So and also the power output is moderate in this case. And one more important fact is that fuels which are used in the form of uh gases. [00:13:10.468] and oxygen they have to be stored in tanks under high pressure. So special equipment is required. Hydrogen is a very good fuel, but storing hydrogen is the biggest challenge [00:13:26.720] for using the fuel cells. So these are certain disadvantages which are there for the fuel cells. [00:13:37.156] What are the applications? Yes, space exploration. All of us are aware that it started a long time back and NASA uses these fuel cells right from the Apollo space program. So they are very useful for. [00:13:53.956] powering the space vehicles. So they are an auxiliary power. They act as auxiliary power generators in space vehicles and also vehicle traction. [00:14:06.340] for cars, buses and commercial vehicles, they are looked upon as the next generation uh power uh generators for cars, buses and commercial vehicles. [00:14:18.768] So vehicle traction, traction means moving, when when you're you using not fuel, but you're using the battery or the galvanic cell is being used, in this case the fuel cell is being used to power the motor to run the machine. [00:14:34.560] So and then of course large scale power generation if that would be possible. Stationary applications uh have been uh done to a great extent, but still mobile applications are still being researched. So if that comes then [00:14:53.220] we are going to be uh very well placed as far as power powering the vehicles is uh concerned because we know that the non-renewable resources like petroleum, etcetera, the fossil fuels.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:09.136] They are depleting fast, so if we could get these fuel cells to power the cars, buses and commercial vehicles and also for large scale power generation, it is going to be very useful in future. [00:15:25.552] Let us do efficiency of a fuel cell. So efficiency of a fuel cell as any efficiency of a machine is thermodynamic

efficiency, which is given by ratio of the work output to the heat input. [00:15:42.704] Any efficiency is always measured by what is the output work done by the input, uh input heat in this case. So what is the work output in the case of a fuel cell? [00:15:55.024] We can say it is the electrical energy which we are electrical work which are which we are getting from the fuel cell, and that is given by the Gibbs free energy. [00:16:06.948] Gives free energy because we know that delta G is equal to minus N FE. [00:16:16.484] So we know that the electrical output which you get, the the electrical energy which we get from the battery is because of the Gibbs free energy. So if we know the Gibbs free energy, we know what is the work output. And what is the heat input in this case? It is the enthalpy change which happens. [00:16:34.176] During the cell reaction. Which reaction are we talking about? We are talking about the combustion of the fuel. So whenever the fuel burns in oxygen or any other oxidant, then the heat which is released is the is the [00:16:50.016] uh heat input that is because the chemical energy is what we are inputting in in uh using as input in the uh fuel cell. So therefore we have [00:16:59.780] Heat input is enthalpy change of the cell reaction and the work output is the Gibbs free energy. So Gibbs free energy delta G as I mentioned just now is a measure of the electrical work done, and enthalpy change delta H is the measure of the heat released during the reaction. So when we calculate [00:17:17.552] The efficiency is given in the form of percentage. That would be it's denoted by the symbol, this uh this symbol percentage that is delta G by delta H. That is the [00:17:29.520] uh work output by the heat input multiplied by 100 and since delta g is given by minus nf e we can write this as minus nf e by delta h into 100. So this is how we calculate the efficiency of a fuel cell. [00:17:46.400] In the next class we will look at hydrogen, oxygen, alkaline fuel cells. Thank you.

EV0000098 · EV0000099 · EV0000100 · EV0000101

Full transcript: 2530 normalized words, 74 timestamped segments, 1074.304 seconds duration. Low-confidence segments flagged: 0.

EV0000086 · EV0000087 · EV0000088 · EV0000089

7 Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2)

This lesson is indexed from the source video titled Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000102 · EV0000103

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.516] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module 5 on energy storage devices and we'll discuss about fuel cells. In today's class, we are going to learn about hydrogen oxygen alkaline fuel cell [00:00:19.844] The principle, construction and working, the advantages and the disadvantages. Alkaline fuel cells as the name suggests, the electrolyte which we are going to

use is going to be an alkali. [00:00:36.708] If you remember when we talked about fuel cells in general, we use a fuel at the anode and an oxidant at the cathode, and there is an electrolyte. [00:00:50.212] So when we talk of alkaline fuel cell, alkali is used as the electrolyte, and the most commonly used alkali is KOH. These are low temperature fuel cells and they operate at about 100 degrees centigrade. [00:01:11.012] Most common oxid which is used at the cathode is oxygen. Oxygen reduction is a very slow process. But if we use alkaline electrolytes, [00:01:29.764] As compared to acid electrolytes, it is found that the oxygen reduction is much more rapid in alkaline electrolytes. And because of that, we need not use noble metal catalysts. [00:01:48.132] Oxygen reduction or hydrogen oxidation. Hydrogen oxidation which happens at the anode or oxygen reduction which happens at the cathode being slow processes we need to use catalysts. [00:02:02.564] But because it is faster, the oxygen reduction is faster in the alkaline electrolyte as compared to acid electrolyte, we can use non noble metal electrocatalysts. But [00:02:19.492] One problem with alkaline fuel cells is that carbon containing fuels cannot be used as when they undergo oxidation, carbon dioxide is formed as a product. [00:02:35.140] Now this reacts with the electrolyte KOH and it forms K2CO3. [00:02:42.756] This K2CO3 is going to block the electrodes, the electrode pathways, and therefore this reduces the efficiency of the cell. The reaction would be carbon dioxide, which is formed by the due to the oxidation of [00:02:59.328] carbon containing fuels with KOH is going to give you K2 CO3 plus H two O. [00:03:08.004] Let us discuss the construction of such a fuel cell. Now we will take the example of hydrogen oxygen fuel cell, which is the most common type of alkaline fuel cell. [00:03:27.236] used and has many applications and the electrolyte obviously is going to be an alkali. So just by saying that we are going to use hydrogen, oxygen, alkaline fuel cell, we can easily make out that [00:03:45.508] At the anode it will be hydrogen which is going to be used. At the cathode we will use oxygen and the electrolyte is going to be an alkali. [00:03:58.916] So we can see here the fuel which we are using is hydrogen gas. The oxidant which we use is oxygen. [00:04:08.260] and the electrolyte is KOH that is 30 to 45% of KOH is used. But one thing we need to note here that the KOH used has to be warm. We'll discuss the reason in the upcoming slides. [00:04:27.332] These are the electroactive materials. Hydrogen and oxygen are the electroactive materials, but we also need electrodes. [00:04:36.356] So the anode that is where hydrogen is being supplied as the fuel, it is porous carbon impregnated with the platinum catalyst and at the cathode we have [00:04:50.128] porous carbon impregnated with silver catalyst. So at the cathode we have the oxygen which is being supplied at the anode we have the fuel which is being supplied and we have electrolyte that is KOH.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.356] Let us look at the diagram. Now here the cell representation is we always start with the anode [00:05:17.892] for any galvanic cell for any cell for that matter we always start with the anode. So what is being used at the anode? Hydrogen because hydrogen has to undergo oxidation and at the anode the process is always oxidation. So hydrogen is used and we know that we are using [00:05:35.664] carbon electrodes which are impregnated with platinum. So we will put the platinum here and then comes the electrolyte. So here we have KOH as the electrolyte and then coming to the cathode side we are using oxygen as the cathodic electroactive material and the [00:05:54.512] Carbon electrode is impregnated with silver. So the cell representation is going to be written like this hydrogen in contact with platinum, which is in contact with KH and that is again in contact with silver and finally it is in contact with oxygen. [00:06:10.820] So let us see how we are going to draw this in a

diagram. So as we know fuel cells there are three compartments. One compartment is the anodic compartment where we let in the fuel. And we also have a [00:06:28.516] exit for the excess fuel and on the cathode side that is the other compartment where we have the cathode here is the entry for air or oxygen in this case and then there is also an exit point for the excess air or oxygen. [00:06:48.480] and in the center we have the electrolyte that is the KOH which is being used. [00:06:56.644] And apart from that we have the electrodes which are shown here that is the anode and this side is the cathode. So in this particular case the anode will be carbon electrode impregnated with platinum and here we have [00:07:12.816] the carbon electrode which is impregnated with silver. [00:07:20.100] Let us understand the working of such a cell. Now at the annual, what do we have? [00:07:27.440] When we say working, we are basically going to understand the reactions which are taking place at the anode and cathode and the overall reaction. So at the anode we know that we are supplying hydrogen as the fuel and then it is coming in contact with the electrolyte. [00:07:45.104] So here we have the electrolyte that is QH. So hydrogen reacts with [00:07:50.176] OH minus and it is going to form water and two electrons are given out here because it's a oxidation reaction. And then at the cathode, these electrons are getting absorbed. So we have oxygen which we are supplying at the [00:08:06.432] cathode here. So oxygen is going to react with water and take up these two electrons. So it's a reduction reaction and it is going to give back two OH minus. So [00:08:18.864] Overall reaction when you add up these two reactions the half cell reactions they are known as the anode and the cathode, we find that it is hydrogen plus half O2 is giving you water. So basically it is just combustion of hydrogen which is taking place in the fuel cell. [00:08:38.564] But what we have done is we have separated the [00:08:43.120] oxidation and the reduction reaction into separate compartments. So as a result what happens is when hydrogen undergoes oxidation electrons are generated. These electrons move through the external wire and they are taken up at the cathode because of the [00:09:00.752] reduction of oxygen happening here. So this way by the separating the reactions into anode and cathode just like we do for any galvanic cell, we have been able to have the stream of electrons flowing and therefore we can generate power from this cell. [00:09:17.792] So what exactly happens, these reactions, how do they take place? So we know that hydrogen, the fuel, [00:09:25.312] is being given from here so let us say it is hydrogen here in this case so hydrogen or any fuel for that matter is given here in this case we are talking of hydrogen oxygen fuel cell so hydrogen is being given here and it diffuses through the anode and [00:09:42.960] Hence, it is very important that the anode has to be [00:09:49.476] porous in nature. So hydrogen diffuses through the anode and it gets adsorbed on the electrode surface. So if it is porous obviously the surface area will be more and since the catalyst is also present over there we can

EV0000106 · EV0000107 · EV0000108 · EV0000109

Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.136] Expedite the reaction. So when this hydrogen gets in and it reacts with the it gets absorbed on the electrode, it reacts with OH minus to form water. So water is getting formed at the anodic reaction, and you can see here that that [00:10:24.816] vent has been provided here for the water to escape. Now here when the uh along with the excess of hydrogen water also escapes from here. In some cells we find that there is a vent provided over here and from here the water escapes. [00:10:44.240] So it doesn't matter, it can be here or here, basically it is getting formed uh during the anodic reaction. So at the exit we find that there is excess of the fuel that is hydrogen and water which is getting uh being removed from here. [00:10:59.460] Now at the cathode, similarly we have oxygen which is coming in from the

[00:11:05.712] uh this vent and then it is going to also diffuse through the cathode. So the cathode also has to be porous in nature. Whenever we make any fuel cells it is very essential that the electrodes have to be porous in nature because they have [00:11:23.440] to absorb the fuel the oxidant and also the catalyst is present on the electrode. So therefore the hydrogen oxygen is getting ads at the [00:11:34.304] uh cathode surface and it gets reduced to OH minus ions. So these OH minus ions are going to move here and again H2 is going to react with the OH minus and water is getting formed. So overall reaction we have hydrogen and oxygen is giving you water. [00:11:53.284] So the product water now which is getting formed if it is not allowed to be removed from the cell then it is going to dilute the cage. [00:12:08.452] Therefore, what we do is we operate the cell at 100 degrees so that the water which is formed from KOH can be it is going to escape as steam. Therefore, [00:12:24.608] Because this water which is getting formed is obviously going to dilute the KH and we have used a particular concentration of KH. And if the KVH gets very diluted, then the efficiency of the reaction is going to fall down. Therefore, we make sure that this excess water. [00:12:42.288] Which is getting formed continuously is removed from the system. Hence, it is very important that water is allowed to go out of the system, and that is why we operate the cell at about 100 degrees centigrade, or in other words, [00:12:59.856] Why we use warm KH so that water escapes as steam and we maintain the concentration of KVH which is required for that for the reaction. And one more interesting fact which you must be familiar with is that during the Apollos phase [00:13:17.424] program the water which was uh coming out of this fuel cell was collected and the astronauts were using this water for drinking. [00:13:30.116] Now the EMF which is generated because we are using this fuel cell basically to generate power. So the voltage which is generated happens to be 1.23 volts. [00:13:44.132] Coming to advantages of this fuel cell, it operates at a low temperature. So because of that, corrosion of equipment which is being used in the fuel cell happens to a very [00:13:59.536] small extent as compared to those fuel cells which are operate at a very high temperature. And alkali which is used as the electrolyte is allowing us to use a non-noble metal [00:14:17.632] catalyst and because of that what happens is the cell is now less expensive. Now if we don't use alkali if we are using acid then we know that we have to use noble metal catalyst only and as I explained earlier oxygen reduction [00:14:36.224] is faster in an alkaline medium as compared to an acidic medium. So we can afford to use non-mobile metal catalysts here and therefore the it will work out to be much less expensive as compared to other fuel cells. [00:14:54.112] And there are certain disadvantages that the reactants must be free from carbon. We also saw this point that if we have carbon containing fuels, then they will undergo oxidation to form carbon dioxide, and this reacts with alkali.

EV0000110 · EV0000111 · EV0000112 · EV0000113

Transcript block 4 (source interval 000900-001199 seconds): [00:15:12.480] And it forms carbonates, which is going to reduce the efficiency of the cell. So it also means that the fuel and the oxidant, suppose we are talking of hydrogen and oxygen, then fuel and oxidant. [00:15:27.008] which have have which are used has has to be free of carbon compounds. And other problem is that since we are using electrolyte which is a liquid, then that could pose some handling problems. So we have advantages as well as disadvantages. [00:15:46.640] And depending on the application where we are going to use this fuel cell, we will uh decide which type of fuel cell is suitable or whether this fuel cell is suitable for that particular [00:16:01.232] application. So in the next class we are going to talk about methanol oxygen fuel cell where we are going to use a polymer membrane as the electrolyte. Thank you.

EV0000114 · EV0000115 · EV0000116 · EV0000117

Full transcript: 2150 normalized words, 65 timestamped segments, 976.277333 seconds duration. Low-confidence segments flagged: 0.

EV0000102 · EV0000103 · EV0000104 · EV0000105

8 CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells

This lesson is indexed from the source video titled CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000118 · EV0000119

Graphs and diagrams

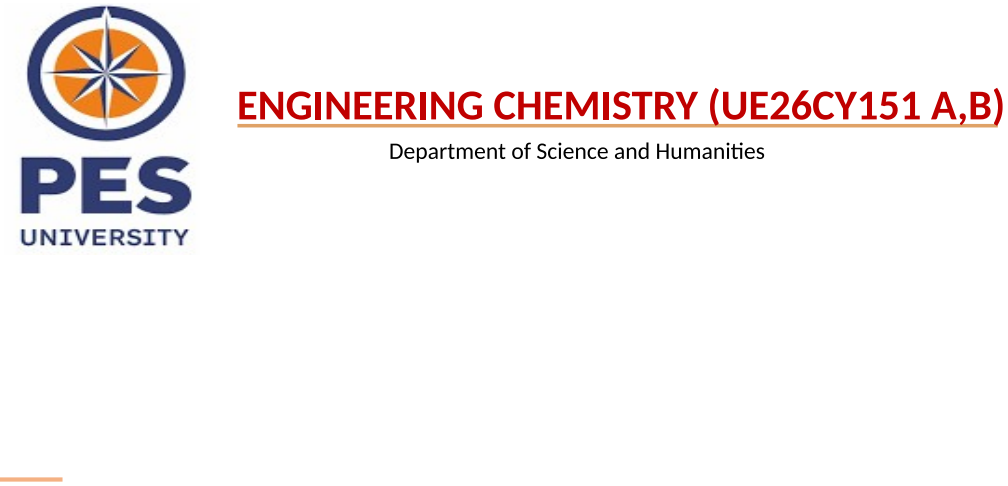

Figure G000059. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 1.

EV0000361

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

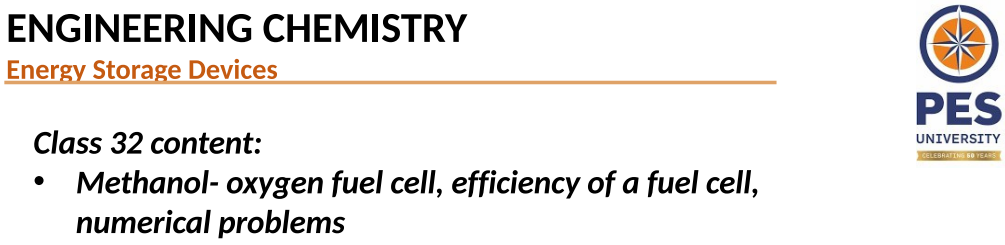

Figure G000060. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 2.

EV0000364

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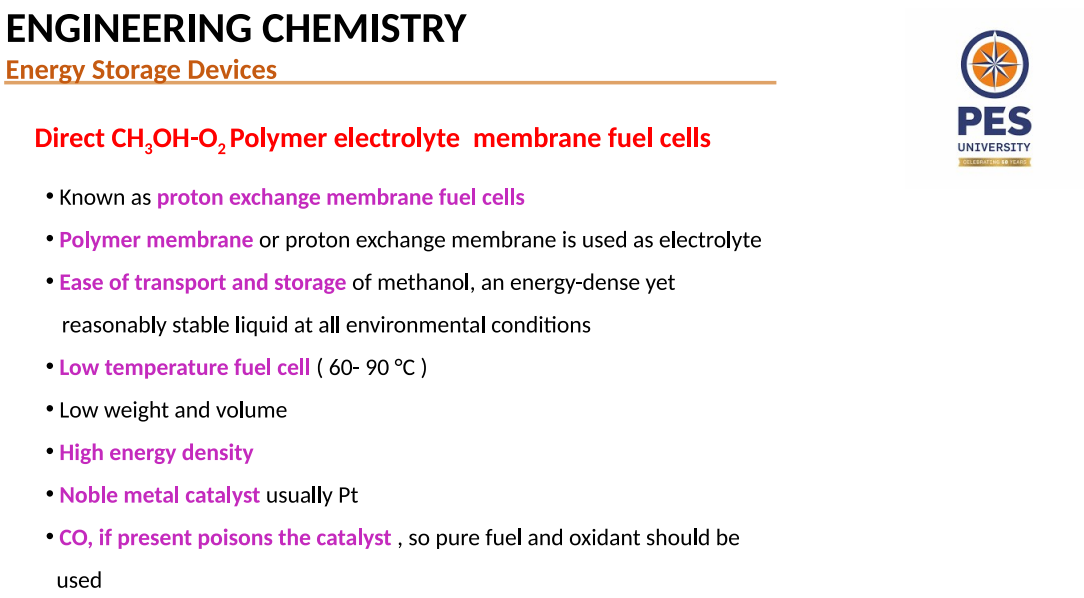

Figure G000061. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 3.

EV0000367

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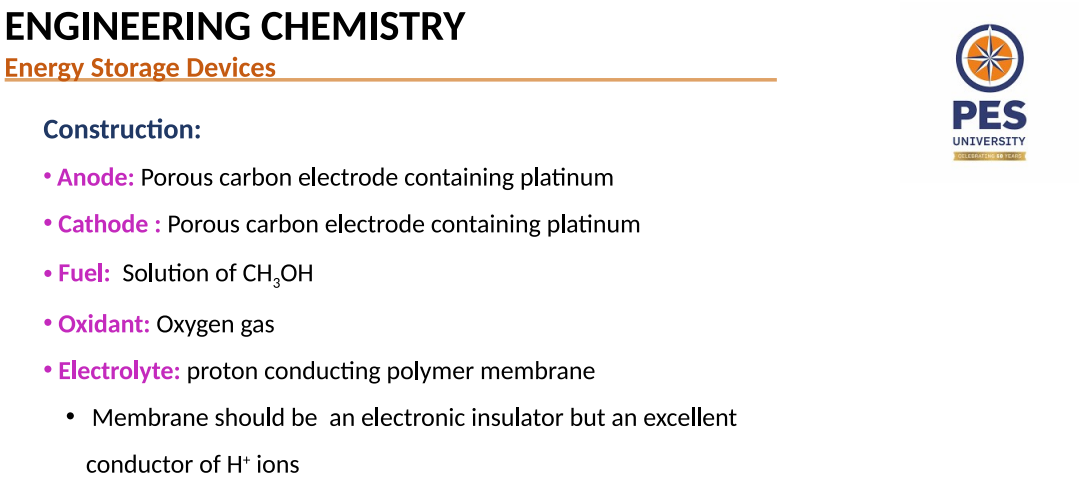

Figure G000062. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 4.

EV0000370

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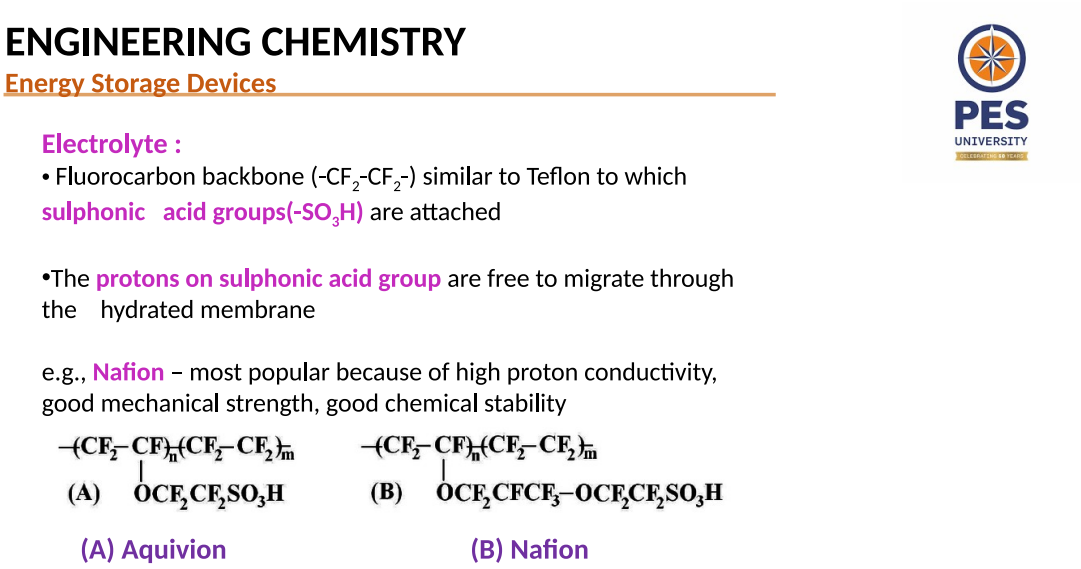

Figure G000063. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 5.

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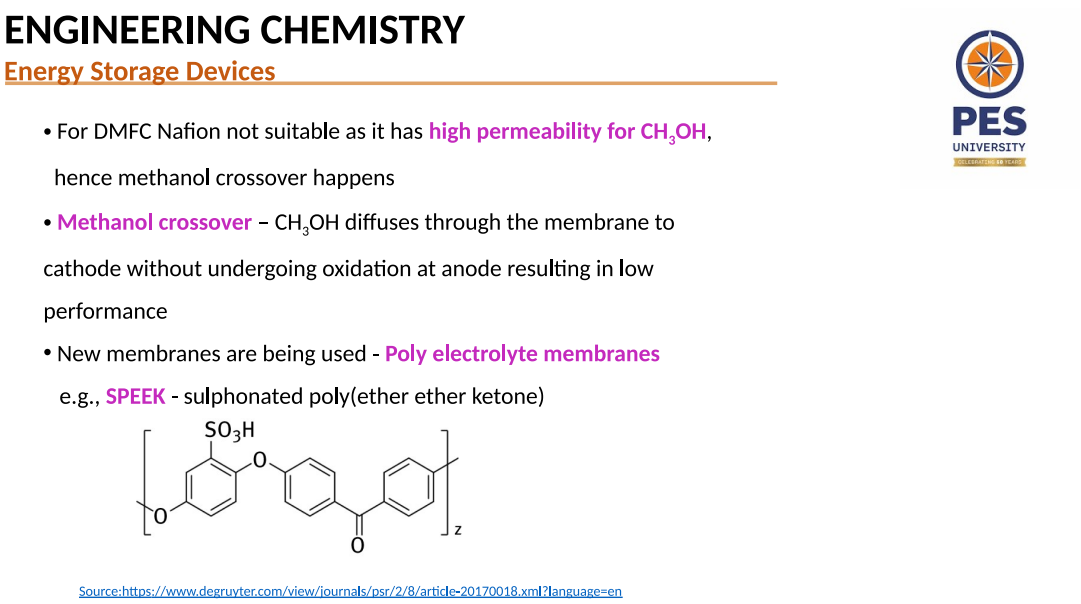

Figure G000064. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 6.

EV0000376

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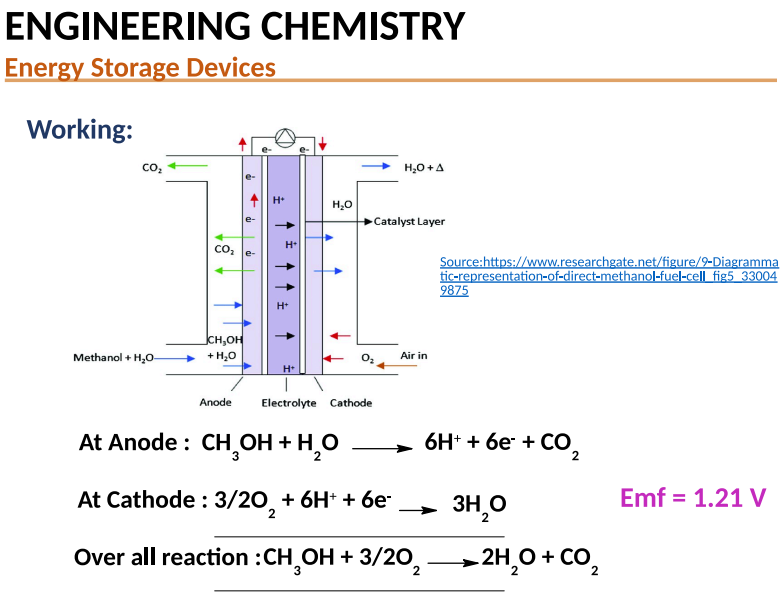

Figure G000065. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 7.

EV0000379

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ENGINEERING CHEMISTRY

Energy Storage Devices

- **Aqueous solution of methanol** is fed at the anode
- Common operating temperatures are in the range **60- 90 °C**
- Oxidation of methanol on a catalyst layer takes place to form carbon dioxide , protons and electrons
- **Protons (H^+) are transported** across the proton exchange membrane – to the cathode where they react with **oxygen** to produce water
- Water is consumed at the anode and produced at the cathode
- Electrons are forced to travel through an external circuit from anode to cathode, providing power to connected devices

Source:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7142913/#:~:text=The%20core%20preparation%20for%20electrolyte,anode%20to%20the%20cathode%20side.

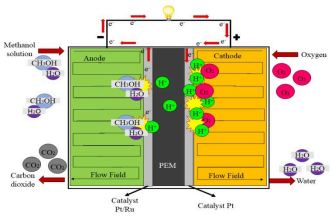

Figure G000066. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 8.

EV0000382

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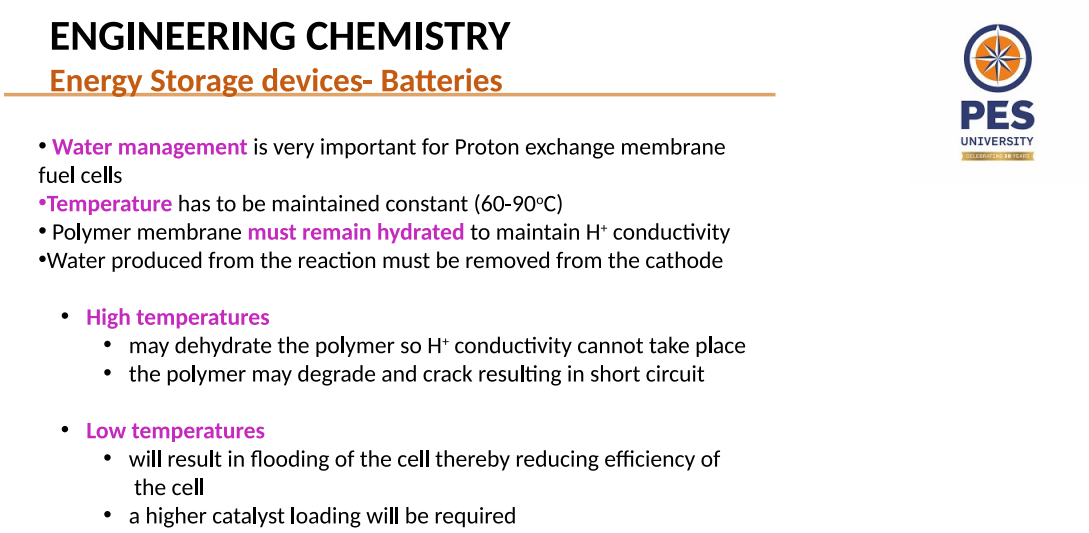

Figure G000067. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 9.

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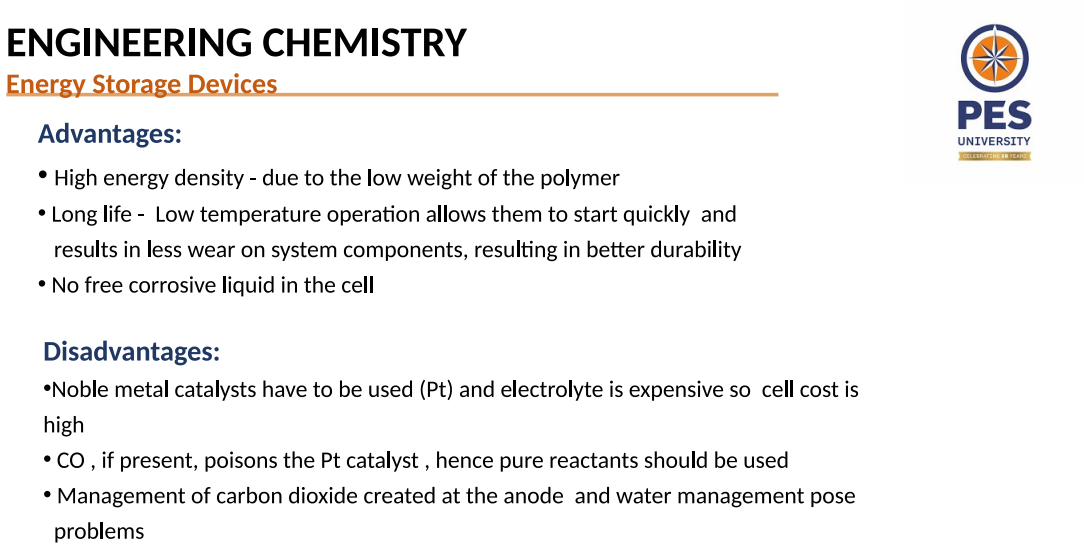

Figure G000068. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 10.

EV0000388

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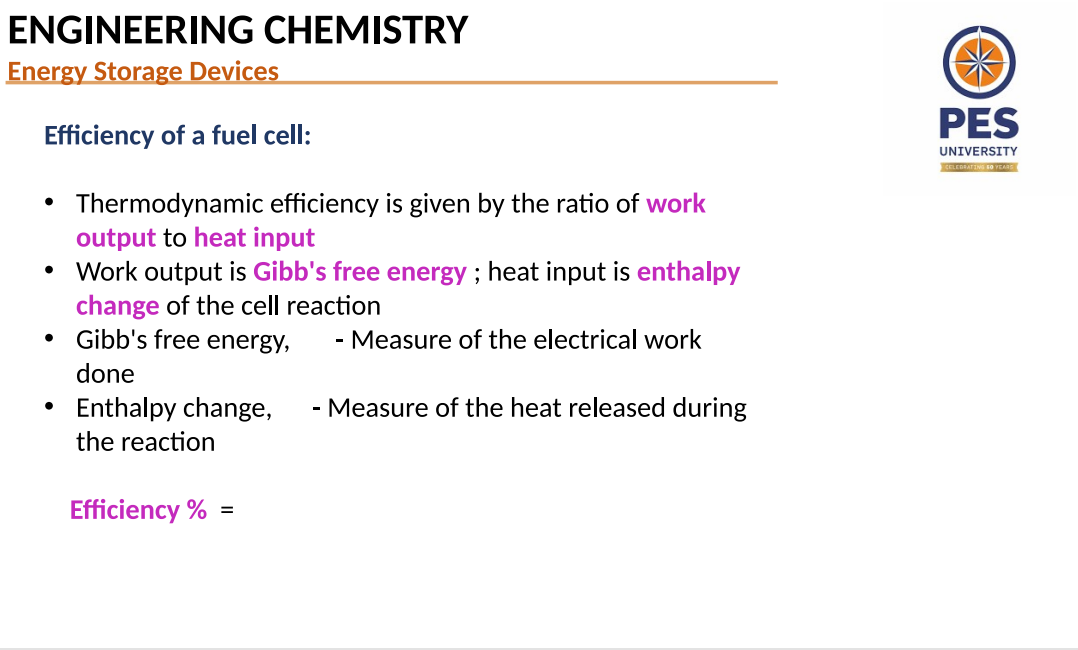

Figure G000069. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 11.

EV0000391

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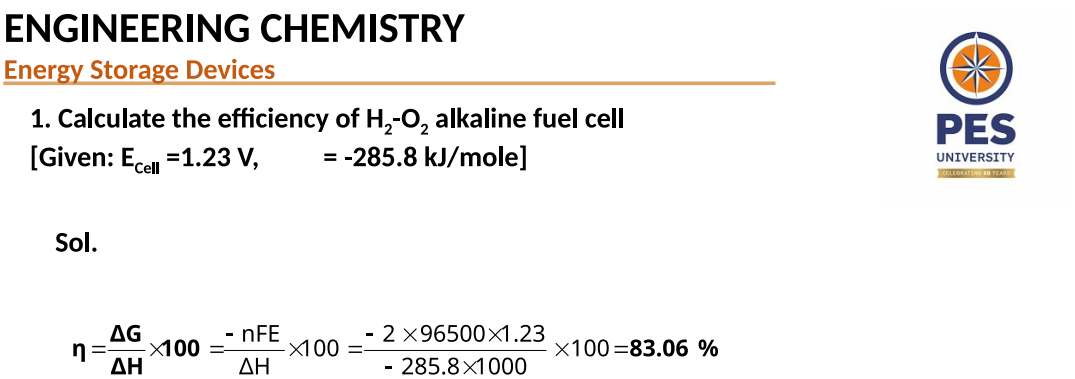

Figure G000070. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 12.

EV0000394

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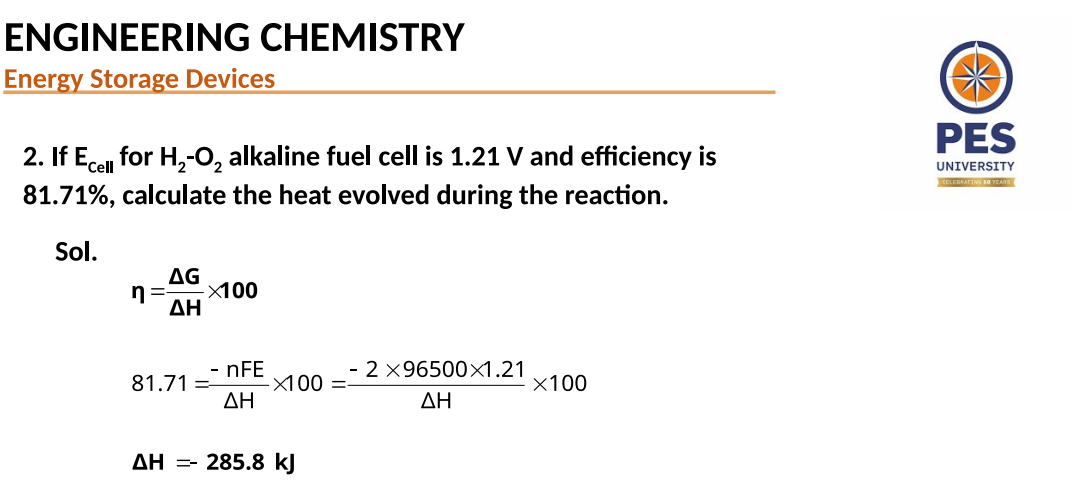

Figure G000071. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 13.

EV0000397

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000072. CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel · source slide page 14.

EV0000400

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000360 · EV0000361 · EV0000362
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Class 32 content: Methanol- oxygen fuel cell, efficiency of a fuel cell, numerical problems EV0000363 · EV0000364 · EV0000365
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Direct CH3OH-O2 Polymer electrolyte membrane fuel cells Known as proton exchange membrane fuel cells Polymer membrane or proton exchange membrane is used as electrolyte Ease of transport and storage of methanol, an energy-dense yet reasonably stable liquid at all environmental conditions Low temperature fuel cell (60- 90 °C) Low weight and volume High energy density Noble metal catalyst usually Pt CO, if present poisons the catalyst , so pure fuel and oxidant should be used EV0000366 · EV0000367 · EV0000368
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Construction: Anode: Porous carbon electrode containing platinum Cathode : Porous carbon electrode containing platinum Fuel: Solution of CH3OH Oxidant: Oxygen gas Electrolyte: proton conducting polymer membrane Membrane should be an electronic insulator but an excellent conductor of H+ ions EV0000369 · EV0000370 · EV0000371

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Electrolyte : Fluorocarbon backbone (-CF2-CF2-) similar to Teflon to which sulphonic acid groups(-SO3H) are attached The protons on sulphonic acid group are free to migrate through the hydrated membrane e.g., Nafion – most popular because of high proton conductivity, good mechanical strength, good chemical stability (A) Aquivion (B) Nafion [Image text, OCR] -(CF-CF)(CF-CF (CF-CF)(CF-CF (A) OCF,CF,SO,H (B) OCF, CFCF,-OCE,CF,SO,H EV0000372 · EV0000373 · EV0000374
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices For DMFC Nafion not suitable as it has high permeability for CH3OH, hence methanol crossover happens Methanol crossover – CH3OH diffuses through the membrane to cathode without undergoing oxidation at anode resulting in low performance New membranes are being used - Poly electrolyte membranes e.g., SPEEK - sulphonated poly(ether ether ketone) Source:https://www.degruyter.com/view/journals/psr/2/8/article-20170018.xml?language=en EV0000375 · EV0000376 · EV0000377
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working: [Image text, OCR] e CO2 HO +△ H+ H2O Catalyst Layer CO2 H+ H+ JCH;OH Methanol + H2O +H2O Air in H Anode Electrolyte Cathode Source:https://www.researchgate.net/figure/9-Diagrammatic-representation-of-direct-methanol-fuel-cell_fig5_330049875 Emf = 1.21 V EV0000378 · EV0000379 · EV0000380
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Aqueous solution of methanol is fed at the anode Common operating temperatures are in the range 60- 90 °C Oxidation of methanol on a catalyst layer takes place to form carbon dioxide , protons and electrons Protons (H+) are transported across the proton exchange membrane – to the cathode where they react with oxygen to produce water Water is consumed at the anode and produced at the cathode Electrons are forced to travel through an external circuit from anode to cathode, providing power to connected devices [Image text, OCR] Methanol solution Anode Cathode Oxygen CH3OH H H2O O2 H+ CH3OH H+ H2O H+ O2 CH3OH H+ H+ H2O H+ CH3OH H+ H2O H+ CO2 2C CO2 CO2 PEM H2O H2C Carbon FlowField FlowField dioxide Water Catalyst Pt/Ru CatalystPt Source:https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7142913/#:~:text=The%20core%20preparation%20for%20electrolyte,anode%20to%20the%20cathode%20side. EV0000381 · EV0000382 · EV0000383
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage devices-Batteries Water management is very important for Proton exchange membrane fuel cells Temperature has to be maintained constant (60-90oC) Polymer membrane must remain hydrated to maintain H+ conductivity Water produced from the reaction must be removed from the cathode High temperatures may dehydrate the polymer so H+ conductivity cannot take place the polymer may degrade and crack resulting in short circuit Low temperatures will result in flooding of the cell thereby reducing efficiency of the cell a higher catalyst loading will be required EV0000384 · EV0000385 · EV0000386

10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages: High energy density - due to the low weight of the polymer Long life - Low temperature operation allows them to start quickly and results in less wear on system components, resulting in better durability No free corrosive liquid in the cell Disadvantages: Noble metal catalysts have to be used (Pt) and electrolyte is expensive so cell cost is high CO , if present, poisons the Pt catalyst , hence pure reactants should be used Management of carbon dioxide created at the anode and water management pose problems EV0000387 · EV0000388 · EV0000389
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Efficiency of a fuel cell: Thermodynamic efficiency is given by the ratio of work output to heat input Work output is Gibb's free energy ; heat input is enthalpy change of the cell reaction Gibb's free energy, - Measure of the electrical work done Enthalpy change, - Measure of the heat released during the reaction Efficiency % = EV0000390 · EV0000391 · EV0000392
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices 1. Calculate the efficiency of H2-O2 alkaline fuel cell [Given: ECell =1.23 V, = -285.8 kJ/mole] Sol. EV0000393 · EV0000394 · EV0000395
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices 2. If ECell for H2-O2 alkaline fuel cell is 1.21 V and efficiency is 81.71%, calculate the heat evolved during the reaction. Sol. EV0000396 · EV0000397 · EV0000398
14. Source slide page 14 (extracted text): THANK YOU Department of Science and Humanities EV0000399 · EV0000400 · EV0000401

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.220] Hello everyone, welcome back to this course on engineering chemistry. Today we'll continue with module 5 on energy storage devices that is on fuel cells. [00:00:14.116] So in today's class we will study about methanol oxygen polymer electrolyte membrane fuel cell, the principle construction and working advantages and disadvantages. [00:00:29.508] So this fuel cell is known as a direct methanol oxygen polymer electrolyte membrane fuel cell. So as you can see from the name itself, that the electrolyte we are going to use in this fuel cell is going to be a polymer. [00:00:48.740] This fuel cell is also known as proton exchange membrane fuel cells, thus telling us that the polymer which we are going to use is [00:01:00.100] capable of exchanging protons. So you can call it as a polymer electrolyte membrane fuel cell, or it can also be referred to as proton exchange membrane fuel cell. [00:01:12.388] So as we can understand the polymer membrane or the proton exchange membrane is being used as the electrolyte. Why is methanol [00:01:24.292] fuel cell so popular these days. If you remember in the hydrogen oxygen fuel cell I mentioned that storage and transport of hydrogen is going to be difficult because it is in the form of a gas. But methanol being a liquid [00:01:40.384] We can easily transport and store the methanol, and hence there is a possibility that we can use it for portable devices. So apart from that, it is an energy dense liquid which is reasonably stable at all environmental conditions. [00:02:00.132] Now one very important characteristic of this fuel cell that it is a low temperature fuel cell and it operates at 60 to 90 degrees centigrade. This fuel cell has low weight and volume. Low weight and volume [00:02:17.060] because we are we are using polymer as the electrolyte and you know polymers are lightweight. So therefore this cell will have very low weight and volume and therefore [00:02:31.968] What is the good characteristic of this fuel cell?

That it is going to have very high energy density. The weight has reduced, therefore the energy density will be very high. [00:02:42.564] But there is one problem we face with this fuel cell, that we have to use only noble metal as catalyst, and usually it is platinum which is employed as the catalyst in these cells. And one more problem which we face that carbon monoxide. [00:03:00.480] If it is present, then it is going to poison the catalyst. We are using platinum as the catalyst. It is going to poison the catalyst, and therefore pure fuel or oxidant only should be used. [00:03:15.140] So let us look at the construction of this fuel cell just like any other galvanic cell we will have anode, cathode, and the electrolyte. So what is the anode being used here? The electrode is a porous carbon electrode containing platinum. And similarly [00:03:32.016] Cathode is also a porous carbon electrode containing platinum. Just now I mentioned that noble metal only should be used, but then we always have to have a porous electrode. [00:03:44.016] because all reactions are going to take place on the electrode. The catalyst is present there, the electrolyte is in contact, and also the fuel or the oxidant is going to come in contact with the electrode. So the electrode plays a very important role in a fuel cell. [00:04:00.992] So at the anode and cathode we are using platinum but in the form of porous carbon electrode which is impregnated with uh platinum. Now the fuel we use here as we are talking about methanol oxygen fuel cell. So obviously the fuel we use is going to be uh that of methanol, but [00:04:20.064] It is a solution of methanol. We never use pure methanol. We are going to use a solution of methanol and the oxidant is oxygen gas. So here we have used a solution. [00:04:33.184] of methanol and the oxidant is a oxygen is the oxygen gas and the electrolyte as we are studying now it is a proton conducting polymer membrane now any electrolyte for that matter [00:04:46.112] What should be the criteria that it has to be an electronic insulator? What does it mean? That it is not going to allow the electrons to move through [00:04:56.016] uh itself that is the electrolyte should be an electronic insulator. Yes, we are using a polymer here, so this membrane is an electronic insulator, but at the same time we want the hydrogen ions to move. That's why we have chosen a proton exchange membrane. So it should be an

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:13.680] Excellent conductor of the hydrogen ions. So which electrolyte or which polymer are we going to choose for this particular cell? Let us see. [00:05:24.900] Electrolyte is usually the most common electrolyte used is a fluorocarbon backbone. It's similar to Teflon. You know Teflon how does it look? It's a CF2, CF2, CF2. You have so many units of CF2. [00:05:40.448] Which are bound together and that is Teflon. So that forms the backbone to which sulfonic acid groups SO3H are attached. Now why did we use sulfonic acid here? Because we know that sulfonic acid has [00:05:56.804] The SO3 minus and H plus it can dissociate in water as SO3 minus and H plus so that this proton now is free to move. So when we have sulfonic acid group. [00:06:11.696] It becomes like this in the presence of water. So this H plus is actually free to move through the electrolyte. So that is why we have chosen a strong backbone of Teflon, but it has these sulfonic acid uh groups which are attached to it. Now [00:06:28.740] The H plus ions obviously are going to be able to migrate through when the membrane is hydrated. So when because when it is hydrated, you will be able to [00:06:41.220] get the H plus ions to move through the electrolyte. Okay. The most common one in any protonic stain membrane electrolyte fuel cell or uh we can call it as a polymer electrolyte fuel cell. It's always napion. It is a very very popular [00:07:01.124] polymer which is used in these kind of fuel cells and uh that's because it has very high proton conductivity that's what we want the diode should move

uh where uh very easily through the membrane and it has good mechanical strength and it also has good chemical stability. [00:07:19.792] So you can see here this is the structure of NAFOM. As I said, there's a backbone of CF2 CF2, and then that is the Teflon backbone, and then here we see the SO3H groups which have been appended onto this uh backbone. Okay, and a similar one similar to NAFON. [00:07:37.552] is acquebn which also has a similar uh structure just that the CF2 CF2 uh [00:07:44.784] branching I mean the CF2 CF2 chain has been reduced. Here it is a longer chain and here you can see it's a smaller chain. Nevertheless Nepheon happens to be the most popular uh membrane which is uh being used. [00:07:59.076] But when we come to DMFC, what is DMFC? Direct methanol fuel cells. That is what we are studying now. Nafheon is not suitable for such a uh fuel cell. [00:08:14.320] where we are using methanol and it's a diet methanol fuel cell. Why? Because along with H plus ion movement naphion is highly permeable to methanol. So [00:08:26.528] What happens because of that? It's allowing the methanol also to move through the membrane and therefore a phenomenon known as methanol crossover happens. So what is methanol crossover? [00:08:39.524] Here you can see the methanol diffuses through the membrane to the cathode. So instead of reacting at the anode undergoing [00:08:48.752] uh oxidation at the anode we find that methanol just crosses over to the cathode and there it is going to combine with the uh oxygen which is not what we want because if we want to generate power from the cell then oxidation of methanol has to happen at the anode but [00:09:06.384] But unfortunately, when we use Nafion, it is found that this methanol crossover happens. So, therefore, that will result in what you will get lower voltage because less number of electrons are being generated. Why? Because there is less oxidation of methanol that is happening at the [00:09:23.152] uh anode so we do not want to use nafion as the membrane and therefore a lot of research is going on so what are the uh what is the other substitute for nafion so many new membranes have been discovered and we find that [00:09:40.784] poly electrolyte membranes seem to serve the purpose and one example of such an electrode electrolyte sorry poly electrolyte membranes uh so one such electrolyte would be it is called speak what does it what does it stand for it's a sulfonated poly [00:09:59.728] Ether ether ketone the structure is given here. So this is known as speak and it is found to be very useful in DMFC. So naphion is not used, instead

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:13.344] Many new membranes are there. One example is SPEEK. Sulfonated polyether ether ketone is used as the membrane. Now let us look at the working of this electrode for this fuel cell. We can see the picture here that [00:10:33.668] We have uh methanol and water which is coming from here. And what would be the reaction? So since here fuel we are letting in fuel from here, so obviously we will have methanol. [00:10:47.844] with water reacts with water a solution of methanol we have used and that is going to generate the H plus and we are interested in these H plus ions and electrons are given out plus CO2. So the electrons obviously will be moving through the [00:11:04.336] uh external wire and H plus is now generated it can move through the electrolyte. [00:11:10.256] Now at the cathode, what is going to happen? We are letting in oxygen. So oxygen is going to react with the H plus which has reached from where? From the anode to the cathode, and it is going to take up the electrons which are coming through the external circuit and form water. [00:11:27.920] So overall reaction would be very simple that methanol undergoes oxidation to give you water and carbon dioxide, and the EMF is found to be around 1.21 volt. So this is what your methanol oxygen cell looks like, and now let us see. [00:11:47.908] exactly what is happening here. So you can see in the picture here an

aqueo solution of methanol is fed at the anode. So we have at the anode an equal solution of methanol plus water [00:12:03.248] A weak solution of methanol is fed. One reason for a weak solution being used is that methanol crossover is minimum when we use a weaker solution. [00:12:14.320] And then the common operating temperatures, as already mentioned, is 60 to 90 degrees centigrade, and the oxidation of methanol on a catalyst layer takes place, and carbon dioxide is [00:12:29.664] Being given out as we saw in the reaction, we get carbon dioxide as a product, and at the same time, protons are being generated, and these protons now are going to migrate through the electrolyte. What is your electrolyte? It is a [00:12:48.208] polymer electrolyte membrane or a proton exchange membrane. So we have these SO3 uh H plus uh SO3 H groups. So your proton can now hop through this provided the membrane is hydrated. If the membrane is hydrated the H plus ions will [00:13:05.920] Just hop through the electrolyte and reach the cathode. Now once it is at the cathode, we have oxygen which is coming in at the cathode. Oxygen reacts with this H plus ions as we saw in the reaction, and then it is going to give out. [00:13:22.800] water. So the water now again is consumed at the [00:13:29.280] Anode the water is being consumed at the anode because we said a weak solution of methanol is being fed over there. So water is consumed at the uh cathode and it is given out at the uh water is consumed at the anode and it is given out at the cathode. Now electrons are forced to [00:13:46.928] Travel through an external circuit. So in the external circuit you will find that the electrons are moving from the anode to the cathode, and that is how it provides uh power to the uh any device which is connected. [00:14:02.896] So this is the working of the polymer electrolyte membrane fuel cell and specially in this case we are using methanol as the fuel and oxygen as the oxidant. [00:14:17.252] Water management in this fuel cell is very very important and the temperature has to be maintained 60 to 90 degrees. We cannot have a higher temperature, we cannot have a lower temperature. Now one important [00:14:35.040] fact which we have to know is that the polymer membrane must remain hydrated to maintain the H plus conductivity. Unless it is hydrated you cannot have the H plus ion moving through because only then the SO3H is going to dissociate. [00:14:52.676] and water which is produced during the reaction it has to be removed from the cathode very frequently.

EV0000126 · EV0000127 · EV0000128 · EV0000129

Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.196] This is called water management. Now what is these are two facts that we have to know that the polymer should remain hydrated at the same time the excess water which is getting formed should be removed from the system. Now when we have high temperature, what is the problem if we use [00:15:21.968] high temperature what will happen the polymer may get dehydrated and therefore H plus conductivity cannot take place and also the polymer may degrade and crack resulting in short circuiting. [00:15:38.244] So this is these are two reasons why high temperatures cannot be employed when we are using uh uh polymer electrolyte uh polymer membrane as the electrolyte. And another reason for this, uh another uh another fact which we have to know is that [00:15:56.016] Neither can low temperatures be used because this will result in flooding the cell because water will not be removed if you keep it at a very low temperature. [00:16:07.824] Water will not be removed and this will result in flooding of the cell and that reduces the efficiency of the cell. And also because the temperature is very low, we will require a higher uh loading of catalyst. So more catalyst will be required. [00:16:26.020] We are using platinum as the catalyst. [00:16:29.136] Uh we have we it will be very expensive. So if you have to use more of the catalyst,

then it is going to make the cell where it is going to cost us a lot. So water management is very very important. Neither can we use a high temperature, neither can we use a low temperature. Therefore [00:16:46.768] So the temperature which it has to operate at is 60 to 90 degrees only. [00:16:55.076] What are the advantages of this fuel cell? So as mentioned it is it has a very high energy density because of the low weight of the polymer. So energy density is very high. [00:17:11.504] And then it has a long life because we are using work making it work at a very low temperature and therefore the with low temperature it will start very quickly. [00:17:24.880] first of all, and it also results in less wear and tear of the system components and therefore a better durability. And then [00:17:34.736] No free corrosive liquid is being used in the cell because the electrolyte is a polymer. Therefore, there is no free corrosive liquid which is being used in the cell. Now what are the disadvantages? [00:17:48.548] Noble metal catalysts have to be used and the electrolyte is also expensive. We are using special electrolytes, special proton exchange membranes have to be used. So this results in high cost of this cell. [00:18:07.088] So one is noble metal catalysts which are expensive and also the electrolyte is also expensive. And if we have CO then carbon monoxide if at all present it is going to poison the platinum catalyst and hence pure reactant should be. [00:18:24.784] Be used. So again, we have to make sure that very pure oxidant and fuel has to be used in this case and CO should not be present. So if it poisons the catalyst, obviously the efficiency of the fuel cell is going to go down. And management of carbon dioxide, which [00:18:42.352] which is created at the anode and water management pose problems. As I mentioned that water shouldn't be present in a right amount, when in the right amount and it should not be in excess. So [00:18:57.808] Uh just now we discussed water management. So it is very important that the cell operate between sixty and ninety degrees centigrade. [00:19:10.180] So this is about the direct methanol oxygen fuel cell. In the next class, we are going to look at another fuel cell that is a solid oxide fuel cell where we are going to use carbon monoxide as the fuel. So it will be our carbon monoxide oxygen. [00:19:29.956] fuel cell using solid oxide as the electrolyte. Thank you.

EV0000130 · EV0000131 · EV0000132 · EV0000133

Full transcript: 2834 normalized words, 79 timestamped segments, 1178.24 seconds duration. Low-confidence segments flagged: 0.

EV0000118 · EV0000119 · EV0000120 · EV0000121

9 Supercapacitors , Ragone plot

This lesson is indexed from the source video titled Supercapacitors , Ragone plot. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000134 · EV0000135

Graphs and diagrams

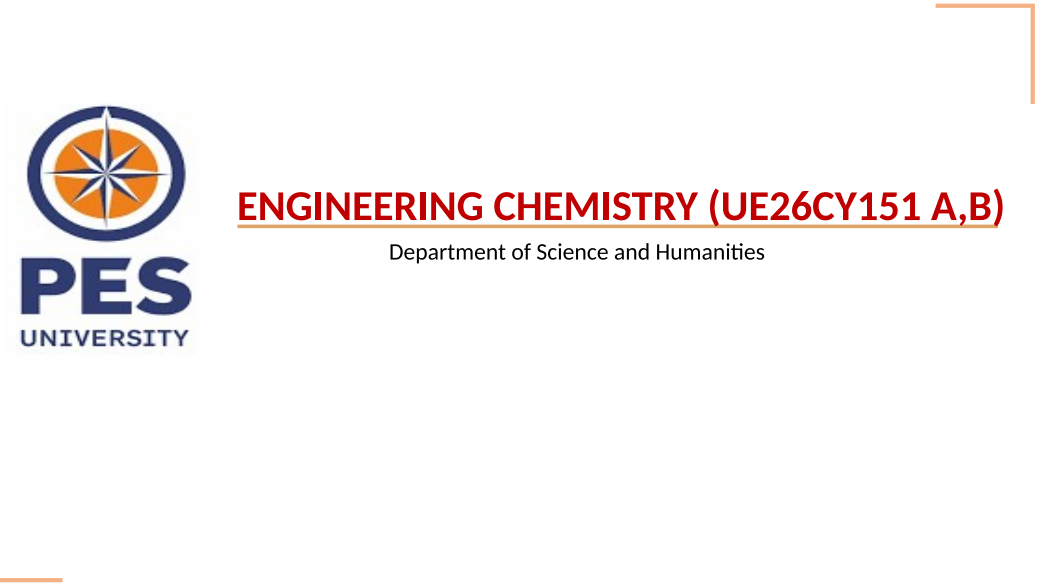

Figure G000073. Supercapacitors construction and working Ragone plot · source slide page 1.

EV0000403

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Figure G000074. Supercapacitors construction and working Ragone plot · source slide page 2.

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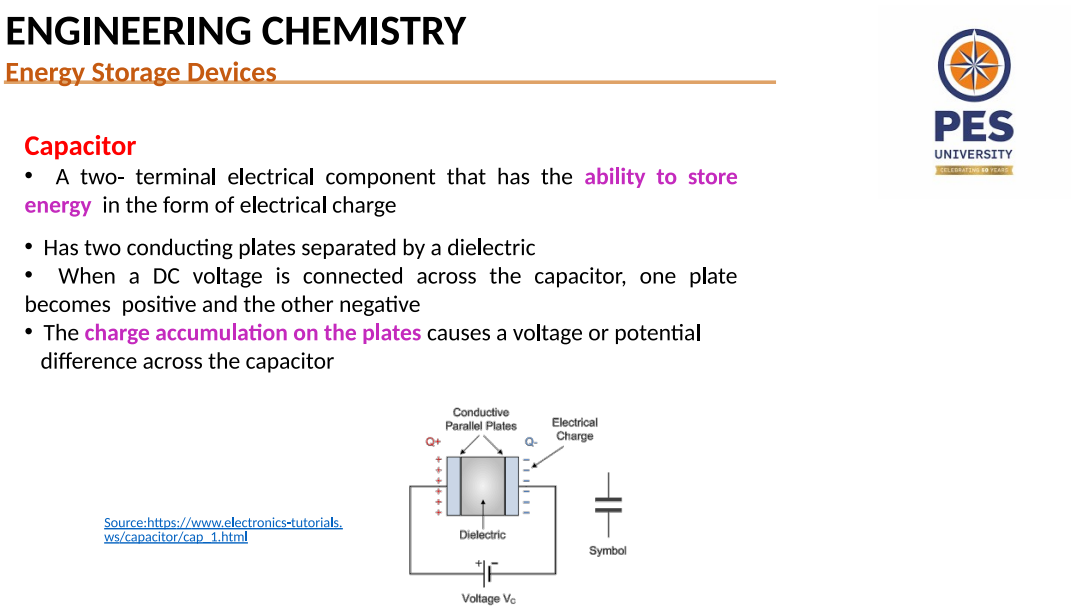

Figure G000075. Supercapacitors construction and working Ragone plot · source slide page 3.

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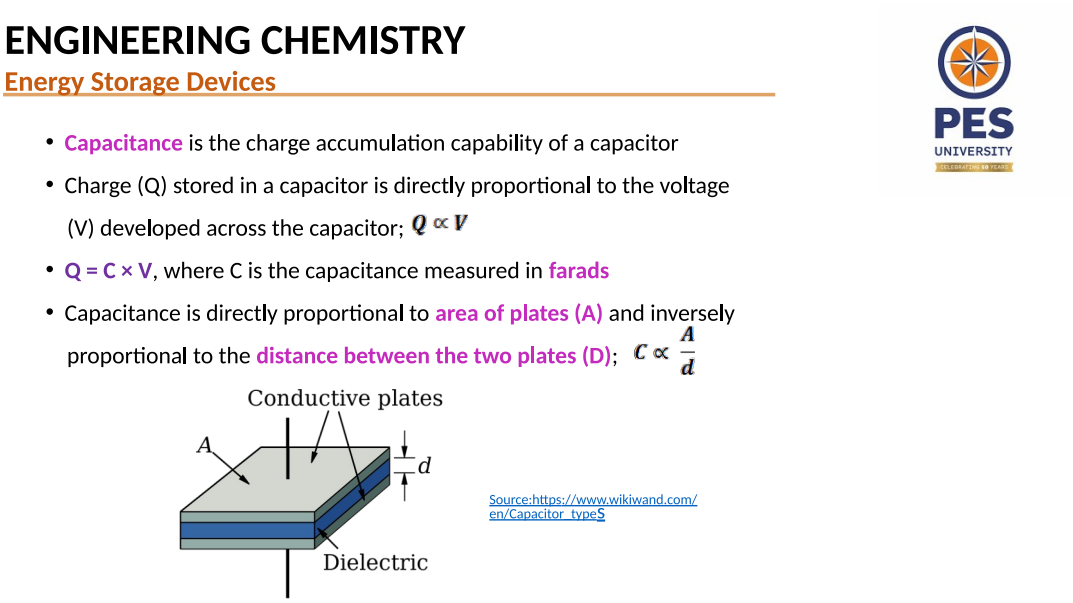

Figure G000076. Supercapacitors construction and working Ragone plot · source slide page 4.

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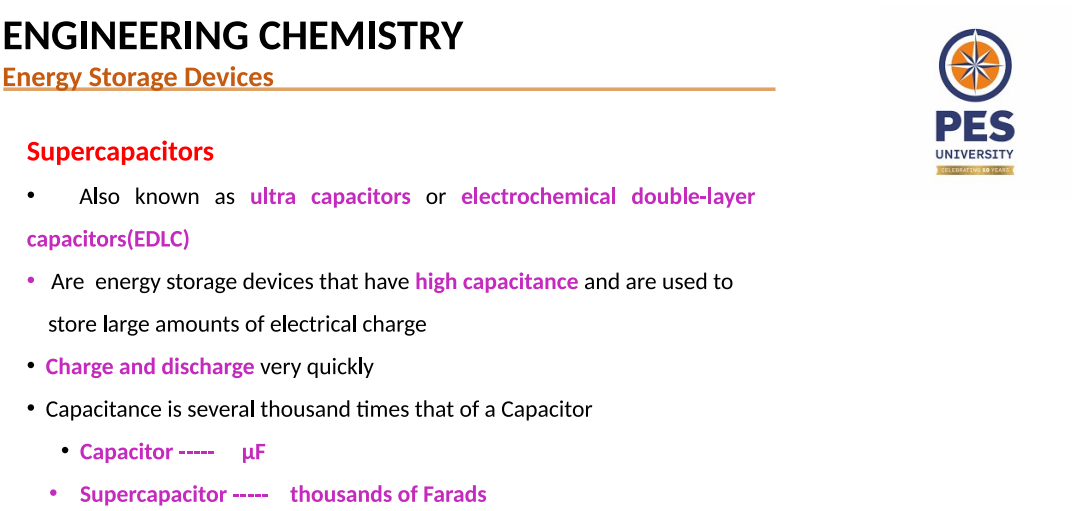

Figure G000077. Supercapacitors construction and working Ragone plot · source slide page 5.

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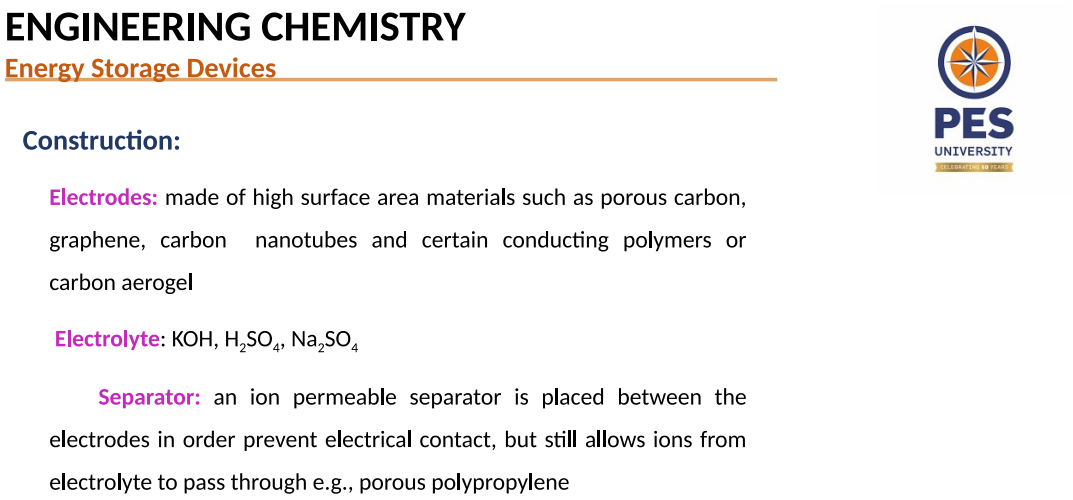

Figure G000078. Supercapacitors construction and working Ragone plot · source slide page 6.

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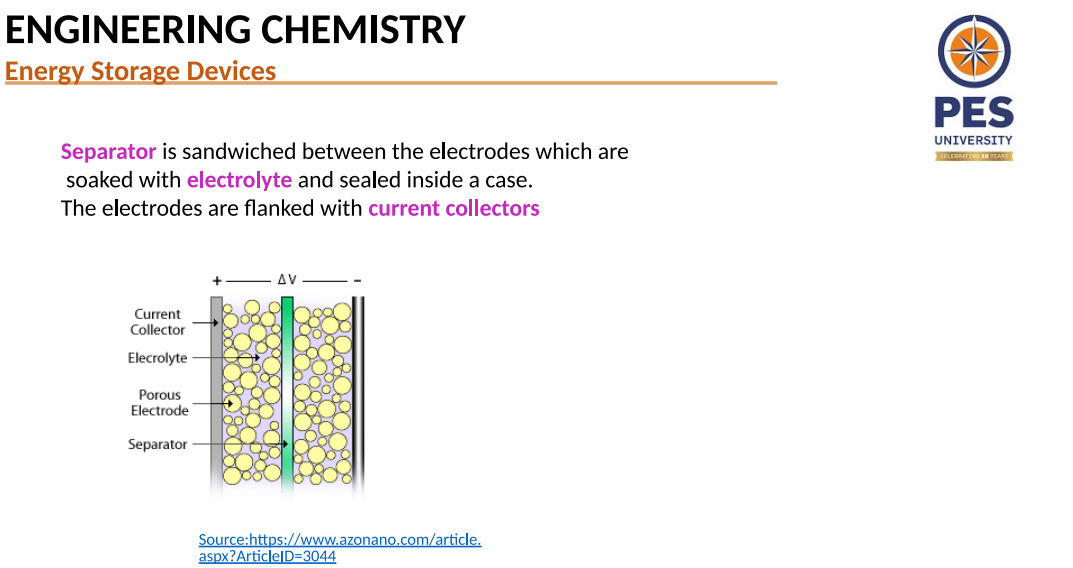

Figure G000079. Supercapacitors construction and working Ragone plot · source slide page 7.

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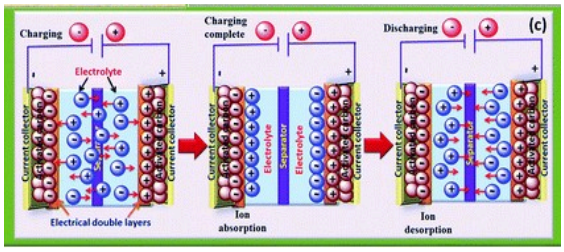
ENGINEERING CHEMISTRY

Energy Storage Devices

Working:

- When a potential is applied, the positive electrode attracts negative ions in the electrolyte, while negative electrode attracts the positive ions
- Formation of electrical double layer at entire electrode /electrolyte interface with a charge separation in nanometer scale



Source:https://pubs.rsc.org/en/content/articlelanding/2019/na/c9na00374f#!divAbstract

Figure G000080. Supercapacitors construction and working Ragone plot · source slide page 8.

EV0000424

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ENGINEERING CHEMISTRY

Energy Storage Devices

- Only absorption and desorption of ions takes place at the electrode during charging and discharging and there are no redox reactions
- Distance between the charged layers in nanorange, d is very small and use of porous electrodes gives very high surface area , A is large

$$C \propto \frac{A}{d}$$; Capacitance is very high

- Formation of the electrical double-layer at each electrode/electrolyte interface, overall capacitance (C_T) is expressed as: $\frac{1}{C_T} = \frac{1}{C_+} + \frac{1}{C_-}$

where C_+ is the capacitance of cathode/electrolyte interface and C_- is the capacitance of anode/electrolyte interface

Figure G000081. Supercapacitors construction and working Ragone plot · source slide page 9.

EV0000427

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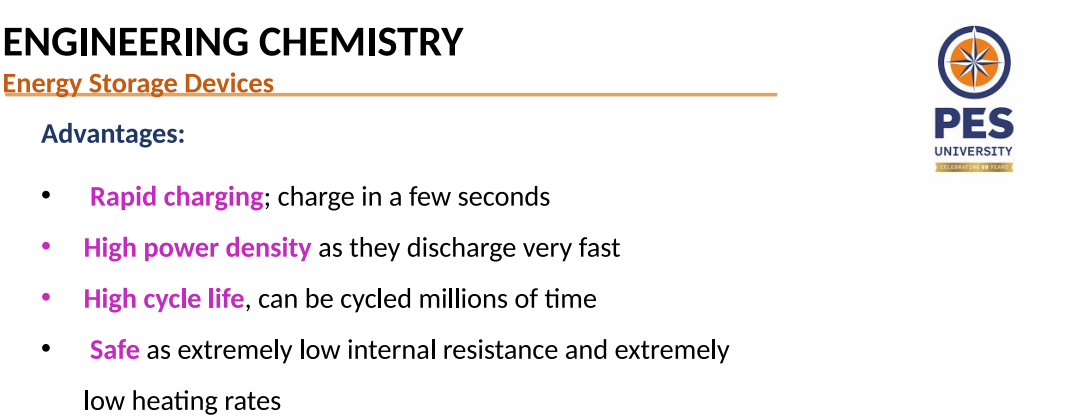

Figure G000082. Supercapacitors construction and working Ragone plot · source slide page 10.

EV0000430

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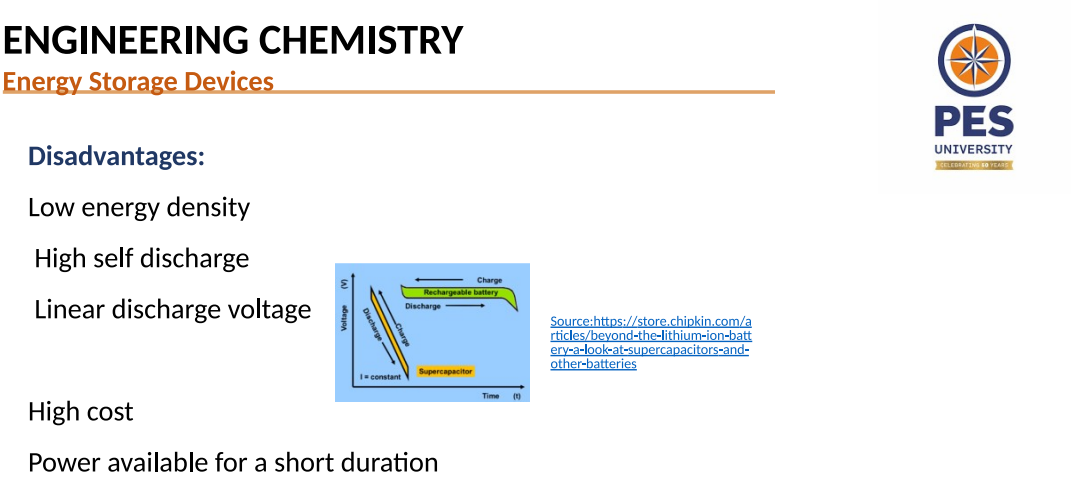

Figure G000083. Supercapacitors construction and working Ragone plot · source slide page 11.

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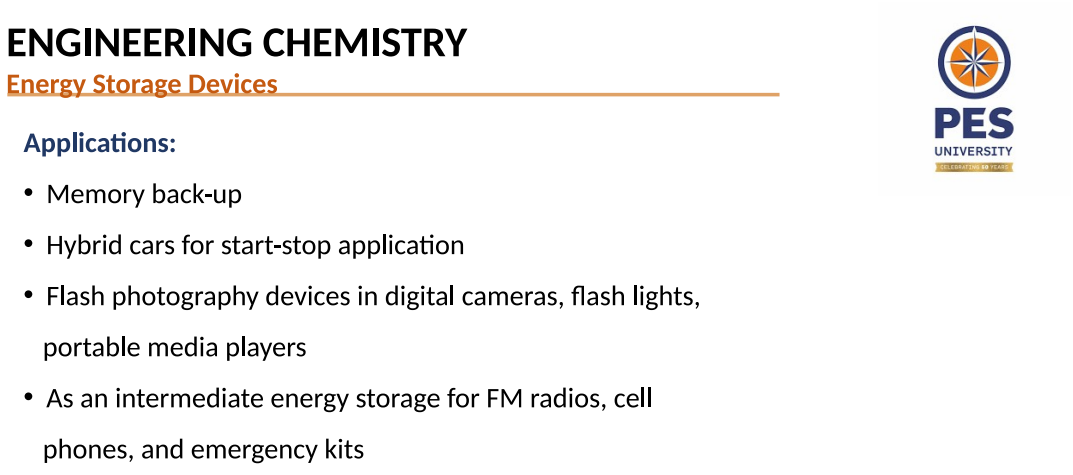

Figure G000084. Supercapacitors construction and working Ragone plot · source slide page 12.

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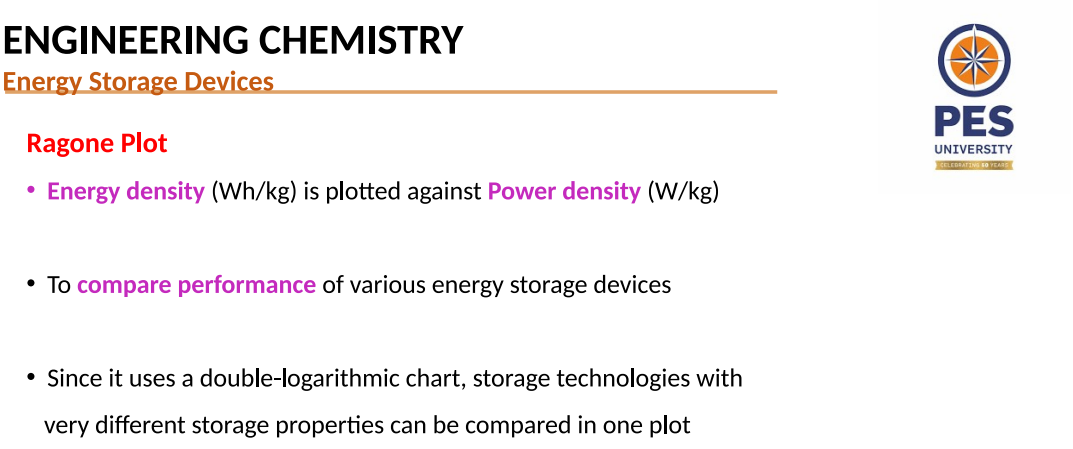

Figure G000085. Supercapacitors construction and working Ragone plot · source slide page 13.

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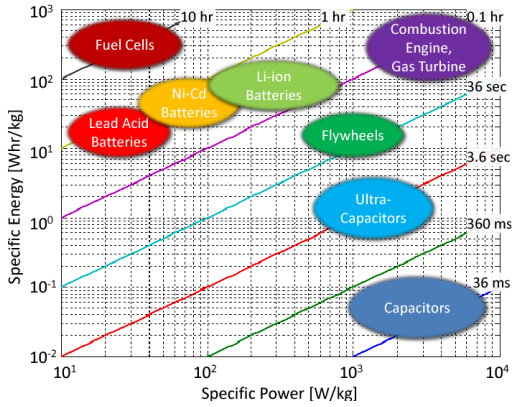
ENGINEERING CHEMISTRY

Energy Storage Devices



Source:https://www.semanticscholar.org/paper/Education-on-vehicle-electrification%3A-Battery-Fuel-Moura-Siegel/7e83f03396b55f2a894461480dda7e9c154f5721/figure/0

Figure G000086. Supercapacitors construction and working Ragone plot · source slide page 14.

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ENGINEERING CHEMISTRY

Energy Storage Devices

- **Fuel cells** have high energy density as the electroactive species can be continuously supplied but the power density is low due to slow kinetics of redox reactions at electrodes
- **Battery systems** offer moderate energy density and power density.
- **Ultracapacitors** (supercapacitors) can deliver very high power density as they can discharge a large amount of charge quickly (because no redox reaction is involved) but energy density is very limited because the charge cannot be stored for a long time

Figure G000087. Supercapacitors construction and working Ragone plot · source slide page 15.

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Figure G000088. Supercapacitors construction and working Ragone plot · source slide page 16.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000402 · EV0000403 · EV0000404
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Class 33 content: Supercapacitors: construction & working, Ragone plot EV0000405 · EV0000406 · EV0000407
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Capacitor A two- terminal electrical component that has the ability to store energy in the form of electrical charge Has two conducting plates separated by a dielectric When a DC voltage is connected across the capacitor, one plate becomes positive and the other negative The charge accumulation on the plates causes a voltage or potential difference across the capacitor Source:https://www.electronics-tutorials.ws/capacitor/cap_1.html EV0000408 · EV0000409 · EV0000410
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Capacitance is the charge accumulation capability of a capacitor Charge (Q) stored in a capacitor is directly proportional to the voltage (V) developed across the capacitor; $Q = C \times V$, where C is the capacitance measured in farads Capacitance is directly proportional to area of plates (A) and inversely proportional to the distance between the two plates (D); Source:https://www.wikiwand.com/en/Capacitor_types EV0000411 · EV0000412 · EV0000413

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Supercapacitors Also known as ultra capacitors or electrochemical double-layer capacitors(EDLC) Are energy storage devices that have high capacitance and are used to store large amounts of electrical charge Charge and discharge very quickly Capacitance is several thousand times that of a Capacitor Capacitor ----- µF Supercapacitor ----- thousands of Farads EV0000414 · EV0000415 · EV0000416
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Construction: Electrodes: made of high surface area materials such as porous carbon, graphene, carbon nanotubes and certain conducting polymers or carbon aerogel Electrolyte: KOH, H2SO4, Na2SO4 Separator: an ion permeable separator is placed between the electrodes in order prevent electrical contact, but still allows ions from electrolyte to pass through e.g., porous polypropylene EV0000417 · EV0000418 · EV0000419
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Separator is sandwiched between the electrodes which are soaked with electrolyte and sealed inside a case. The electrodes are flanked with current collectors Source:https://www.azonano.com/article.aspx?ArticleID=3044 EV0000420 · EV0000421 · EV0000422
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working: When a potential is applied, the positive electrode attracts negative ions in the electrolyte, while negative electrode attracts the positive ions Formation of electrical double layer at entire electrode /electrolyte interface with a charge separation in nanometer scale [Image text, OCR] (a) EDLC Pseudocapacitor Hybrid (b) 00000000 + 00000000 ++++++++ +++++? 000000 000 hyte Elec Electrolyte Eleetrleal double layers Puedecapacssce Redox EDLC Redox Pseudocapacltauce Charging Charging complete Discharging (c) Electrolyte +++++++ 00000000 +++++++++ 9 Electrical double layers len lou absorption desorption Source:https://pubs.rsc.org/en/content/articlelanding/2019/na/c9na00374f#!divAbstract EV0000423 · EV0000424 · EV0000425
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Only absorption and desorption of ions takes place at the electrode during charging and discharging and there are no redox reactions Distance between the charged layers in nanorange, d is very small and use of porous electrodes gives very high surface area , A is large ; Capacitance is very high Formation of the electrical double-layer at each electrode/electrolyte interface, overall capacitance (CT) is expressed as: where C+ is the capacitance of cathode/electrolyte interface and C− is the capacitance of anode/electrolyte interface EV0000426 · EV0000427 · EV0000428
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages: Rapid charging; charge in a few seconds High power density as they discharge very fast High cycle life, can be cycled millions of time Safe as extremely low internal resistance and extremely low heating rates EV0000429 · EV0000430 · EV0000431
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Disadvantages: Low energy density High self discharge Linear discharge voltage High cost Power available for a short duration Source:https://store.chipkin.com/articles/beyond-the-lithium-ion-battery-a-look-at-supercapacitors-and-other-batteries EV0000432 · EV0000433 · EV0000434

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Applications: Memory back-up Hybrid cars for start-stop application Flash photography devices in digital cameras, flash lights, portable media players As an intermediate energy storage for FM radios, cell phones, and emergency kits EV0000435 · EV0000436 · EV0000437
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Ragone Plot Energy density (Wh/kg) is plotted against Power density (W/kg) To compare performance of various energy storage devices Since it uses a double-logarithmic chart, storage technologies with very different storage properties can be compared in one plot EV0000438 · EV0000439 · EV0000440
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices [Image text, OCR] 103 10hr 0.1hr Combustion Fuel Cells Engine, Gas Turbine 102 Li-ion Ni-Cd 36sec Batteries [a Batteries Lead Acid Flywheels Batteries 101 3.6sec Ultra- 100 Capacitors 360.ms 10-1 36m Capacitors 10-2 101 102 103 104 SpecificPower[w/kg] Source:https://www.semanticscholar.org/paper/Education-on-vehicle-electrification%3A-Battery-Fuel-Moura-Siegel/7e83f03396b55f2a894461480dda7e9c154f5721/figure/0 EV0000441 · EV0000442 · EV0000443
15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Fuel cells have high energy density as the electroactive species can be continuously supplied but the power density is low due to slow kinetics of redox reactions at electrodes Battery systems offer moderate energy density and power density. Ultracapacitors (supercapacitors) can deliver very high power density as they can discharge a large amount of charge quickly (because no redox reaction is involved) but energy density is very limited because the charge cannot be stored for a long time EV0000444 · EV0000445 · EV0000446
16. Source slide page 16 (extracted text): THANK YOU Department of Science and Humanities EV0000447 · EV0000448 · EV0000449

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module five on energy storage devices on supercapacitors and this will be the last class of this module. [00:00:13.604] So we will talk about the principle construction and working advantages, disadvantages and applications of supercapacitor, and then we will discuss the Raguni plot. [00:00:24.996] Before we talk about supercapacitors, let us understand what is a capacitor and how is a supercapacitor different from a capacitor. So a capacitor [00:00:35.780] has two conducting plates which are separated by a dielectric and they are electric components that have the ability to store energy in the form of electrical charge. So a capacitor can hold charge between its plates. [00:00:53.552] Now, how does it work? When we use a power source, the negative terminal is [00:01:01.296] attached to one of the plates so that gets the negative charge and the positive terminal when it is attached to the other plate that that plate gets the positive charge. So we have a positive plate and a negative plate which is separated by a dielectric medium. [00:01:19.844] So the charge accumulation of uh on these plates causes a voltage or a potential difference across the capacitor. [00:01:29.828] Now an important property of a capacitor is the charge accumulation capability, and that is known as capacitance. [00:01:41.220] The charge stored in a capacitor is directly proportional to the voltage

developed across the capacitor. So we can say Q is directly proportional to V. So if we write it as Q is equal to C multiplied by V, [00:01:57.664] Then C is known as the capacitance and the unit used is ferrats. [00:02:04.772] So this capacitance now is directly proportional to the area of the plates. Now we understood the capacitor, the construction, we have two plates which are separated by the dielectric medium. So the area of these two plates. [00:02:22.272] will decide the capacitance of the capacitance because capacitance is directly proportional to the area of the plate as you can see in the picture here. This is the area of the plate and it is inversely proportional to the distance between the two plates. [00:02:39.620] So when you want a high capacitance, we understand that the area of the plate should be high and the distance should be low so that we have a very good capacitor and it can store a lot of energy. [00:02:56.964] What are supercapacitors then? They are also known as ultra capacitors or electrochemical double layer capacitors, E, D, and C. [00:03:07.332] So these supercapacitors, as we can understand from the name, they have very high capacitance and are used to store a large amount of electrical charge. [00:03:20.496] Now how much higher is the capacitance as compared to a capacitor? If we know that capacitor is usually in the capacitance is given in microfarads, but for a supercapacitor it will be listed in thousands of ferrats. [00:03:38.224] you can understand that the capacitance has gone up to such a high extent and they also charge and discharge very quickly. This is for any capacitor and of course it is applicable to supercapacitor also that they charge and discharge very quickly. [00:03:56.164] Let us look at the construction. Why is the capacity so much higher than that of a capacitor? So we have like any other battery. [00:04:07.744] We have electrodes, we have the electrolyte and we have a separator. So the electrodes here they are made of high surface area materials such as porous carbon, graphene, carbon nanotubes and certain conducting polymers or carbon energy. [00:04:24.708] So the electrodes which we choose for a supercapacitor are going to be made of [00:04:30.980] Very important high surface area materials. Now the electrolyte is usually KOH, H2SO4, NE2SO4, and so many other electrolytes are being explored now. The separator is an iron permeable separator and it's [00:04:49.892] As in batteries, it is placed between the electrodes, and what is the role of a separator? It is always to prevent electrical contact between the anode and the cathode, but it is going to allow the ions to

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.128] uh move from the one electrode to the electrode to the other electrode and the example for this would be porous polypropylene. [00:05:13.668] the separator. How exactly are these components placed? You can see in this picture. The separator is sandwiched between the electrodes. So we have [00:05:24.384] The high surface area electrode which are highly porous. So you can see the two electrodes over here, which are highly porous in nature, and then we have the separator which is sandwich between these two porous electrodes. And then this whole [00:05:41.156] system is now placed in a case and then electrolyte is poured over it. Now when you pour the electrolyte, so where does this electrolyte go? It goes through all this porous uh electrodes which we have placed over here. And then of course the electrodes we need [00:05:58.480] Just like in a battery, we need current collectors. So we have these current collectors which are flanking the [00:06:05.360] two electrodes. So the porous electrodes are here, the separator is here, the electrolyte is poured onto the electrodes, and then we have this uh current collectors which are placed on either side of the electrode. So this is the construction. Let us look at the working [00:06:24.868] When a potential is applied, so the positive electrode, one of the electrode obviously will become positive based on the terminal of the external

source. So that positive electrode now [00:06:38.884] is going to attract the negative ions from the electrolyte. So remember we have placed an electrolyte there and this electrode the moment it you give make it positive it is going to attract the negative ions from the electrode. [00:06:56.080] And similarly, the negative electrode which is connected to the negative terminal is going to become is going to get attract the positive ions. So what is the result now? The negative electrode has attracted the [00:07:12.144] positive ions and the positive electrode has attracted the negative ions so as a result we have the formation of an electric double layer [00:07:23.664] at the electrode so here as well as here so here the electrode is negative it is attracting positive ions and on this side we have the positive electrode which is attracting the negative ions so basically let us look at this electric double layer [00:07:39.264] Now this electric double layer is the distance between the two the positive and the negative layer is just nanometers. So the charge separation now between the two layers is [00:07:54.880] the the positive and the negative layer is in the nanometer ring. So the electric double layer now is responsible for the capacitor having uh the a supercapacitor having such a high uh [00:08:10.692] capacitance. So and it it and this double layer now because you made a porous electrode it is formed at the entire electrode electrolyte interface and plus the charge separation is just nanometer range. So [00:08:28.576] Now we have what does it mean that we have reduced the D to such a small extent, then [00:08:37.924] What is happening here? Only absorption and desorption. So when you are this is this happens when you are charging the supercapacitor. Now when you're using it to deliver power, then these ions just get desorbed from the surface. So while you are charging, they get absorbed to the surface. [00:08:57.552] They get they form the double layer and then when you are discharging then these ions are getting away from the uh electrode surface. So basically it is just the absorption and desorption of ions which take place and there is absolutely no redox reaction. [00:09:15.216] that is happening here. So because of this [00:09:20.176] We find that this supercapacitor has many advantages, which we will discuss now. So we have to remember there is no redox reaction involved at all. It is just this absorption and desorption which happens during the charging and the disorganization. [00:09:37.872] charging process uh now what has happened the distance between the charge lay has become into the nano range as I explained just now so D is very small and [00:09:51.524] Also we used porous electrodes. So because of the porous electrode, what do we get? We have a very high surface area. The moment you make a material porous, we know that it is going to get in get a very high surface area. So A is very large. So what

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.712] We know about capacitance C is proportional to A by D. That means [00:10:13.312] If the area is very high and the D is very very small, in fact here it's in the nanometer scale. So definitely the capacitance is going to be very high for such a device and that is why we see that it's rated at thousands of ferrats. [00:10:31.364] Now this formation of electric double layer of course takes place at both the electrode electrolyte interface and therefore at each of these electrodes we have the capacitance. So when you want the total capacit, we have a capacitance for each of these electrodes. [00:10:49.136] And the total capacitance would be 1 by C T is equal to 1 over C plus plus 1 over C minus. [00:10:59.044] What are the advantages of these supercapacitors? Rapid charging. They charge very, very quickly. Why? Because it just involves the movement of these ions towards the electrodes and forming of the double layer. And very high. [00:11:15.600] power density because power density depends on how fast the energy is getting discharged and we know that they can discharge also

very quickly because there is no redox reaction that is taking place. It's only desorption of ions that is required. [00:11:31.472] And of course high cycle life they can be cycled millions of times as compared to your battery, which the best batteries have about five hundred to thousand is very high uh cycle life, but here we have [00:11:47.652] uh device which can be cycled several time because again there is no redox reaction that has to take place. It is just the absorption and desorption of ions which can happen as many times like we can say it's almost unlimited cycle life. [00:12:02.576] And yes, compared to batteries, they are much much safer as the internal resistance is very low, and when the internal resistance is low, we know that the heating rate is also going to be low. Then there are certain disadvantages also. What is it that they have very low energy density. [00:12:23.204] Low energy density. What is the reason that it the reason is they discharge very quickly, so obviously they cannot retain or hold the charge for a long time. So energy density is low, and of course, high self-dcharge because it just involves absorption and disorption of ions. So there is a high [00:12:43.056] High self discharge. Now this is very important. Linear discharge voltage. If you remember for a battery when we draw voltage versus time, the [00:12:56.676] line looks like this the graph looks like this what happens it remains steady for a long time and then there is a drop in the voltage when we say the battery is dead. So it can give you a steady voltage for a long time. [00:13:12.720] Whereas for a supercapacitor, the drop is linear. So this is known as linear discharge characteristics. So here we see that the drop is linear. So therefore the [00:13:27.172] volta is going to drop very quickly with time, whereas in a battery we find that we can get a steady voltage for a reasonable amount of time. Now another reason is of course the high cost of [00:13:41.060] This supercapacitor because of the electrodes which are you being used, very highly porous uh electrodes have to be used. They are very expensive. And then the power is available for a very short duration because they discharge very quickly. Where would such [00:14:00.656] Batteries be useful or supercapacitors be useful. They are used for memory backup. Then hybrid cars for start stop applications in flash photography where you need high power for a very short period of time. So that is the perfect place for [00:14:18.352] to for these super capacitors. So flash photography in digital cameras. [00:14:23.152] flashlights, portable media players, and also as an intermediate energy storage for FM radios, cell phones and emergency kits. So basically wherever we need a quick burst of energy, then supercapacitors are very very useful. [00:14:43.332] Now we come to the last part where we talk about [00:14:48.420] comparison of performance of all kinds of energy storage devices. We have talked about different types of batteries, we talked about fuel cells, we talked about supercapacitor. So can we have a comparison of all these

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.432] Energy storage devices comparison compare the performance of these and that can be done when we plot energy density. [00:15:13.536] that is given in watt hour per kg versus the power density that is watt per kg of a particular device and such a plot is known as a raguni plot. So ragoni plot basically it is a plot which gives you [00:15:29.232] Energy density versus power density for a whole range of energy storage devices and why is it possible to use um to plot all these energy different energy storage uh uh devices? That's because it's a double log chart. [00:15:46.624] So storage technology technologies with different storage properties can be compared on one plot and such a plot is known as a Ragoni plot. Let us have a look at this. So you can see here energy density or we can call it specific energy and versus power density that is specific power. [00:16:04.448] So where are these different

storage devices appearing in this plot? So you can see capacitors here and we are more interested in the ultra capacitors. So ultra capacitors if you see they have high power density whereas low energy density. [00:16:23.780] The reason that they discharge very quickly, so they can give you a very high power density. They have very high power density, but they don't retain the charge for a long time, hence they have low energy density. The reason being that there is no redox reaction. [00:16:41.552] is simple absorption and desorption of ions that has to take place and therefore they charge as well as discharge very quickly. So they cannot retain energy for a long time. So the energy density is low while the power density is very high. Now just directly opposite we [00:16:59.120] can see these fuel cells. We also studied about fuel cells now fuel cells have very high energy density but low spow density. Now the reason for this we can understand is that fuel cells [00:17:15.844] The reactions have to take place, and the reactions are usually very slow. If you remember, we always use catalysts. So therefore the reactions are slow, and hence the few if our fuel cells the power density is going to be very low, whereas the energy density is high because we can [00:17:33.616] can supply the fuel and the energy and it can work as long as you supply the fuel so the energy density is very high. Batteries of course come somewhere in between they are occupying this position, solid acid battery, nickel cadmium battery, lithium ion battery, lithium ion battery [00:17:51.184] Has an advantage because of lithium being used, which is a lightweight metal. So the energy density is again high for a lithium ion battery. And of course, when you talk of combustion engines, we know that we can keep on supplying the gasoline and therefore the [00:18:08.752] energy density is going to be high and the reactions are very very quick where we know that chain reactions happen. So therefore the power density is also very high. So a summary of all this is here. So fuel cells will have high energy density and whereas they have [00:18:26.368] uh low power density batteries will have moderate energy density and power density ultra capacitors or supercapacitors they can deliver very high power uh power therefore power density is very high whereas the uh [00:18:41.572] Energy density is going to be low for a ultra capacitor or a supercapacitor. So in the this is the end of this module, and in the next class, we are going to talk about. [00:19:01.796] Then uh next topic and that is corrosion. Thank you.

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Full transcript: 2707 normalized words, 78 timestamped segments, 1146.986667 seconds duration. Low-confidence segments flagged: 0.

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10 Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2))

This lesson is indexed from the source video titled Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000150 · EV0000151

Graphs and diagrams

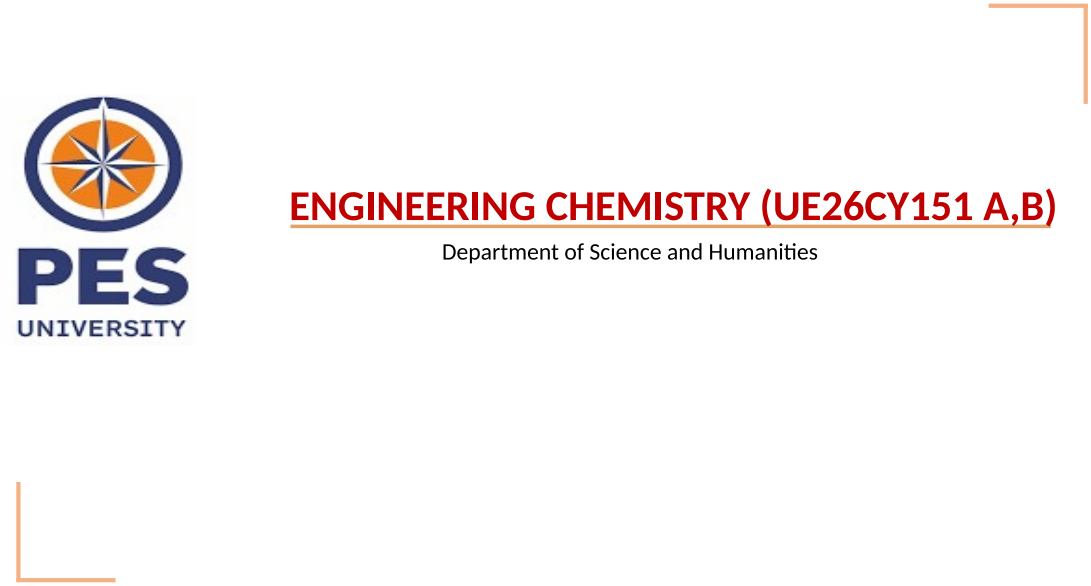

Figure G000115. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 1.

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Figure G000116. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 2.

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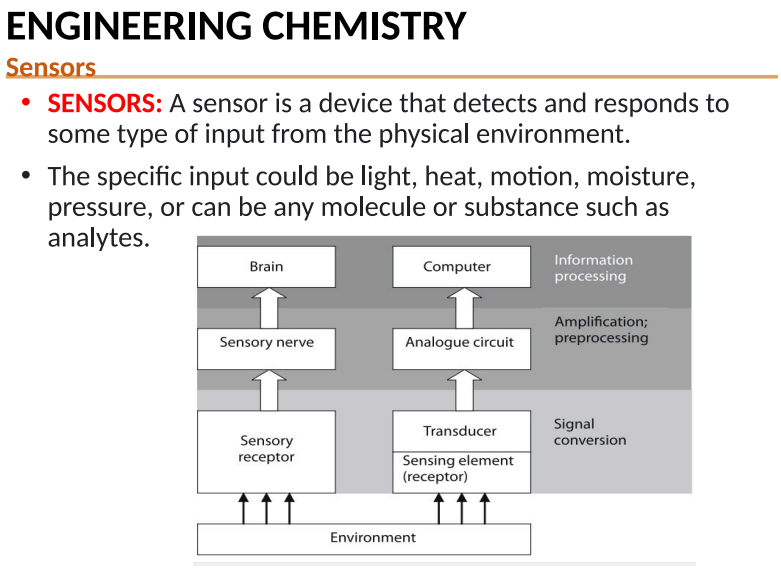

Figure G000117. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 3.

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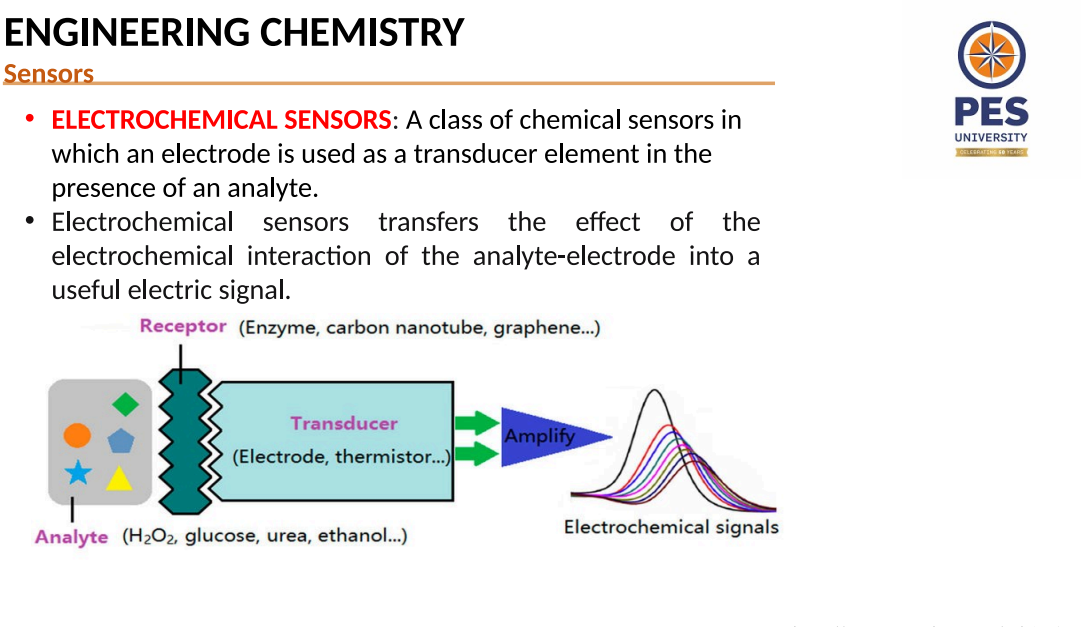

Figure G000118. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 4.

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Figure G000119. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 5.

EV0000539

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ENGINEERING CHEMISTRY

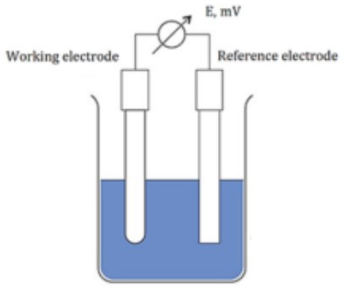
Sensors

TYPES OF ELECTROCHEMICAL SENSORS

1. POTENTIOMETRIC SENSORS

- These sensors detect changes in the concentration of a substance by observing how the concentration changes the electrical potential produced by a redox reaction at the electrode.
- The redox reaction takes place on the electrode surface:

  Oxidant + ne^- → Reduced product
- It measures the potential difference (voltage) between an indicator electrode and a reference electrode under conditions of negligible current flow, and relates this potential to the activity or concentration of a specific ion or analyte in a solution.



$E_{cell} = E_o + RT/nF \log(\frac{C_o}{C_R})$

At 298K, $E_{cell} = \frac{0.0591}{n} \log(\frac{C_o}{C_R})$

Figure G000120. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 6.

EV0000542

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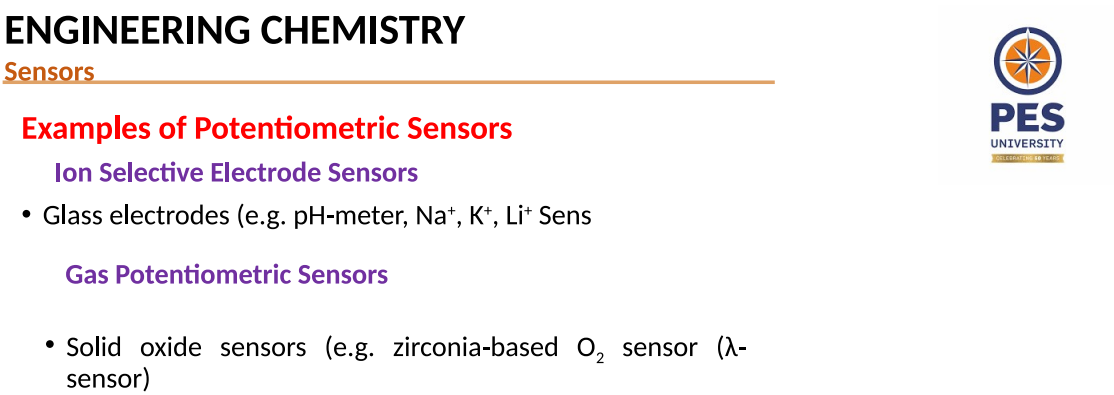

Figure G000121. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 7.

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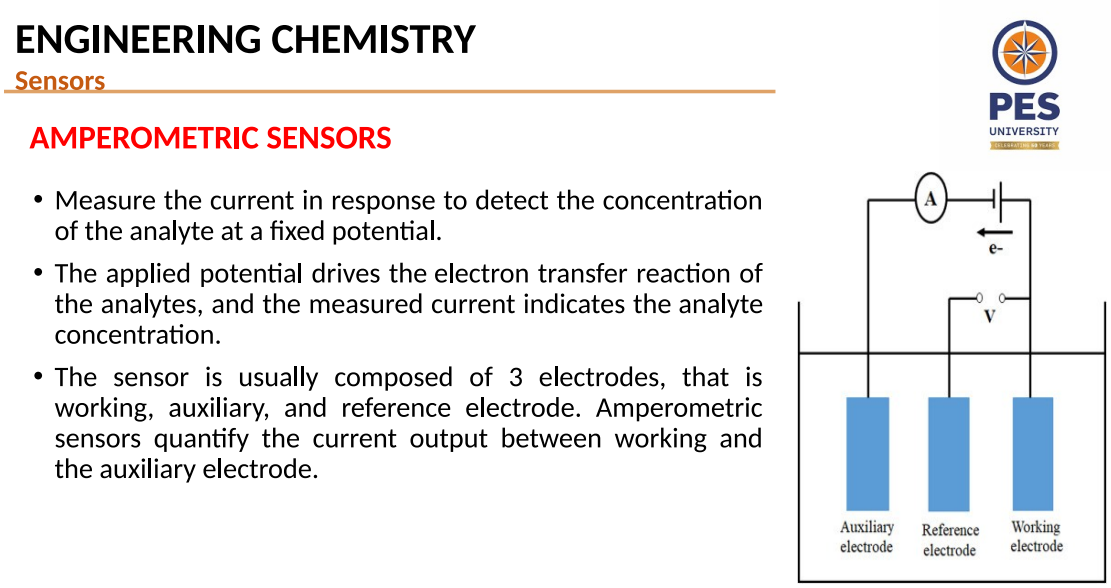

Figure G000122. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 8.

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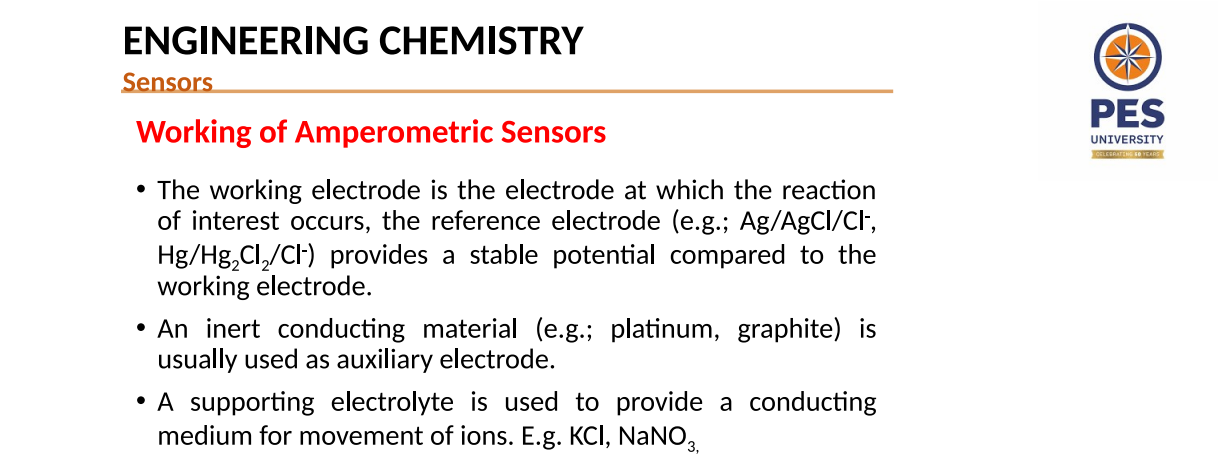

Figure G000123. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 9.

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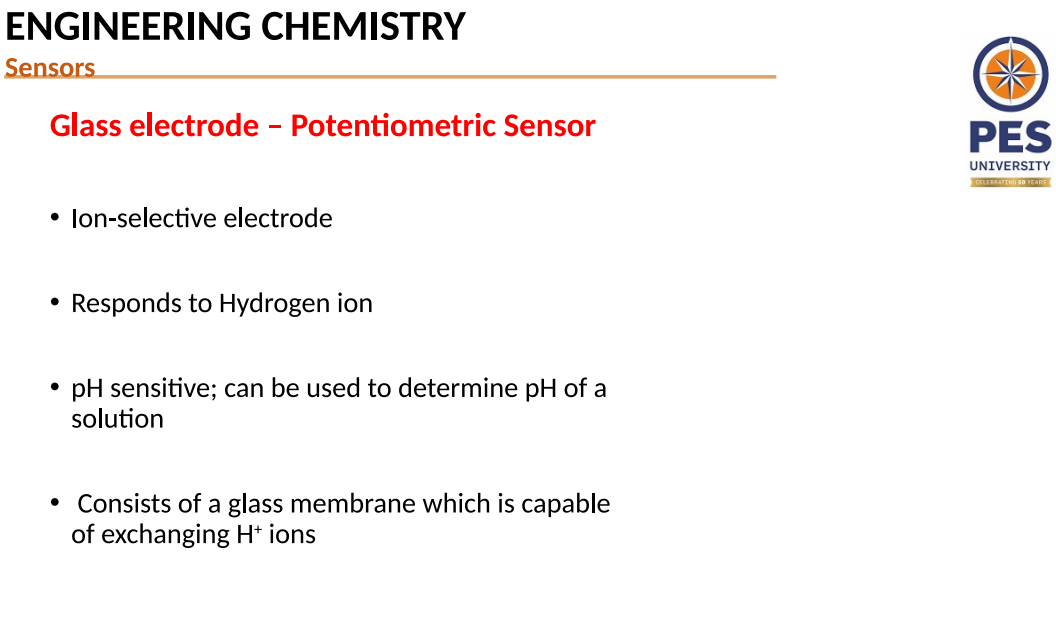

Figure G000124. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 10.

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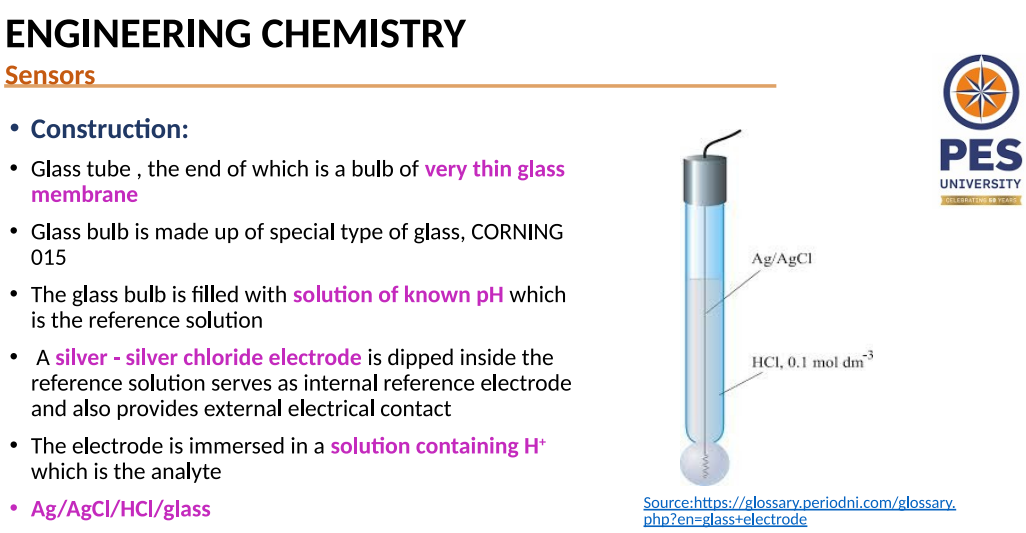

Figure G000125. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 11.

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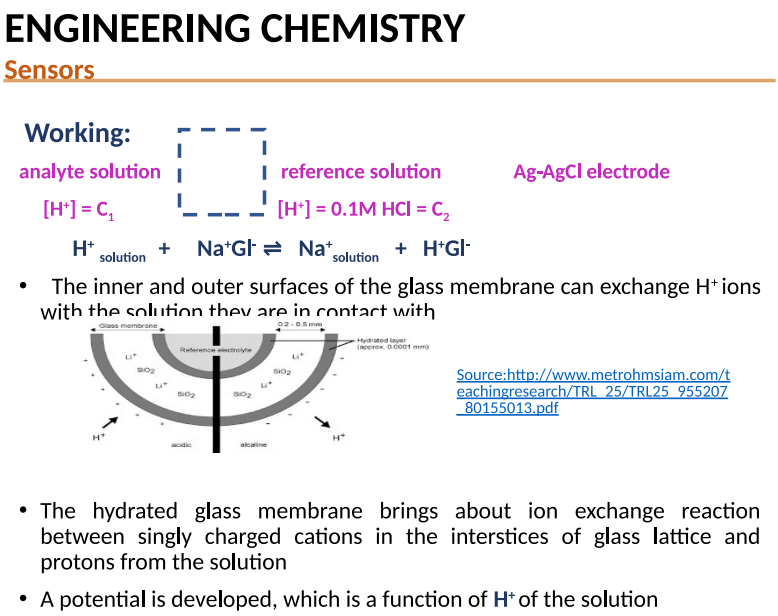

Figure G000126. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 12.

EV0000560

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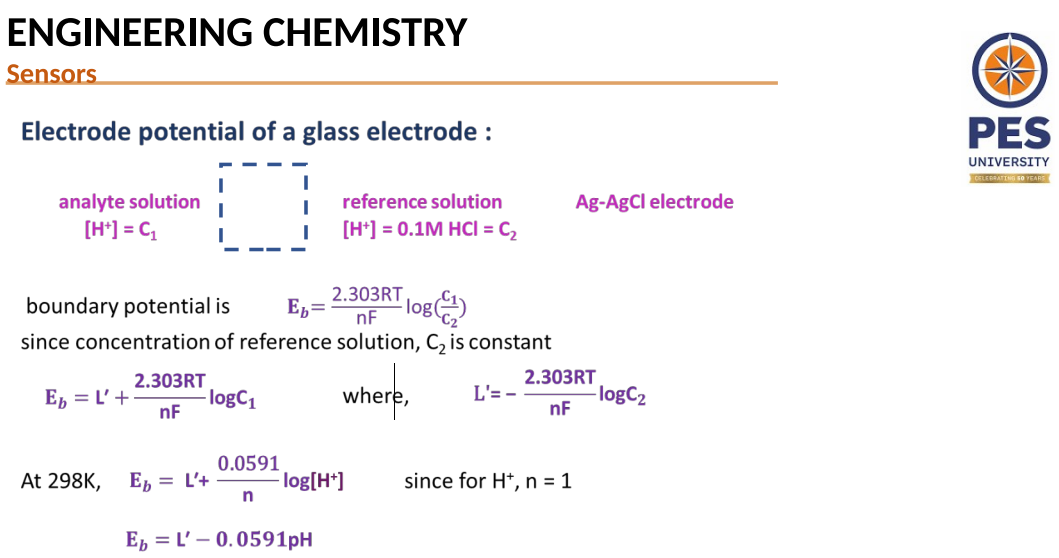

Figure G000127. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 13.

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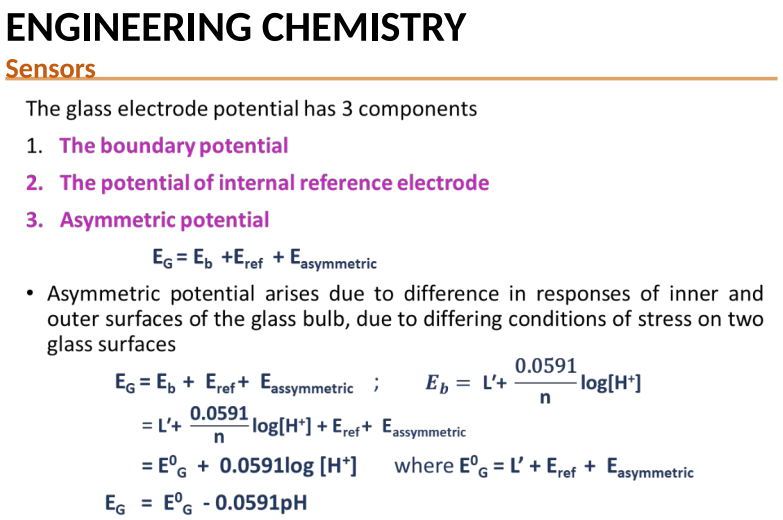

Figure G000128. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 14.

EV0000566

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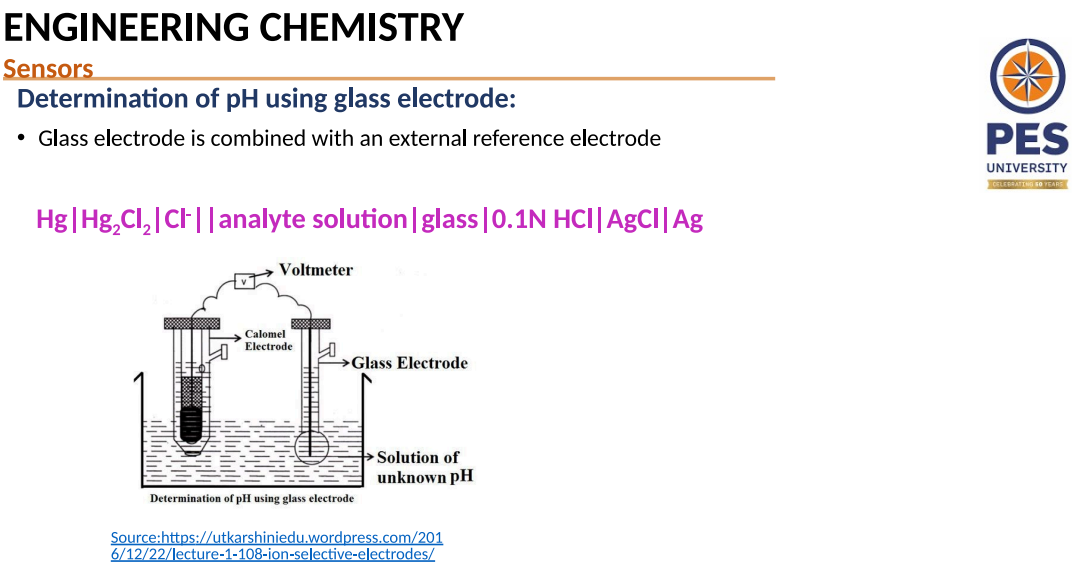

Figure G000129. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 15.

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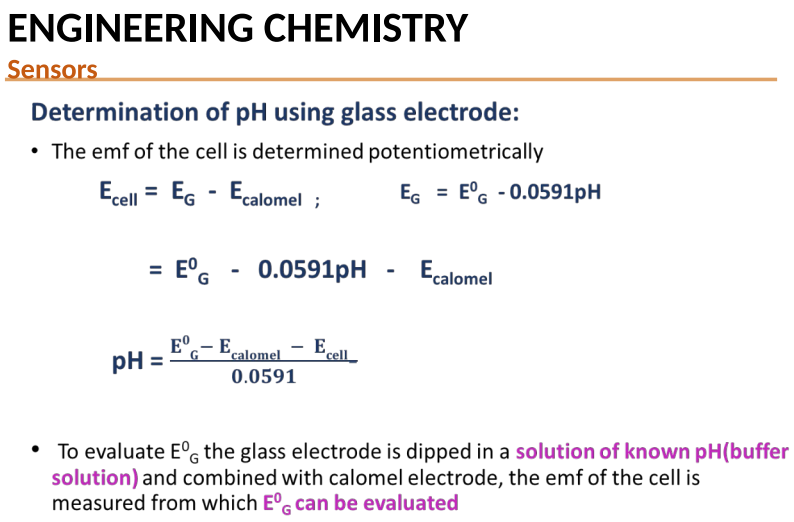

Figure G000130. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 16.

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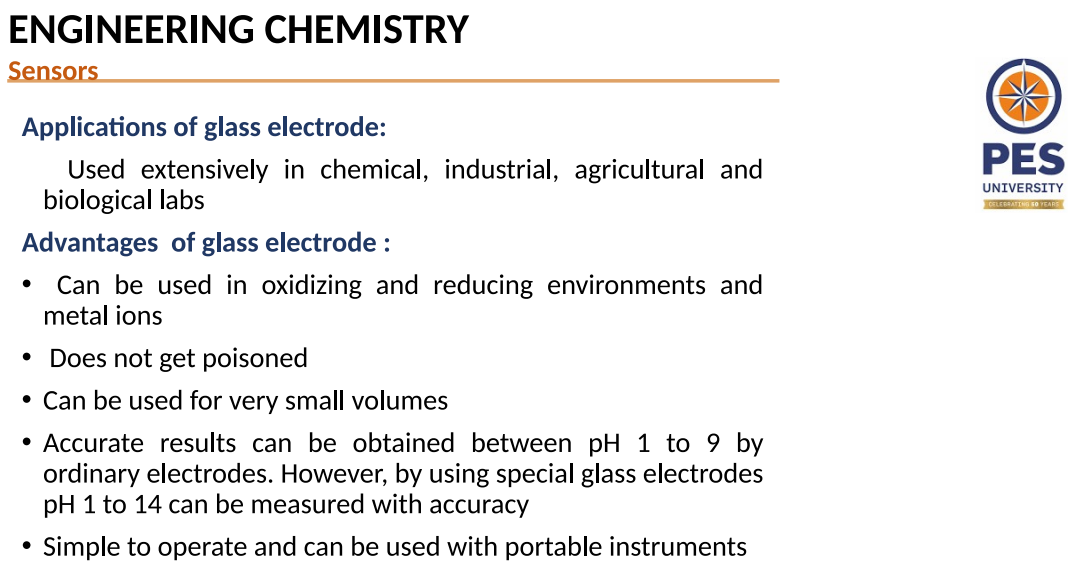

Figure G000131. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 17.

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ENGINEERING CHEMISTRY

Sensors

Disadvantages of glass electrode:

- Because of high resistance of glass, a simple potentiometer cannot be used. It requires sensitive potentiometer for emf measurements
- Glass membrane is very delicate, hence has to be handled carefully
- At very high pH levels usually over a pH of 9 , Alkaline error is observed

$$H^+_{solution} + Na^+Gl^- \rightleftharpoons Na^+_{solution} + H^+Gl^-$$

When the Sodium ion level is relatively high, some of the H^+ ions in the gel layer around the sensitive electrode membrane are replaced by Na^+ ions
The electrode may eventually respond to Na^+ instead of H^+ ions, giving a false lower pH value than the actual value

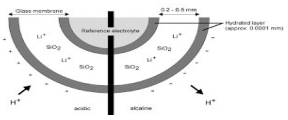

Figure G000132. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 18.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

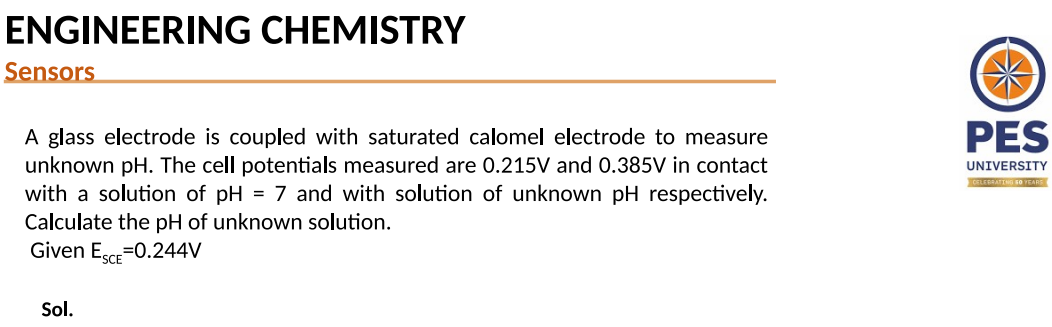

Figure G000133. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 19.

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Figure G000134. Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode · source slide page 20.

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000526 · EV0000527 · EV0000528
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Sensors Class 36 content: Sensors: Types of Electrochemical sensors, pH sensor (Glass electrode) EV0000529 · EV0000530 · EV0000531
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Sensors SENSORS: A sensor is a device that detects and responds to some type of input from the physical environment. The specific input could be light, heat, motion, moisture, pressure, or can be any molecule or substance such as analytes. [Image text, OCR] G 口 Chemical Sensors (17).pdf - Foxit PDF Reader ? 口 File Home Comment View Form Protect Share Help Tell me. · QFind h T 国 自 T T B Hand Select 9 SnapShot Cipboard Zoom Page Fit Reflow Rotate Typewriter Highlight Link Bookmark Fle Image Audio Fil & Option· View* Attachment Annotation & Video Sign Start UE22CY151.pdf ELECTRO CHEM SENS... Chemical Sensors (17).pdf Merge and split PDFs 口 D Information Brain Computer processing ? Amplification; Sensorynerve Analogue circuit preprocessing Signal Transducer Sensory conversion receptor Sensing element (receptor) Environment ☒ 9/273 >☒ ☒☒☒☒ + 178.00% K 7 8 Type here to search ENG 14:40 28°C 17-08-2022 Signal processing in living organisms and in intelligent machines EV0000532 · EV0000533 · EV0000534

4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Sensors ELECTROCHEMICAL SENSORS: A class of chemical sensors in which an electrode is used as a transducer element in the presence of an analyte. Electrochemical sensors transfers the effect of the electrochemical interaction of the analyte-electrode into a useful electric signal. [Image text, OCR] Receptor (Enzyme, carbon nanotube, graphene...) Transducer Amplify (Electrode,thermistor... Electrochemical signals Analyte (H2O2, glucose, urea, ethanol...) https://www.researchgate.net/publication/335398612_Electrochemical_Sensors_Fabricated_by_Electrospinning_Technology_An_Overview EV0000535 · EV0000536 · EV0000537
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Sensors Types of Electrochemical Sensors Potentiometric Sensors Amperometric Sensors EV0000538 · EV0000539 · EV0000540
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Sensors TYPES OF ELECTROCHEMICAL SENSORS POTENTIOMETRIC SENSORS These sensors detect changes in the concentration of a substance by observing how the concentration changes the electrical potential produced by a redox reaction at the electrode. The redox reaction takes place on the electrode surface: Oxidant + ne- ☒ Reduced product It measures the potential difference (voltage) between an indicator electrode and a reference electrode under conditions of negligible current flow, and relates this potential to the activity or concentration of a specific ion or analyte in a solution. [Image text, OCR] E,mV Working electrode Reference electrode EV0000541 · EV0000542 · EV0000543
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Sensors Examples of Potentiometric Sensors Ion Selective Electrode Sensors Glass electrodes (e.g. pH-meter, Na+, K+, Li+ Sens Gas Potentiometric Sensors Solid oxide sensors (e.g. zirconia-based O2 sensor (λ-sensor) EV0000544 · EV0000545 · EV0000546
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Sensors AMPEROMETRIC SENSORS [Image text, OCR] Auxiliary Reference Working electrode electrode electrode Measure the current in response to detect the concentration of the analyte at a fixed potential. The applied potential drives the electron transfer reaction of the analytes, and the measured current indicates the analyte concentration. The sensor is usually composed of 3 electrodes, that is working, auxiliary, and reference electrode. Amperometric sensors quantify the current output between working and the auxiliary electrode. EV0000547 · EV0000548 · EV0000549
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Sensors Working of Amperometric Sensors The working electrode is the electrode at which the reaction of interest occurs, the reference electrode (e.g.; Ag/AgCl/Cl-, Hg/Hg2Cl2/Cl-) provides a stable potential compared to the working electrode. An inert conducting material (e.g.; platinum, graphite) is usually used as auxiliary electrode. A supporting electrolyte is used to provide a conducting medium for movement of ions. E.g. KCl, NaNO3, EV0000550 · EV0000551 · EV0000552
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Sensors Glass electrode – Potentiometric Sensor Ion-selective electrode Responds to Hydrogen ion pH sensitive; can be used to determine pH of a solution Consists of a glass membrane which is capable of exchanging H+ ions EV0000553 · EV0000554 · EV0000555

11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Sensors Construction: Glass tube , the end of which is a bulb of very thin glass membrane Glass bulb is made up of special type of glass, CORNING 015 The glass bulb is filled with solution of known pH which is the reference solution A silver - silver chloride electrode is dipped inside the reference solution serves as internal reference electrode and also provides external electrical contact The electrode is immersed in a solution containing H+ which is the analyte Ag/AgCl/HCl/glass Source:https://glossary.periodni.com/glossary.php?en=glass+electrode EV0000556 · EV0000557 · EV0000558
12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Sensors Working: analyte solution reference solution Ag-AgCl electrode [H+] = C1 [H+] = 0.1M HCl = C2 H+ solution + Na+Gl- ⇌ Na+solution + H+Gl- The inner and outer surfaces of the glass membrane can exchange H+ ions with the solution they are in contact with The hydrated glass membrane brings about ion exchange reaction between singly charged cations in the interstices of glass lattice and protons from the solution A potential is developed, which is a function of H+ of the solution [Image text, OCR] Giass memorene — 0.2-0.5mm Hydrated layer Rederence electrclyte (approx. 0.0001 mm) SiO2 $i02 Li * $iO2 SO2 ? H* H* acific akcaline Source:http://www.metrohmsiam.com/teachingresearch/TRL_25/TRL25_955207_80155013.pdf EV0000559 · EV0000560 · EV0000561
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Sensors EV0000562 · EV0000563 · EV0000564
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Sensors EV0000565 · EV0000566 · EV0000567
15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Sensors Determination of pH using glass electrode: Glass electrode is combined with an external reference electrode Hg|Hg2Cl2|Cl-||analyte solution|glass|0.1N HCl|AgCl|Ag [Image text, OCR] Voltmeter Calomel Electrode A >GlassElectrode Solutionof unknownpH Determination ofpHusingglasselectrode Source:https://utkarshiniedu.wordpress.com/2016/12/22/lecture-1-108-ion-selective-electrodes/ EV0000568 · EV0000569 · EV0000570
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Sensors EV0000571 · EV0000572 · EV0000573
17. Source slide page 17 (extracted text): ENGINEERING CHEMISTRY Sensors Applications of glass electrode: Used extensively in chemical, industrial, agricultural and biological labs Advantages of glass electrode : Can be used in oxidizing and reducing environments and metal ions Does not get poisoned Can be used for very small volumes Accurate results can be obtained between pH 1 to 9 by ordinary electrodes. However, by using special glass electrodes pH 1 to 14 can be measured with accuracy Simple to operate and can be used with portable instruments EV0000574 · EV0000575 · EV0000576

18. Source slide page 18 (extracted text): ENGINEERING CHEMISTRY Sensors Disadvantages of glass electrode: Because of high resistance of glass, a simple potentiometer cannot be used. It requires sensitive potentiometer for emf measurements Glass membrane is very delicate, hence has to be handled carefully At very high pH levels usually over a pH of 9 , Alkaline error is observed H+ solution + Na+Gl- ⇌ Na+solution + H+Gl- When the Sodium ion level is relatively high, some of the H+ ions in the gel layer around the sensitive electrode membrane are replaced by Na+ ions The electrode may eventually respond to Na+ instead of H+ ions, giving a false lower pH value than the actual value EV0000577 · EV0000578 · EV0000579
19. Source slide page 19 (extracted text): ENGINEERING CHEMISTRY Sensors A glass electrode is coupled with saturated calomel electrode to measure unknown pH. The cell potentials measured are 0.215V and 0.385V in contact with a solution of pH = 7 and with solution of unknown pH respectively. Calculate the pH of unknown solution. Given ESCE=0.244V Sol. EV0000580 · EV0000581 · EV0000582
20. Source slide page 20 (extracted text): THANK YOU Department of Science and Humanities EV0000583 · EV0000584 · EV0000585

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.484] Hello, everyone. Welcome back to this course on engineering chemistry. We continue with module 4 on electrochemical equilibria. [00:00:09.668] And in today's class we are going to study the glass electrode, which is an example for ion selective electrode. We look at the construction, the working and how to use the glass electrode to determine pH of a solution. [00:00:24.420] So what is a glass electrode? It's an ion selective electrode which responds to H plus ions. And therefore we can say that it is [00:00:34.000] pH sensitive. So if we want to determine the pH of the solution, then the ion selective electrode which we are going to use is going to be the glass electrode. So like any other ion selective electrode, there has to be a membrane which is going to [00:00:51.488] be able to exchange these H plus ions in this case because we want to determine pH of the solution and here it is the glass membrane. [00:01:01.744] So the most important constituent of R ion selective electrode, the glass electrode, is the glass membrane which is capable of exchanging the H plus ions. Let us look at the construction of this glass electrode. So we have [00:01:20.292] a glass tube, and the end of which [00:01:23.424] We have a bulb. Now, this bulb is the most important part of the glass electrode because this is made of a very thin membrane of glass, and this glass is of a specific type, and that is corning 015. [00:01:41.072] It is used to make this glass. [00:01:44.416] The bulb the glass bulb is made up of coning 015, and this is a very very thin membrane. So we have the membrane which is the most important constituent and it is made of special type of glass. Now inside [00:02:02.064] We need a reference solution. Now, since we are going to measure the H plus ion concentration, the best choice would be it would be to have an HCl. [00:02:14.128] Solution. So the H Cl solution, since it is an internal reference solution, so this is the internal reference solution. And that is [00:02:26.800] HCl of 0.1 molar concentration. So we know the concentration of the H plus ions inside, and that happens to be 0.1 molar. And in and in this solution, there is the internal reference electrode that is the silver silver chloride. [00:02:44.432] Electrode. So that completes the glass electrode. So we have the glass membrane. [00:02:51.840] And then we have the internal reference solution of HCl of 0.1 molar concentration, and then we have the internal reference electrode, which is the silver silver chloride electrode. So we this is glass electrode. Now we want to determine the

concentration of H plus in the [00:03:09.552] Analyte solution. So what do we do now? We have to place this glass electrode in a solution which contains H plus, the concentration of which is not known to us, and therefore using this glass electrode we are going to find out. [00:03:25.504] the concentration of the H plus ion in the analy solution. So what would be the electrode representation for the glass electrode? Now we can see here we will write it as AG, AG C L and H Cl. [00:03:41.284] If you write only this, then that would mean that we are talking about the silver silver chloride electrode. But since glass electrode and we know it is only the glass which is capable of exchanging these H plus ions and not the silver silver chloride electrode. [00:03:59.056] Therefore, we have to mention the glass here. So we have AG, AGCL, LCL, and glass. That is the electrode representation. Let us look at the working of this glass electrode. Now we know that. [00:04:14.244] This is the schematic representation, just like we have for the ion selective electrode, we have the membrane here, and on one side of the membrane we have the reference solution. In this case, it is 0.1 molar HCl. Let us say the concentration is C2, and then we have the internal reference electrode which is placed in [00:04:32.016] inside the reference solution and that is the silver silver chloride electrode. Now coming to the other side of the membrane we have the analyte solution where we have H plus of concentration C1. [00:04:45.124] Now we have been talking about the glass membrane being able to exchange the H plus ions. Now, what is that reaction? That is H plus solution plus Na plus glass. Glass

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.036] If you look at the structure happens to be Na plus Gl minus, and that is in equilibrium with Na plus solution. [00:05:08.528] Plus h plus gl minus. So when there is exchange, what is happening? What can you understand from this reaction? That NA plus Gl minus, which is glass, is being able to exchange this H plus. So H plus I comes into the glass now, and N [00:05:26.224] Plus goes into the solution. This is the exchange reaction which is taking place at the membrane, at the surface of the why is this possible? First let us understand that. [00:05:43.492] If you look at the structure of glass, stru uh glass is actually a three-dimensional network of silica tetra hetra. So we have a full three-dimensional structure of SiO44 minus. [00:06:01.328] And since an open network kind of structure we have this interstitial space which are filled with cations. So we have four minus charge that is balanced with the help of cations. Now which cations will be there? It is going to depend on how the glasses. [00:06:18.960] Is made if it's sodium silicate, we are going to have sodium ice. If it's alumino silicate, we can have aluminium ice, calcium silicate, calcium ice. If it's alumino, potassium silicate, we'll have both aluminum as well as K plus ice. So it depends on the composition of the glass which is being used. [00:06:35.568] So we have this silica tetrahedra SiO44 minus this interstitial spaces is where we have these cations. Now when this glass electrode [00:06:48.368] This membrane comes in contact with the solution containing H plus ions. What happens is [00:06:56.228] The outer lay of this glass membrane are going to swell up. When they absorb this solution and they swell up. So when they swell up, it's easy for the ions to [00:07:11.040] be exchange. So that is how the exchange reaction is happening. So what happens when you put it in an H plus ion concentration? These H plus ions will go in and the sodium ions will come out. Now there could be sodium ions there could be calcium ions there could be aluminium ions in the glass [00:07:27.344] But mostly the divalent and the trivalent are held very tight. So therefore, it's only the monovalent ions which are found to be mobile. So you can see that in the acidic medium we find that. [00:07:41.968] The H plus ion now you can see in the picture over here, since it is acidic

medium, we find that H plus ion moves into this gel layer and the NA plus ion is going to come into the solution. So this is the exact [00:07:57.984] picture of what is happening. What do we mean by exchange of these ions? So the hydrated glass membrane there is an ion exchange reaction and the singly charged ions which are occupying the interstitial space of the glass lattice are going to be exchanged with the proton [00:08:15.664] From the solution, and based on the concentration of H plus ion, a potential is going to be developed at the membrane. At the membrane, we have a potential which is getting formed. So let us see if we can. [00:08:33.060] Get an expression for the potential of the glass electrode. So if we have the the same if we take the same schematic representation, we have this membrane, and here we have the reference solution. [00:08:47.568] Uh where we know it is 0.1 molar HCl. Again, this is internal reference electrode that is AG ADCl electrode. On the other side of the membrane, we have the analyte solution where we have H plus concentration is equal to C1. Now this membrane. [00:09:02.896] surface and this membrane surface are in contact with different concentration of H plus ions. Now however thin the membrane is made it is [00:09:14.848] invariably in contact with two different solutions of the H plus ions. So the inner surface is going to be in contact in contact with [00:09:26.096] The concentration C2, the outer membrane is in contact with the concentration C1. Now, because of this, at the outer membrane we get a potential E1, at the inner membrane we get a potential E2. Now the [00:09:43.236] At the boundary now, therefore, at the membrane boundary, we have a difference in the potential that is E1 minus E2. So in this case, a boundary potential is getting generated. Why? Because the membrane [00:09:59.664] However thin it is, it has an inner surface and it has an outer surface. The inner surface is in contact with one concentration of H plus, the outer surface is in contact with a different concentration of H plus. So the boundary potential expression is given by E boundary is equal to 2.303 Rq over N F log of C1 by.

EV0000154 · EV0000155 · EV0000156 · EV0000157

Transcript block 3 (source interval 000600-000899 seconds): [00:10:17.360] C2. Now C2 is a constant because it is an internal reference solution. We already know it is 0.1 molar H Cl. So we can now write that in the form of a constant. So let us take [00:10:29.604] Say the constant is L dash plus 2.303 RT over N F log of C1. Now what is L dash here? It is minus of 2.303 R T over N F log of C2. Now at 298 Kelvin, this expression now becomes L dash plus 0.05. [00:10:47.376] nine one over n log of h plus now since [00:10:51.968] For H plus I n n is equal to 1. Your expression now for the boundary potential we got L dash minus 0.0591 p h. So we see that the boundary potential is dependent on the pH of the solution. [00:11:11.204] Now, this is not the only potential which exists in the glass electrode. So, in the ion selective electrode, what did we have? [00:11:20.208] We had the boundary potential plus the potential due to the internal reference electrode, which has been placed there to provide the electrical contact. So that potential the reference electrode also has a potential. So that is going to get added up to the boundary potential, but in the case of glass electrode. [00:11:37.728] Apart from this boundary potential and the internal reference electrode potential, we have a third potential coming up that is known as the asymmetric potential. So, therefore, the sum of boundary potential, reference electrode potential, internal reference electrode potential, and the asymmetric potential potential. [00:11:55.376] Potential will give us the total glass electrode potential. Now, what is this asymmetry potential? Let us understand this. [00:12:06.724] Now, the membrane, how much thin it is, we know that the inner and outer surfaces are bound to have different structure. Now, [00:12:18.800] We know that the glass bulb is curved, so the radius of curvature itself is different for the inner and the outer

membrane. So what happens because of that? There is a small potential. Now the stress is different in the inner and outer membrane, and due to this difference in the structure, it gives rise to a potential. [00:12:36.528] Which is known as asymmetry potential. Now, how do I know that this potential actually exists? Now, if I take a solution. [00:12:43.824] outside the analyte solution as the same concentration as the reference solution, we should not be able to measure any other potential except the potential due to the internal reference electrode. But we find that we even though both the concentrations are same, there exists a very small potential which is known [00:13:01.552] As the asymmetric potential, and that is due to the membrane itself, and it is [00:13:08.112] Time dependent. Now it what does it mean that it keeps changing with time. So e glass is e boundary plus e reference plus e asymmetry. Now just now we got an expression for the e boundary that is L dash plus 0.0591 over n log of h plus. So we substitute and [00:13:25.744] We get this expression. Now we know that L dash E reference as well as E asymmetry. [00:13:34.144] are all constants. So we can club them together into a constant called and we refer to as E0 G. So here we have E0 G is equal to L dash plus E reference plus the E asymmetric, and therefore we get the final expression for the glass electrode that is EG is [00:13:51.792] equal to e naught g minus 0.0591 ph so if I know if I can know what is EG, E0 G being constant, I can [00:14:03.984] Find out what is the pH of the solution. But since this is a single electrode, we know that single electrode potentials cannot be measured. Therefore, we need to make a combination, and now what we have is we have to combine it with the external reference solution. [00:14:21.744] solution using a solid bridge. So external reference solution sorry external reference electrode. So we have to have an elect electrode another electrode and that will be known as the external reference electrode. [00:14:39.168] So we have external reference electrode, the analyte solution, and the glass electrode. So both the external reference electrode as well as the glass electrode are now [00:14:49.232] put into the analyte solution are dipped into the analyte solution. You can see here this is the calomel electrode, which is the external reference electrode, and this is the glass electrode. So how do we write

EV0000158 · EV0000159 · EV0000160 · EV0000161

Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.624] cell representation for this combination. And why are we doing this? Because we want to see how to use the glass electrode to determine pH of the solution. So what do we have here? We have the [00:15:14.660] External reference electrode that is the calomide electrode HE2Cl2 Cl minus. Now there's a salt bridge which is connecting here. So we have the analyte solution, and on this side, what do we have? Now if you look here, this electrode. [00:15:30.624] Through the solid here, through the porous grid here, it is in contact with the analy solution. Now analyde solution is in contact with the glass membrane. That in turn is in contact with the solution, that is the internal reference solution, and that is in contact with the silver superchloride electrode. So that's how we write this. [00:15:48.272] We have galomel electrode which is diped in analyte that is in contact with the glass which is in contact with 0.1 normal HCl or 0.1 molar HCl and that is in contact with the silver chloride and then silver. So this is how we write the cell representation. Let us find an expression now. [00:16:08.484] Using the glass electrode, how are we determining the pH? We know that we have made a combination and we can measure the EMF of this cell. This can be measured potentiometrically. So E cell is equal to [00:16:21.680] E right minus E left, so E glass minus E calomel. E glass just now we got this expression that it is E naught G minus 0.0591 PH and when we substitute we get this expression E naught G minus 0.0591 PH minus E calomel and if you

rearrange we get [00:16:39.312] the expression for pH. So use if we can measure E cell, we know what is E calomel. [00:16:45.984] E naught G is fixed for a particular glass electrode and then using this expression I can determine the pH of the solution. So all we have to do is combine the glass electrode with an external reference electrode and measure the E cell. [00:17:02.736] Now E naught G, E naught G value is not fixed for [00:17:08.452] All the glass electrodes. I mean what I mean is E0 G is not the same for all the glass electrode. Therefore, before using any glass electrode, I need to measure what is E0 G for that particular glass electrode. So what do I do for that? Using the same expression or the same setup, I can do it. [00:17:26.224] Do I do now? I take a solution of known pH. If I know pH and I know what is e calomel, I can measure E cell, I can determine what is E0 G. So this is how we are going to determine E0 G first, and that can be used for the to determine the [00:17:45.472] Uh further once we know what is E0 G, then we can determine pH of any solution. So how do we uh keep the pH constant? That is with the help of a buffer solution. We uh we use a uh solution of known pH, usually a constant pH like buffer solution, and we find out E0 G and [00:18:03.120] And further, knowing ERG we can just measure the E cell and get the values of the various solution the hydrogen and concentration of different solution. [00:18:15.492] Now, applications of the glass electrode. It is used extensively in chemical, industrial, agricultural, and biological labs. What are the advantages? It can be used in oxidizing and reducing environments. [00:18:28.992] Which was not possible with the help of hydrogen electrode. Hydrogen electrode can also be used to determine H plus, but it cannot be used in oxidizing and reducing environment environments. And metal ions, of course, we know it's an ion selective electrode. It is going to [00:18:43.200] Uh be sensitive or detect only the H plus ions and not the other metal ions, and it does not get poison unlike the hydrogen electrode, and it can be used for very small volumes. And one of the most important uh [00:18:58.564] Advantage is that we get accurate results up to a pH of nine. [00:19:04.944] However, beyond the pH of 9, we need special type of glass electrodes which can measure up to 14 pH with a good amount of accuracy. And they are of course very simple to operate and can be used with portable instruments. [00:19:23.204] What are the disadvantages of glass electrode? Now we know that there is a membrane. [00:19:29.488] And whenever there is a membrane, it offers very high resistance. The same thing happens with the glass electrode also. It is common for any ion selective electrode. So glass membrane offers very high resistance and simple potentiometers cannot be used, so we have to use sensitive. [00:19:46.656] uh potentiometers for EMF measurements and secondly the glass membrane which we are using is a very very thin membrane so it is very delicate and hence has to be handled very carefully.

EV0000162 · EV0000163 · EV0000164 · EV0000165

Transcript block 5 (source interval 001200-001499 seconds): [00:20:00.560] Now one very important thing which happens beyond a pH of 9 is known as alkaline error. So we have to understand what is this alkaline error? [00:20:13.284] Just now I mentioned up to pH of 9 it gives very good accurate results, but beyond the pH of 9, we see some kind of error in the normal uh in the glass electrode, and that electric is referred to as the alkaline error. So what happens is we know that this is the exchange. [00:20:31.056] reaction which normally happens at uh in when there is a glass electrode in contact with h plus ions. Now beyond the pH of 9 the concentration of H plus has become very less. [00:20:45.220] And if sodium ions are more in the solution, then the reverse reaction happens. It's an equilibrium. So this [00:20:53.600] now is more and the reverse reaction happens. So as a result what will happen? The NA plus ion gets into the uh into the glass and the H plus ion is now

given out. So you can see in alkaline medium here we find that the H plus ion is coming out and N A plus ion is [00:21:11.280] Going into that means the electrode becomes responsive to Na plus ions. Therefore, the pH now which you measure is going to be lower than the [00:21:23.168] Actual pH. So what you measure will be lower because now it is responding to NA plus ions also. So you get a false pH value, which will be lower than the actual pH value. So this is known as alkaline error, which happens beyond a pH of 9. [00:21:41.284] So that is about the glass electrode and in the next class we are going to look at another ion selective electrode that is the fluoride ion selective electrode. Thank you.

EV0000166 · EV0000167 · EV0000168 · EV0000169

Full transcript: 3416 normalized words, 93 timestamped segments, 1315.306667 seconds duration. Low-confidence segments flagged: 0.

EV0000150 · EV0000151 · EV0000152 · EV0000153

11 Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1)

This lesson is indexed from the source video titled Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1). Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000170 · EV0000171

Graphs and diagrams

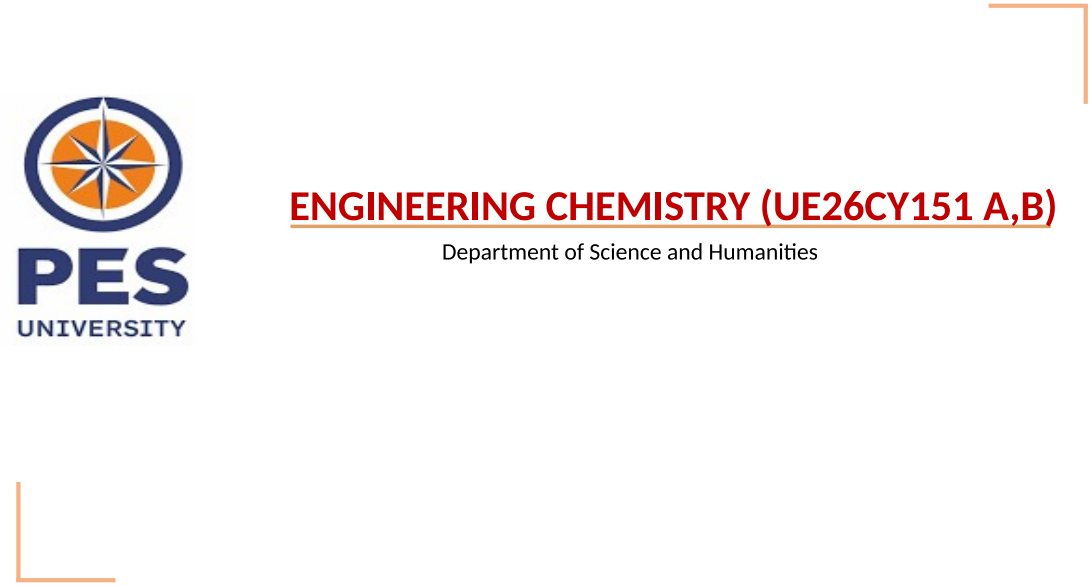

Figure G000135. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 1.

EV0000587

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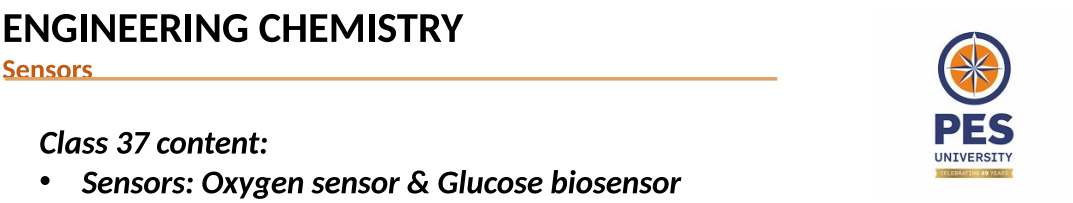

Figure G000136. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 2.

EV0000590

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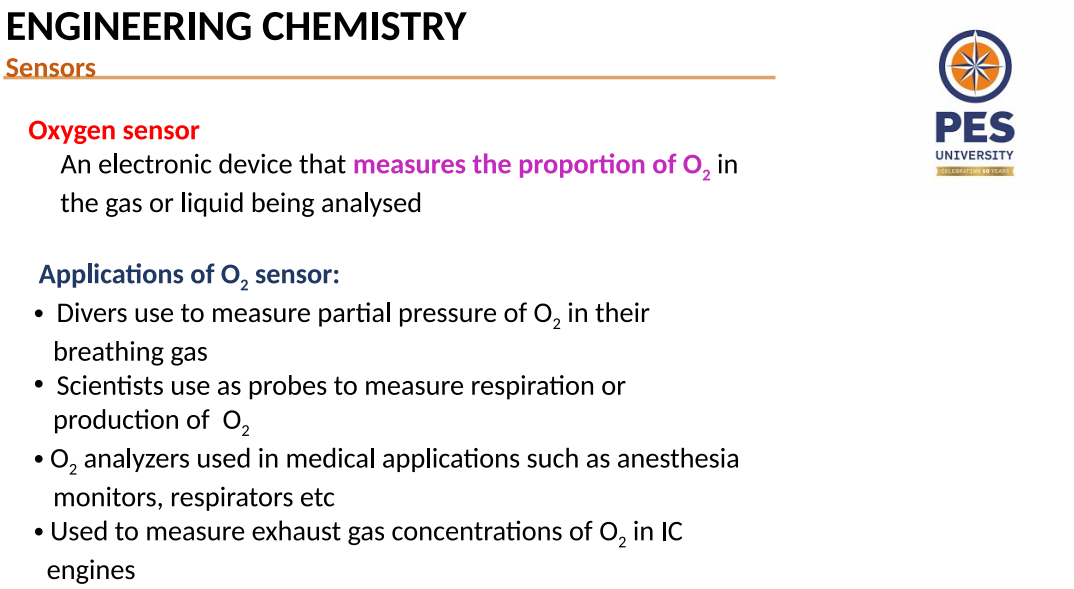

Figure G000137. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 3.

EV0000593

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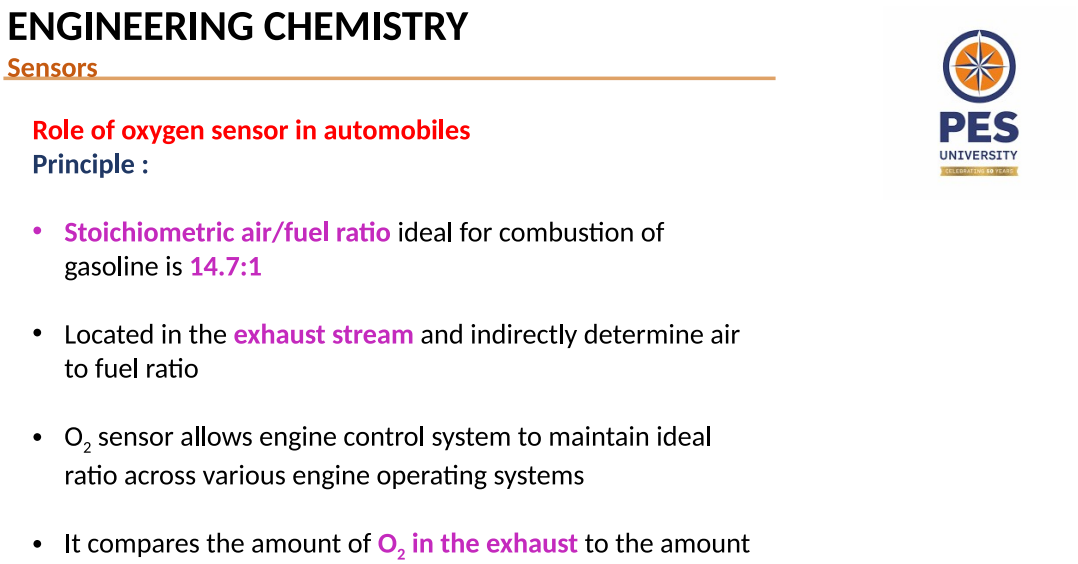

Figure G000138. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 4.

EV0000596

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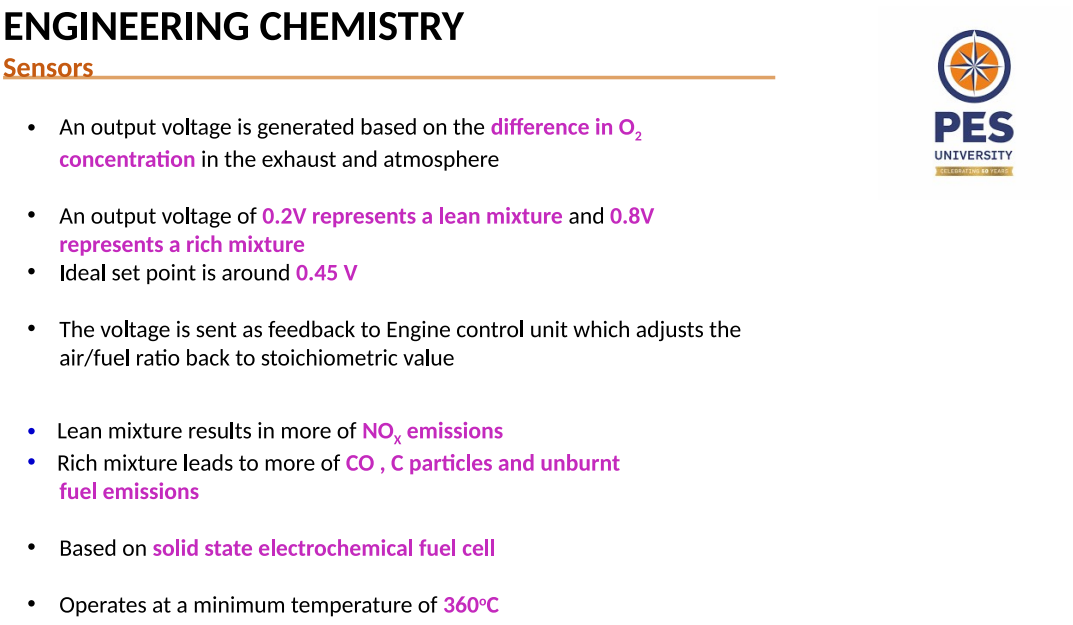

Figure G000139. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 5.

EV0000599

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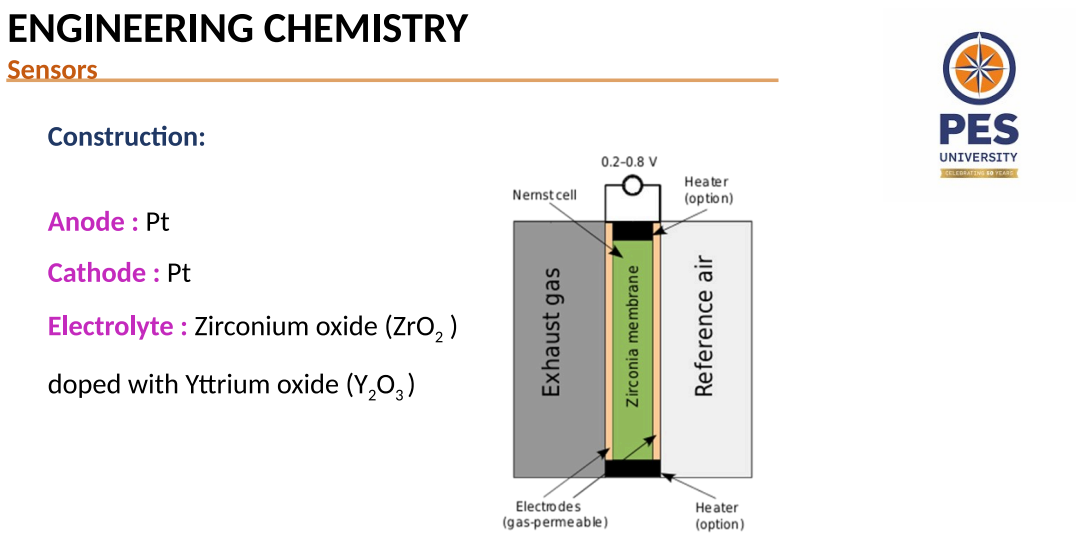

Figure G000140. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 6.

EV0000602

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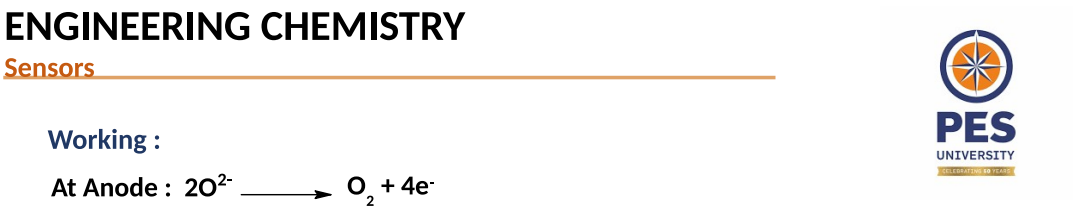

Figure G000141. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 7.

EV0000605

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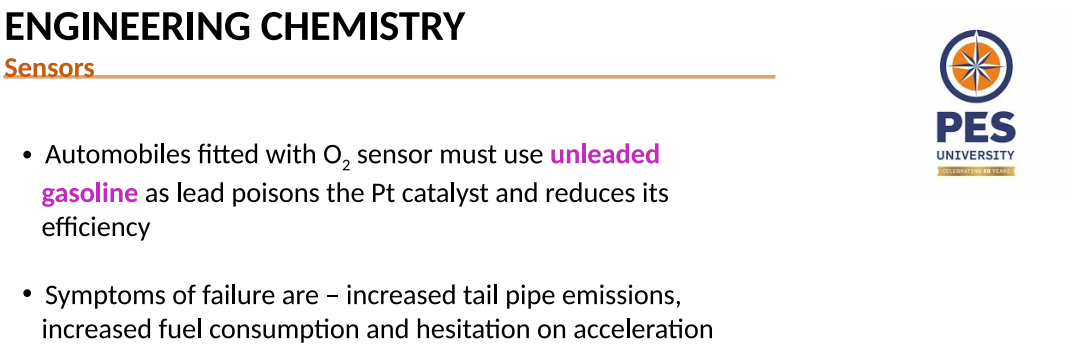

Figure G000142. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 8.

EV0000608

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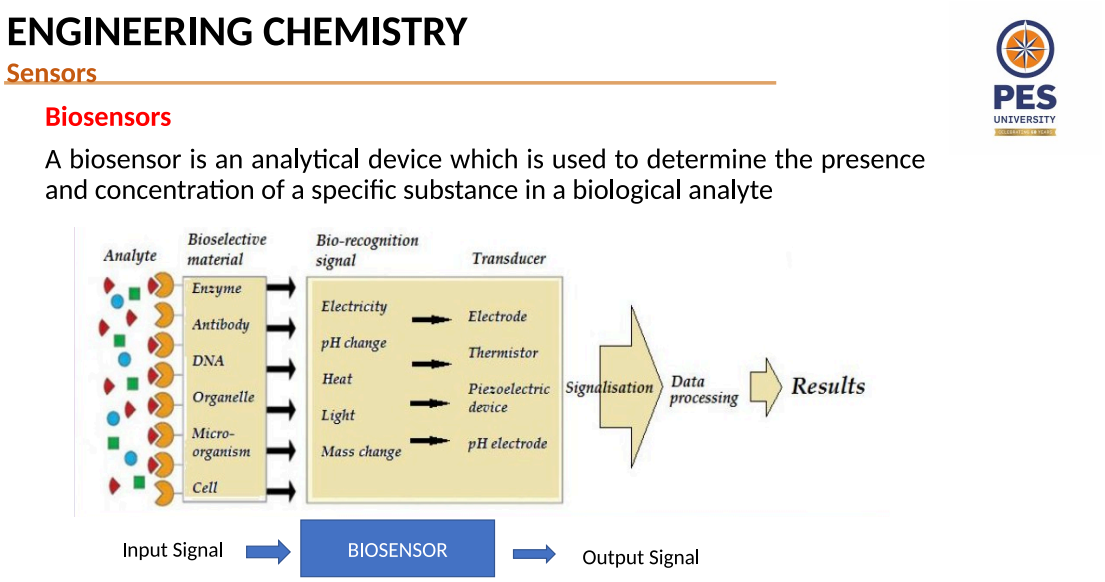

Figure G000143. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 9.

EV0000611

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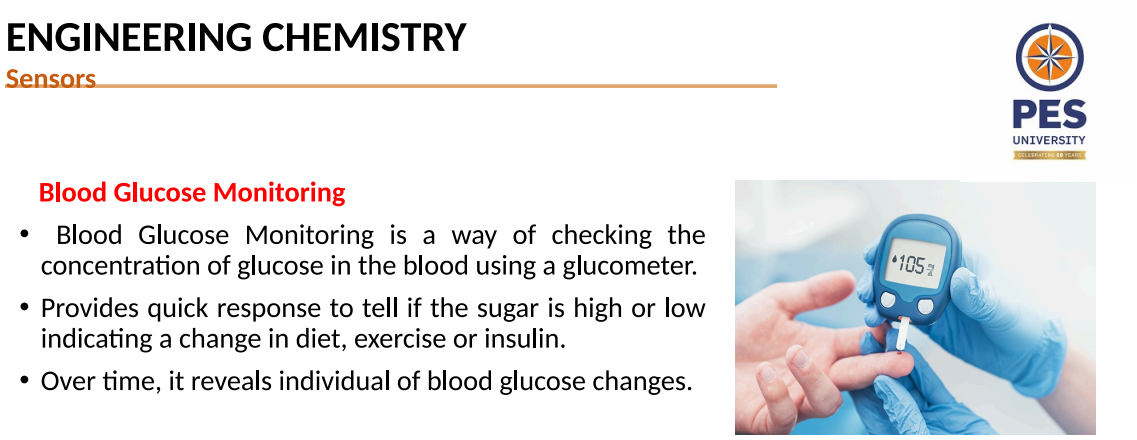

Figure G000144. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 10.

EV0000614

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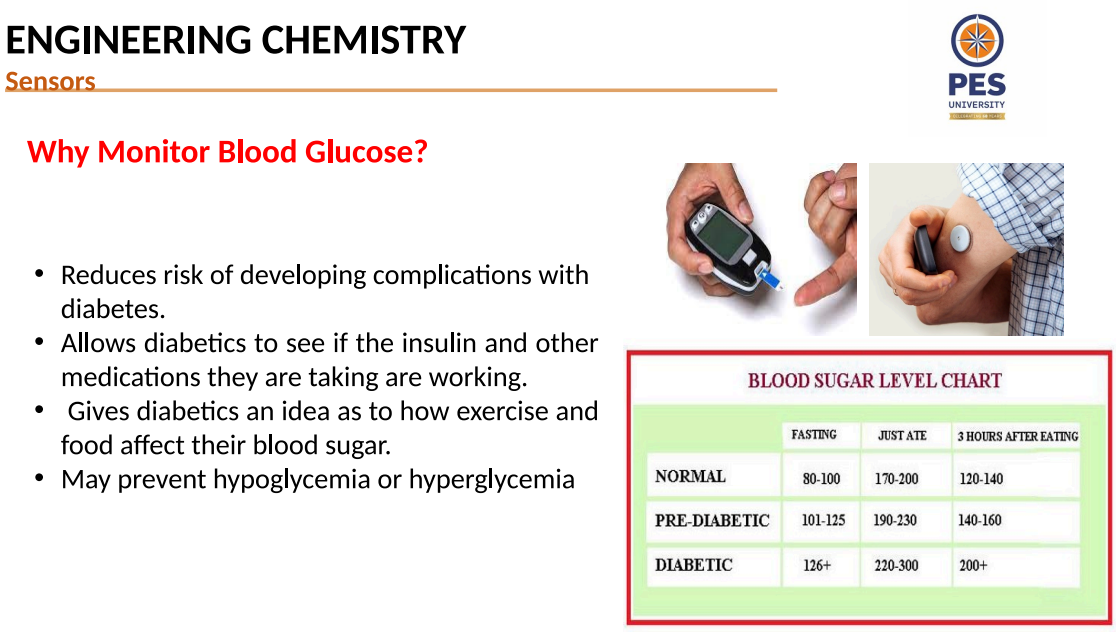

Figure G000145. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 11.

EV0000617

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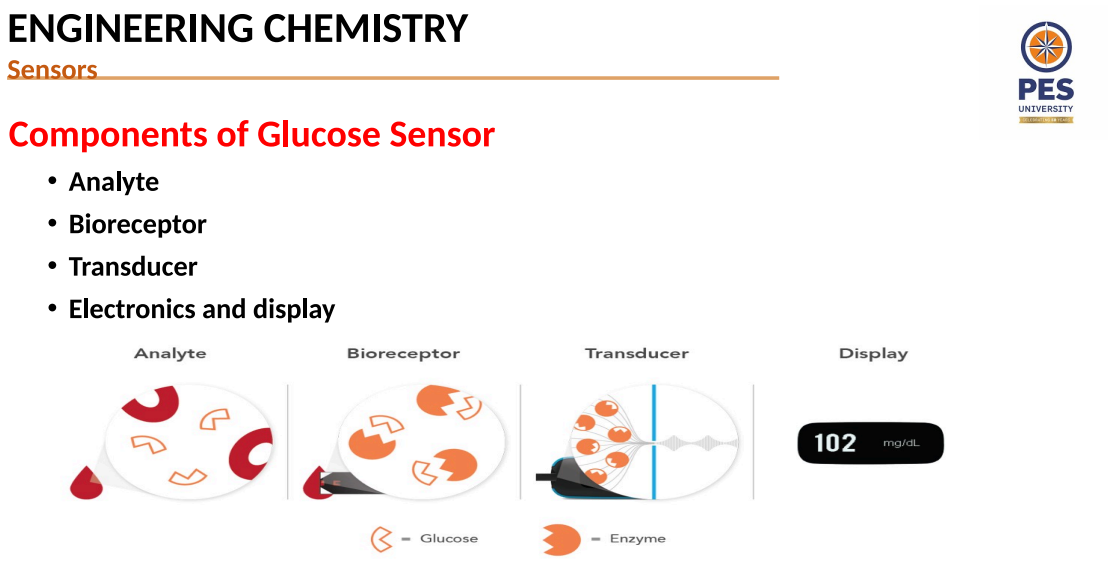

Figure G000146. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 12.

EV0000620

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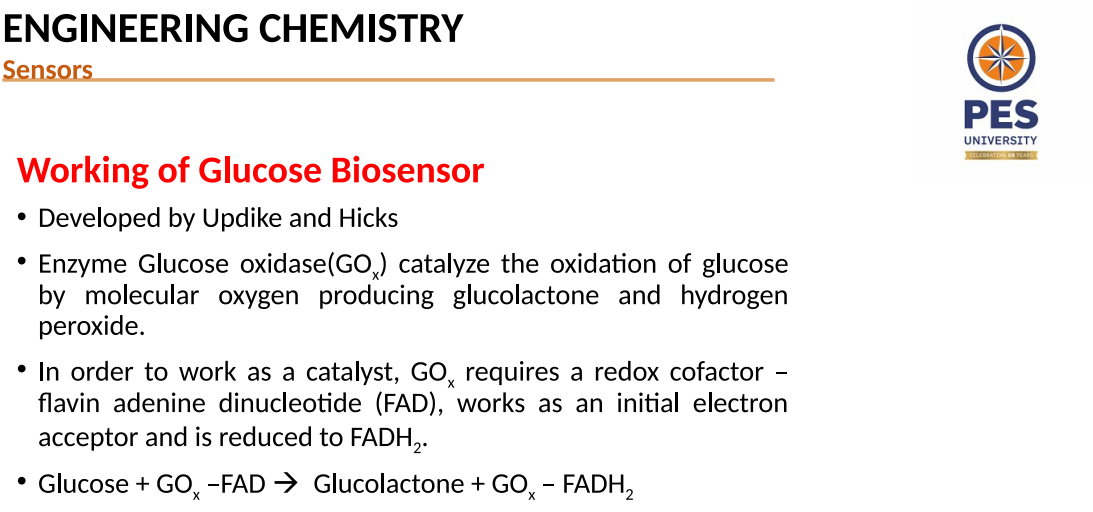

Figure G000147. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 13.

EV0000623

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ENGINEERING CHEMISTRY

Sensors

Working of Glucose Biosensor

- The cofactor is regenerated by reacting with oxygen, leading to the formation of hydrogen peroxide
- $GOx - FADH_2 + O_2 \rightarrow GOx - FAD + H_2O_2$
- Hydrogen peroxide is oxidized at a platinum electrode.
- The number of electron transfers, at electrode surface is directly proportional to the number of glucose molecules present in the blood.
- $H_2O_2 \rightarrow 2H^+ + O_2 + 2e^-$

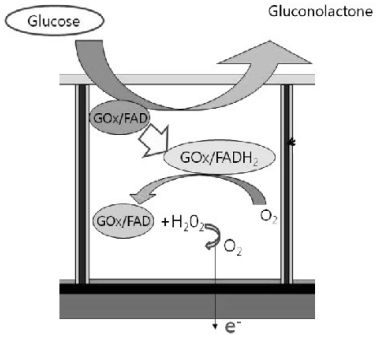

Figure G000148. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 14.

EV0000626

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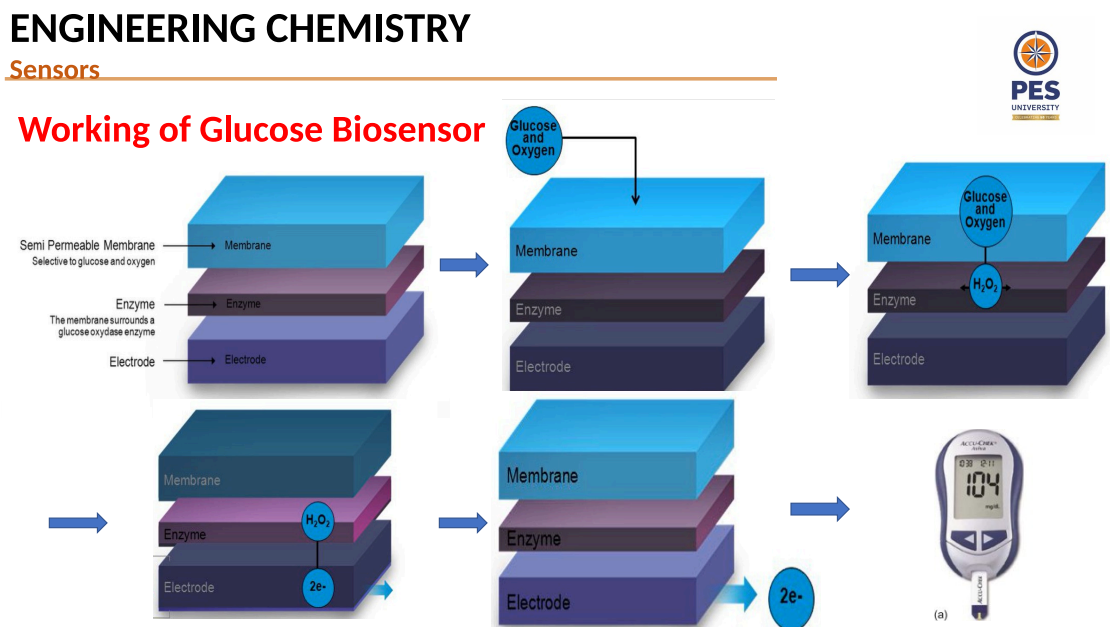

Figure G000149. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 15.

EV0000629

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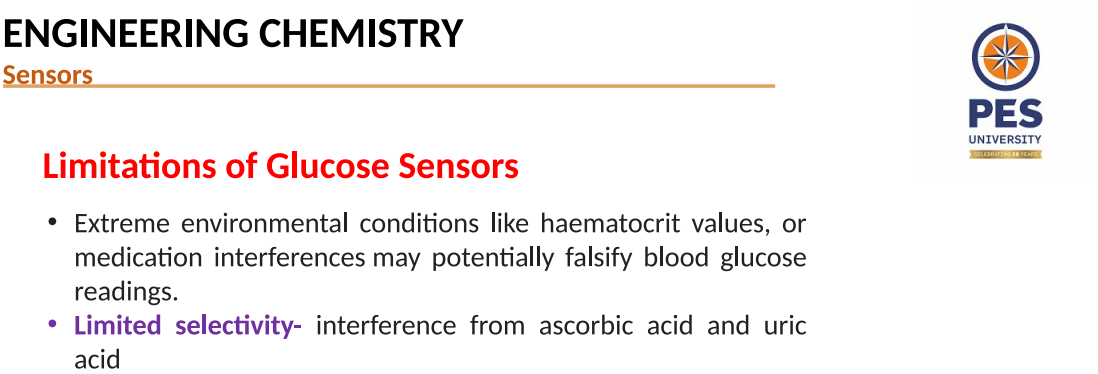

Figure G000150. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 16.

EV0000632

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Figure G000151. Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor · source slide page 17.

EV0000635

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000586 · EV0000587 · EV0000588
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Sensors Class 37 content: Sensors: Oxygen sensor & Glucose biosensor EV0000589 · EV0000590 · EV0000591
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Sensors Oxygen sensor An electronic device that measures the proportion of O2 in the gas or liquid being analysed Applications of O2 sensor: Divers use to measure partial pressure of O2 in their breathing gas Scientists use as probes to measure respiration or production of O2 O2 analyzers used in medical applications such as anesthesia monitors, respirators etc Used to measure exhaust gas concentrations of O2 in IC engines EV0000592 · EV0000593 · EV0000594
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Sensors Role of oxygen sensor in automobiles Principle : Stoichiometric air/fuel ratio ideal for combustion of gasoline is 14.7:1 Located in the exhaust stream and indirectly determine air to fuel ratio O2 sensor allows engine control system to maintain ideal ratio across various engine operating systems It compares the amount of O2 in the exhaust to the amount of O2 in the atmosphere EV0000595 · EV0000596 · EV0000597

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Sensors An output voltage is generated based on the difference in O2 concentration in the exhaust and atmosphere An output voltage of 0.2V represents a lean mixture and 0.8V represents a rich mixture Ideal set point is around 0.45 V The voltage is sent as feedback to Engine control unit which adjusts the air/fuel ratio back to stoichiometric value Lean mixture results in more of NOX emissions Rich mixture leads to more of CO , C particles and unburnt fuel emissions Based on solid state electrochemical fuel cell Operates at a minimum temperature of 360oC EV0000598 · EV0000599 · EV0000600
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Sensors Construction: Anode : Pt Cathode : Pt Electrolyte : Zirconium oxide (ZrO2) doped with Yttrium oxide (Y2O3) EV0000601 · EV0000602 · EV0000603
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Sensors Working : Cell voltage is given by where P1 and P2 are the partial pressures of O2 in reference air and exhaust gas respectively EV0000604 · EV0000605 · EV0000606
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Sensors Automobiles fitted with O2 sensor must use unleaded gasoline as lead poisons the Pt catalyst and reduces its efficiency Symptoms of failure are – increased tail pipe emissions, increased fuel consumption and hesitation on acceleration EV0000607 · EV0000608 · EV0000609
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Sensors Biosensors A biosensor is an analytical device which is used to determine the presence and concentration of a specific substance in a biological analyte [Image text, OCR] ElectrochemicalBiosensors A biosensor is an analytical device which is used to determine the presence and concentration of a specific substance in a biological analyte. Bioselective Bio-recognition Analyte material signal Transducer Enzyme Electricity Antibody Electrode pHchange Thermistor DNA Heat Piezoelectric Signalisation Data Results Organelle processing Light device Micro- Masschange pHelectrode organism Cell Output signals Input signals Biosensor Share 41/62 BIOSENSOR Input Signal Output Signal EV0000610 · EV0000611 · EV0000612
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Sensors Blood Glucose Monitoring Blood Glucose Monitoring is a way of checking the concentration of glucose in the blood using a glucometer. Provides quick response to tell if the sugar is high or low indicating a change in diet, exercise or insulin. Over time, it reveals individual of blood glucose changes. [Image text, OCR] 105 EV0000613 · EV0000614 · EV0000615
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Sensors Why Monitor Blood Glucose? Reduces risk of developing complications with diabetes. Allows diabetics to see if the insulin and other medications they are taking are working. Gives diabetics an idea as to how exercise and food affect their blood sugar. May prevent hypoglycemia or hyperglycemia [Image text, OCR] G192.168.254.1:8090 -(X E Glucose Sensor - an o X ☐ G Gmail What Is a Normal Bloc X Gblood sugar level in ta X tPrintable Blood Sugar X + C wordtemplatesonline.net/blood-sugar-level-charts/ E ☆ ☐ Paused ★Bookmarks M f (38,034 unread) - h... Google Scholar http://www.google... Sci-Hub:removing... YouTube Maps sci hub Paraphrasing Tooll. G Gmail WTO BUSINESS PERSONAL REAL ESTATE 240 210 BLOODSUGARLEVELCHART FASTING JUSTATE 3HOURSAFTEREATING NORMAL 80-100 170-200 120-140 PRE-DLABETIC 101-125 190-230 140-160 DLABETIC 126+ 220-300 200+ Blood Sugar Chart 05 CLOSEX (play slideshow). 8 O Type here to search 22℃ENG 10:23 X 19-11-2022 EV0000616 · EV0000617 · EV0000618

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Sensors Components of Glucose Sensor Analyte Bioreceptor Transducer Electronics and display [Image text, OCR] How the Glucose Sensor Wor X How the Glucose Sensor WorX GCheng, E. Wang, and X. Yang.× G components in glucose sensc X @ How Does a Glucose Sensor X + × → C agamatrix.com/blog/glucose-sensors/#:˜:text=For%20a%20glucose%20biosensor%2C%20the,that%20specifically%20recognizes.. E ☆ G 口 Paused Update Bookmarks ? ? (38,034 unread) - h... Google Scholar Ghttp://www.google... Sci-Hub: removing... ? YouTube Maps sci hub Paraphrasing Tool |... Analyte Bioreceptor Transducer Display 102 mg/dL Glucose ：Enzyme Chemical Sensors (...,df 皇 engproc-06-00017..zip UE22CY151.pdf Chemical Sensors (..pdf UE22CY151.pdf Show all X 8 Type here to search 22℃ENG 18:45 21-08-2022 EV0000619 · EV0000620 · EV0000621
13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Sensors Working of Glucose Biosensor Developed by Updike and Hicks Enzyme Glucose oxidase(GOx) catalyze the oxidation of glucose by molecular oxygen producing glucolactone and hydrogen peroxide. In order to work as a catalyst, GOx requires a redox cofactor –flavin adenine dinucleotide (FAD), works as an initial electron acceptor and is reduced to FADH2. Glucose + GOx –FAD ☒ Glucolactone + GOx – FADH2 EV0000622 · EV0000623 · EV0000624
14. Source slide page 14 (extracted text): ENGINEERING CHEMISTRY Sensors Working of Glucose Biosensor The cofactor is regenerated by reacting with oxygen, leading to the formation of hydrogen peroxide GOx – FADH2 + O2 ☒ GOx – FAD + H2O2 Hydrogen peroxide is oxidized at a platinum electrode. The number of electron transfers, at electrode surface is directly proportional to the number of glucose molecules present in the blood. H2O2 ☒ 2H+ +O2 + 2e- [Image text, OCR] Gluconolactone Glucose GOX/FAD GOx/FADH2 GOX/FAD +H02= O e EV0000625 · EV0000626 · EV0000627

15. Source slide page 15 (extracted text): ENGINEERING CHEMISTRY Sensors [Image text, OCR] How the Glucose SensorX How the Glucose Sensor Wor X G Cheng, E. Wang, and X. Yang. X G components in glucose sensc × @ How Does a Glucose Sensor X → C slideplayer.com/slide/4214795/ ☆ G 口 Paused Update Bookmarks ? ? (38,034 unread) - h.. Google Scholar http://www.google.... Sci-Hub: removing... YouTube Maps sci hub Paraphrasing Tool |.. Glucose glucose and and Oxygen ome in th the xidase he first Membrane eaction e Enzyme Electrode Chemical Sensors (.pdf 皇 engproc-06-00017...zip UE22CY151.pdf Chemical Sensors (...pdf UE22CY151.pdf Show all X 8 O Type here to search 22℃ENG 18:56 21-08-2022 Working of Glucose Biosensor [Image text, OCR] How the Glucose Sensor X How the Glucose Sensor Wor X G Cheng, E. Wang, and X. Yang. X G components in glucose sensc X @ How Does a Glucose Sensor × → C slideplayer.com/slide/4214795/ ☆ G 口 Paused Update Bookmarks ? ? (38,034 unread) - h... Google Scholar Ghttp://www.googl.... Sci-Hub: removing... ? YouTube ? Maps sci hub Paraphrasing Tool |... SemiPermeableMembrane Membrane Selectivetoglucoseandoxygen Enzyme Enzyme The membrane surrounds a glucoseoxydase enzyme Electrode Electrode Chemical Sensors (.pdf 皇 engproc-06-00017...zip UE22CY151.pdf Chemical sensors (..pdf UE22CY151.pdf Show all X 8 O Type here to search 22℃ENG 18:55 21-08-2022 [Image text, OCR] How the Glucose Sensor X How the Glucose Sensor Wor X G Cheng, E. Wang, and X. Yang. X G components in glucose sensc × @ How Does a Glucose Sensor X → C slideplayer.com/slide/4214795/ ☆ G 口 Paused Update ★Bookmarks ? (38,034 unread) - h.. Google Scholar G http://www.google... Sci-Hub: removing... YouTube Maps sci hub Paraphrasing Tool |... ontactwitn e oxidase Glucose and Oxygen nd oxygen Membrane rted into H2O2 Peroxide Enzyme d gluconic Electrode Chemical Sensors (..pdf 皇 engproc-06-00017..zip UE22CY151.pdf Chemical Sensors (...pdf UE22CY151.pdf Show all X 8 O Type here to search 22℃ENG 18:58 21-08-2022 [Image text, OCR] How the Glucose Sensor X How the Glucose Sensor Wor X G Cheng, E. Wang, and X. Yang. X G components in glucose sensc × @ How Does a Glucose Sensor X → C slideplayer.com/slide/4214795/ ☆ G 司 口 Paused Update Bookmarks ? ? (38,034 unread) - h... Google Scholar G http://www.google... Sci-Hub: removing... YouTube Maps sci hub Paraphrasing Tool |.. ent Membrane ed to a Enzyme value 2e- er value Electrode he pump ISIG Chemical Sensors (.pdf 皇 engproc-06-00017..zip UE22CY151.pdf Chemical Sensors (..pdf UE22CY151.pdf Show all X 8 O Type here to search ENG 19:01 22°C 21-08-2022 [Image text, OCR] How the Glucose SensorX How the Glucose Sensor Wor X G Cheng, E. Wang, and X. Yang. X G components in glucose sensc × @ How Does a Glucose Sensor X → C slideplayer.com/slide/4214795/ ☆ G 司 口 Paused Update ★Bookmarks ? (38,034 unread) - h... Google Scholar G http://www.google... Sci-Hub: removing... YouTube Maps sci hub Paraphrasing Tool |.. is applied to ode, causing own into: Membrane gen H202 Enzyme n rons Electrode 2e- more glucose in your body, the more H2O2 generated. ThemoreH2O2,themorecurrentgenerated. Chemical Sensors (..,pdf engproc-06-00017...zip UE22CY151.pdf Chemical Sensors (...pdf UE22CY151.pdf Show all X 8 O Type here to search 22℃ENG 18:59 21-08-2022 EV0000628 · EV0000629 · EV0000630
16. Source slide page 16 (extracted text): ENGINEERING CHEMISTRY Sensors Limitations of Glucose Sensors Extreme environmental conditions like haematocrit values, or medication interferences may potentially falsify blood glucose readings. Limited selectivity- interference from ascorbic acid and uric acid EV0000631 · EV0000632 · EV0000633
17. Source slide page 17 (extracted text): THANK YOU Department of Science and Humanities EV0000634 · EV0000635 · EV0000636

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Hello everyone, welcome back to this course on engineering chemistry. We will continue with module five on energy storage devices and will discuss oxygen sensor. Firstly, we will understand what is an oxygen sensor and look at some applications of oxygen sensor. [00:00:18.976] and specifically how is an oxygen sensor used in automobiles the principle construction and working [00:00:28.740] Let's define what is an oxygen sensor. It is an electronic device that measures the proportion of oxygen in the gas or liquid being analyzed. So what would be the applications of such a sensor? First and foremost. [00:00:48.964] Divers use it to measure the partial pressure of oxygen in their breathing gas. We know that in the atmosphere we have about twenty percent of oxygen, which is the right proportion for us to breathe. [00:01:05.284] Now divers when they go deep sea they need to carry cylinders of oxygen and there they are using that oxygen for breathing and hence the partial pressure of oxygen has to be maintained at the same value what is the [00:01:23.056] There in the atmosphere. Hence monitoring the partial pressure of oxygen is very important, and therefore they use oxygen sensors. They are equipped, the deep sea divers gas cylinders carrying oxygen are equipped with oxygen sensors. [00:01:42.884] Another application is when scientists use as probes to measure respiration or production of oxygen. Oxygen analyzers used in [00:01:57.572] Many applications such as anesthesia monitors and respirators have to use these oxy sensors to determine the partial pressure of oxygen which is being used in these medical appliances. Then [00:02:18.276] It is also used to measure the exhaust gas concentrations of oxygen in IC engines, that is in internal combustion engines. So these are some of the applications for oxygen sensor, and now we are going to study the last application. [00:02:36.048] Mentioned here, that is to measure the exhaust gas concentration of oxygen in IC engines. [00:02:45.252] So what is the role of an oxygen sensor in automobiles? So first let us understand the principle based on which the oxygen sensor works and specifically in automobiles. When fuel burns [00:03:04.644] the ideal ratio for combustion of air is to fuel would be fourteen point seven is to one. [00:03:16.004] Why do we need this ratio? Because when combustion is complete, we know that the there will be only carbon dioxide and water which are going to be let out into the environment. [00:03:33.376] into the atmosphere. But if the ratio is not at the stechometric air fuel ratio, then there is going to be other gases which will be let out into the atmosphere. [00:03:46.560] uh atmosphere which are harmful to us. And therefore, if we want pollution to be minimum, then [00:03:55.680] Th it is the stechometric air fuel ratio which should be used in the IC engines. And of course this ratio is for gasoline. So if we want minimum pollution [00:04:10.276] in the atmosphere due to burning of these fuels that is gasoline then the ratio to be maintained is 14.7 is to one. [00:04:24.356] Now these oxygen sensors they are located in the exhaust stream and they indirectly determine the air to fuel. [00:04:35.152] ratio so we know that we require the air to fuel ratio should be 14.7 is to one. So these oxygen sensors are helping us to maintain this air fuel ratio. [00:04:49.188] Now the oxygen sensor therefore allows the engine control system to maintain this ideal ratio across various engine operating systems. We know that when an automobile is moving then

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:04.848] The engine operating conditions keep changing based on whether it is moving slow, whether it is moving fast, whether it's going on a uh terrain which is not smooth. So every time the operating conditions are changing and the oxygen sensor therefore allows [00:05:22.608] the engine control system to maintain the

ideal ratio across all these conditions. So how does it achieve that? It is going to compare the amount of oxygen in the exhaust to the amount of oxygen in the atmosphere. [00:05:40.420] Now just now I mentioned that it is placed in the exhaust stream. And we know that in the exhaust stream some amount of oxygen is going to be [00:05:51.556] let out after the combustion reaction has happened. So this oxygen it is going to compare with the atmospheric oxygen. Let us see further what does it do. Now based on the difference in the oxygen concentration [00:06:09.796] It is going to give an output voltage. So this output voltage is getting generated due to the difference in concentration in the exhaust and the atmosphere. Now if the output voltage is [00:06:25.520] Point two, then that represents a lean mixture. What do we mean by lean mixture? A lean mixture would be one which is lean in the fuel. That means that there is less of fuel and more of oxygen. So [00:06:41.104] the voltage which is generated is 0.2 volts. Whereas if it is a rich mixture, rich mixture meaning it is rich in the fuel, there's lot of fuel and less of uh oxygen. The required amount of oxygen is not available and in that case the [00:06:58.336] voltage is 0.8 volts. Now this is understandable because if it is a lean mixture that means there is more oxygen in the exhaust team. So when you compare it with the atmospheric oxygen the difference is going to be less whereas if it [00:07:16.080] It is a rich mixture, there is going to be less oxygen in the exhaust stream, and when you compare it with the atmospheric oxygen, the difference is going to be higher. So a lean mixture gives a smaller value 0.2 volts, whereas a rich mixture is going to give you a 0.8 volt. [00:07:34.752] As the output voltage. The ideal set point would be around 0.45 volts. Now, once this voltage is generated, it is sent back as a feedback to the engine control unit. Now this engine control unit based on the voltage. [00:07:54.688] which it is receiving the signal voltage voltage signal which it is receiving it is going to adjust the air fuel ratio back to the stechometric value. [00:08:06.276] So continuously the feedback is being sent to the engine control unit, which is going to now adjust the air-fuel ratio so that most of the times the end the [00:08:19.296] engine is working at stechometric value and what is the result of that that there is going to be [00:08:27.264] very less pollution. That means complete combustion. We are just making sure that there is complete combustion and therefore only benign gases are let out into the uh atmosphere or less harmful gases are [00:08:41.632] given out into the atmosphere. So we the oxygen sensor we can say is going to help us reduce pollution. Now [00:08:51.760] What happens when there is a lean mixture? When we have a lean mixture that means the there is less of fuel and more of oxygen. So basically there is an oxidizing atmosphere. So definitely carbon containing fuels would get completely oxided. [00:09:09.504] But something else happens here. The emissions now contain NOx gases like NO, NO2, etc. Now why does that happen? These NOx gases, if we want to get rid of them, because they also cause pollution, we know uh smog or acid rain is the [00:09:29.040] uh result of these NOx emissions. So these NOx gases if you want to get rid of them then we have to provide a reducing atmosphere where these will get converted into nitrogen. [00:09:42.416] But we are providing an oxidizing atmosphere because we have a lean mixture. There's a lot of oxygen and hence we find that NOPS emissions will increase if we have a lean mixture. Whereas if it is a rich mixture, that means the amount of oxygen is not sufficient to cause the combustion

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.028] harmful gases which are going to be let out into the atmosphere and like carbon monoxide and then also some carbon particles will be there and some unburnt fuel. So we will have these kind of emissions into the atmosphere if it is a rich mixture. So [00:10:22.800] So we understand that neither should it be a

lean mixture nor should it be a rich mixture, and therefore the engine needs to work at [00:10:33.744] the stechometric value that is 14.7 is to one and the oxygen sensor is helping us to do maintain that stechometric value. [00:10:44.708] Now we said that it is going to oxygen sensor is going to generate an output voltage when it compares the amount of oxygen the exhaust and the atmosphere. [00:10:58.320] So what is so it's very obvious that since voltage is being generated, it is a kind of galvanic cell which is working over here. So what type of galvanic cell do we have? The cell is a solid state electrochemical fuel cell. [00:11:15.940] Just now we discussed about different types of fuel cells and in that one of them was the solid state fuel cell. So solid state fuel cell meaning that the electrolyte is a solid. So the [00:11:30.468] Oxygen sensor is basically an electrochemical fuel cell which is going to use solid as the electrolyte and we also know that when we use solid as electrolyte then we need then the cell will operate at a high temperature. So oxygen sensor is [00:11:48.256] requires a minimum temperature of 360 degrees centigrade. So it is going to operate only at higher because we are using solid as the electrolyte. Let us study the construction of the oxygen sensor. [00:12:06.564] The anode here is platinum. Since it's a fuel cell, we know that there is going to be an anode, there is going to be cathode, and we have [00:12:17.680] the exhaust gas on one side and the reference air on the other side because we are going we the oxy sensor has to compare the exhaust gas with the reference air. So the the [00:12:32.228] Electrodes which are used here are platinum electrodes. So anode is also platinum and cathode is also made of platinum. And the electrolyte, as I mentioned just now, it's a solid state. [00:12:44.880] Electrochemical fuel cell. So the electrolyte is zirconium oxide uh oxide which is doped with yetrium oxide. [00:12:53.124] Now this is the most common uh solid electrolyte which is used and in oxygen sensor. So here we have this zilconium membrane which is doped with Y2A3, which is being used as the electrolyte, and one side we have the exhaust gas, the other side we have the [00:13:10.896] Reference air and of course these electrodes which are platinum. So this is the construction and based on the difference between the concentration of the exhaust gas and the reference air, we are going to get a voltage which will be varying between point. [00:13:28.464] two to point eight. So it is like a concentration cell. [00:13:36.772] Let us study the working of this oxygen sensor. What happens is we have oxygen at the cathode. Now oxygen is going to take up four electrons and undergo reduction. So it is going to give you two O2 minus [00:13:54.468] Now these oxide ions have to move through the electrolyte and reach the anode. That is the reason why we chose zirconium oxide doped with Y2O. [00:14:12.848] Three because we know that when zirconium oxide is duped with Y2O three, it creates oxide ion vacancies in the electrolyte. And since [00:14:24.032] we have oxide ion which is formed at the ca at cathode and this oxide ion has to jump through the electrolyte and reach the anode so we have to provide this kind of electrolyte which is a solid electrolyte zirconium oxide doped with y2o3 [00:14:41.712] So when this oxide ion now reaches the anode, then oxidation happens here and 2O2 minus is going to again become O2 plus 4 electrons. So these are the reactions which happen at the anode and cathode, and we can see that both [00:14:59.280] the reactions are similar because it is like a concentration cell. Both sides at the anode as well as the cathode we have the oxygen which is present at the cathode we have reduction happening giving you O2 minus these O2 minus ions are moving towards the anode

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:16.848] And there they undergo oxidation, giving you O2 plus 4 electrolyte electrons. And the electrolyte is zirconium oxide doped with Y2O3. So we can easily justify the use of this. [00:15:30.596] Solid electrolyte here because we

need an electrolyte which is going to facilitate the movement of oxide ions. [00:15:39.280] What would be the cell voltage? The cell voltage can be given by the Nernst equation. Now, since it is a concentration cell, we know that there will be no E naught, E naught value cancels values cancel for the anode and the cathode reactions, and therefore we have two. [00:15:57.040] point three zero three r t over four f four is because of the number of electrons which are getting exchanged in these reactions so log of p1 by p2 now p1 is the partial pressure of oxygen in the [00:16:13.200] reference air that is in the atmosphere and P2 is the partial pressure of oxygen in the [00:16:20.368] exhaust. So E cell is given by 2.303 RT over 4F log of P1 by P2. So P1 and P2 stand for the partial pressures of oxygen in the reference and the exhaust. [00:16:38.244] Now automobiles which are fitted with oxygen sensor they must use unleaded gasoline. [00:16:45.892] The reason for that is lead is going to poison the platinum catalyst. We know that we have used platinum at the anode and cathode. So if we use leaded gasoline, then it is go the lead which is present in this is going to [00:17:03.664] Poison the platinum catalyst, and that is going to reduce the efficiency of the oxygen sensor. Now, what is leaded gasoline? Now, in gasoline, to prevent knocking, there is [00:17:22.180] a substance which is used which is known as the anti knock agent and earlier it was tetraethyl lead T E L which was being used as an anti knock agent to prevent knocking in the machine. Now [00:17:38.880] When tetraethylaid is used because if the [00:17:42.608] automobile is equipped with an oxygen sensor, this lead is going to poison the platinum and this this will reduce the efficiency of the oxygen sensor. And we know that oxygen sensor is playing a very important role and therefore [00:17:59.072] Nowadays, unleaded gasoline is being used because most of the automobiles are equipped with oxygen sensor. So unleaded gasoline is used, meaning that tetraethyl lead is now replaced with another antinoch. [00:18:16.784] Like for example methyl tertiary butyl ether. Now the symptoms of failure. [00:18:23.840] For an oxygen sensor would be increased tailpipe emissions and increased fuel consumption and hesitation on acceleration. We should understand that the oxygen sensor has to have a very good response time because continuously [00:18:41.520] it should monitor what is the amount of oxygen in the exhaust, compare it and give it compare it with the atmospheric oxygen and give a feedback to the engine control unit so that it adjusts the [00:18:56.548] in uh the air fuel ratio to stepometric value, the ideal value, and then we will ensure that there is minimum pollution in the [00:19:08.868] atmosphere. So it is essential that all automobiles be equipped with this oxygen sensor. So in the next class we are going to talk about another set of energy storage devices and that is supercapacitors. [00:19:28.516] Thank you.

EV0000182 · EV0000183 · EV0000184 · EV0000185

Full transcript: 2560 normalized words, 79 timestamped segments, 1169.984 seconds duration. Low-confidence segments flagged: 0.

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12 Source slide deck: Li ion battery LFP batteries

This entry preserves the complete slide deck titled Li ion battery LFP batteries as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_03, to avoid an unsupported match. EV0000294 · EV0000295

Graphs and diagrams

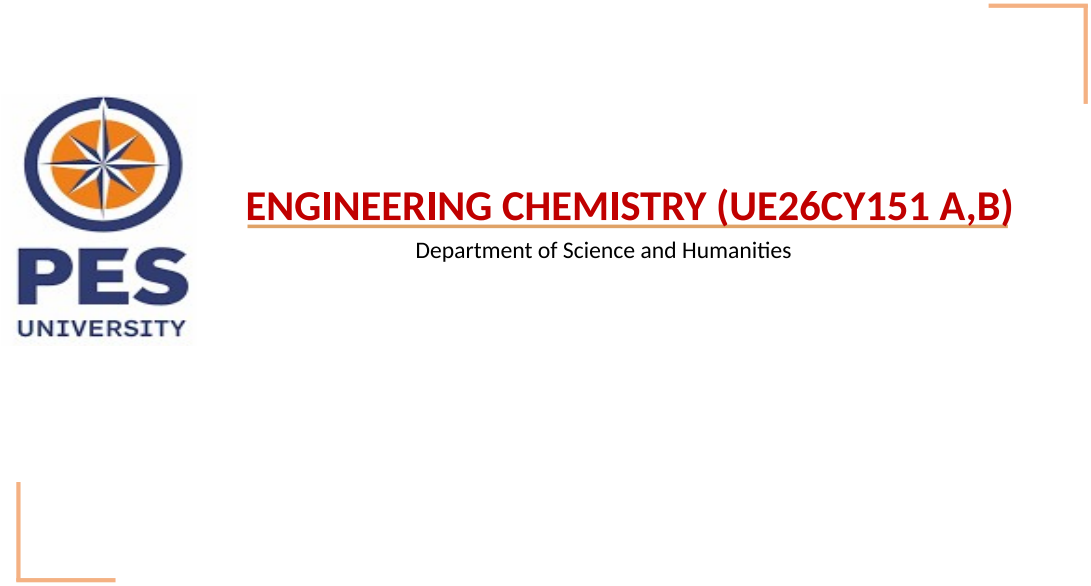

Figure G000037. Li ion battery LFP batteries · source slide page 1.

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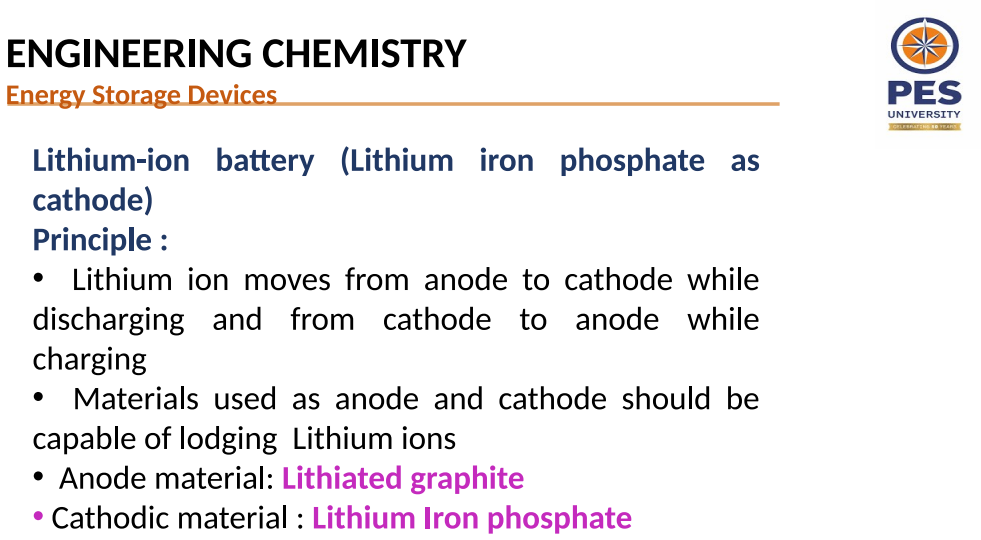

Figure G000039. Li ion battery LFP batteries · source slide page 3.

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ENGINEERING CHEMISTRY

Energy Storage Devices

Construction

Component	Material	Function
Anode (negative electrode)	Graphite, C_6	Stores lithium ions during charging
Cathode (positive electrode)	Lithium iron phosphate, $LiFePO_4$	Source and host of lithium ions
Electrolyte	Lithium salt such as $LiPF_6$ dissolved in organic carbonate solvents	Conducts Li^+ ions between electrodes
Separator	Microporous polymer, commonly polyethylene (PE) or polypropylene (PP)	Prevents direct electrical contact while allowing Li^+ transport
Current collector—anode	Copper (Cu) foil	Conducts electrons from/to the graphite electrode
Current collector—cathode	Aluminium (Al) foil	Conducts electrons from/to the $LiFePO_4$ electrode
Cell casing	Steel, aluminium, or polymer pouch	Provides mechanical protection

Figure G000040. Li ion battery LFP batteries · source slide page 4.

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ENGINEERING CHEMISTRY

Energy Storage Devices

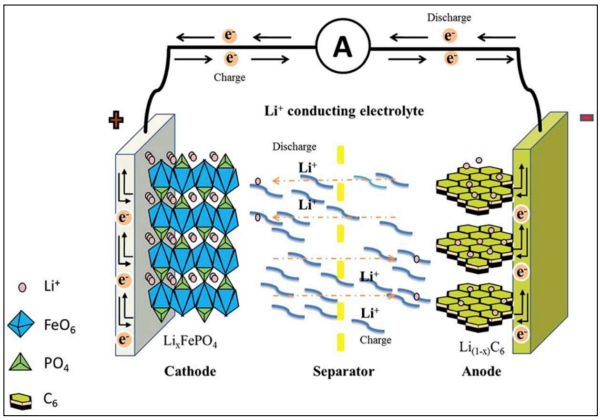

Figure G000041. Li ion battery LFP batteries · source slide page 5.

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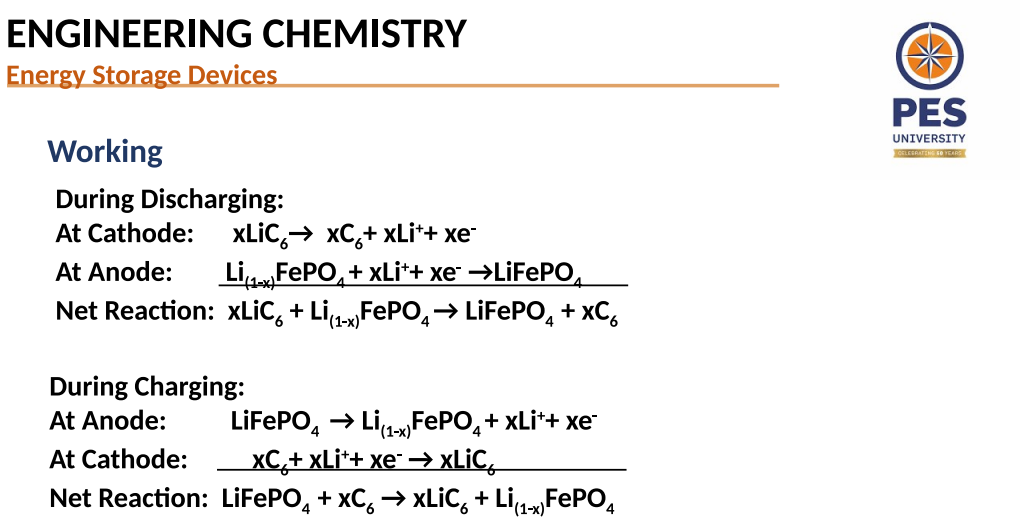

Figure G000042. Li ion battery LFP batteries · source slide page 6.

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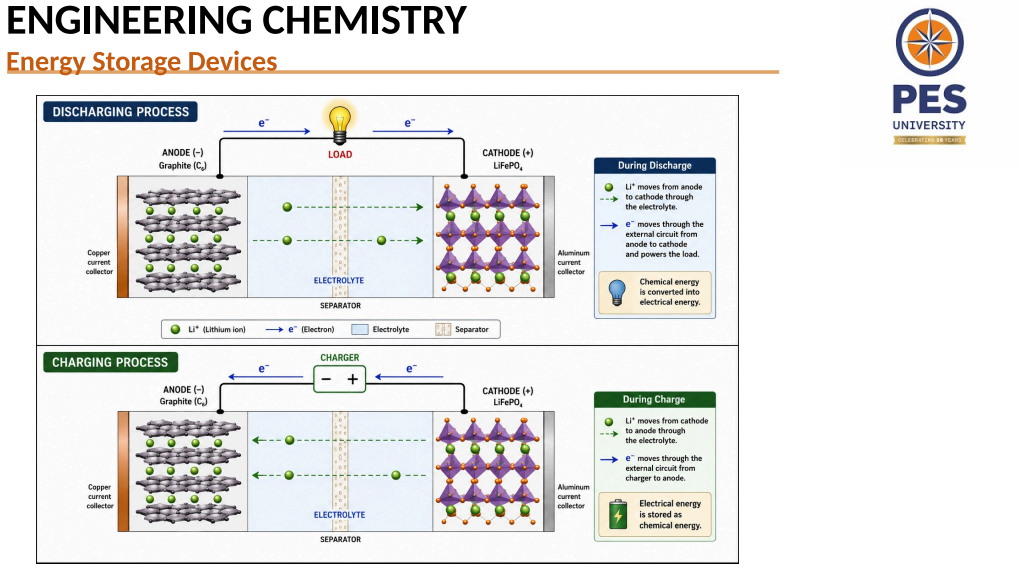

Figure G000043. Li ion battery LFP batteries · source slide page 7.

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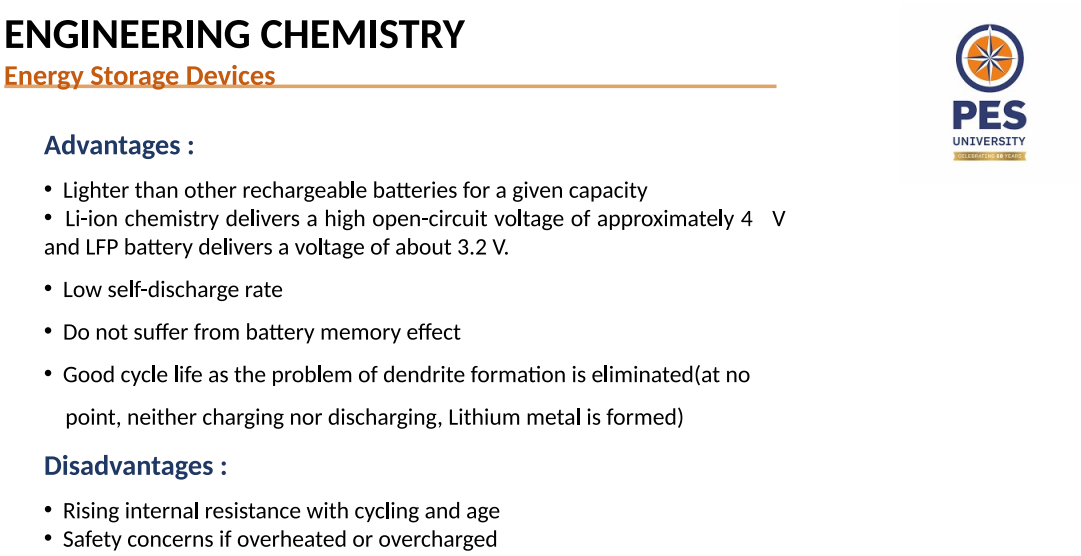

Figure G000044. Li ion battery LFP batteries · source slide page 8.

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Figure G000045. Li ion battery LFP batteries · source slide page 9.

EV0000319

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000046. Li ion battery LFP batteries · source slide page 10.

EV0000322

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Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000294 · EV0000295 · EV0000296
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Class 30 content: Lithium-ion battery –(Lithium iron phosphate as cathode) EV0000297 · EV0000298 · EV0000299
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Lithium-ion battery (Lithium iron phosphate as cathode) Principle : Lithium ion moves from anode to cathode while discharging and from cathode to anode while charging Materials used as anode and cathode should be capable of lodging Lithium ions Anode material: Lithiated graphite Cathodic material : Lithium Iron phosphate EV0000300 · EV0000301 · EV0000302
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Construction Component | Material | Function Anode (negative electrode) | Graphite, C_6 | Stores lithium ions during charging Cathode (positive electrode) | Lithium iron phosphate, $LiFePO_4$ | Source and host of lithium ions Electrolyte | Lithium salt such as $LiPF_6$ dissolved in organic carbonate solvents | Conducts Li^+ ions between electrodes Separator | Microporous polymer, commonly polyethylene (PE) or polypropylene (PP) | Prevents direct electrical contact while allowing Li^+ transport Current collector—anode | Copper (Cu) foil | Conducts electrons from/to the graphite electrode Current collector—cathode | Aluminium (Al) foil | Conducts electrons from/to the $LiFePO_4$ electrode Cell casing | Steel, aluminium, or polymer pouch | Provides mechanical protection EV0000303 · EV0000304 · EV0000305

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices [Image text, OCR] Discharge A Charge Li+conductingelectrolyte + Discharge Lit Li+ Lit FeO6 LiFePO4 Charge Li(l1-x)C6 PO4 Cathode Separator Anode EV0000306 · EV0000307 · EV0000308
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Working During Discharging: At Cathode: xLiC6→ xC6+ xLi++ xe- At Anode: Li(1-x)FePO4 + xLi++ xe- →LiFePO4 Net Reaction: xLiC6 + Li(1-x)FePO4 → LiFePO4 + xC6 During Charging: At Anode: LiFePO4 → Li(1-x)FePO4 + xLi++ xe- At Cathode: xC6+ xLi++ xe- → xLiC6 Net Reaction: LiFePO4 + xC6 → xLiC6 + Li(1-x)FePO4 EV0000309 · EV0000310 · EV0000311
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices [Image text, OCR] DISCHARGING PROCESS ANODE (-) LOAD CATHODE (+) Graphite (C) LiFePO4 During Discharge Litmoves from anode to cathode through the electrolyte. emovesthrough the external circuit from anodetocathode Copper Aluminum andpowerstheload. current current collector collector ELECTROLYTE Chemical energy is converted into electrical energy. SEPARATOR Li+ (Lithium ion) Separator →e(Electron) Electrolyte CHARGINGPROCESS CHARGER ANODE (-) CATHODE (+) Graphite (C) LiFePO4 During Charge Li*movesfrom cathode to anode through the electrolyte. emovesthroughthe external circuitfrom charger to anode. Copper Aluminum current current Electrical energy collector collector ELECTROLYTE is stored as chemical energy. SEPARATOR EV0000312 · EV0000313 · EV0000314
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Advantages : Lighter than other rechargeable batteries for a given capacity Li-ion chemistry delivers a high open-circuit voltage of approximately 4 V and LFP battery delivers a voltage of about 3.2 V. Low self-discharge rate Do not suffer from battery memory effect Good cycle life as the problem of dendrite formation is eliminated(at no point, neither charging nor discharging, Lithium metal is formed) Disadvantages : Rising internal resistance with cycling and age Safety concerns if overheated or overcharged EV0000315 · EV0000316 · EV0000317
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices [Image text, OCR] Camera GPS Minibus Mobile Laptop Torch-light Bluetooth Toy Helicopter TabletPC EV0000318 · EV0000319 · EV0000320
10. Source slide page 10 (extracted text): THANK YOU Department of Science and Humanities EV0000321 · EV0000322 · EV0000323

SHORT-NOTE RECAP

Source deck preserved: Li ion battery LFP batteries. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000294

13 Source slide deck: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys

This entry preserves the complete slide deck titled Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_03, to avoid an unsupported match. EV0000450 · EV0000451

Graphs and diagrams

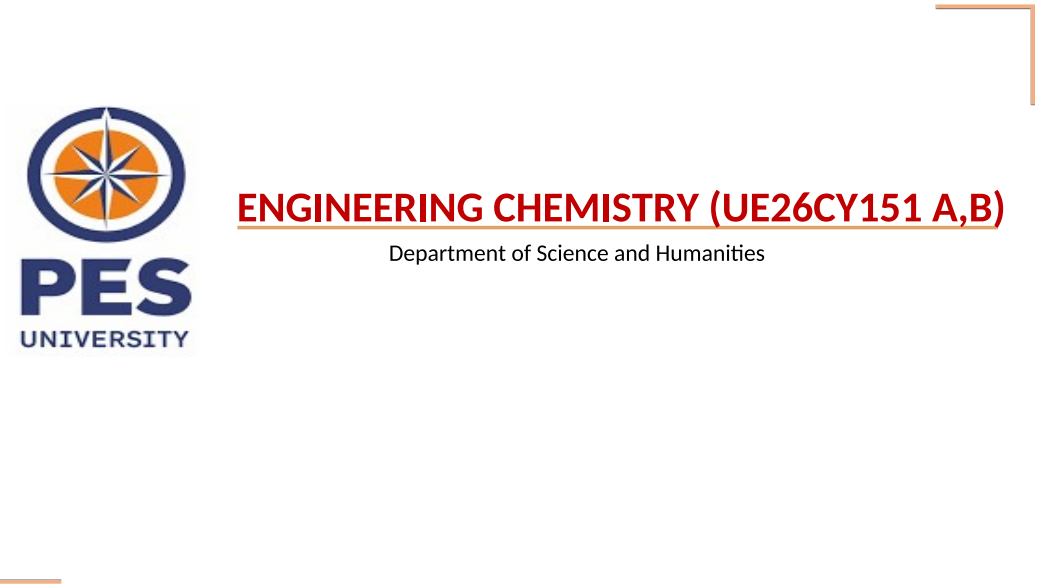


Figure G000089. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 1.

EV0000451

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

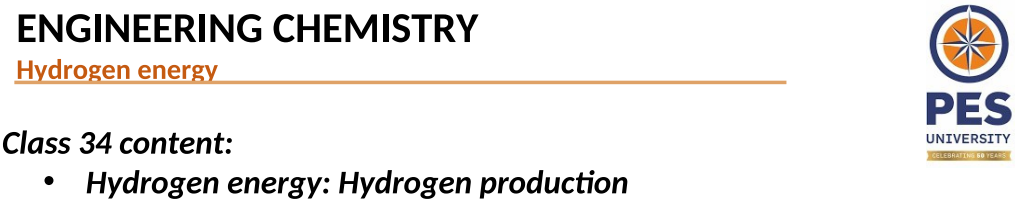


Figure G000090. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 2.

EV0000454

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

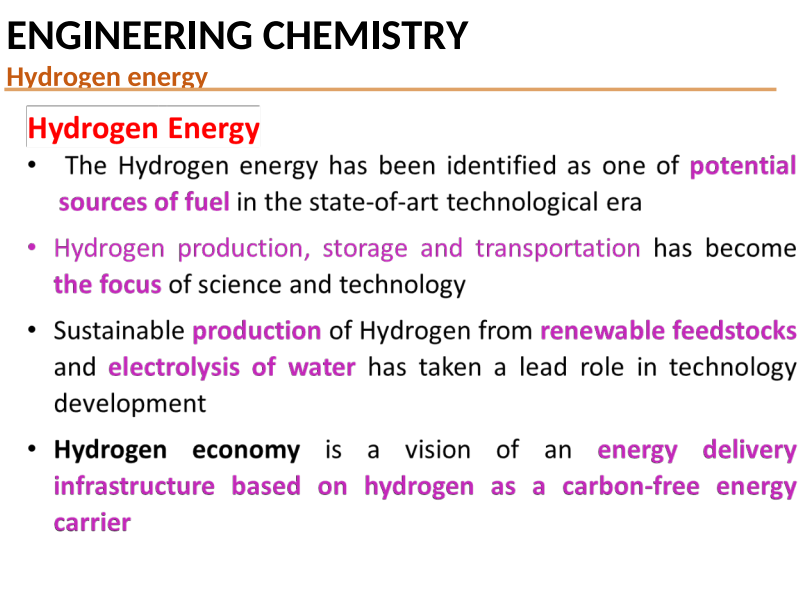


Figure G000091. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 3.

EV0000457

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

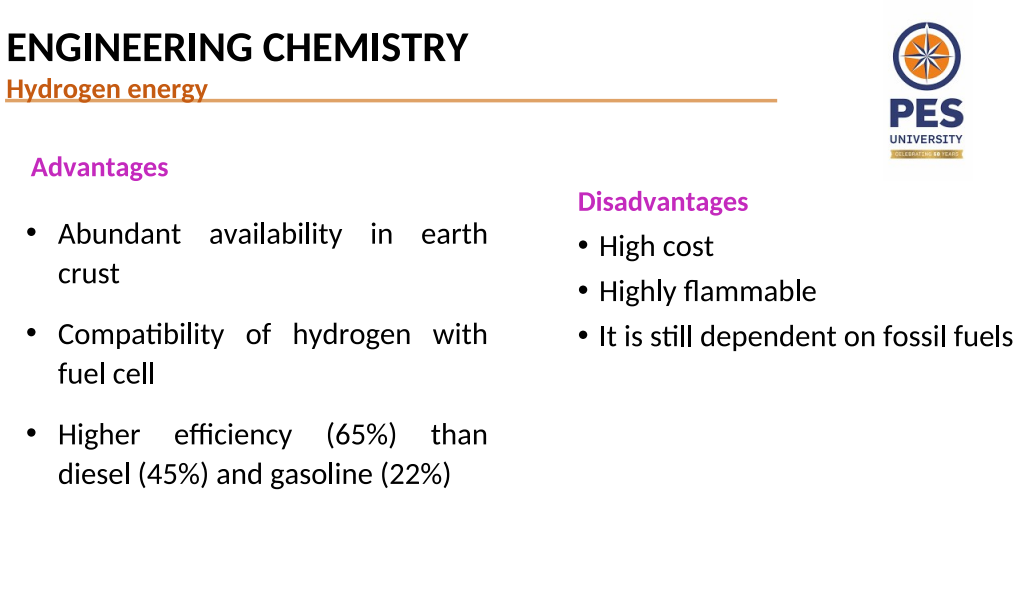


Figure G000092. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 4.

EV0000460

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

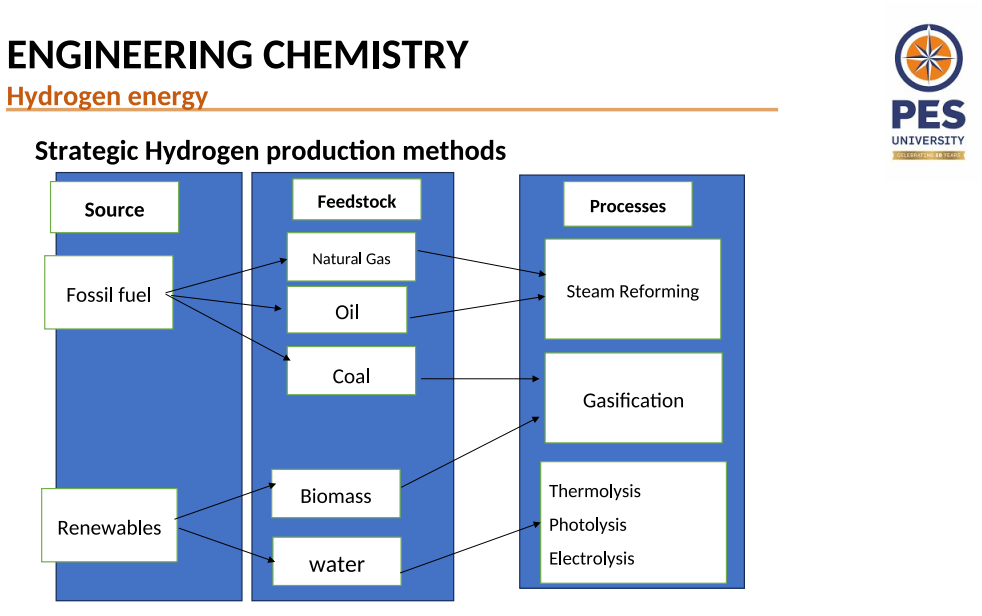


Figure G000093. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 5.

EV0000463

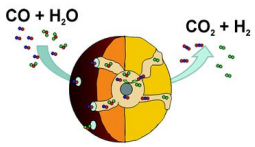
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Hydrogen energy

Hydrogen Production methods

- **Steam reforming** (steam methane reforming –Grey Hydrogen)
  - Natural gas and steam is used to produce Hydrogen
  - Most widespread process for generation of hydrogen
- **Gasification process** (Black Hydrogen)
  - Hydrogen is produced from coal by gasification process
  - Coal is subjected to gasification using a controlled supply of oxygen and steam to produce synthesis gas (syngas), mainly containing CO and H_2.
  - The CO in the syngas undergoes the water-gas shift reaction with steam to produce additional H_2 and CO_2.
  - The resulting gas mixture is then purified to obtain pure H_2.



Schematic representation of water – gas shift reaction

6

Figure G000094. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 6.

EV0000466

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

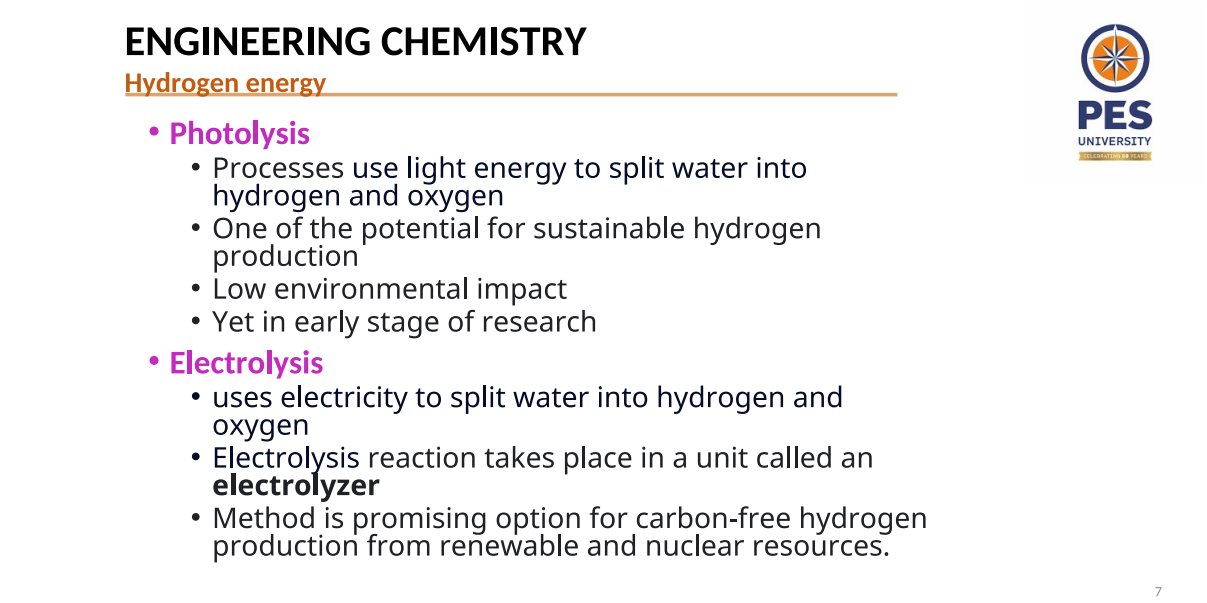


Figure G000095. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 7.

EV0000469

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

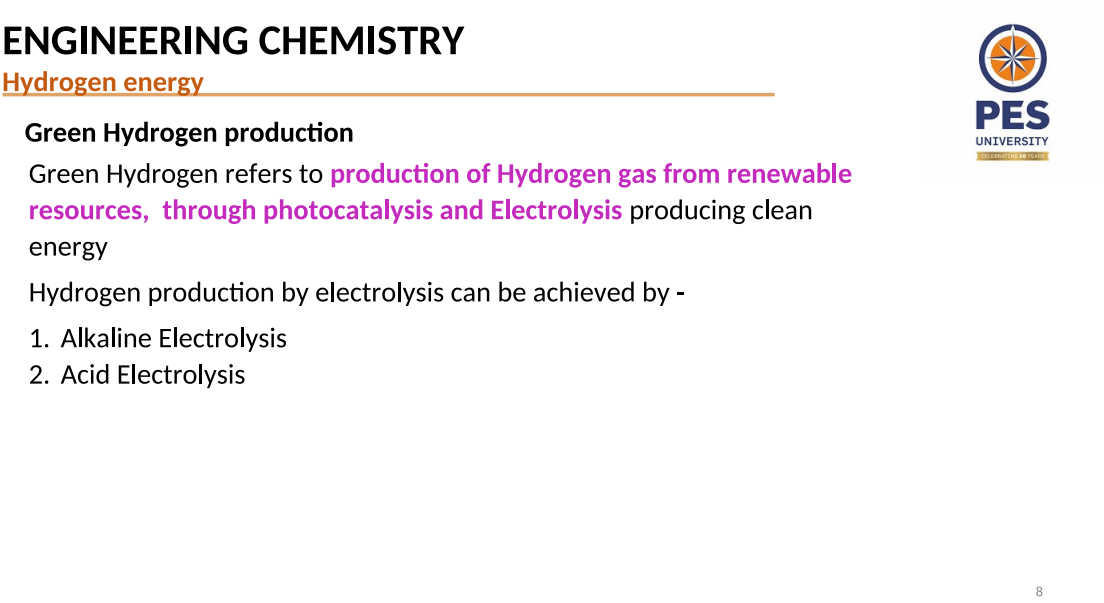


Figure G000096. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 8.

EV0000472

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

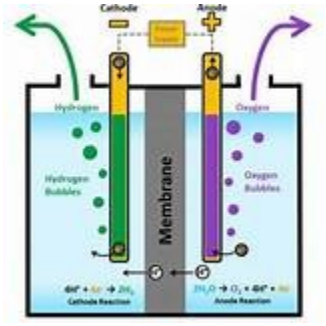
ENGINEERING CHEMISTRY

Hydrogen energy

Alkaline Electrolysis

Construction

- Anodic materials: Ni and its alloys
- Cathodic materials: Cu, Pt
- Electrolyte : KOH, NaOH (20-40 wt %)
- Catalyst: Ni-Co-Au, RuO_2
- Operation temperatures are between 343 – 363 K
- Operating pressure up to 3 Mpa



9

Figure G000097. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 9.

EV0000475

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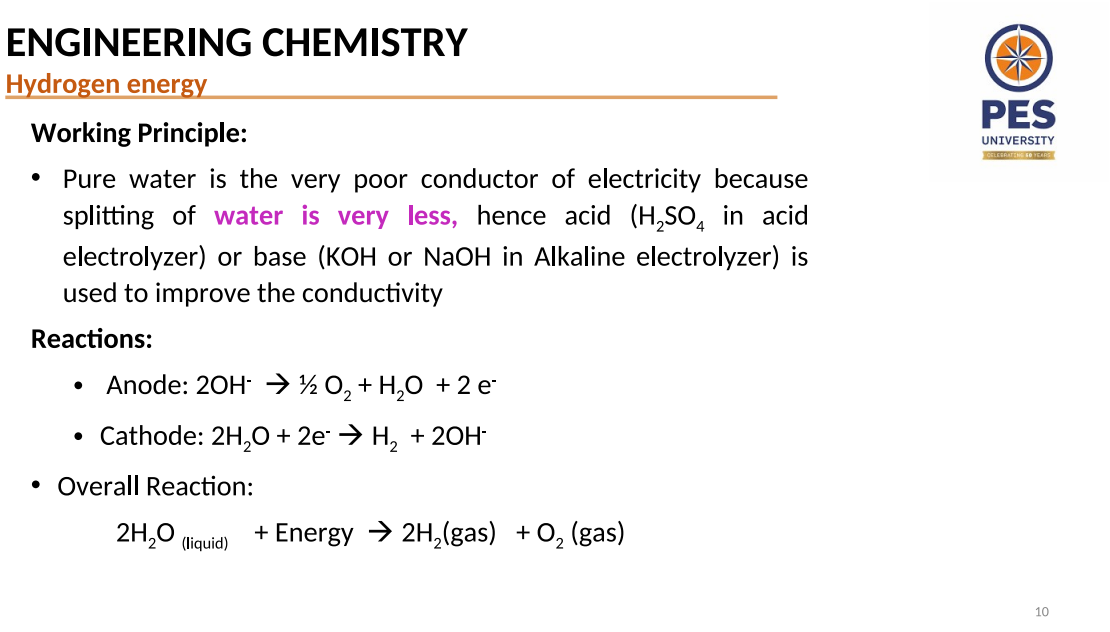


Figure G000098. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 10.

EV0000478

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

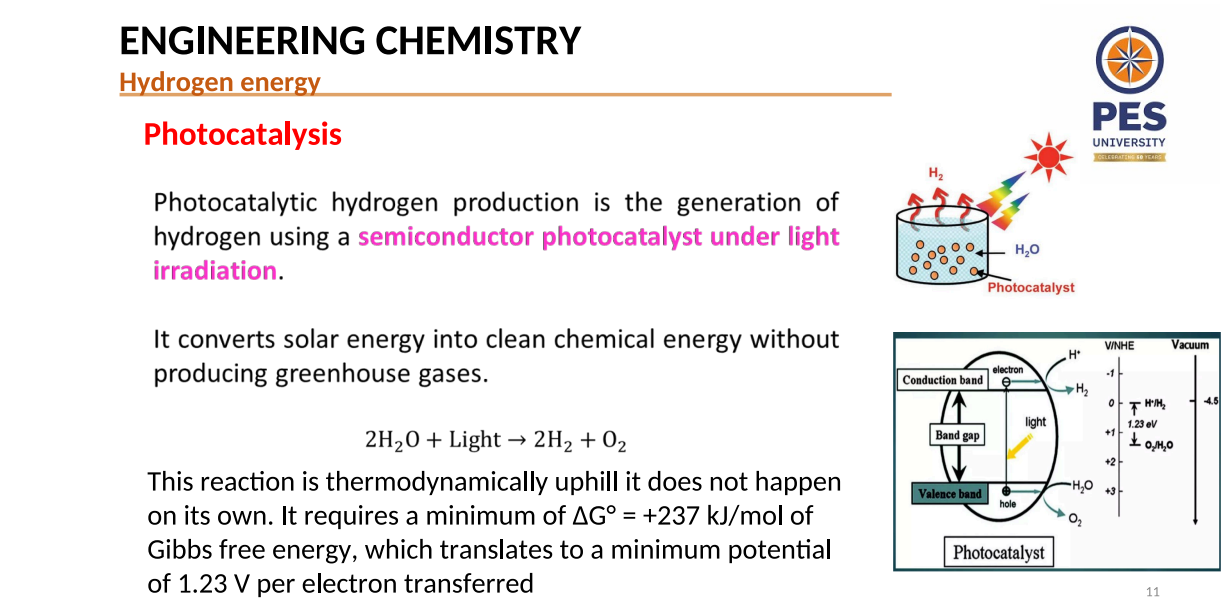


Figure G000099. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 11.

EV0000481

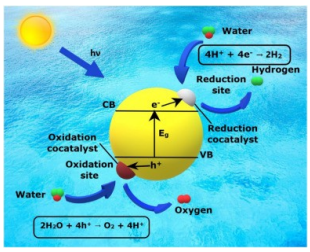
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

Hydrogen energy

Mechanism of Water Splitting

- **Photon absorption:** TiO_2 absorbs a photon (hν ≥ Eg).
- **Charge generation:** Electrons get excited from VB to CB, leaving holes in the VB.
- **Separation & migration:** Charge carriers (electrons and holes) migrates to the surface of the photocatalyst.
- **Surface redox:** H^+ reduced to H_2; water oxidised to O_2.



https://www.tandfonline.com/doi/full/10.1080/01614940.2024.2446476

Oxidation in valance band : $2\,H_2O + 4\,h^+ \rightarrow O_2 + 4\,H^+$

Reduction in conduction band : $4\,H^+ + 4\,e^- \rightarrow 2\,H_2$

Overall reaction : 2 H2O → 2 H2 + O2

12

Figure G000100. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 12.

EV0000484

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

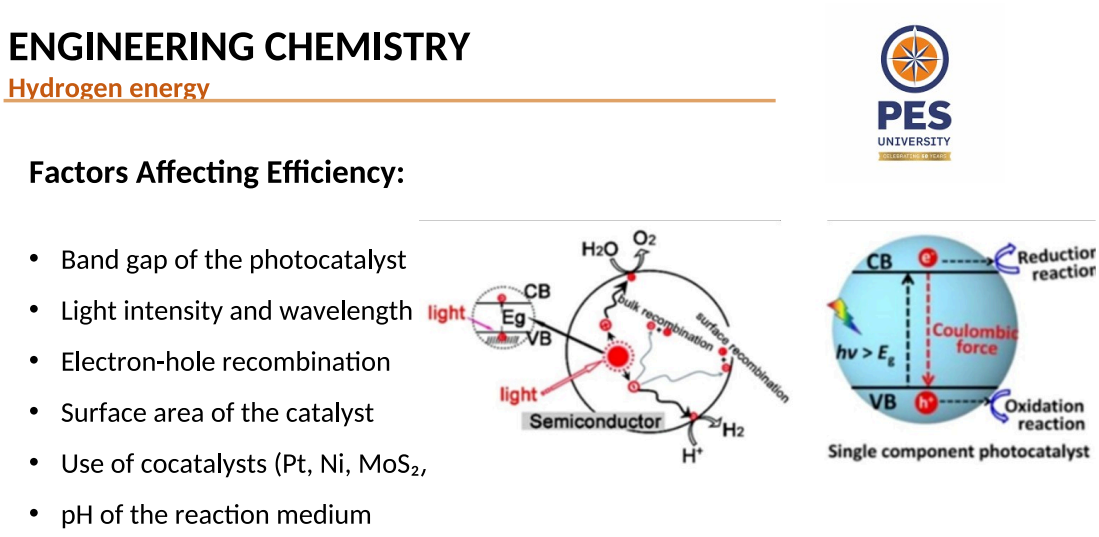


Figure G000101. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 13.

EV0000487

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000102. Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys · source slide page 14.

EV0000489

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY Department of Science andA,B) Humanities EV0000450 · EV0000451 · EV0000452
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Class34content: •Hydrogenenergy:Hydrogen production EV0000453 · EV0000454 · EV0000455
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy 3 EV0000456 · EV0000457 · EV0000458
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Advantages Disadvantages • Abundant availability in •High cost earth crust •Highly flammable • Compatibility of hydrogen •It is still dependent on fossil fuels with fuel cell • Higher efficiency (65%) than diesel (45%) and gasoline (22%) 4 EV0000459 · EV0000460 · EV0000461
5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Strategic Hydrogen production methods Feedstock Source Processes Natural Gas Steam Fossil fuel Reforming Oil Coal Gasification Thermolysis Biomass Renewable Photolysis s water Electrolysis 5 EV0000462 · EV0000463 · EV0000464

6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Hydrogen Production methods • Steam reforming (steam methane reforming –Grey Hydrogen) • Natural gas and steam is used to produce Hydrogen • Most widespread process for generation of hydrogen • Gasification process (Black Hydrogen) • Hydrogen is produced from coal by gasification process • Coal is subjected to gasification using a controlled supply of oxygen and steam to produce synthesis gas (syngas), mainly containing CO and H_2. • The CO in the syngas undergoes the water-gas shift reaction with steam to produce additional H_2 and CO_2. Schematic representation of • The resulting gas mixture is then purified to obtain pure H_2. water – gas shift reaction 6 EV0000465 · EV0000466 · EV0000467

7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy •Photolysis • Processes use light energy to split water into hydrogen and oxygen • One of the potential for sustainable hydrogen production • Low environmental impact • Yet in early stage of research •Electrolysis • uses electricity to split water into hydrogen and oxygen • Electrolysis reaction takes place in a unit called an electrolyzer • Method is promising option for carbon-free hydrogen production from renewable and nuclear resources. 7 EV0000468 · EV0000469 · EV0000470

8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy reen Hydrogen production Green Hydrogen refers to production of Hydrogen gas from renewable resources, through photocatalysis and Electrolysis producing clean energy Hydrogen production by electrolysis can be achieved by - 1.Alkaline Electrolysis 2.Acid Electrolysis 8 EV0000471 · EV0000472 · EV0000473

9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Alkaline Electrolysis Construction •Anodic materials: Ni and its alloys •Cathodic materials: Cu, Pt • Electrolyte : KOH, NaOH (20-40 wt %) •Catalyst: Ni-Co-Au, RuO2 •Operation temperatures are between 343 – 363 K •Operating pressure up to 3 Mpa 9 EV0000474 · EV0000475 · EV0000476

10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Working Principle: • Pure water is the very poor conductor of electricity because splitting of water is very less, hence acid (H2SO4 in acid electrolyzer) or base (KOH or NaOH in Alkaline electrolyzer) is used to improve the conductivity Reactions: • Anode: 2OH- ☒ ½ O2 + H2O + 2 e- • Cathode: 2H2O + 2e- ☒ H2 + 2OH- • Overall Reaction: 2H2O (liquid) + Energy ☒ 2H2(gas) + O2 (gas) 10 EV0000477 · EV0000478 · EV0000479

11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Photocatalysis This reaction is thermodynamically uphill it does not happen on its own. It requires a minimum of $\Delta G^\circ = +237$ kJ/mol of Gibbs free energy, which translates to a minimum potential of 1.23 V per 11 electron transferred EV0000480 · EV0000481 · EV0000482

12. Source slide page 12 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Mechanism of Water Splitting • Photon absorption: TiO2 absorbs a photon ($h\nu \geq Eg$). • Charge generation: Electrons get excited from VB to CB, leaving holes in the VB. • Separation & migration: Charge carriers (electrons and holes) migrates to the surface of the photocatalyst. • Surface redox: H^+ reduced to H2; water oxidised to O2. Oxidation in valance band : 2 H2O + 4 h^+ → O2 + 4 https://www.tandfonline.com/doi/full/10 .1080/01614940.2024.2446476 H^+ Reduction in conduction band : 4 H^+ + 4 e^- → 2 H2 Overall reaction : 2 H2O → 2 H2 + O2 12 EV0000483 · EV0000484 · EV0000485

13. Source slide page 13 (extracted text): ENGINEERING CHEMISTRY Hydrogen energy Factors Affecting Efficiency: •Band gap of the photocatalyst •Light intensity and wavelength •Electron-hole recombination •Surface area of the catalyst •Use of cocatalysts (Pt, Ni, MoS_2) •pH of the reaction medium ## Page / slide 14 (OCR) THANKYOU Department of Science and Humanities PES UNIVERSITY EV0000486 · EV0000487 · EV0000488
14. Source slide page 14: no readable `.slides.md` text was extracted. The original rendered page is included as the adjacent source figure for visual inspection. EV0000489

SHORT-NOTE RECAP

Source deck preserved: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000450

14 Source slide deck: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides

This entry preserves the complete slide deck titled Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_03, to avoid an unsupported match. EV0000490 · EV0000491

Graphs and diagrams

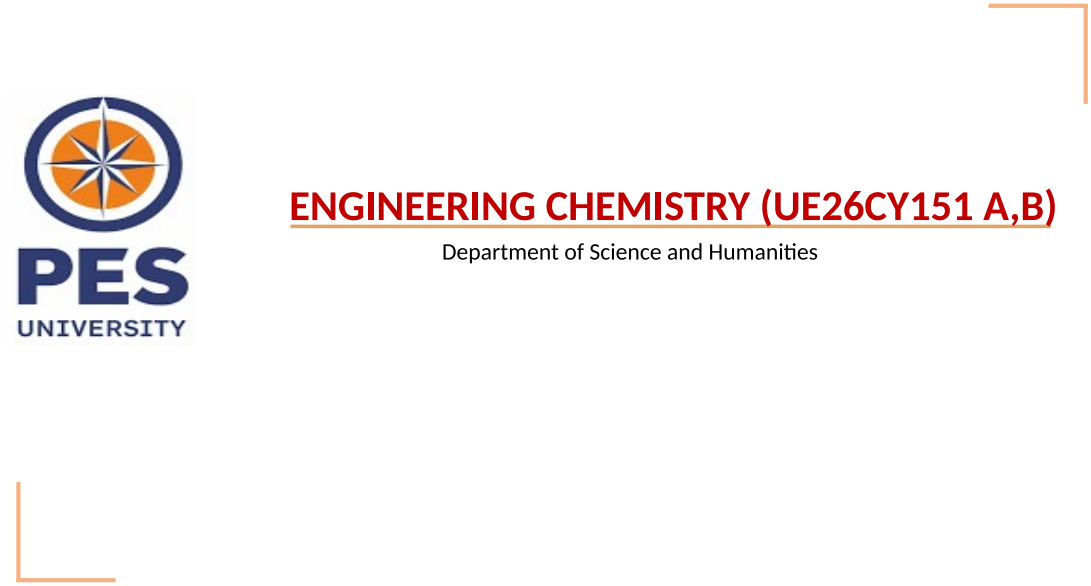


Figure G000103. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 1.

EV0000491

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

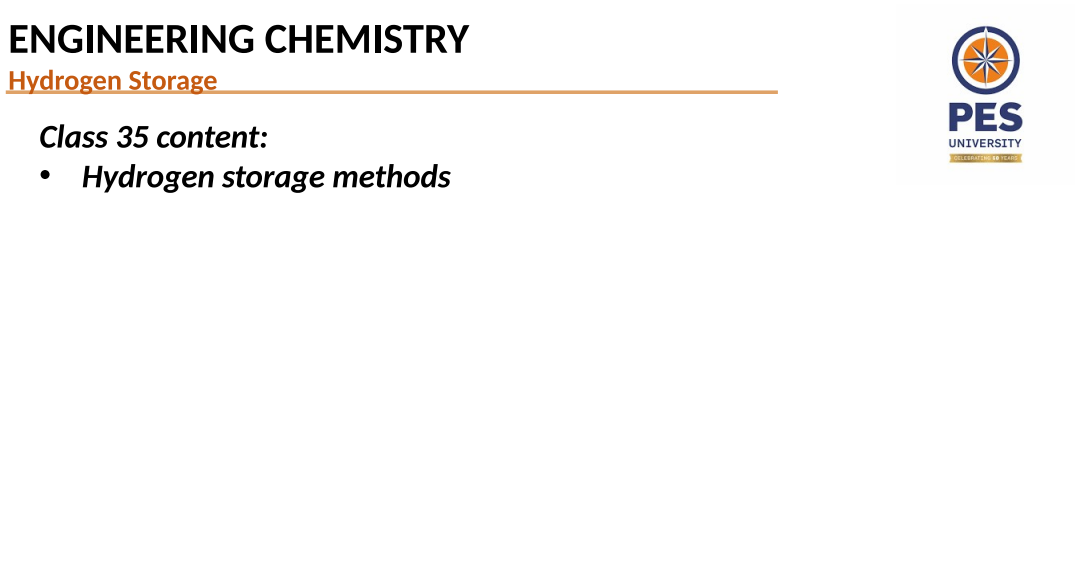


Figure G000104. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 2.

EV0000494

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

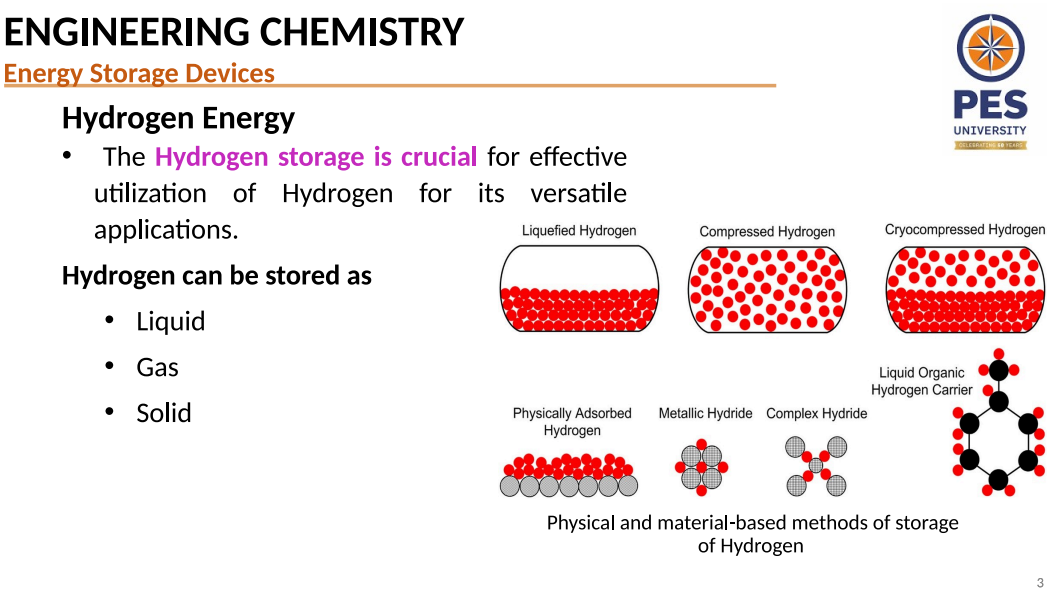


Figure G000105. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 3.

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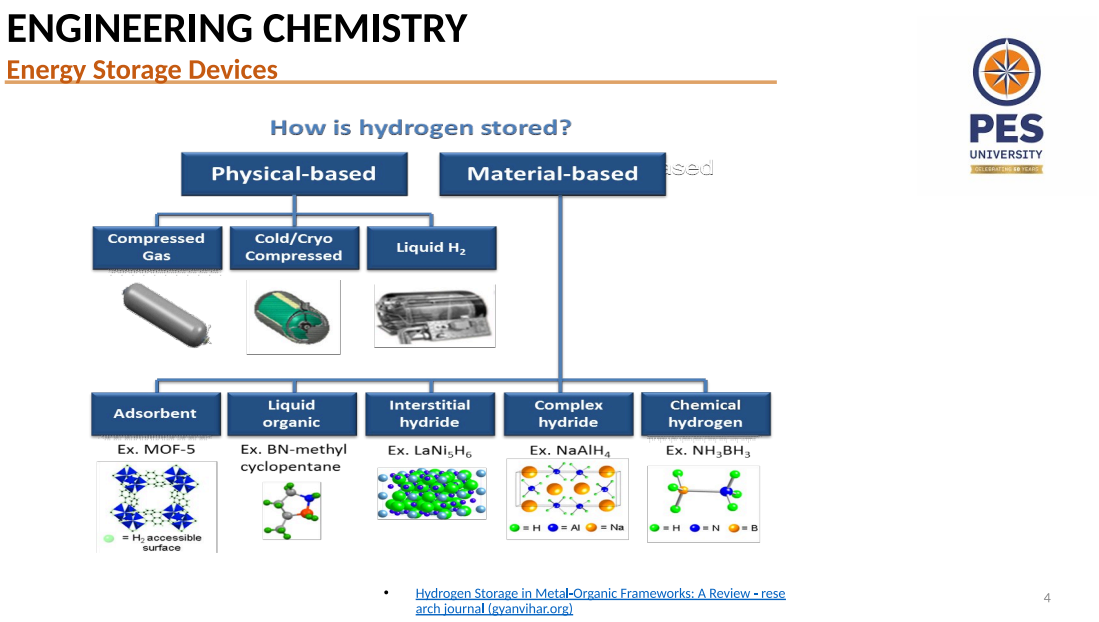

Figure G000106. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 4.

EV0000500

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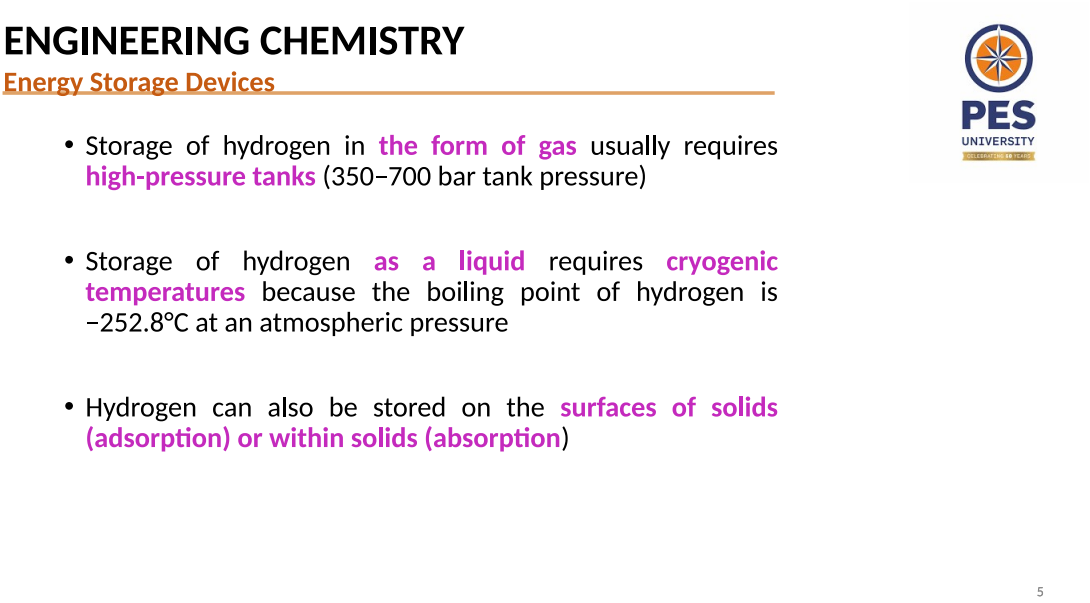

Figure G000107. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 5.

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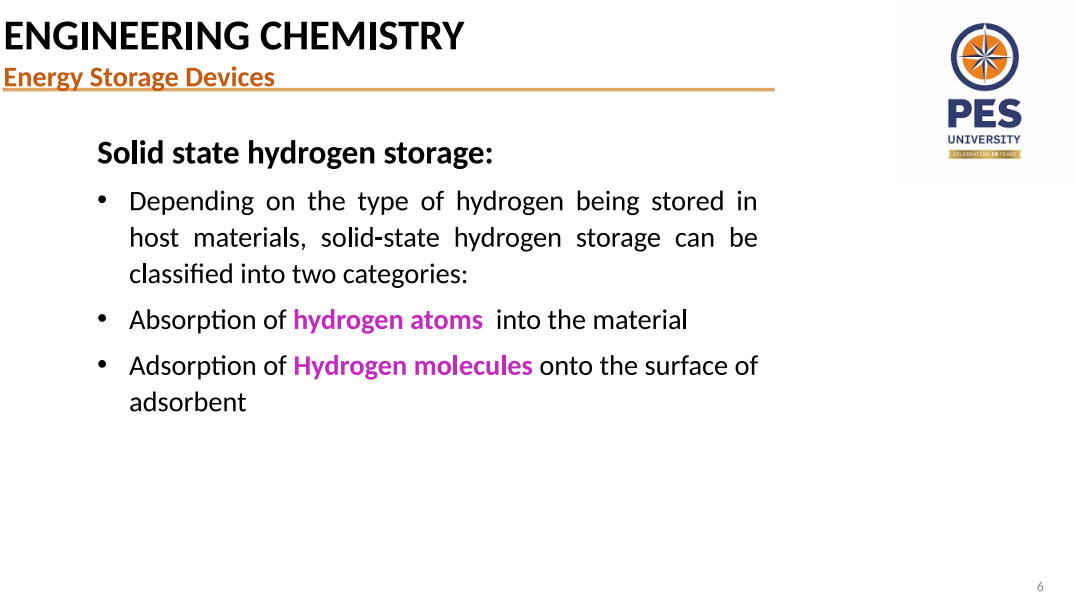


Figure G000108. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 6.

EV0000506

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

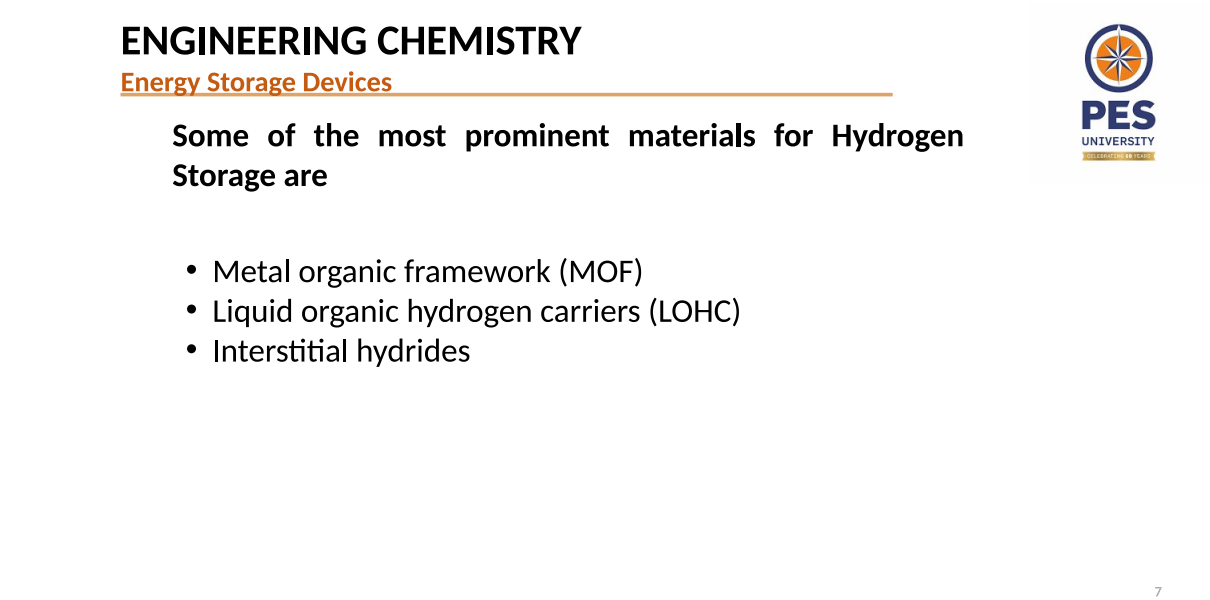


Figure G000109. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 7.

EV0000509

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

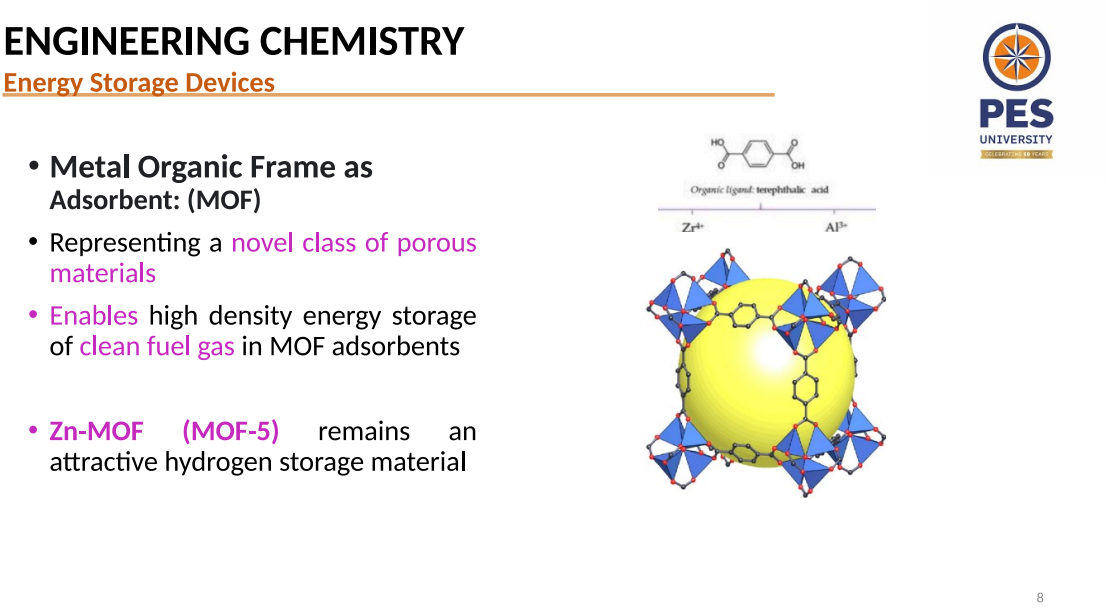


Figure G000110. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 8.

EV0000512

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

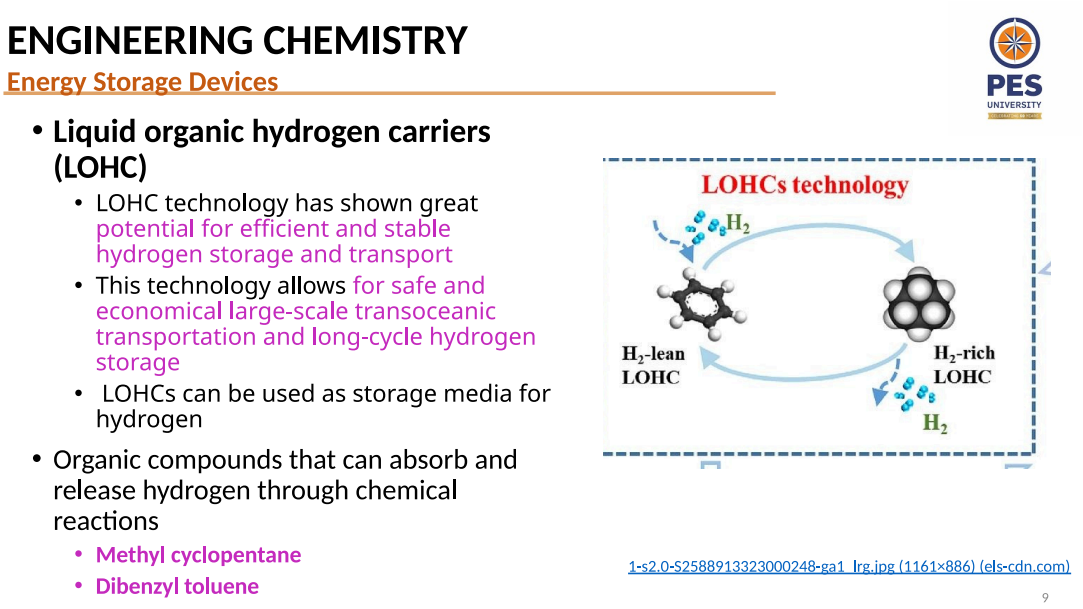


Figure G000111. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 9.

EV0000515

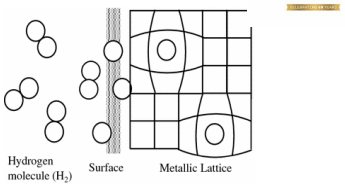
How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING CHEMISTRY

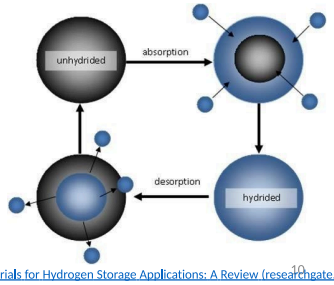
Energy Storage Devices

Interstitial hydride

- Contain a larger quantity of hydrogen than the same volume of liquid hydrogen
- Have advantage in the amount of hydrogen on a weight basis
- Certain interstitial hydrides are very suitable for hydrogen storage and transportation
- Example: $LaNi_5H_6$



Solid Hydrogen Storage Materials: Interstitial Hydrides | SpringerLink



Nanomaterials for Hydrogen Storage Applications: A Review (researchgate.net)

10

Figure G000112. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 10.

EV0000518

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

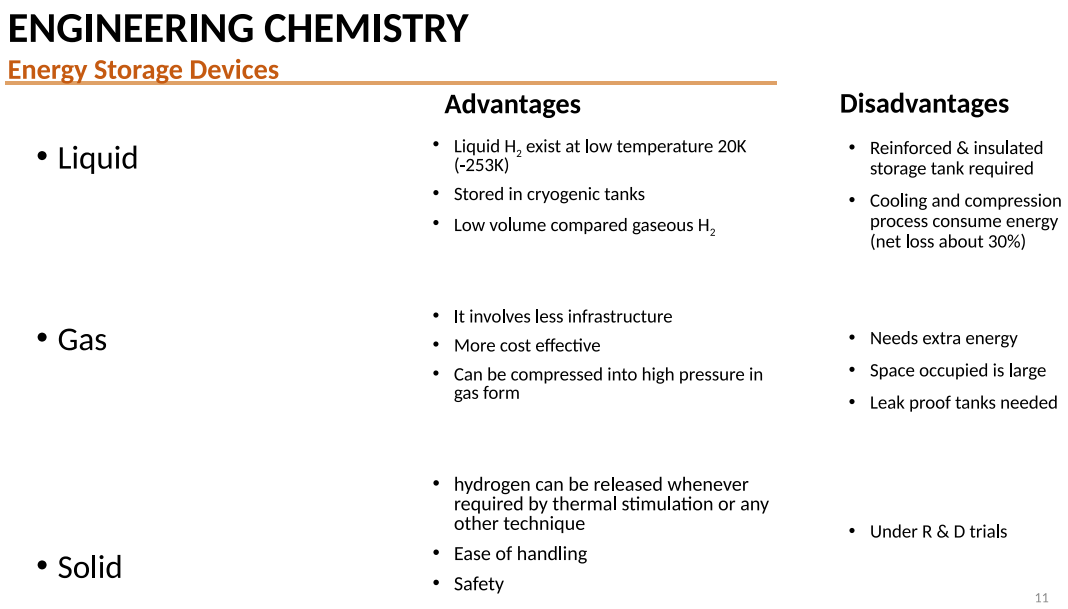


Figure G000113. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 11.

EV0000521

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000114. Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides · source slide page 12.

EV0000524

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000490 · EV0000491 · EV0000492
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Hydrogen Storage Class 35 content: Hydrogen storage methods 2 EV0000493 · EV0000494 · EV0000495
3. Source slide page 3 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Hydrogen Energy The Hydrogen storage is crucial for effective utilization of Hydrogen for its versatile applications. Hydrogen can be stored as Liquid Gas Solid [Image text, OCR] Liquefied Hydrogen Compressed Hydrogen Cryocompressed Hydrogen Liquid Organic Hydrogen Carrier PhysicallyAdsorbed Metallic Hydride Complex Hydride Hydrogen Physical and material-based methods of storage of Hydrogen 3 3 EV0000496 · EV0000497 · EV0000498
4. Source slide page 4 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices [Image text, OCR] How is hydrogen stored? ased Physical-based Material-based Compressed Cold/Cryo Liquid H2 Gas Compressed Liquid Interstitial Complex Chemical Adsorbent organic hydride hydride hydrogen =H = AI = Na =H =H2accessible surface Hydrogen Storage in Metal-Organic Frameworks: A Review - research journal (gyanvihar.org) 4 [Speaker notes] BN –BORON NITRIDE EV0000499 · EV0000500 · EV0000501

5. Source slide page 5 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Storage of hydrogen in the form of gas usually requires high-pressure tanks (350–700 bar tank pressure) Storage of hydrogen as a liquid requires cryogenic temperatures because the boiling point of hydrogen is −252.8°C at an atmospheric pressure Hydrogen can also be stored on the surfaces of solids (adsorption) or within solids (absorption) 5 5 EV0000502 · EV0000503 · EV0000504
6. Source slide page 6 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Solid state hydrogen storage: Depending on the type of hydrogen being stored in host materials, solid-state hydrogen storage can be classified into two categories: Absorption of hydrogen atoms into the material Adsorption of Hydrogen molecules onto the surface of adsorbent 6 EV0000505 · EV0000506 · EV0000507
7. Source slide page 7 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Some of the most prominent materials for Hydrogen Storage are Metal organic framework (MOF) Liquid organic hydrogen carriers (LOHC) Interstitial hydrides 7 7 EV0000508 · EV0000509 · EV0000510
8. Source slide page 8 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Metal Organic Frame as Adsorbent: (MOF) Representing a novel class of porous materials Enables high density energy storage of clean fuel gas in MOF adsorbents Zn-MOF (MOF-5) remains an attractive hydrogen storage material 8 EV0000511 · EV0000512 · EV0000513
9. Source slide page 9 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Liquid organic hydrogen carriers (LOHC) LOHC technology has shown great potential for efficient and stable hydrogen storage and transport This technology allows for safe and economical large-scale transoceanic transportation and long-cycle hydrogen storage LOHCs can be used as storage media for hydrogen Organic compounds that can absorb and release hydrogen through chemical reactions Methyl cyclopentane Dibenzyl toluene [Image text, OCR] LOHCstechnology Critical elements 8H Catalyst H,-lean H,-rich Reactor LOHC LOHC Application Biomass-based LOHCs Main systems Introduce H Conversion CHs ofbiomass CH Directionally Aromatic N-doped regulate compounds compounds Basic structure 1-s2.0-S2588913323000248-ga1_lrg.jpg (1161×886) (els-cdn.com) 9 EV0000514 · EV0000515 · EV0000516
10. Source slide page 10 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Interstitial hydride Contain a larger quantity of hydrogen than the same volume of liquid hydrogen Have advantage in the amount of hydrogen on a weight basis Certain interstitial hydrides are very suitable for hydrogen storage and transportation Example: LaNi5H6 Solid Hydrogen Storage Materials: Interstitial Hydrides | SpringerLink [Image text, OCR] absorption unhydrided desorption hydrided 10 Nanomaterials for Hydrogen Storage Applications: A Review (researchgate.net) EV0000517 · EV0000518 · EV0000519
11. Source slide page 11 (extracted text): ENGINEERING CHEMISTRY Energy Storage Devices Disadvantages Advantages Liquid H2 exist at low temperature 20K (-253K) Stored in cryogenic tanks Low volume compared gaseous H2 It involves less infrastructure More cost effective Can be compressed into high pressure in gas form hydrogen can be released whenever required by thermal stimulation or any other technique Ease of handling Safety Reinforced & insulated storage tank required Cooling and compression process consume energy (net loss about 30%) Needs extra energy Space occupied is large Leak proof tanks needed Under R & D trials Liquid Gas Solid 11 EV0000520 · EV0000521 · EV0000522

12. Source slide page 12 (extracted text): THANK YOU Department of Science and Humanities
EV0000523 · EV0000524 · EV0000525

SHORT-NOTE RECAP

Source deck preserved: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000490

15 Source slide deck: Experiential learning Based on Hydrogen storage

This entry preserves the complete slide deck titled Experiential learning Based on Hydrogen storage as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Chemistry, Unit_03, to avoid an unsupported match. EV0000637 · EV0000638

Graphs and diagrams

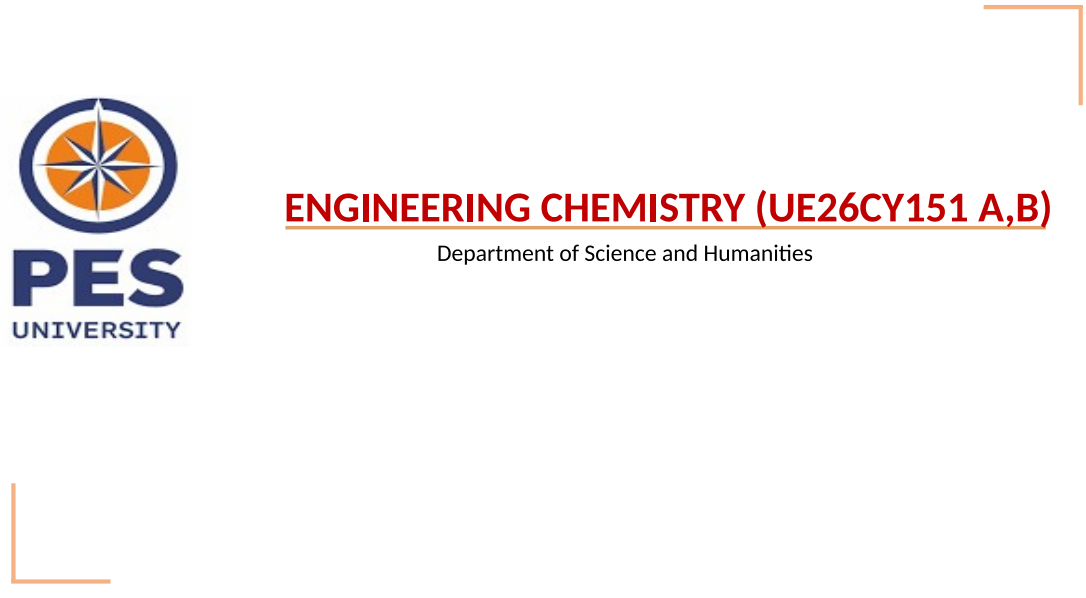


Figure G000152. Experiential learning Based on Hydrogen storage · source slide page 1.

EV0000638

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000153. Experiential learning Based on Hydrogen storage · source slide page 2.

EV0000641

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000154. Experiential learning Based on Hydrogen storage · source slide page 3.

EV0000644

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): ENGINEERING CHEMISTRY (UE26CY151 A,B) Department of Science and Humanities EV0000637 · EV0000638 · EV0000639
2. Source slide page 2 (extracted text): ENGINEERING CHEMISTRY Experiential Learning – UNIT 3 Term paper (Review article) - Hydrogen storage methods EV0000640 · EV0000641 · EV0000642
3. Source slide page 3 (extracted text): THANK YOU Department of Science and Humanities EV0000643 · EV0000644 · EV0000645

SHORT-NOTE RECAP

Source deck preserved: Experiential learning Based on Hydrogen storage. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is candidate-based; this index does not claim a manual content review.

EV0000637

16 Historical question-source pages · unit mapping candidates

This section preserves historical paper/question-bank pages whose extracted terms matched Engineering Chemistry, Unit_03 (Energy storage devices, hydrogen energy, and sensors) above the dataset's candidate threshold. The association has not been manually confirmed, source text/OCR may be incomplete, and the pages are not rewritten or presented as solved questions. EV0000646 · EV0000647

Important points and why

1. Archive page 50 from Chem_Extra_Topics__1-HfOeZNwo.pdf; unit match candidate score 0.5833; source text/OCR may be partial. OCR/sparse page: visual confirmation is required. Principle Features and Content e) 1. Prevent Waste The primary goal is to synthesize the target product with little to no by- products or waste. Waste treatment adds to costs. 2. Atom Economy Maximize the incorporation of reactants into the final product. Aim for 100% atom economy using addition/rearrangement reactions. 3. Safer Synthesis Minimize or avoid hazardous substances in synthesis. Example: Use CO, instead of toxic phosgene in polyurethane synthesis. 4. Safe Products Products should be effective but with reduced toxicity. Minimize harmful side effects on users and the environment. 5. Design Benign Chemicals Use safer solvents/reaction conditions. Avoid harmful solvents like chloroform; prefer water, CO,, ionic liquids, or solvent-free systems. 6. Design for Energy Efficiency Reduce energy intensity by using catalysts (e.g., bio-catalysts) and modern methods like microwave or ultrasound irradiation. 7. Use of Renewable Feedstock Prefer renewable raw materials like biomass over non-renewable sources such as coal and petroleum. 8. Reduce Derivatives Avoid unnecessary derivatization steps (e.g., blocking/protecting groups) to minimize waste and enhance atom efficiency. 9. Catalysis Use catalysts to accelerate reactions, reduce energy use, and achieve high yields with recyclable, selective reactions. 10. Design for Degradation Products should degrade into harmless substances after use to prevent environmental accumulation (e.g., pesticides, polymers). 11. Pollution Prevention Employ real-time monitoring and control of processes to prevent hazardous substances. Recycle unreacted chemicals. 12. Accident Prevention Choose chemicals and processes to minimize risks of accidents like fires, explosions, and toxic releases. Equip facilities with safety measures. EV0000646
2. Archive page 4 from 10__1vFS6E1bKV.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (5.0 Marks) (4.0 Marks) (5.0 Marks) (4.0 Marks) 3.a. Calculate the capacity (Ah), energy density (Wh/Kg) and electricity storage density (Ah/Kg) of Zn-air battery if 3.6g of Zn is stored in the battery weighing 75g. Given data: Voltage available from the battery = 1.39V and gram molar mass of Zn = 65.38 --(5M) 3.b. For Lithium-ion battery, provide the following details: (4M) Anode: Cathode: Reaction during Charging: Any two revalant disvantages: 3.c. For H2-O2 alkaline fuel cell, provide the following details: (5M) Anode: Cathode: Any two relevant advantahes: Why fuels feed should be free form CO ? 3.d. i. Working of Glucose Biosensor involves "Enzyme Glucose oxidase(GOx) catalyze the oxidation of glucose by molecular oxygen producing glucolactone and hydrogen peroxide" briefly explain using the reactions involved (2M) ii. Calculate the efficiency of the H -O fuel cell, if its cell potential is 1.235V (Given the enthalpy of formation of water is -285.83kj/mole, n=2). (2M) 2 2 2 EV0000647

3. Archive page 3 from 11__1zfeA5C-J7.pdf; unit match candidate score 0.8333; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 2.d. Give any two factors which affect conductance of an electrolyte. 3.a. (i) Why should a polymer electrolyte membrance fuel cell be operated in the temperature range of 60 - 90 C ? (2 marks) (ii) Why can't carbon containing fuels be used in an alkaline fuel cell? (2 marks) (iii) Why is it that expensive catalyst need not be used in a solid oxide fuel cell? (2 marks) 3.b. (i) Calculate the energy density in Wh/Kg and power density in W/Kg for a battery which weighs 3 Kg. The battery has 120g of Zn as anode and discharges a constant current for 1800 minutes. The voltage output of the battery is 1.4V, the number of electrons involved in the redox reaction is 2 and atomic mass of ziinc is 65 and Faraday's constant is 96500 C/mol. (4 marks) (ii) Calculate the efficiency of H -O alkaline fuel cell, Given: E =1.20 V , no. of electrons involved in the redox reaction is 2 and enthalpy change accompanying the reaction ΔH = -285.8 kJ/mole. (2marks) 3.c. (i) How does an amperometric glucose biosensor work and give us the the concentration of glucose in the blood? Explain with relevant reactions. (No diagram required). 4 marks (ii) Give two points of difference between a battery and a supercapacitor. 2 marks 0 2 2 oCell EV0000648
4. Archive page 4 from 1__1rBrYYqw3s.pdf; unit match candidate score 1.3333; source text/OCR may be partial. (6.0 Marks) (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.b. i) With the help of neat labeled diagram explain the construction and working of H -O alkaline fuel cell. ii) Calculate the potential of the H -O alkaline fuel cell if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole, F=96,500 c/mole.) 3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b) Li-ion battery (c) Supercapacitor ii) Mention any two materials used for hydrogen storage. 3.d. i) Discuss the working principle of glucose biosensor with relevant reactions. ii) Mention any one limitation of glucose sensor. 3.e. 50 cm of given hard water sample requires 15.6 cm of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. 2 2 2 2 3 3 EV0000649
5. Archive page 3 from 2__11pj9jsL0V.pdf; unit match candidate score 1.25; source text/OCR may be partial. (6.0 Marks) (5.0 Marks) (2.0 Marks) i) Name the components of the H -O alkaline fuel cell. Calculate the efficiency of fuel the cell if ΔH = -285.5 kJ/mol and E = 1.21 V. ii) Draw a Ragone plot and indicate the position of the following energy conversion/storage devices: fuel cells, batteries, and supercapacitors. (7.0 Marks) 2.c. What is the galvanization? Mention the steps involved in the galvanization process and its advantages. 2.d. For the electrochemical phenomenon of corrosion. i). Write two reactions at the cathode surface in the presence of O ii). Write the formation of corrosion product if iron is exposed to ambient conditions. (Give reactions) Justify the following, i) Sodium meta arsenite is a cathodic inhibitor. 2.e. What are the factors affecting the conductance of an electrolyte? 3.a. 2. 2 2 formation (H2O) ocell EV0000650
6. Archive page 4 from 2__11pj9jsL0V.pdf; unit match candidate score 1.0833; source text/OCR may be partial. (5.0 Marks) (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.b. For a Zinc-air battery : i) Draw a neat labeled diagram ii) Write the reactions taking place at the anode and cathode ii) Why do zinc-air batteries have high energy density? 3.c. What is the sensor? Mention the FIVE types of sensors with suitable examples. 3.d. Fill in the following table for the alkaline electrolysis process. (Hydrogen storage and production) Charge carrier: Anodic materials:

   Cathodic materials: Electrolyte materials Catalyst materials: 3.e. What is intrinsic viscosity ? How is it related to the molecular mass of a polymer? EV0000651

7. Archive page 4 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 4/7 (3.0 Marks) (7.0 Marks) (7.0 Marks) 2.e. Discuss, with the help of Henderson-Hasselbalch equation, under which situation pH and pKa become equal 3.a. (a) Discuss the construction and working principles of Zn-air battery with a labelled diagram and the relevant chemical reactions (5M) (b) Calculate the electricity storage density (Ah/kg) of a Lithium battery which has 2.0 g of lithium as anodic material. The total weight of the battery is 65 g. (Given : Atomic mass of lithium is 7) (2M) 3.b. Discuss the construction and working principles of a H -O alkaline fuel cell with a clearly labelled diagram and the chemical reactions. Mention two disadvantages of H -O alkaline fuel cell (5M + 2M) 2 2 2 2 EV0000652
8. Archive page 5 from 3__1MwY1CTl_j.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 3/25/25, 3:51 PM ESA - Dec 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 5/7 (4.0 Marks) (4.0 Marks) (3.0 Marks) 3.c. (a) If the E for H -O alkaline fuel cell is 1.23 V and efficiency is 83%, calculate the heat evolved (in kilo Joules) during the reaction (2M) (b) Why do supercapacitors have an enormous capacitance? (2M) 3.d. Discuss the construction and working principles of the oxygen sensor placed in the automobile exhaust stream with labelled diagram and chemical reactions 3.e. (a) Define viscosity coefficient of a liquid (1M) (b) Write Poiseuille's equation and explain all the terms (2M) oCell 2 2 EV0000653
9. Archive page 4 from 4__1eKMFaJ2lx.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 4/7 (2.0 Marks) (6.0 Marks) (6.0 Marks) 2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method. ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? 3.a. i) what are reserve batteries ? Give an example. ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: F = 96500 C/mol, atomic mass of lithium = 7) 3.b. i) With the help of neat labeled diagram explain the construction and working of H -O alkaline fuel cell. ii) Calculate the potential of the H -O alkaline fuel cell if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole, F=96,500 c/mole.) 2 2 2 2 EV0000654
10. Archive page 5 from 4__1eKMFaJ2lx.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 5/7 (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b) Li-ion battery (c) Supercapacitor ii) Mention any two materials used for hydrogen storage. 3.d. i) Discuss the working principle of glucose biosensor with relevant reactions. ii) Mention any one limitation of glucose sensor. 3.e. 50 cm of given hard water sample requires 15.6 cm of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. 3 3 EV0000655

11. Archive page 4 from 5__1pIeTzocnU.pdf; unit match candidate score 0.9167; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 4/7 (2.0 Marks) (6.0 Marks) (6.0 Marks) 2.e. What are the two factors on which the absorbance of light of a particular wavelength by a substance in solution varies with? 3.a. i) Give any two reasons why secondary batteries have limited cycle life. ii) A battery using 1.7 g of lithium as anodic material gives a steady voltage of 3.6 V and lasts for 2 hours. If the weight of the battery is 50 g, then calculate the power density and energy density of the battery. [Given: Atomic mass of lithium is 7, F= 96500 C/mol] 3.b. i) If the E Cell for H -O alkaline fuel cell is 1.23 V the heat evolved during the reaction ΔH = -285.8 kJ/mole, calculate the efficiency of the fuel cell. (ii) Why can't carbon-containing fuels be used in an alkaline fuel cell? (iii) Porous carbon materials are used as electrode materials in supercapacitors. Give reason. EV0000656
12. Archive page 5 from 5__1pIeTzocnU.pdf; unit match candidate score 0.6667; source text/OCR may be partial. 11/11/24, 11:34 AM ESA -JAN 2024 - UE23CY151A (set- 2) about:blank 5/7 (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.c. i) What is a biosensor? Mention the components required for the construction of glucose sensors. ii) Give any two points of difference between a potentiometric and amperometric sensor. 3.d. Mention the five different types of materials used for hydrogen storage. 3.e. Give any two advantages of the bottom-up solution combustion synthesis method for the synthesis of nanomaterials. EV0000657
13. Archive page 4 from 6__1H6QH_PVNf.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 4/7 (2.0 Marks) (6.0 Marks) (6.0 Marks) 2.e. (i) Mention the role of sulphuric acid in the estimation of ferrous iron by potentiometric titration method. ii) Why pH is equals to pKa at half equivalence point in the pH metric titration of weak acid vs strong base ? 3.a. i) what are reserve batteries ? Give an example. ii) Calculate the capacity (Ah), energy density (Wh/kg) and power density (W/kg) of a battery if 2.4 g of lithium is stored in the battery and the battery lasts for 120 mins. The battery weighs 115 g and gives steady voltage of 3.4 V. (Given: F = 96500 C/mol, atomic mass of lithium = 7) 3.b. i) With the help of neat labeled diagram explain the construction and working of H -O alkaline fuel cell. ii) Calculate the potential of the H -O alkaline fuel cell if its efficiency is 84.25 %. (Given the enthalpy of formation of water is -285.83 kJ/mole, F=96,500 c/mole.) 2 2 2 2 EV0000658
14. Archive page 5 from 6__1H6QH_PVNf.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/27/25, 11:47 AM ESA June 2024 - UE23CY151B/UE23CY151A (set- 1) about:blank 5/7 (6.0 Marks) (5.0 Marks) (2.0 Marks) 3.c. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b) Li-ion battery (c) Supercapacitor ii) Mention any two materials used for hydrogen storage. 3.d. i) Discuss the working principle of glucose biosensor with relevant reactions. ii) Mention any one limitation of glucose sensor. 3.e. 50 cm of given hard water sample requires 15.6 cm of 0.075 M EDTA for complete reaction in the estimation of hardness by complexometry. Calculate the total hardness of water. 3 3 EV0000659

15. Archive page 3 from 7__16pEjcSTVO.pdf; unit match candidate score 0.75; source text/OCR may be partial. (2.0 Marks) (6.0 Marks) (6.0 Marks) (6.0 Marks) 2.d. Give two factors that affect the conductance of an electrolyte. 3.a. For a zinc - air battery- (i) Give its anodic, cathodic and net cell reaction (ii) Give the reason for its high energy density (iii) What would happen if carbondioxide enters the cell (iv) Give its one important application 3.b. (i) Calculate the efficiency of H -O alkaline fuel cell Given: E =1.20 V, Enthaphy of formation of water ΔH = -285.8 kJ/mole (ii) Why is the capacitance of a supercapacitor much higher than that of a normal capacitor? 3.c. (i) How does an oxygen sensor respond to a lean mixture? Why should the stiochiometric air to fuel ratio be maintained at 14.7:1? (ii) Calculate the capacity of a lithium battery which stores 2.0 g of lithium, Given : Atomic mass of lithium is 7 and number of electrons involved in the redox reaction is 1 and Faraday's constant is 96500C/mol. 2 2 Cell EV0000660

16. Archive page 4 from 8__1015pA2nzY.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/25/25, 4:03 PM ESA - Dec 2024 - UE22CY151B/UE22CY151A (set- 1) about:blank 4/7 (4.0 Marks) (5.0 Marks) (4.0 Marks) 3.b. What are supercapacitors? Write one advantage, disadvantage and application of supercapacitors 3.c. i) Discuss the working principle of glucose biosensor with relevent reactions. ii) Mention any one limitation of glucose sensor. 3.d. i) Mark the regions in which the following energy storage devices appear in the Ragone plot. a) Fuel cell (b)Li-ion battery (c) Supercapacitor EV0000661

17. Archive page 4 from Chem-EC[2025]__1ZtWpoG9Dv.pdf; unit match candidate score 1.0833; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 2) about:blank 4/6 (6.0 Marks) (6.0 Marks) (7.0 Marks) 3.b. (i) Give the Nernst equation corresponding to the output voltage generated in an 'Oxygen Sensor' and explain the terms. What are the inferences one can draw from the output voltage values and how is it useful. (4 marks) (ii) Why is Polymer electrolyte membrane fuel cell operated in the temperature range of 60 -90 C. (2 marks) 3.c. (i) Give the synthesis of hydrogen using the steam reforming process with methane as the starting material. (Mention the conditions of temperature, pressure and catalyst used in the reaction). (4 marks) (ii) How are 'Liquid organic hydrogen carriers (LOHC)' used in storage and transport of hydrogen gas. Give one example of LOHC. (2 marks) 3.d. (i) Of all the energy storage / conversion devices discussed, give a suitable one for the following applications- (4 marks) Hearing aids Mobiles and laptops Space exploration In hybrid cars for start-stop application (ii) Calculate the viscosity average molecular weight of polystyrene (atactic) from the following data: Intrinsic Viscosity =0.73, K = 11.5x10−5 and Mark Houwink parameter (a) is 0.73. (3 marks) 0 EV0000662

18. Archive page 4 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 4/7 (3.0 Marks) (7.0 Marks) (6.0 Marks) 2.e. i) Why colorimetric estimation of copper is done at 620 nm? (1M) ii) Draw the nature of the graphs in the determination of pKa value of weak acid. (2M) 3.a. For a Zn-air battery i) What are the materials used as anode, cathode, electrolyte and separator? (2M) ii) Write the reactions taking place at anode and cathode when the battery is discharging. (2M) iii) Define energy density of a battery and calculate energy density of zinc-air battery in Wh/kg. If 6.6g of Zn is stored in the battery weighing 36.8g. (Given; voltage of the given Zinc-air battery is 1.3 V and gram molar mass of zinc is 65.38) (3M) 3.b. i) Explain the construction and working of H -O alkaline fuel cell. (4M) ii) Calculate the efficiency of the H -O alkaline fuel cell. (Given; the enthalpy of formation of water is -285.83 kj/mole, E =1.19V). (2M) 2 2 2 2 cell EV0000663

19. Archive page 5 from Chem-RR[2025]__1xN30tqzNj.pdf; unit match candidate score 0.8333; source text/OCR may be partial. 3/25/25, 10:57 AM ESA - Jan 2025 - UE24CY151A (set- 1) about:blank 5/7 (4.0 Marks) (6.0 Marks) (2.0 Marks) 3.c. i) Mention the advantages and Limitations of Steam reforming method of hydrogen production. (2M) ii) "Liquid organic hydrogen carriers (LOHC) can be used as storage media for hydrogen" Justify the statement with example. (2M) 3.d. What is an oxygen sensor? Give the construction, working of oxygen sensor and explain its role in automobiles. 3.e. Calculate the viscosity average molecular weight of a polymer solution by the following data. K=45.3x10 , a=0.64, intrinsic viscosity=0.75 -5 EV0000664

SHORT-NOTE RECAP

19 historical archive pages are included as question-source references. Inspect the original paper before relying on OCR or unit placement; no answers are newly authored here.

EV0000646

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9979; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part I) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__46f678f584cf1e30.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9980; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Batteries: Components and working of a battery, classification of batteries – primary, secondary, and reserve batteries (Part II)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__fdb0b572df73e22f.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Numericals_Energy storage devices transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/028_Numericals_Energy_storage_devices__e2ecdc69f7d2645d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Zinc -air battery transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.
Zinc -air battery transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/029_Zinc_-air_battery__50608371ec91d5a3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9976; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_1__b31208265c8084c7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9972; only the JSON segmentation supplies the displayed timestamps.
Fuel cells: Principle and working, H2-O2 alkaline fuel cell (Part 2) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell_Part_2__7423df6b25117f2d.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9968; only the JSON segmentation supplies the displayed timestamps.
CH3OH-O2 polymer electrolyte fuel cell, efficiency of fuel cells, numerical problems based on efficiency of fuel cells transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__0efd131a5b8eae34.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Supercapacitors , Ragone plot transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9994; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Supercapacitors , Ragone plot transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/033_Supercapacitors_Ragone_plot__d04b49e0a94293ee.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Types of electrochemical sensors, pH sensor – ion-selective electrode - glass electrode (Part 2)) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/036_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_-_glass_electrode_Part_2__070afd1335845586.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9973; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Slide page text: Batteries Components and working of a battery classification of batteries primary secondary and reserve batter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Batteries Components and working of a battery classification of batteries primary secondary and reserve batter	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Batteries Components and working of a battery classification of batteries primary secondary and reserve batter (converted PDF; original hash d9c42f3a686ef40d0ad83248232738371df224c923330cc30c846a0fe086e2ba)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Battery characteristics Numericals based on battery characteristics	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Battery characteristics Numericals based on battery characteristics	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Battery characteristics Numericals based on battery characteristics (converted PDF; original hash d969bdbbc833f7c51be9da5bbb4bfe7b42a2df103a0bab422c5e056b6f825668)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Modern batteries Zn air battery Li metal batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Modern batteries Zn air battery Li metal batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Modern batteries Zn air battery Li metal batteries (converted PDF; original hash bdc805e3e05fab8565464d114d48cf4c1b7038bda5f4a1e5861ff359c1b7d021)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Li ion battery LFP batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Li ion battery LFP batteries	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Li ion battery LFP batteries (converted PDF; original hash e7d3c3ad6b1affd5f4b6b070ba5abc52a531d271f5ffaa87ba3cc92ac0bc6425)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Fuel cells Principle and working H2 O2 alkaline fuel cell	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Fuel cells Principle and working H2 O2 alkaline fuel cell	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Fuel cells Principle and working H2 O2 alkaline fuel cell (converted PDF; original hash bf5f43fd3bc7b37ccd89e3b36ad6afdb975dd6a975cc6895f1df8efd31603657)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for CH3OH O2 polymer electrolyte fuel cell efficiency of fuel cells numerical problems based on efficiency of fuel (converted PDF; original hash 6248c89e8ebd35030a04cc8e9da52a8570ca4eca0fa27c7945a1729440651644)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Supercapacitors construction and working Ragone plot	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Supercapacitors construction and working Ragone plot	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Supercapacitors construction and working Ragone plot (converted PDF; original hash 91f6dd1e8f435dea9f720bbbc2eb56a2bfa46fd1a26c6dd5229b080cf303d6d3)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Hydrogen production and storage Hydrogen production methods Gasification steam reforming photolysis electrolys (converted PDF; original hash bab821775eabfe2b59e29b4273d5a699b07265c9a08ef00d7ffe97ec6df36ef3)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Hydrogen storage methods physical and material based MOFs LOHC Interstitial hydrides (converted PDF; original hash 52d83559f0c2af524e16a30cc9ba96ba11682dcdd8566e52ead1c28dfa273bc6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Sensors Types of electrochemical sensors pH sensor ion selective electrode glass electrode (converted PDF; original hash 91c0f334bd8f387fe672ecac12ac8855df52bc5c66dcb7560d2ce9d78698005e)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Gas sensor O2 sensor in automobile exhaust stream Biosensors Blood glucose sensor (converted PDF; original hash 7f4ee6571c560e964fab20c01797fd6cf48c0ce32ee23b3a61dce064c5c46476)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Experiential learning Based on Hydrogen storage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Experiential learning Based on Hydrogen storage	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Experiential learning Based on Hydrogen storage (converted PDF; original hash 67f035536e4cb2e700cb78ee432824ffb622d60f3e7a2a975fd8d831efdbcf55)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Extracted historical pages: Chem_Extra_Topics__1-HfOeZNwo.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Extra/Chem_Extra_Topics__1-HfOeZNwo.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 10__1vFS6E1bKV.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/10__1vFS6E1bKV.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 11__1zfeA5C-J7.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/11__1zfeA5C-J7.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: 1__1rBrYYqw3s.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/1__1rBrYYqw3s.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 2__11pj9jsL0V.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/2__11pj9jsL0V.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 3__1MwY1CTl_j.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/3__1MwY1CTl_j.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 4__1eKMFaJ2lx.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/4__1eKMFaJ2lx.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 5__1pIeTzocnU.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/5__1pIeTzocnU.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 6__1H6QH_PVNf.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/6__1H6QH_PVNf.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 7__16pEjcSTVO.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/7__16pEjcSTVO.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: 8__1015pA2nzY.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2024/8__1015pA2nzY.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.
Extracted historical pages: Chem-EC[2025]__1ZtWpoG9Dv.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-EC[2025]__1ZtWpoG9Dv.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Source	Path	Use
Extracted historical pages: Chem-RR[2025]__1xN30tqzNj.pdf	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/First_Year_8enterprise/Files/Chemistry/Papers/ESA Papers/2025/Chem-RR[2025]__1xN30tqzNj.pdf	Candidate-matched page text excerpt; original PDF bytes, hash and path remain in datasets/question-archive.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.
EV0000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0099. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0099. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0099. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0099. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000022	SRC00002/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0112. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000023	SRC00003/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0112. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00004/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0112. Sidecar variants compare at similarity 0.9979; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000025	SRC00005/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0100 to s0112. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000030	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000032	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000037	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000038	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000039	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000040	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000041	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0082. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000042	SRC00006/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000043	SRC00007/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000044	SRC00008/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0100. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000045	SRC00009/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0083 to s0100. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000046	SRC00006/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0115. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000047	SRC00007/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0115. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000048	SRC00008/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0115. Sidecar variants compare at similarity 0.9980; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000049	SRC00009/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0101 to s0115. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000050	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000051	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000052	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000053	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000054	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000055	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000056	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000057	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000058	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000059	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000060	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000061	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000062	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000063	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000064	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000065	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00010/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000067	SRC00011/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000068	SRC00012/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0084. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000069	SRC00013/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0084. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000070	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000071	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000072	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000073	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000074	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000075	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000076	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000077	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0044. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000078	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000079	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000081	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0045 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00014/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00015/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00016/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0079. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00017/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000086	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000088	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000092	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000093	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000094	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000095	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0062. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000100	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0074. Sidecar variants compare at similarity 0.9976; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000101	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0063 to s0074. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 02	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 03	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 04	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9972; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 05	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 06	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 07	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 08	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 0.9972; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 09	SRC00025/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 10	SRC00022/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 11	SRC00023/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 12	SRC00024/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 0.9972; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 13	SRC00025/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 14	SRC00022/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 15	SRC00023/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000116	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0065. Sidecar variants compare at similarity 0.9972; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000117	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0065. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000118	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000119	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000120	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9968; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000121	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000122	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000123	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000124	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 0.9968; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000125	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000126	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000127	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000128	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 0.9968; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000129	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001130	SRC00026/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001131	SRC00027/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001132	SRC00028/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 0.9968; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001133	SRC00029/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001134	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001135	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001136	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 37	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 38	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 39	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 40	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 41	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 42	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 43	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 44	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 45	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 46	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 47	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 48	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 0.9994; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 49	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0078. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 50	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000151	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000152	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000153	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000154	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000155	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000156	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0023 to s0043. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000158	SRC00034/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00035/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00036/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00037/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0044 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00034/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000163	SRC00035/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00036/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0085. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 65	SRC00037/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0065 to s0085. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 66	SRC00034/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0093. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 67	SRC00035/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0093. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 68	SRC00036/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0093. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 69	SRC00037/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0086 to s0093. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 70	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 71	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000172	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9973; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000175	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000176	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 0.9973; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000177	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 79	SRC00039/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 80	SRC00040/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 0.9973; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 81	SRC00041/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 82	SRC00038/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 83	SRC00039/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 84	SRC00040/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 0.9973; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 85	SRC00041/t00090 0-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000186	SRC00042/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000187	SRC00044/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000188	SRC00043/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000189	SRC00042/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000190	SRC00044/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000191	SRC00043/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000192	SRC00042/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000193	SRC00044/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000194	SRC00043/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000195	SRC00042/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000196	SRC00044/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000197	SRC00043/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00001 98	SRC00042/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00001 99	SRC00044/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 00	SRC00043/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 01	SRC00042/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 02	SRC00044/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 03	SRC00043/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000204	SRC00042/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000205	SRC00044/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000206	SRC00043/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000207	SRC00042/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000208	SRC00044/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000209	SRC00043/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000210	SRC00042/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000211	SRC00044/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000212	SRC00043/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000213	SRC00042/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000214	SRC00044/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/027_Batteries_Components_and_working_of_a_battery_classification_of_batteries_primary_secondary_and_reserve_batter__a7dbe956e2f9e64b.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000215	SRC00043/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000216	SRC00045/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000217	SRC00047/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000218	SRC00046/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000219	SRC00045/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000220	SRC00047/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000221	SRC00046/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000222	SRC00045/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000223	SRC00047/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000224	SRC00046/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000225	SRC00045/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000226	SRC00047/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000227	SRC00046/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000228	SRC00045/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000229	SRC00047/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000230	SRC00046/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000231	SRC00045/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000232	SRC00047/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000233	SRC00046/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000234	SRC00045/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000235	SRC00047/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000236	SRC00046/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000237	SRC00045/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000238	SRC00047/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000239	SRC00046/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000240	SRC00045/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000241	SRC00047/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000242	SRC00046/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000243	SRC00045/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000244	SRC00047/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000245	SRC00046/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000246	SRC00045/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000247	SRC00047/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000248	SRC00046/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000249	SRC00045/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000250	SRC00047/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000251	SRC00046/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000252	SRC00045/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000253	SRC00047/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000254	SRC00046/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000255	SRC00045/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000256	SRC00047/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/028_Battery_characteristics_Numericals_based_on_battery_characteristics__68997fc498ccec08.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000257	SRC00046/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000258	SRC00048/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000259	SRC00050/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000260	SRC00049/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000261	SRC00048/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000262	SRC00050/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000263	SRC00049/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000264	SRC00048/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000265	SRC00050/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000266	SRC00049/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000267	SRC00048/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000268	SRC00050/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000269	SRC00049/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 70	SRC00048/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 71	SRC00050/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 72	SRC00049/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 73	SRC00048/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 74	SRC00050/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 75	SRC00049/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 76	SRC00048/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 77	SRC00050/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 78	SRC00049/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 79	SRC00048/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 80	SRC00050/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 81	SRC00049/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000282	SRC00048/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000283	SRC00050/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000284	SRC00049/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000285	SRC00048/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000286	SRC00050/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000287	SRC00049/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000288	SRC00048/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000289	SRC00050/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000290	SRC00049/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000291	SRC00048/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000292	SRC00050/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/029_Modern_batteries_-_Zn-air_battery_Li_metal_batteries__d6291cfbaed760d2.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000293	SRC00049/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000294	SRC00051/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000295	SRC00053/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000296	SRC00052/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000297	SRC00051/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000298	SRC00053/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000299	SRC00052/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000300	SRC00051/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000301	SRC00053/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000302	SRC00052/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000303	SRC00051/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000304	SRC00053/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000305	SRC00052/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000306	SRC00051/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000307	SRC00053/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000308	SRC00052/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000309	SRC00051/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000310	SRC00053/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000311	SRC00052/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000312	SRC00051/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000313	SRC00053/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00052/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000315	SRC00051/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000316	SRC00053/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000317	SRC00052/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000318	SRC00051/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000319	SRC00053/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000320	SRC00052/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000321	SRC00051/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000322	SRC00053/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/030_Li-ion_battery_LFP_batteries__6f51933ca02c511c.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000323	SRC00052/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003324	SRC00054/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003325	SRC00056/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003326	SRC00055/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003327	SRC00054/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003328	SRC00056/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003329	SRC00055/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003330	SRC00054/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003331	SRC00056/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003332	SRC00055/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003333	SRC00054/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003334	SRC00056/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003335	SRC00055/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 36	SRC00054/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 37	SRC00056/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 38	SRC00055/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 39	SRC00054/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 40	SRC00056/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 41	SRC00055/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000342	SRC00054/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000343	SRC00056/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000344	SRC00055/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000345	SRC00054/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000346	SRC00056/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000347	SRC00055/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003348	SRC00054/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003349	SRC00056/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003350	SRC00055/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00003351	SRC00054/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003352	SRC00056/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003353	SRC00055/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 54	SRC00054/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 55	SRC00056/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 56	SRC00055/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 57	SRC00054/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 58	SRC00056/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/031_Fuel_cells_Principle_and_working_H2-O2_alkaline_fuel_cell__4f9d70e7c5fe6ffc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 59	SRC00055/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 60	SRC00057/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 61	SRC00059/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 62	SRC00058/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 63	SRC00057/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 64	SRC00059/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV00003 65	SRC00058/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 66	SRC00057/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 67	SRC00059/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 68	SRC00058/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 69	SRC00057/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00003370	SRC00059/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003371	SRC00058/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003372	SRC00057/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003373	SRC00059/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003374	SRC00058/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003775	SRC00057/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003776	SRC00059/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003777	SRC00058/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003778	SRC00057/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003779	SRC00059/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000380	SRC00058/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000381	SRC00057/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000382	SRC00059/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000383	SRC00058/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000384	SRC00057/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000385	SRC00059/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000386	SRC00058/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000387	SRC00057/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000388	SRC00059/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000389	SRC00058/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003390	SRC00057/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003391	SRC00059/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003392	SRC00058/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00003393	SRC00057/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003394	SRC00059/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000395	SRC00058/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000396	SRC00057/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000397	SRC00059/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000398	SRC00058/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000399	SRC00057/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000400	SRC00059/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/032_CH3OH-O2_polymer_electrolyte_fuel_cell_efficiency_of_fuel_cells_numerical_problems_based_on_efficiency_of_fuel__6b09eb5e1d59f7e7.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000401	SRC00058/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000402	SRC00060/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000403	SRC00062/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000404	SRC00061/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000405	SRC00060/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000406	SRC00062/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000407	SRC00061/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000408	SRC00060/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000409	SRC00062/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000410	SRC00061/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000411	SRC00060/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000412	SRC00062/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000413	SRC00061/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000414	SRC00060/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000415	SRC00062/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000416	SRC00061/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000417	SRC00060/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000418	SRC00062/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00061/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00060/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000421	SRC00062/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000422	SRC00061/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000423	SRC00060/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000424	SRC00062/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000425	SRC00061/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000426	SRC00060/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000427	SRC00062/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00061/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000429	SRC00060/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000430	SRC00062/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000431	SRC00061/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000432	SRC00060/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000433	SRC00062/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00061/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000435	SRC00060/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000436	SRC00062/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00061/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00060/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000439	SRC00062/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00061/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000441	SRC00060/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000442	SRC00062/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00061/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000444	SRC00060/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00062/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00061/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000447	SRC00060/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000448	SRC00062/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/033_Supercapacitors_-_construction_and_working_Ragone_plot__4434b0781e5c877d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00061/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000450	SRC00063/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000451	SRC00065/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00064/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000453	SRC00063/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000454	SRC00065/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00064/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00063/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00065/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000458	SRC00064/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000459	SRC00063/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV00004060	SRC00065/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004061	SRC00064/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00004062	SRC00063/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004063	SRC00065/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV00004064	SRC00064/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00004065	SRC00063/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000466	SRC00065/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000467	SRC00064/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000468	SRC00063/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000469	SRC00065/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000470	SRC00064/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000471	SRC00063/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000472	SRC00065/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000473	SRC00064/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000474	SRC00063/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000475	SRC00065/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000476	SRC00064/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000477	SRC00063/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000478	SRC00065/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000479	SRC00064/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000480	SRC00063/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000481	SRC00065/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000482	SRC00064/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000483	SRC00063/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.

Evidence	Source unit	Status	Decision
EV0000484	SRC00065/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000485	SRC00064/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000486	SRC00063/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000487	SRC00065/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.
EV0000488	SRC00064/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000489	SRC00065/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.ppt; visual page included as vector figure. Human layout/content review is pending.

Evidence	Source unit	Status	Decision
EV0000490	SRC00066/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000491	SRC00068/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000492	SRC00067/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000493	SRC00066/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000494	SRC00068/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000495	SRC00067/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 96	SRC00066/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 97	SRC00068/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 98	SRC00067/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 99	SRC00066/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 00	SRC00068/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 01	SRC00067/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000502	SRC00066/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000503	SRC00068/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000504	SRC00067/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000505	SRC00066/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000506	SRC00068/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000507	SRC00067/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000508	SRC00066/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000509	SRC00068/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000510	SRC00067/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000511	SRC00066/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000512	SRC00068/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000513	SRC00067/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000514	SRC00066/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000515	SRC00068/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000516	SRC00067/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000517	SRC00066/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000518	SRC00068/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000519	SRC00067/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000520	SRC00066/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000521	SRC00068/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000522	SRC00067/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000523	SRC00066/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000524	SRC00068/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/035_Hydrogen_storage_methods_physical_and_material-based_MOFs_LOHC_Interstitial_hydrides__2f8f6a5c8a28f09d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000525	SRC00067/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000526	SRC00069/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000527	SRC00071/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000528	SRC00070/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000529	SRC00069/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000530	SRC00071/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000531	SRC00070/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000532	SRC00069/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000533	SRC00071/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000534	SRC00070/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000535	SRC00069/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000536	SRC00071/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000537	SRC00070/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000538	SRC00069/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000539	SRC00071/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000540	SRC00070/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000541	SRC00069/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000542	SRC00071/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000543	SRC00070/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000544	SRC00069/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000545	SRC00071/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000546	SRC00070/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000547	SRC00069/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000548	SRC00071/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000549	SRC00070/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 50	SRC00069/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 51	SRC00071/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 52	SRC00070/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 53	SRC00069/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 54	SRC00071/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 55	SRC00070/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000556	SRC00069/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000557	SRC00071/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000558	SRC00070/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000559	SRC00069/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000560	SRC00071/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000561	SRC00070/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000562	SRC00069/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000563	SRC00071/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000564	SRC00070/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000565	SRC00069/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000566	SRC00071/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000567	SRC00070/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000568	SRC00069/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000569	SRC00071/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000570	SRC00070/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000571	SRC00069/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000572	SRC00071/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000573	SRC00070/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000574	SRC00069/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000575	SRC00071/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000576	SRC00070/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000577	SRC00069/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000578	SRC00071/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000579	SRC00070/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000580	SRC00069/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000581	SRC00071/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000582	SRC00070/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV0000583	SRC00069/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000584	SRC00071/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/036_Sensors_Types_of_electrochemical_sensors_pH_sensor_ion-selective_electrode_glass_electrode__b69609c2ebcb9c89.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000585	SRC00070/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000586	SRC00072/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000587	SRC00074/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000588	SRC00073/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000589	SRC00072/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000590	SRC00074/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000591	SRC00073/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000592	SRC00072/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000593	SRC00074/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000594	SRC00073/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000595	SRC00072/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000596	SRC00074/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000597	SRC00073/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000598	SRC00072/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000599	SRC00074/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000600	SRC00073/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000601	SRC00072/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000602	SRC00074/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000603	SRC00073/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000604	SRC00072/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000605	SRC00074/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000606	SRC00073/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000607	SRC00072/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000608	SRC00074/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000609	SRC00073/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000610	SRC00072/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000611	SRC00074/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000612	SRC00073/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000613	SRC00072/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000614	SRC00074/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000615	SRC00073/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000616	SRC00072/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000617	SRC00074/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000618	SRC00073/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000619	SRC00072/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000620	SRC00074/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000621	SRC00073/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000622	SRC00072/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000623	SRC00074/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000624	SRC00073/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000625	SRC00072/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000626	SRC00074/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000627	SRC00073/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000628	SRC00072/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000629	SRC00074/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000630	SRC00073/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000631	SRC00072/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000632	SRC00074/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000633	SRC00073/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000634	SRC00072/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000635	SRC00074/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor__de570046a3f3911a.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000636	SRC00073/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000637	SRC00075/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000638	SRC00077/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000639	SRC00076/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000640	SRC00075/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000641	SRC00077/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000642	SRC00076/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000643	SRC00075/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000644	SRC00077/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/Slides/038_Experiential_learning-_Based_on_Hydrogen_storage__28061e60e4e1cbfc.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000645	SRC00076/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000646	SRC00078/q50	covered	Original archive PDF page 50; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV00006047	SRC00079/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006048	SRC00080/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV00006049	SRC00081/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006050	SRC00082/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV00006051	SRC00082/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006052	SRC00083/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006053	SRC00083/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV00006054	SRC00084/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006055	SRC00084/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV00006056	SRC00085/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006057	SRC00085/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV00006058	SRC00086/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006059	SRC00086/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.
EV00006060	SRC00087/q3	covered	Original archive PDF page 3; candidate unit association is text-based and unconfirmed.
EV00006061	SRC00088/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV00006062	SRC00089/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.

Evidence	Source unit	Status	Decision
EV0000663	SRC00090/q4	covered	Original archive PDF page 4; candidate unit association is text-based and unconfirmed.
EV0000664	SRC00090/q5	covered	Original archive PDF page 5; candidate unit association is text-based and unconfirmed.

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